

Airy amplitudes for relaxed trees, compacted trees and finite-language automata

Positive amplitudes and expansions to every fixed order,
binary and every fixed arity (OEIS A082161 and relatives)

A merged research report built from six AI-assisted manuscripts
(batch 77, manuscripts 62, 15, 18, 66, 29 and 48)
and three AI-assisted bundle reports (batch 102, Reports 100, 101 and 103; Part V)

Manuscripts dated 2 October 2026; merged 2 October 2026; Part V added 5 October 2026

Abstract

Relaxed trees, compacted trees and minimal deterministic automata accepting finite languages form one family of triangular recurrences indexed by the arity, or alphabet size, k . Their counts grow like $(n!)^{k-1}$ times an exponential, times $\exp(cn^{1/3})$ with c a multiple of the largest Airy zero $z = -2.338107410459767\dots$, times a power of n . For $k = 2$ the three counts are OEIS A082161, A254789 and A331120; for $k = 3$ the relaxed counts are A082162, for $k = 4$ A102102. The order of growth was known (Elvey Price, Fang and Wallner; Ghosh Dastidar and Wallner); the existence of a limiting amplitude was open. This report prints six manuscripts as four Parts. Part I (relaxed binary trees) and Part II (compacted binary trees) prove convergence to a positive amplitude and a Poincaré expansion to every fixed order in $n^{-1/3}$, with three explicit logarithmic corrections. Part I uses an exact Jacobi symmetrization; Part II treats the compacted recurrence as a signed delay whose scalar drift cancels, and shows that the $n^{-1/3}$ term cancels in the compacted-to-relaxed ratio. Part III proves the same for every fixed $k \geq 3$ and all three models, with common first two corrections, the first model-dependent defects, eventual decrease of the normalized ratios and a cancellation identity. Its ternary section gives the $k = 3$ base proof, the third coefficient and a second route to the automaton amplitude; its comparison section gives the earliest, Θ -level form of the positive completed-run comparison, with an explicit constant uniform in k and n . Part IV proves the binary automaton theorem, with power $n^{7/8}$, by the method of Part II. Every Part inverts its expansion with Lambert W and separates smooth inverses from integer thresholds. The amplitudes are proved positive but are not evaluated; no transseries, convergent series, uniformity in k or certified decimal is claimed. The report is AI-assisted and unrefereed, and none of it is formalized.

[5 October 2026, batch 102.] Part V adds three bundle reports that prove the binary theorems of Parts I, II and IV again, in even time, with a positive semidefinite two-step Jacobi product and three-lag signed memory. New with it: a proof that the relaxed amplitude is positive which uses no published lower bound (Lemma 72.1); three further logarithmic coefficients for relaxed and compacted trees and one for automata; ratio expansions through n^{-3} (trees) and $n^{-7/3}$ (automata); the next explicit term of the automaton inverse; and uncertified amplitude estimates, among them 76.438 for automata.

Contents

Part 0. Guide: the family, its normalizations and this merge	8
What the report proves	8
One family, indexed by the arity	9
Normalization dictionary	10
Notation: one letter, several meanings	11
Sources and provenance	12
How the Parts depend on one another	14
Status, scope and non-claims	14
Instances of results elsewhere in the repository; neighbouring reports	15
Conventions of the merge	16
I Relaxed binary trees	16
1 Statement and scope	17
2 Exact recurrence and contractive symmetrization	18
3 The coarse Airy limit and a uniform spectral gap	19
4 Quantitative estimates from a finite quasimode	20
5 Convergence of the scalar amplitude	21
6 A boundary smoothing estimate	23
7 The limiting constant and its strict positivity	23
8 Formal profiles at every finite order	24
9 Approximate solutions in the normalized norm	26
10 A scalar bootstrap to every finite order	26

11 Coefficient extraction and computational checks	27
12 The rational coefficient field	28
13 Asymptotic inversion and discrete thresholds	29
14 Reproducible computation and numerical limitations	36
15 Further research questions	37
 II Compacted binary trees	 38
16 Statement and scope	39
17 Exact positive path models	40
18 The shared Jacobi evolution	42
19 A globally bounded signed delay	43
20 Cancellation of the scalar drift	44
21 Convergence of the scalar amplitude	45
22 Formal profiles at every finite order	46
23 Global residuals and the zero-limit bootstrap	47
24 The endpoint and the common positive amplitude	48
25 Explicit coefficients and their rational field	49
26 Inverse asymptotics and discrete thresholds	50
27 Cancellation in the compacted to relaxed ratio	52
28 Reproducibility and numerical limitations	53
29 Further questions	54
 III Fixed arity $k \geq 3$	 54
30 Results and indexing	55

31	Exact arrays and boundary harmonics	57
32	Positive forms and the singular Airy limit	58
33	Evolving profiles and a positive amplitude	60
34	Recursion and transfer to all finite orders	61
35	Compacted trees and automata as finite memory perturbations	62
36	Inverse counting scales and integer thresholds	65
37	Checks scope and further research	67
	<i>The ternary case (source 66)</i>	
38	Statements and normalizations	68
39	Exact transfer and the boundary weight	69
40	The singular Airy limit	71
41	An evolving Airy profile	73
42	Tracking and the relaxed amplitude	74
43	Polynomial Airy recursion to every finite order	75
44	Positive defects and the automaton amplitude	77
45	Inverse asymptotics and integer thresholds	80
46	Verification and scope	82
	<i>Orders of growth by the completed-run comparison (source 29)</i>	
47	Statement and state convention	82
48	Exact recurrence and positive completed runs	83
49	Summable exclusion and its consequences	84
50	A logarithmic inverse enclosure	85
51	What remains open and a boundary obstruction	86
52	Verification and literature scope	87

IV	Binary finite-language automata	88
53	Statement and mathematical scope	88
54	Exact positivity and the initial boundary	89
55	The shared Jacobi evolution	90
56	A globally bounded signed delay	91
57	Cancellation of the scalar drift	92
58	Convergence of the scalar amplitude	92
59	Formal profiles at every finite order	93
60	Global residuals and the zero-limit bootstrap	94
61	The endpoint and the common positive amplitude	95
62	Explicit coefficients and their rational field	96
63	Selected smooth inverse and discrete thresholds	96
64	Verification and numerical limitations	97
65	Literature boundary and further questions	98
V	Second routes: an even-time Jacobi product and signed memory	99
66	Reading Part V: sources, second routes and notation	99
	<i>Report 100: relaxed binary trees</i>	
67	Result and relation to earlier work	105
68	The exact finite Jacobi product	107
69	A quantitative frozen Airy theorem	109
70	Localization and drift of the Perron vector	111
71	A killed lazy walk and boundary smoothing	112
72	Tracking the Perron direction and proving positivity	114

73	The amplitude and its convergent representation	115
74	An exact polynomial Airy coefficient algorithm	116
75	From formal profiles to all algebraic orders	118
76	Explicit diagonal coefficients and numerical checks	121
77	Rigorous inverse asymptotics	124
78	Further questions and proof boundaries	127
79	Reproducibility and provenance	128
	<i>Report 101: compacted binary trees</i>	
80	The counting problem and the result	129
81	The exact signed memory recurrence	130
82	The relaxed Jacobi estimates used here	132
83	Global bounds for the two gauged memory operators	133
84	Normalization and convergence of the scalar amplitude	134
85	Boundary smoothing and the positive amplitude	136
86	A forced signed memory lemma	137
87	An exact polynomial Airy coefficient algorithm	138
88	Compatible profiles with arbitrarily small genuine residuals	139
89	Explicit coefficients and the scope of finite certificates	141
90	Rigorous inverse asymptotics	143
91	Reproducibility and limits of the result	145
	<i>Report 103: minimal binary automata</i>	
92	The result and its published starting point	147
93	The exact counting array and its startup	148
94	The relaxed Jacobi estimates used by the proof	150

95	The exact signed three lag equation	151
96	Convergence of the leading amplitude	153
97	A convergent characterization of the amplitude	155
98	An exact formal construction at every depth	156
99	Compatible carriers and genuine residuals	158
100	Forced signed memory and the all orders theorem	160
101	Explicit coefficients and numerical diagnostics	162
102	Rigorous inverse asymptotics	163
103	Further questions and boundaries of the proof	167
104	Reproducibility and source record	168
105	Further questions and research (Part V)	169
	Appendix	172
A	Self-contained Jacobi estimates	172

Part 0. Guide: the family, its normalizations and this merge

This guide was written when the six manuscripts were merged. Everything after it, from Part I on, is the manuscripts' text, except for the passages marked **[write, 2 October 2026]** and the changes listed in “Conventions of the merge” below.

What the report proves

- **Part I** (source 62, relaxed binary trees, A082161). There is $\gamma > 0$ with $a_n = \gamma n! 4^n e^{3zn^{1/3}} n(1 + \sum_{k \leq M} c_k n^{-k/3} + O_M(n^{-(M+1)/3}))$ for every M , $c_k \in \mathbb{Q}[z]$; logarithmic corrections $\frac{53z^2}{90}$, $\frac{44z}{27}$, $\frac{141}{140} - \frac{1304z^3}{42525}$ (equations (F1)–(F3) in Section 1). Also the rational coefficient field, and Lambert- W inverses with discrete threshold brackets.
- **Part II** (source 15; compacted binary trees, A254789). By Theorem 16.1, $c_n = \gamma_c n! 4^n e^{3zn^{1/3}} n^{3/4}(1 + \sum \mathbf{c}_k n^{-k/3} + O_M)$ with $\gamma_c > 0$ and logarithmic corrections $\frac{53z^2}{90}$, $\frac{271z}{216}$, $\frac{393}{1120} - \frac{1304z^3}{42525}$; exact positive four-state, three-state and renewal models and their symmetry obstructions; the ratio corollary (27.2), in which the $n^{-1/3}$ term of $\log(c_n/r_n)$ cancels; inverses. Appendix A is its self-contained proof of the shared Jacobi estimates.
- **Part III** (sources 18, 66 and 29, every fixed arity $k \geq 3$). Theorem 30.1: positive amplitudes A_k^R, A_k^C, A_k^B and expansions to every fixed order for relaxed, compacted and automaton counts, with the common corrections (30.7)–(30.8). Theorem 30.2: the first model-dependent defects $(q/k)^k/(k-2) \cdot n^{-(k-2)}$ and half of it, and eventual decrease of both normalized ratios. The cancellation identity (35.4) and an inverse. The ternary section (source 66): the $k = 3$ base proof, Theorems 38.1 and 38.2 with the explicit c_3 and scalar coefficients $\mathfrak{s}_3, \dots, \mathfrak{s}_6$, the explicit ternary weight with $1 \leq \omega \leq 3/2$, and the bridge-tail route to the automaton amplitude. The comparison section (source 29): Theorem 47.1, the comparison $P_{k,n} \leq B_n/(2^{n-1}R_n) \leq 1$ with $P_{k,n} = \prod_{j=2}^n (1 - 1/(2j^{k-1}))$ and $P_k \geq (2\sqrt{2}/\pi) \sin(\pi/\sqrt{2})$; a uniform finite-level approximation; a Θ -only inverse enclosure; and the boundary obstruction that the weights of the ternary section and of Lemma 31.1 resolve.
- **Part IV** (source 48, binary finite-language automata, A331120). Theorem 53.1: $b_n = \gamma n! 8^n e^{3zn^{1/3}} n^{7/8}(1 + \sum \mathbf{c}_k n^{-k/3} + O_M)$ with $\gamma > 0$ and logarithmic corrections $\frac{53z^2}{90}$, $\frac{623z}{432}$, $\frac{3497}{4480} - \frac{1304z^3}{42525}$; a nonnegative renewal representation; a selected smooth inverse.

[write, 5 October 2026, batch 102] Part V (bundle Reports 100, 101, 103; relaxed trees, compacted trees, automata, $k = 2$) proves the theorems of Parts I, II and IV a second time, in even time $n = N/2$. Report 100 proves the relaxed amplitude positive from the exact recurrence alone (Lemma 72.1; Remark 66.1 applies it to Part I) and gives the relaxed logarithmic coefficients b_4, b_5, b_6 , the normalized ratio $r_n/(4nr_{n-1})$ through n^{-3} , a sharper endpoint law and a telescoping series for the projection constant. Report 101 proves the compacted b_4, b_5, b_6 ; with Corollary 66.3 the compacted ratio holds through n^{-3} . Report 103 gives the automaton $d_4 = -5429z^2/19440$, c_4, σ_7 , the ratio $b_n/(8nb_{n-1})$ through $n^{-7/3}$, the $X^{-1/3}$ term of the inverse and the uncertified amplitude estimate 76.438. Section 66.2 maps every result to its counterpart here.

One family, indexed by the arity

In the coordinates of Part III (Section 31), fix $k \geq 2$, put $q = k - 1$, and for $m \geq 1$, $x \geq qm$ let

$$\begin{aligned} r(x, m) &= r(x, m - 1) + (m + 1)r(x - 1, m), \\ c(x, m) &= c(x, m - 1) + (m + 1)c(x - 1, m) - (m - 1)c(x - k, m - 1), \\ b(x, m) &= 2b(x, m - 1) + (m + 1)b(x - 1, m) - mb(x - k, m - 1), \end{aligned}$$

with base rows $r(x, 0) = c(x, 0) = b(x, 0) = 1$ for $x \geq 0$, the automaton auxiliary value $b(-1, 0) = 1$, and all other missing values zero. The relaxed, compacted and automaton counts are $R_n = r(qn, n)$, $C_n = c(qn, n)$ and $B_n = b(qn, n)$, where n counts internal nodes of the trees and transient states of the automata (each automaton has one further, rejecting, sink). At $k = 2$ the three recurrences are, letter for letter, Part I's $r_{n,m}$, Part II's (16.1) and Part IV's (53.1). The initial terms below were recomputed from these recurrences when the report was written.

k	model	OEIS	$n = 0, 1, 2, \dots$	Part
2	relaxed	A082161	1, 1, 3, 16, 127, 1363, 18628, 311250, ...	I
2	compacted	A254789	1, 1, 3, 15, 111, 1119, 14487, 230943, ...	II
2	automata	A331120	1, 1, 6, 60, 900, 18480, 487560, 15824880, ...	IV
3	relaxed	A082162 ^a	1, 1, 7, 139, 5711, 408354, 45605881, ...	III
3	compacted	—	1, 1, 7, 133, 5299, 371329, 40898599, ...	III
3	automata	— ^b	1, 1, 14, 532, 42644, 6011320, 1330452032, ...	III
4	relaxed	A102102	1, 1, 15, 1000, 189035, 79278446, ...	III
4	automata	— ^b	1, 1, 30, 3900, 1460700, 1220162880, ...	III

^aThe entry starts at $n = 1$; the sources adjoin $R_0 = 1$. The array A128249 lists the relaxed counts of all arities (Section 30).

^bSource 29's bounded search verified no OEIS identifier for automata over $k \geq 3$ letters; A082162 counts relaxed ternary trees, not ternary automata.

Where the method changes. After normalization, the compacted and automaton recurrences are the relaxed one plus a delayed term (Part III, equation (35.1)) whose relative size per step is $O_k(i^{-(k-1)})$ (Section 35.2); its cumulative effect on the logarithm is of order $n^{-(k-2)}$ (Theorem 30.2). For $k \geq 3$ this is summable, the completed-run products P_C , P_B of (35.2) are positive, and the three models share the scale $\mathcal{G}_{k,n} = (n!)^q (k^k/q^q)^n e^{\gamma^{1/3}} n^{(2k-1)/3}$ (automata with an extra 2^n). For $k = 2$ the delay is of order N^{-1} and not summable: it changes the power of n instead of adding a correction. Parts II and IV cancel the resulting scalar drift, $1/(4N)$ and $1/(8N)$, against the factors $N^{-1/4}$ and $N^{-1/8}$, which gives $n^{3/4}$ and $n^{7/8}$ in place of the relaxed n . Source 29 notes the same threshold from the other side: at $k = 2$ its lower product tends to zero like $n^{-1/2}$. Source 48's remark on the exclusion family $(\alpha m + \beta)$, with heuristic power $1 - \alpha/4$, is a heuristic it states without proof.

The binary and fixed-arity coefficients agree where they should. With $k = 2$, $q = 1$, Part III's general formulas (30.7)–(30.8) give $c_{2,1} = 53z^2/90$ and $c_{2,2} = z(2809z^3 + 26400)/16200$, exactly Part I's c_1, c_2 in (F3), and with $k = 3$ they give the c_1, c_2 of Theorem 38.2. Both reductions were checked symbolically (difference zero). Source 18's proof note `18-arity-all-orders-proof.md` observes the $k = 2$ reduction of $c_{k,1}$ itself. Part III claims nothing for $k = 2$, and must not be read as doing so: at $k = 2$ the compacted and automaton second coefficients differ from the relaxed one ($\frac{271z}{216}$ and $\frac{623z}{432}$ against $\frac{44z}{27}$ in the logarithm; the differences $-\frac{3}{8}z$ and $-\frac{3}{16}z$ are in ratio two, as Theorem 30.2 has it for $k \geq 3$, while the n^{-1} differences $-\frac{21}{32}$ and $-\frac{29}{128}$ are not).

Normalization dictionary

Notion	Parts I, II, IV (sources 62, 15, 48) and source 29	Part III (sources 18, 66)
Largest Airy zero	$z = -2.338107410459767\dots$	a (also a_1 ; a_m the m th zero)
Scaled zero	$a := 2^{-1/3}z$ (not the zero itself; source 29 has none)	$\lambda := a/B$, $B = (2/q)^{1/3}$; at $k = 2$, λ is Parts I, II, IV's a
Airy profile	$F(x) = \text{Ai}(z + 2^{1/3}x)$, $F'' = 2(x + a)F$	$f(x) = \text{Ai}(a + Bx)$, $f'' = (2/q)(x + \lambda)f$; source 66: $B = 1$, $f(x) = \text{Ai}(a + x)$
Time, height	$N = n + m$, $j = n - m$; endpoint $N = 2n$, $j = 0$	$i = x + m$, $j = x - qm$; endpoint $i = kn$, $j = 0$
Normalized array	$d_{N,j} = r/((N + j)/2)!$ (I); $e_{N,j}$ (II, IV)	$d_i(j) = q^{2xr}(x, m)/x!$; at $q = 1$ this is Part I's $d_{N,j}$
Stretched exponent	$3zn^{1/3}$	$\gamma n^{1/3}$, $\gamma = 3(kq/2)^{1/3}a$; source 29: β_k
Power of n	n (I), $n^{3/4}$ (II), $n^{7/8}$ (IV)	n^η , $\eta = (2k - 1)/3$ (source 29: α_k)
Amplitude	γ (I, IV), γ_c and γ_r (II)	A_k^R , A_k^C , A_k^B (18); C_R , C_B (66)
Index n	internal nodes (I, II); transient states (IV, 29)	internal nodes (R_n , C_n); transient states (B_n)

Source 29 also counts by total states $N = n + 1$ (its Remark on total states), which shifts the power of n to $(2 - k)/3$. Source 48's array $c_{n,m} = b_{n,m}/2^m$ is a normalized automaton array, not Part II's compacted count.

[**write, 5 October 2026, batch 102**] Part V uses a third normalization: the Airy zero is a or a_1 (as in Part III), $\kappa = \varkappa = 2^{2/3}a$ is twice the a of Parts I, II, IV, time is halved ($n = N/2$, $\varepsilon = n^{-1/3}$, half-grid $x_k = (k + \frac{1}{2})\varepsilon$), and its scalar coefficients are $\sigma_m = 2s_m$. Its own table of conventions is Section 66.3.

One polynomial–Airy identity. Every Part solves its formal profiles with the same identity. For $L = \partial_x^2 - cx - \ell$ and polynomials P, Q ,

$$L(PF + QF') = (P'' + 2(cx + \ell)Q' + cQ)F + (2P' + Q'')F',$$

which reduces $L(PF + QF') = RF + SF'$ to a triangular polynomial equation for Q . Parts I, II and IV use $c = 2$, $\ell = 2a$ (equation (F28) and (22.5)); source 66 uses $c = 1$, $\ell = a$ ((43.3)); source 18's (34.2) is the case $c = 2/q$, $\ell = 2\lambda/q$ after division by $q/2$. The batch-77 report on tree-child networks, [SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a213863-tree-child-networks](#), proves the same identity with general c, ℓ for d -combining tree-child networks and uses the same zero-limit projection bootstrap; the two reports neither cite nor depend on each other.

[**write, 5 October 2026, batch 102**] That report now also prints bundle Reports 104 and 102 (its Parts III and IV), of the same delivery series as Part V's Reports 100, 101 and 103. They use the same lemma family as Report 100: positive Jacobi products, the killed half-line kernel, the forced-product bootstrap, and the restart comparison of Lemma 72.1 for positivity. Report 102's "companion relaxed-tree article", which it says it does not need, is presumably Report 100. The reports stay separate.

Notation: one letter, several meanings

No symbol of any source was renamed: each Part, and each source section of Part III, keeps its source's letters. The sources use one letter for several notions and one notion under several letters. The table fixes the meaning in each Part and prints the tempting false reading beside the true one.

[**write, 5 October 2026, batch 102**] The table covers Parts I–IV. Part V has its own (Section 66.3); the readings most likely to mislead there are a (the Airy zero itself), S_n (a diagonal matrix, not the Jacobi matrix), $\lambda_n \approx 4$, and Report 103's d_j (logarithmic) and c_j (multiplicative).

Symbol	Meaning and where	Do not read it as
z, a	See the dictionary above. In Part I, a_n is the counting sequence and a_1^{Ai} the Airy zero.	Parts I, II, IV's a is <i>not</i> the Airy zero; Part III's a is.
γ	Parts I and IV: the positive amplitude. Part II: γ_c , with γ_r the relaxed amplitude of Part I. Part III: $\gamma = 3(kq/2)^{1/3}a < 0$, the coefficient of $n^{1/3}$ (source 66: $\gamma = 3 \cdot 3^{1/3}a$). Part I's (F23): γ_E is Euler's constant.	Part III's γ is not an amplitude.
k, q	Part III: the arity $k \geq 3$ and $q = k - 1$. Parts I, II, IV: k is an order index ($c_k n^{-k/3}$) or a run length ($w_m(k)$, $W_m(k)$); q is $p \log x + C$ in the inverse sections, q_0 a limiting projection, q an exponent in the bootstraps. Source 66: p_j, q_j are row probabilities.	In Parts I, II, IV, k is never an arity.
$c, \mathbf{c}$	Part II: $c_{n,m}, c_n$ compacted counts. Part IV: $c_{n,m} = b_{n,m}/2^m$. Part III: $c(x, m)$, C_n compacted k -ary counts, $c_i(t)$ column sums, $c_{k,r}^X$ coefficients. Relative coefficients: c_k (I), $\mathbf{c}_k$ (II, IV), c_r (66).	Part IV's $c_{n,m}$ is not a compacted count.
b, B	$b_{N,j}$ Jacobi edge weights and $b_N = \langle g_N, B_N g_{N-3} \rangle$ (II, IV), B_N the normalized delay (II, IV). Part IV: $b_{n,m}$, b_n automaton counts. Part III: $b(x, m)$, B_n automaton counts, $B = (2/q)^{1/3}$. Source 66: b_i a scalar projection, B a polynomial in $Af + Bf'$ and the weighted adjoint $B = \omega^{-1}T^*\omega$. Parts I, II: $B = 53z^2/90$ in the inverse sections.	Part IV uses b three ways, as its source did.
λ	Parts I, II, IV: λ_N , the top eigenvalue of S_N (≈ 2). Part III: $\lambda = a/B$. Source 29: $\lambda_k = C_k^{1/q}/e$.	Not the same quantity in any two of these.
R, P	Part III: R_n relaxed counts; Parts II, IV: $R_N = P_N N^{-1/4}$, resp. $P_N N^{-1/8}$; Part I: $R(a, \varepsilon)$, R_m . P_N spectral product (I, II, IV); P_C, P_B, P_i (18); P_3, P_i (66); $P_{k,n}, P_k$ (29); $p = 3/2$ (I) and $p = 5/4$ (II) in the inverses, p also the residual order.	
C, A	C : matching constant A_∞/q_0 (I; C_* in II, IV), $\log(\gamma\sqrt{2\pi})$ in Part I's inverse, compacted counts C_n (III), amplitudes C_R, C_B (66), growth constant $C_k = 2k^k/q^q$ (29). A : amplitudes A_k^X and growth constant k^k/q^q (III), $3z$ in the inverses (I, II), completed-run arrays $A(n, m)$, $A(x, m)$ (IV, 29), A_∞ limiting projections.	Source 29's C_k is a growth constant, not an amplitude.

Symbol	Meaning and where	Do not read it as
η, α, β	$\eta = (2k - 1)/3$ (18; α_k in 29); η_N (II, IV); η_i (66); $\eta_k = \alpha_k + q/2$ (29). α_N, α_i projection errors; $\beta(x, \varepsilon)$ delay coefficients (II, IV); $\beta = (7k - 6)/6$, β^X (18); β_k (29); β_i (66); (α, β) exclusion parameters (48).	
ω, h	ω : a primitive cube root of unity (I, II, IV); the weight ω_j (III). h : h_N parity vector, h_k amplitude coefficients, $h(n)$ inverse model (I, II); $h_j = j + 1$ (III).	
$\mathcal{G}, \mathfrak{s}, \sigma$	$\mathcal{G}_k(a, x)$ normalized profiles (I); $\mathcal{G}_{k,n}$ the common scale (III). s_m ratio coefficients (I, II, IV); $\sigma(\varepsilon)$ scalar series and $\sigma_{i,m}$ singular values (III); source 66's $\mathfrak{s}(\varepsilon)$ is Part III's σ at $k = 3$.	
D, Q, L, U, W	D_N gauge, Q_N contraction (I, II, IV); D_i delay and Q_i kernel (III); $D = q(w + 1)$ slope (III); D_N total-state count (29). L_N delay (II, IV); L_q (III); $L = \log y$ (III). U upward shift (II, IV); $U_i(j)$ up coefficients (III, 29). W_0 Lambert; W_N transverse error (I, II, IV); W_m run weights (IV, 29, 66).	
Thresholds	$N(X)$ (I), $N_c(Y)$ (II), $N(y)$ (18), $\nu(y)$ (66), $I_k(X)$ (29), “the first integer index with $b_n \geq y$ ” (IV): the least n with count at least the target.	Not the inverse of a smooth model.

Sources and provenance

The report was built from six manuscripts, all delivered in the arrival commit 096ee7b87 of 2 October 2026 as batch 77. Five were placed in this directory by commit 34f1acd4b (cluster P3; archive 29 was handed over to that cluster by cluster P4) and source 48 by commit aa7345800 (cluster P2). All six are dated 2 October 2026. The author line is empty in sources 62, 15, 29 and 48 and reads “Research report” in sources 18 and 66; no tool or person is named. Each calls itself an independently checked research report and disclaims publication and peer review.

Part	Source	Archive (.zip) / main file	Lines	pp.	Pin	Labels
I	62	relaxed-binary-trees-reproducibility / relaxe d-binary-trees.tex	1221	23	none	air:rb:
II	15	compacted-binary-trees-reproducibility / comp acted-binary-trees.tex	1274	21	none ^a	air:cb:
III	18	fixed-arity-airy-package / fixed-arity-airy- expansions.tex	515	12	none	air:ka:
III	66	ternary-airy-amplitudes-result / ternary-airy- amplitudes.tex	656	14	none	air:t3:
III	29	larger-alphabet-automata-report / larger-alph abet-automata.tex	344	6	none ^a	air:ka:al:
IV	48	oeis-dfa-finite-language-report / dfa-finite- languages.tex	967	16	a68b926cf	air:dfa:

^aRecords a search of the repository, not a continuation pin; see below.

Pins. No source continues a repository path. Source 48’s overlap receipt cites repository files at commit a68b926cf. Source 15’s provenance note (shipped as 15-compacted-src-literature-and-provenance.md) records a repository search at commit 89daa64f that found no A254789

material, and source 29’s raw search receipt (not shipped; see the README) refers to commit 4b874cea0. All three commits are ancestors of the placement commits. Those searches were right when made: before this batch the repository had no report on any of these sequences. This report is now that material; the sentences in the Parts that report the empty searches are kept, with a note.

Where the merge had to choose.

1. *Base and order.* Source 62 is the base: source 15 cites it, source 18’s formulas reduce to it at $k = 2$, and it treats the binary problem that Elvey Price, Fang and Wallner left open. Part II follows because its ratio corollary uses Part I. Part III collects every fixed $k \geq 3$. Part IV is last because its source was placed by a later commit, into the Part number reserved for it; mathematically it is the automaton twin of Part II.
2. *One Jacobi appendix.* Source 15’s appendix “Self-contained Jacobi estimates” is printed once, as Appendix A. Source 48’s appendix is the same text with one word changed (“DFA theorem” for “compacted theorem”) and is not reprinted. Source 62’s own Sections 2–5, which prove the same estimates in the base text’s own words, are kept; the appendix is a second, self-contained write-up of them.
3. *Part IV prints what is specific to automata.* Source 48’s Sections 3–8 and 10 follow Part II line by line. Five passages are word for word Part II’s: the definitions and Jacobi lemma of Section 3, the finite-memory lemma, the formal-equation paragraph of Section 7, the bootstrap of Section 8.2 and the rational-field paragraph of Section 10. They are replaced by pointers. Every display in which the automaton differs is printed: the renewal model, the delay coefficient with its $m = 1$ term, the seeds, the drift $1/(8N)$, the exponents, the scalar and logarithmic coefficients, the amplitude formula and the inverse.
4. *Source 66 inside Part III.* Source 66 is the $k = 3$ case of source 18 and the base proof that 18 generalizes: 18’s proof notes say so and cite 66’s proof file by hash (SHA-256 18cdbc4e..., the file shipped as 66-ternary-amplitude-proof-candidate.md). Its two theorems are kept in its section, marked as the $k = 3$ instance of Theorems 30.1 and 30.2, because the proof printed after them is that base proof. Its own additions are named there.
5. *Source 29 inside Part III.* The positive completed-run comparison appears three times in the sources: as source 29’s main theorem (every $k \geq 3$, explicit finite- n constant, Θ only), as step (35.2) of source 18’s proof (every $k \geq 3$, infinite product) and as step (44.1) of source 66’s ($k = 3$). It is stated as a theorem once, Theorem 47.1, credited to source 29 as its earliest form; the two proof steps carry pointers to it. Source 29’s other results are printed in full.
6. *Nothing else is omitted* except the title blocks, the tables of contents and the separate bibliographies (merged at the end). Each Part, and each source section of Part III, opens with its source’s abstract as delivered.

[write, 5 October 2026, batch 102] Part V was added from bundle Reports 100, 101 and 103 of batch 102 (arrival commit 60f54ea06, placement commit 6ab1f1979, both 5 October 2026; the reports are dated 2 October 2026). The author line of Reports 100 and 101 is “Research prepared for private review”, that of Report 103 “Research prepared for Vladimir”. Report 100 records a repository collision check at 4b874cea0, the commit to which source 29’s search receipt also refers; Reports 101 and 103 embed Report 100’s archive unchanged and name no repository path. None of the three knew of Parts I–IV. Their provenance, and where Part V had to choose, are in Sections 66.1 and 66.4.

How the Parts depend on one another

- Part I is self-contained apart from the Elvey Price–Fang–Wallner lower bound [1], used for positivity.
- Part II’s main theorem is self-contained through Appendix A and the same lower bound. Only its ratio corollary (Section 27) uses Part I’s (F2).
- Part III uses neither Part I nor Part II. Its only external input is the relaxed lower bound of Ghosh Dastidar and Wallner [3]. Its general proof extends the ternary proof of source 66; the comparison of source 29 is a Θ -level statement that the general theorem upgrades.
- Part IV uses Part II’s Section 18, its finite-memory Lemma 20.1, its bootstrap and Appendix A, together with the relaxed array of Part I (for domination) and the Elvey Price–Fang–Wallner automaton lower bound [2].

[write, 5 October 2026, batch 102] Part V uses no earlier Part. Its Report 100 needs no external analytic input at all: Lemma 72.1 replaces the published lower bound, and by Remark 66.1 Part I’s positivity step may rest on that lemma instead of [1]. Report 101 uses Report 100’s Sections 2–5 and the compacted lower bound of [1]; Report 103 uses Report 100’s Sections 2–5, the comparison $b_n \geq 2^{n-1}c_n$ of [2] and the same compacted lower bound.

Status, scope and non-claims

The six manuscripts are AI-assisted and unrefereed; each disclaims publication and peer review, and their audits are research verification. Their finite computations (exact recurrences, symbolic coefficient recursions, floating diagnostics) check algebra and indexing and are not proofs of the analytic steps. Nothing in this report is formalized in Lean or Rocq, and placing it in this repository confers no formal status.

The union of the sources’ non-claims, all kept in the Parts:

- No amplitude is evaluated or enclosed. The decimals 166.9506 and 166.9515 (Part I), 173.12105 (Part II) and the normalized ratios of Part IV are evidence, not enclosures; Part III gives no amplitude values at all.
- Every expansion is a Poincaré expansion to each fixed order. No convergence of the series, Gevrey bound, optimal truncation, exponentially small term or transseries is claimed, and no canonical real interpolation of an integer sequence.
- Part III is not uniform in k and proves eventual, not everywhere, decrease of the normalized ratios.
- Inverses: smooth-model inverses, interpolation inverses and integer thresholds are kept apart; unspecified O -constants give eventual statements, not certified finite thresholds.
- The growth orders are due to Elvey Price, Fang and Wallner [1, 2] and to Ghosh Dastidar and Wallner [3]; the sources claim only the amplitudes, the expansions and what follows from them. Their literature and repository checks are bounded searches, not priority claims.

- Source 66’s “no general-alphabet theorem” and “no compacted ternary result”, source 29’s “no limiting amplitude”, source 62’s “nothing on compacted trees” and source 15’s higher-arity question are statements about each source alone; other Parts of this report now supply those results, as notes in the Parts say.

[**write, 5 October 2026, batch 102**] Part V keeps all non-claims of its three reports and adds uncertified evidence: 166.9520924395 and the extrapolation 166.9520895742200 (relaxed), 173.1267049959 and 173.1267048501473 (compacted), all at or near $n = 3000$ after six corrections, and 76.438 for automata ($G_{3000}^{(4)} = 76.4383234197$). None is an enclosure. Report 100’s “no compacted amplitude theorem” and its conditional compacted inverse are statements about Report 100 alone; Report 101 and Part II supply both.

A correction of the literature that this report has not verified. Sources 18 and 66 state that the proof of the high-path lemma of [3] (Lemma 14 of arXiv v1, Lemma 15 of the published version) contains a false pointwise bound $U_i(j) \leq q$ ($U_i(j) \leq 2$ for $k = 3$; source 66 gives $U_4(1) = 8/3$). For the coefficient $U_i(j)$ as defined in Part III the inequality does fail near the boundary. Whether the published text prints it was not checked against the paper when this report was written. Neither source uses the bound: source 18 avoids it, and source 66 repairs the lemma’s conclusion.

Instances of results elsewhere in the repository; neighbouring reports

Every inverse section is an instance of results proved in the repository’s transseries volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`. No novelty is claimed for these methods.

- The Lambert anchors $x \log(4x/e) = T$ (Parts I, II), $X(\log X + \log 8 - 1) = Y$ (Part IV) and $qn_0 \log n_0 + (\log A - q)n_0 = L$ (Part III, sources 18, 66, 29) solve the dominant block exactly, as in theorem `p0:thm:lambert-core`.
- The explicit corrections of (I10)–(I11), (26.5), (36.1), (45.1), (50.4) and (63.2) are reversions of a perturbed core, as in theorem `p0:thm:perturbed-inversion`.
- The threshold brackets (I18), (26.9) and (50.5), and the rules $N = \lceil x \rceil$ of sources 18 and 66, are the exact-rounding and separation steps of theorem `p0:thm:staircase`. Its order-theoretic parts are formalized as `Fabius.staircase_ceil` and `Fabius.staircase_separation` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; these declarations verify the general rounding step only, nothing specific to this report.

[**write, 5 October 2026, batch 102**] The inverse sections of Part V (Sections 77, 90, 102) are instances of the same three theorems: the anchors $x_0 = Y/W_0(4Y/e)$ and $X = Y/W_0(8Y/e)$, the corrected inverses z_α , z , Z and (102.11), and the one-comparison threshold rules.

Neighbouring reports. The tree-child network report named above shares the polynomial–Airy lemma family. The batch-77 report `SetTheory/Cardinals/docs/reports/automata-and-formal-languages/accessible-and-strong-automata` counts other automata (accessible and strongly connected deterministic automata, in the Korshunov–Liskovets regime of inverse-power corrections); it has no stretched exponential and shares no lemma with this report. The Airy law of `a290268-unbounded-deficits`, in the same collection directory, is an additive fold from an integral representation, unrelated to the Airy zero here.

Conventions of the merge

- Every label carries the report prefix `air:` and the sub-prefix of the table above; guide labels are `air:guide:`, Part labels `air:part:`, the appendix `air:app:jacobi`. Source 62’s section labels were generated from its section titles and keep those names after the prefix. The new section labels added to sources 15, 18, 66, 29 and 48 have the form `air:(sub-prefix)sec:...`
- Sections are numbered through the report. Source 62’s Section k is Section k (Sections 1–15). Source 15’s Section k is Section $k + 15$; source 18’s is $k + 29$; source 66’s is $k + 37$; source 29’s is $k + 46$; source 48’s is $k + 52$. Theorems keep their position within their section, so source 15’s Theorem 1.1 is Theorem 16.1. Equations are numbered within sections; source 62’s tagged equations (F1)–(F36), (G1)–(G5) and (I1)–(I19) keep their tags. Source 62’s references “§ k ” are now references to its sections, which kept their numbers.
- Citations are merged into one bibliography. The companion-article entry of source 15 (**relaxed**) is replaced by references to Part I; its annotation said that the companion was used only for the relaxed all-orders input of source 15’s Section 12 (Section 27 here), that all shared Jacobi estimates are proved in the appendix, and that no publication or peer-review status is implied. Sources 18 and 66 cite Ghosh Dastidar and Wallner as DW, the others as **gdw**; source 29’s **binary** is source 48’s **dfa**. Source 62’s numbered list of references is merged into the bibliography, its annotations kept.
- Sentences in the Parts about “the accompanying archive”, “the package”, “**replay.py**”, “the SHA-256 manifest”, “the README”, “**dependencies/relaxed/**” or “this standalone article” describe the source’s delivered archive, which survives in commit 096ee7b87. Which of its files this report ships, and under which names, is listed in the README. PDFs, checksum manifests, the members’ manuscripts and READMEs and a few duplicate or empty files are not shipped.

[**write, 5 October 2026, batch 102**] Part V follows the same conventions with the label prefix `air:jp:` (sub-prefixes `air:jp:rb:`, `air:jp:cb:`, `air:jp:dfa:` for Reports 100, 101, 103). Its Sections are numbered on from Section 65 and stand before the appendix: Section 66 is its guide, Report 100’s Section k is Section $k + 66$, Report 101’s is $k + 79$, Report 103’s is $k + 91$, and Section 105 is its question register. No earlier number or label changed. Notes added then are marked [**write, 5 October 2026, batch 102**]; sentences of Part V about “the accompanying archive” or “package” describe the delivered archives in commit 60f54ea06.

Part I

Relaxed binary trees

Source. Batch-77 manuscript 62, archive **relaxed-binary-trees-reproducibility.zip**, main file **article/relaxed-binary-trees.tex** (1,221 lines; 23-page PDF), titled “Asymptotic amplitude and all orders for relaxed binary trees”, dated 2 October 2026, empty author line. The base of this report: its text is printed in full, with label prefix `air:rb:`. Its numbered list of references is merged into the bibliography at the end.

Abstract of the source.

The number of relaxed binary trees is known to have order $n!4^n \exp(3zn^{1/3})n$, where z is the largest negative Airy zero. We give a spectral argument for convergence to a positive limiting amplitude and for a full expansion in powers of $n^{-1/3}$. An exact time-dependent gauge turns the counting recurrence into a symmetric Jacobi operator followed by a diagonal contraction. A finite Airy quasimode, the continuum spectral gap, and summable ground-mode drift establish norm tracking. A Catalan kernel estimate recovers the boundary coordinate. Polynomial–Airy approximate solutions and a zero-limit projection bootstrap then transfer every finite formal order to the exact sequence. The known lower bound is used to prove strict positivity of the amplitude. We also derive Lambert W inverse expansions and distinguish them from certified discrete threshold decisions. The accompanying reproducibility archive checks the exact algebra and numerical evidence; its amplitude estimates are not rigorous enclosures.

1 Statement and scope

Let a_n be OEIS A082161, the number of relaxed binary trees with n internal nodes. Its defining triangular recurrence is

$$r_{n,m} = r_{n,m-1} + (m+1)r_{n-1,m}, \quad r_{n,0} = 1, \quad r_{n,m} = 0 \ (m > n), \quad a_n = r_{n,n}.$$

Let $z = a_1^{\text{Ai}} = -2.338107410459767\dots$ be the largest real zero of Ai . (This Airy zero is denoted by z throughout, to avoid confusing it with the counting sequence.) Set

$$a = 2^{-1/3}z, \quad F(x) = \text{Ai}(z + 2^{1/3}x).$$

Theorem. There is a constant $\gamma > 0$ such that, for each integer $M \geq 0$,

$$a_n = \gamma n!4^n e^{3zn^{1/3}} n \left(1 + \sum_{k=1}^M c_k n^{-k/3} + O_M(n^{-(M+1)/3}) \right). \quad (\text{F1})$$

The coefficients belong to $\mathbb{Q}[z]$ and are determined by the polynomial–Airy recursion in §8. The first logarithmic coefficients are

$$\log \frac{a_n}{\gamma n!4^n e^{3zn^{1/3}} n} = \frac{53z^2}{90} n^{-1/3} + \frac{44z}{27} n^{-2/3} + \left(\frac{141}{140} - \frac{1304z^3}{42525} \right) n^{-1} + O(n^{-4/3}). \quad (\text{F2})$$

Consequently

$$\begin{aligned} c_1 &= 53z^2/90, \\ c_2 &= z(2809z^3 + 26400)/16200, \\ c_3 &= (1042139z^6 + 28444320z^3 + 30836700)/30618000. \end{aligned} \quad (\text{F3})$$

Numerically these are 3.2193061325653844, 1.3717168742447035, and -5.306490818002809 . No rigorous numerical enclosure or elementary evaluation of γ is asserted.

[write, 5 October 2026, batch 102] Part V (Report 100) proves this theorem a second time and continues (F2): with z for its a , the next logarithmic coefficients are $b_4 = -422z^2/1215$, $b_5 = z(717281z^3 - 298772235)/344452500$ and $b_6 = (2250428z^3 - 20962395)/41334300$, (76.1), and the normalized ratio $a_n/(4na_{n-1})$ is expanded through n^{-3} in (76.2).

The counting recurrence, the Θ estimate, and the conjectural limiting amplitude and all-orders shape are due to Andrew Elvey Price, Wenjie Fang, and Michael Wallner, Compacted binary trees admit a stretched exponential [1], especially Theorem 1.1 and §3.4. Their paper explicitly leaves existence of the amplitude open and reports the fit 166.95208957. The contribution here is a convergence/stability argument for the relaxed recurrence. It does not extend without further work to compacted trees, whose recurrence is different. It proves a Poincaré expansion to every finite algebraic order, not an exponentially improved transseries or a canonical interpolation between integers.

2 Exact recurrence and contractive symmetrization

[write, 2 October 2026] Sections 2–5 prove the gauge, gap, quasimode and projection estimates that Parts II and IV also use. Source 15 proves the same estimates again, self-contained and with the explicit matrices of Part II, in its appendix, printed as Appendix A; source 48 ships a verbatim copy of that appendix. Both write-ups are kept: this one is the base text, the appendix the version the later Parts cite.

Put

$$d_{N,j} = \frac{r_{(N+j)/2, (N-j)/2}}{((N+j)/2)!}$$

when $0 \leq j \leq N$ and $j \equiv N \pmod{2}$, and zero otherwise. Direct substitution gives

$$d_{0,j} = \mathbf{1}_{j=0}, \quad d_{N,-1} = 0, \quad d_{N,j} = \frac{N-j+2}{N+j} d_{N-1,j-1} + d_{N-1,j+1}. \quad (\text{F4})$$

In particular $a_n = n!d_{2n,0}$. Values outside the admissible support are interpreted as zero.

For $N \geq 1$, define on $0 \leq j \leq N+1$

$$D_N(j)^2 = \frac{\Gamma(N+2)\Gamma(N+1)}{\Gamma(N-j+2)\Gamma(N+j+1)}, \quad D_N(0) = 1,$$

and $v_N(j) = d_{N,j}/D_N(j)$. All vectors are padded with zeros in $\ell^2(\mathbb{Z}_{\geq 0})$. Define S_N to be the symmetric Jacobi matrix on $0, \dots, N+1$, padded by zero, with edge weights

$$b_{N,j} = \sqrt{\frac{N-j+2}{N+j}}, \quad 1 \leq j \leq N+1,$$

and define a diagonal contraction by

$$Q_N(j) = \sqrt{1 - \frac{j(j-1)}{N(N+1)}} \quad (0 \leq j \leq N+1), \quad Q_N(j) = 0 \quad (j > N+1).$$

The factorial identities

$$\frac{D_N(j)}{D_N(j-1)} = b_{N,j}, \quad \frac{D_{N-1}(j)}{D_N(j)} = Q_N(j)$$

on the support where the second ratio is needed establish the exact identity

$$v_N = S_N Q_N v_{N-1}. \quad (\text{F5})$$

Both b and Q lie in $[0, 1]$. If J denotes the half-line adjacency matrix, then $0 \leq S_N Q_N \leq J$ entrywise. If λ_N is the largest eigenvalue of S_N , then $\|S_N Q_N\| \leq \lambda_N$. The finite Jacobi matrix is irreducible, so its top eigenvector ψ_N can be taken positive and of norm one. Its spectrum is symmetric about zero because it exchanges the two parity subspaces. In particular its operator norm is λ_N .

3 The coarse Airy limit and a uniform spectral gap

Write $\varepsilon = N^{-1/3}$ and $x_j = (j+1)\varepsilon$. For vectors supported on the finite interval $0, \dots, N+1$, the quadratic form is exactly

$$\begin{aligned} \langle v, (2I - S_N)v \rangle &= \sum_{j=0}^N b_{N,j+1} |v_{j+1} - v_j|^2 + |v_0|^2 \\ &\quad + \sum_{j=1}^N (2 - b_{N,j} - b_{N,j+1}) |v_j|^2 + (2 - b_{N,N+1}) |v_{N+1}|^2. \end{aligned} \quad (\text{F6})$$

Here $|v_0|^2$ is the first Dirichlet-gradient interval, with fictitious value $v_{-1} = 0$; no nonphysical edge coefficient $b_{N,0}$ is introduced.

For $1 \leq j \leq N+1$,

$$1 - b_{N,j} = \frac{2(j-1)}{(N+j)(1+b_{N,j})} \geq \frac{j-1}{N+j}.$$

It follows from (F6) that, for universal positive constants c, C ,

$$\varepsilon^{-2} \langle v, (2I - S_N)v \rangle \geq c \sum_{j=0}^{N+1} x_j |v_j|^2 - C\varepsilon \|v\|^2. \quad (\text{F7})$$

On each fixed x window the edge weights tend uniformly to one, and

$$\varepsilon^{-2} (2 - b_{N,j} - b_{N,j+1}) \longrightarrow 2x_j$$

uniformly there. Interpolate linearly with value $\varepsilon^{-1/2} v_j$ at x_j and zero at zero. Add one zero node immediately after the final finite node and extend by zero thereafter; the endpoint term in (F6) controls this final interpolation interval. The gradient terms in (F6) give local H^1 bounds for any sequence with bounded scaled energy. Equation (F7) gives uniformly small L^2 tails outside a large fixed window. Local compactness, followed by this tail bound, gives strong L^2 compactness. The boundary interval enforces the trace zero at zero. The resulting liminf form is

$$\int_0^\infty (|f'(x)|^2 + 2x|f(x)|^2) dx.$$

For completeness, convergence of norms in this interpolation follows first on a fixed window from the local gradient bound and the Riemann-sum identity, and then globally by (F7). Sampling smooth compactly supported functions with zero boundary value gives the reverse, limsup bound. Applying the elementary min–max principle in both directions therefore gives convergence of each fixed low eigenvalue of $\varepsilon^{-2}(2I - S_N)$ to that of

$$\mathcal{H} = -\frac{d^2}{dx^2} + 2x, \quad f(0) = 0.$$

The latter eigenvalues are $\mu_k = -2^{2/3}z_k$, where $z_1 = z > z_2 > \dots$ are the negative Airy zeros. Thus, if $\lambda_{2,N}$ is the second-largest positive eigenvalue for large N ,

$$\lambda_N = 2 + 2a\varepsilon^2 + o(\varepsilon^2), \quad \lambda_N - \lambda_{2,N} \sim 2^{2/3}(z - z_2)\varepsilon^2. \quad (\text{F8})$$

As a map between opposite parity subspaces, S_N has singular values equal to its positive eigenvalues (and possibly additional zeros). Hence, after deleting its top singular direction, its norm divided by λ_N is at most $1 - cN^{-2/3}$ for all sufficiently large N . This parity formulation is essential: the negative eigenvalue $-\lambda_N$ is not a second slow direction within the occupied parity space.

4 Quantitative estimates from a finite quasimode

The Airy equation reads $F'' = 2(x + a)F$. Define

$$\begin{aligned} F_2(x) &= \frac{2(-a + 2x)}{15}F(x) + \frac{x(2a + x)}{15}F'(x), \\ F_3(x) &= \frac{F(x) - xF'(x)}{3}, \\ f_N(x) &= F(x) + \varepsilon^2 F_2(x) + \varepsilon^3 F_3(x), \\ \hat{\lambda}_N &= 2 + 2a\varepsilon^2 + 3\varepsilon^3 + \frac{13a^2}{15}\varepsilon^4 + \frac{5a}{3}\varepsilon^5. \end{aligned} \quad (\text{F9})$$

Choose a fixed smooth cutoff χ , equal to one on $[0, 1]$ and zero on $[2, \infty)$, and sample $\chi(j/N^{1/2})f_N(x_j)$. Let $\hat{f}_N$ denote its ℓ^2 normalization. Changing or omitting the cutoff in local calculations produces an error smaller than every inverse power of N .

Here is an explicit way to check and bound the quasimode residual. At a sample point the two edge coefficients are

$$b_-(x, \varepsilon) = \sqrt{\frac{1 - x\varepsilon^2 + 3\varepsilon^3}{1 + x\varepsilon^2 - \varepsilon^3}}, \quad b_+(x, \varepsilon) = \sqrt{\frac{1 - x\varepsilon^2 + 2\varepsilon^3}{1 + x\varepsilon^2}}.$$

Expand $b_-f_N(x-\varepsilon) + b_+f_N(x+\varepsilon) - \hat{\lambda}_N f_N(x)$ through degree five in ε , reducing every derivative using $F'' = 2(x+a)F$. Every coefficient vanishes. At the left boundary the omitted sample is zero because $F(0) = F_2(0) = F_3(0) = 0$. On the support of the cutoff, $x\varepsilon^2 = O(N^{-1/2})$; Taylor remainders are

bounded by $C\varepsilon^6$ times a fixed polynomial in x multiplying Airy decay. All such polynomial–Airy envelopes are square-integrable. Derivatives of the cutoff contribute only $\exp(-cN^{1/4})$ times a power of N . Since the unnormalized sampled norm is asymptotic to $\varepsilon^{-1/2}\|F\|_{L^2}$, this proves

$$\|(S_N - \widehat{\lambda}_N)\widehat{f}_N\| = O(\varepsilon^6). \quad (\text{F10})$$

The Airy gap (F8) identifies the nearby eigenvalue with λ_N ; projecting the residual on the complement of ψ_N then gives

$$\lambda_N = \widehat{\lambda}_N + O(\varepsilon^6), \quad \|\psi_N - \widehat{f}_N\| = O(\varepsilon^4). \quad (\text{F11})$$

The sign is fixed by the positive leading Airy profile. Several useful consequences follow without a separate quantitative localization theorem:

$$\begin{aligned} \|\psi_N - \psi_{N-1}\| &= O(N^{-1}), \\ \|(I - Q_N)\psi_N\| &= O(N^{-4/3}), \\ \psi_N(0) &= \sqrt{2} N^{-1/2}(1 + O(N^{-1/3})). \end{aligned} \quad (\text{F12})$$

For the first, the explicitly normalized samples in (F9) have ℓ^2 derivative $O(N^{-1})$ with respect to continuous N ; the two approximation errors in (F11) are smaller. For the second, use $1 - \sqrt{1-t} \leq t$ for $0 \leq t \leq 1$, the finite polynomial moments of the sampled Airy profile, and (F11). For the endpoint, the pointwise error is at most the ℓ^2 error, and

$$I := \int_0^\infty F(x)^2 dx = 2^{-1/3} \text{Ai}'(z)^2, \quad \frac{F'(0)}{\sqrt{I}} = \sqrt{2}.$$

A Riemann-sum estimate with an integrable derivative gives a relative $O(\varepsilon)$ normalization error; the endpoint Taylor error is $O(\varepsilon^2)$ and the normalized quasimode error divided by the leading endpoint is $O(\varepsilon^{5/2})$. These identities therefore verify the quantitative endpoint estimate in (F12). Standard Airy decay needed above follows, for example, directly from the decaying solution of the Airy ODE; only its polynomially weighted integrability and exponential tail are used.

5 Convergence of the scalar amplitude

Set

$$P_N = \prod_{i=1}^N \lambda_i, \quad u_N = v_N/P_N, \quad M_N = S_N Q_N / \lambda_N.$$

Then $u_N = M_N u_{N-1}$, $\|M_N\| \leq 1$, and $\|u_N\| \leq 1$. Let E_p be projection onto height parity p . The positive eigenvector has equal squared mass on the two parities: this follows by taking the scalar product of each half of $S_N \psi_N = \lambda_N \psi_N$ with that half. Therefore

$$g_N = \sqrt{2} E_{N \bmod 2} \psi_N, \quad h_N = \sqrt{2} E_{(N-1) \bmod 2} \psi_N$$

are unit vectors, with $S_N h_N = \lambda_N g_N$. By (F12),

$$\|h_N - g_{N-1}\| = O(N^{-1}), \quad \|(I - Q_N)h_N\| + \|(I - Q_N)g_{N-1}\| = O(N^{-4/3}). \quad (\text{F13})$$

The second estimate for g_{N-1} uses the first estimate only for its approximation by explicit Airy samples, rather than the crude $O(N^{-1})$ difference: direct use of (F11) at $N - 1$ gives the stated $O(N^{-4/3})$ estimate with Q_N . It also follows by observing that $Q_N \geq Q_{N-1}$ pointwise on the old support and treating its negligible eigenvector tail.

Decompose

$$u_N = A_N g_N + w_N, \quad w_N \perp g_N.$$

Write $\Pi_N = I - |g_N\rangle\langle g_N|$ on the occupied parity subspace. The singular gap from §3 gives, for every vector y on the preceding parity,

$$\|\Pi_N M_N y\| \leq (1 - cN^{-2/3})\|Q_N y\| \leq (1 - cN^{-2/3})\|y\|. \quad (\text{F14})$$

Indeed $\Pi_N S_N = S_N(I - |h_N\rangle\langle h_N|)$ on that parity; this projection removes the top singular direction, regardless of whether y is perpendicular to h_N . Also, (F13) implies

$$\|M_N g_{N-1} - g_N\| = O(N^{-1}).$$

Applying (F14) to w_{N-1} and the latter estimate to the scalar term yields

$$\|w_N\| \leq (1 - cN^{-2/3})\|w_{N-1}\| + CN^{-1}. \quad (\text{F15})$$

Iteration gives $\|w_N\| = O(N^{-1/3})$. One general estimate used here and later is

$$\sum_{k=N_0}^N k^{-r} \prod_{i=k+1}^N (1 - ci^{-2/3}) = O_r(N^{2/3-r}). \quad (\text{F16})$$

To prove it, split at $N/2$. The earlier half is exponentially small in $N^{1/3}$ times a polynomial; on the later half bound the product by $\exp(-c'(N - k)N^{-2/3})$ and sum a geometric series. A finite initial segment makes no difference.

The scalar equation is

$$A_N = t_N A_{N-1} + \langle k_N, w_{N-1} \rangle, \quad t_N = \langle Q_N h_N, g_{N-1} \rangle, \quad k_N = Q_N h_N - g_{N-1}. \quad (\text{F17})$$

The unit-vector identity $\langle h, g \rangle = 1 - \|h - g\|^2/2$ and (F13) show

$$t_N = 1 + O(N^{-4/3}), \quad \|k_N\| = O(N^{-1}). \quad (\text{F18})$$

Since $|A_N| \leq 1$, (F15), (F17), and (F18) give

$$A_N - A_{N-1} = O(N^{-4/3}).$$

Thus

$$A_N = A_\infty + O(N^{-1/3}). \quad (\text{F19})$$

Positivity gives $A_N \geq 0$ and hence $A_\infty \geq 0$; strict positivity is established in §7.

6 A boundary smoothing estimate

Norm tracking alone cannot read a coordinate whose leading size is $N^{-1/2}$. A short final block of the original positive kernel provides the missing smoothing.

Let $U(N, k) = M_N \cdots M_{k+1}$. For $0 \leq \ell \leq N^{2/3}$, positivity and $S_i Q_i \leq J$ give

$$\|e_0^T U(N, N - \ell)\|_2 \leq \frac{\|e_0^T J^\ell\|_2}{\lambda_{N-\ell+1} \cdots \lambda_N}.$$

The squared numerator is the number of nonnegative walks of length 2ℓ returning to zero:

$$\|e_0^T J^\ell\|_2^2 = (J^{2\ell})_{0,0} = \frac{1}{\ell+1} \binom{2\ell}{\ell} \leq C 4^\ell (\ell+1)^{-3/2}.$$

Equation (F11) implies

$$\frac{2^\ell}{\lambda_{N-\ell+1} \cdots \lambda_N} \leq \exp(C\ell N^{-2/3}) \leq C'$$

over the same range. Therefore

$$\|e_0^T U(N, N - \ell)\|_2 \leq C(\ell+1)^{-3/4}. \quad (\text{F20})$$

The exact error recurrence for w_N is

$$w_N = M_N w_{N-1} + f_N^{\text{err}}, \quad f_N^{\text{err}} = A_{N-1} M_N g_{N-1} - A_N g_N, \quad \|f_N^{\text{err}}\| = O(N^{-1}).$$

Use Duhamel over $L = \lfloor N^{2/3} \rfloor$ steps. Since all times in this block are comparable with N , (F15) and (F20) imply

$$\begin{aligned} |w_N(0)| &\leq CL^{-3/4} \|w_{N-L}\| + CN^{-1} \sum_{\ell=0}^{L-1} (\ell+1)^{-3/4} \\ &= O(N^{-5/6}). \end{aligned} \quad (\text{F21})$$

For even N , (F12) gives $g_N(0) = 2N^{-1/2}(1 + O(N^{-1/3}))$.

7 The limiting constant and its strict positivity

From (F11),

$$\log \lambda_N = \log 2 + aN^{-2/3} + \frac{3}{2}N^{-1} + O(N^{-4/3}).$$

Summation gives

$$P_N = \kappa 2^N e^{3aN^{1/3}} N^{3/2} (1 + O(N^{-1/3})), \quad \kappa > 0. \quad (\text{F22})$$

If desired, a convergent expression is

$$\log \kappa = a\zeta(2/3) + \frac{3}{2}\gamma_E + \sum_{N=1}^{\infty} \left[\log(\lambda_N/2) - aN^{-2/3} - \frac{3}{2N} \right]. \quad (\text{F23})$$

Here γ_E is Euler's constant, not the counting amplitude. This is an existence formula; it does not by itself give efficient certified numerical evaluation.

Equations (F19)–(F22), for even N , now give

$$d_{N,0} = 2\kappa A_{\infty} 2^N e^{3aN^{1/3}} N + O\left(2^N e^{3aN^{1/3}} N^{2/3}\right). \quad (\text{F24})$$

The known lower half of the Elvey Price–Fang–Wallner Θ theorem states that $d_{2n,0}$ is bounded below by a positive constant times $4^n e^{3zn^{1/3}} n$ for all sufficiently large n . If $A_{\infty} = 0$, (F24) contradicts that bound. Hence $A_{\infty} > 0$ and

$$\gamma = 4\kappa A_{\infty} > 0. \quad (\text{F25})$$

Taking $N = 2n$ in (F24) proves (F1) with $M = 0$. This completes the amplitude-convergence argument, using the explicit prior lower-bound theorem and the elementary Airy facts used above.

[write, 5 October 2026, batch 102] The prior lower bound is no longer needed here. Part V's Lemma 72.1 (Report 100) proves the relaxed amplitude positive from the exact recurrence alone, by restarting the positive contractive evolution from its Perron vector, and Remark 66.1 shows that (F24) and that lemma give $A_{\infty} = \gamma_r/(4\kappa) > 0$ and $\gamma = \gamma_r$. With this substitution the relaxed theorem rests on no external analytic input (Report 100 alone already proves it so), and the Elvey Price–Fang–Wallner relaxed Θ theorem itself becomes a consequence.

8 Formal profiles at every finite order

Work now with the original recurrence (F4), rather than the symmetric gauge. Seek

$$Z_{N,j} = H_N \sum_{k=0}^K \varepsilon^k G_k(x_j), \quad G_0 = F, \quad G_k(0) = G'_k(0) = 0 \quad (k \geq 1),$$

with

$$\frac{H_N}{H_{N-1}} \sim 2 \left(1 + \sum_{m \geq 2} s_m \varepsilon^m \right). \quad (\text{F26})$$

At the previous time the two arguments and powers are

$$y_{\pm} = (x \pm \varepsilon)(1 - \varepsilon^3)^{-1/3}, \quad \varepsilon_{N-1}^k = \varepsilon^k(1 - \varepsilon^3)^{-k/3},$$

and the downward coefficient is

$$\frac{1 - x\varepsilon^2 + 3\varepsilon^3}{1 + x\varepsilon^2 - \varepsilon^3}.$$

After the leading orders, the coefficient at degree m has the form

$$\mathcal{L}G_{m-2} - 2s_m F + R_m(x)F + S_m(x)F', \quad \mathcal{L} = \frac{d^2}{dx^2} - 2(x + a), \quad (\text{F27})$$

where R_m, S_m are known polynomials over $\mathbb{Q}[a]$, formed from earlier orders. In particular $s_2 = a$. This form follows by Taylor expansion: the new profile first appears through the second-order spatial operator; its lower, zeroth-order terms cancel between the two sides of (F4). Time-change terms involving this new profile enter only at later degree.

To solve $\mathcal{L}(PF + QF') = RF + SF'$, compute

$$\mathcal{L}(PF + QF') = (P'' + 4(x + a)Q' + 2Q)F + (2P' + Q'')F'.$$

Eliminating $P' = (S - Q'')/2$ reduces the problem to

$$-\frac{1}{2}Q''' + 4(x + a)Q' + 2Q = R - \frac{1}{2}S'. \quad (\text{F28})$$

On polynomials of degree at most d , the operator on the left is upper triangular by degree with nonzero diagonal entries $4j + 2$, $0 \leq j \leq d$; hence it is invertible. The new scalar s_m adds $2s_m$ to the forcing R , thus adds exactly s_m to the constant term of Q . Choose it uniquely to enforce $Q(0) = 0$, which is the boundary condition. Integrating P' leaves one free constant; choose it so $P(0) + Q'(0) = 0$, which enforces the derivative normalization. This proves existence and uniqueness at every formal order within the stated normalization, not merely consistency of finitely many guessed coefficients.

[**write, 2 October 2026**] This polynomial–Airy identity, with its triangular reduction, is used by every Part of this report and by the tree-child network report; see “One polynomial–Airy identity” in Part 0.

The first scalar coefficients are

$$s_2 = a, \quad s_3 = \frac{4}{3}, \quad s_4 = \frac{29a^2}{270}, \quad s_5 = -\frac{23a}{81}, \quad s_6 = \frac{74385 - 8783a^3}{85050}. \quad (\text{F29})$$

For example,

$$G_1 = -\frac{2}{3}x^2F, \quad G_2 = \frac{2(2a + 15x^4 + 111x)}{135}F + \frac{x(-8a + 51x)}{270}F'.$$

The amplitude factor can be chosen as a genuine smooth function of N with finite logarithmic expansion

$$\log H_N = N \log 2 + 3aN^{1/3} + \frac{4}{3} \log N + \sum_{k=1}^J h_k N^{-k/3}. \quad (\text{F30})$$

Expanding its exact one-step ratio matches any prescribed finite number of coefficients in (F26). At degree ε^{k+3} , the new h_k has coefficient $-k/3$, so the construction is again triangular and nonsingular. Crucially the leading multiplicative constant in (F30) is fixed to one for every truncation. This avoids an artificial truncation-dependent amplitude.

9 Approximate solutions in the normalized norm

Fix any $p > 1$. Take enough orders in §8, multiply the sampled profile by the cutoff $\chi(j/N^{1/2})$, and restrict to the occupied parity. Call the result Z_N . Choose the truncation so that its local formal residual starts at a power $N^{-p'}$ with $p' \geq p$. Define

$$z_N = D_N^{-1} Z_N / P_N.$$

Then

$$\|z_N\| = O(1), \quad r_N := z_N - M_N z_{N-1} = O(N^{-p}) \quad \text{in } \ell^2. \quad (\text{F31})$$

Here are the needed global bounds. On $j \leq 2N^{1/2}$, the product formula for D_N gives $D_N^{-1} \leq C$ uniformly: $-\log D_N(j) \leq Cj^2/N \leq C'$. On any fixed scaled window $D_N^{-1} = 1 + O(N^{-1/3}(1+x)^2)$. Taylor remainders in the rational coefficients and the time dilation are bounded by N^{-p} times a fixed polynomial–Airy envelope on the cutoff support. Cutoff errors decay like $\exp(-cN^{1/4})$ times a polynomial, also after multiplication by D_N^{-1} . Finally,

$$H_N / P_N \sim \kappa^{-1} N^{-1/6},$$

while a parity-restricted sampled profile has norm asymptotic to $N^{1/6} \sqrt{I/2}$. These bounds prove (F31) by direct summation. The same reasoning proves

$$\langle g_N, z_N \rangle \longrightarrow q_0 := \kappa^{-1} \sqrt{I/2} > 0. \quad (\text{F32})$$

Indeed, after the natural sample normalization, both profiles converge to the same positive Airy function. Neither (F31) nor (F32) needs a pointwise estimate on the unknown solution.

10 A scalar bootstrap to every finite order

Set

$$C = A_\infty / q_0, \quad e_N = u_N - C z_N.$$

Then $e_N = M_N e_{N-1} - C r_N$ and the ground projection tends to zero. Decompose

$$e_N = \alpha_N g_N + W_N, \quad W_N \perp g_N, \quad \alpha_N \longrightarrow 0.$$

The exact same estimates (F14), (F17), and (F18) now give

$$\begin{aligned} \|W_N\| &\leq (1 - cN^{-2/3})\|W_{N-1}\| + CN^{-1}|\alpha_{N-1}| + CN^{-p}, \\ |\alpha_N - \alpha_{N-1}| &\leq CN^{-4/3}|\alpha_{N-1}| + CN^{-1}\|W_{N-1}\| + CN^{-p}. \end{aligned} \tag{F33}$$

Suppose initially that $\alpha_N = O(N^{-q})$, where $q \geq 0$. Formula (F16) gives

$$\|W_N\| = O(N^{-q-1/3} + N^{2/3-p}).$$

Substitute this into the second inequality of (F33). Its right-hand side is summable because $p > 1$; summing from $N + 1$ to infinity and using the zero limiting projection gives

$$|\alpha_N| = O(N^{-q-1/3} + N^{1-p}). \tag{F34}$$

The smaller tail $O(N^{2/3-p})$ has been absorbed by $O(N^{1-p})$. Start with $q = 0$, which holds since u_N, z_N are bounded. Finitely many repetitions of (F34) reach

$$\alpha_N = O(N^{1-p}), \quad \|W_N\| = O(N^{2/3-p}), \quad \|e_N\| = O(N^{1-p}). \tag{F35}$$

There is no boundary-at-infinity assumption beyond the already proved, explicit limit (F32) and the choice of C . No inverse operator with a small gap is silently assumed uniformly bounded.

Apply Duhamel directly to the inhomogeneous e recurrence on the final block $L = \lfloor N^{2/3} \rfloor$. By (F20), (F31), and (F35),

$$\begin{aligned} |e_N(0)| &\leq CL^{-3/4}N^{1-p} + CN^{-p} \sum_{\ell=0}^{L-1} (\ell+1)^{-3/4} \\ &= O(N^{1/2-p}) + O(N^{1/6-p}) = O(N^{1/2-p}). \end{aligned} \tag{F36}$$

The main endpoint is of size $N^{-1/2}$, so the relative endpoint error is $O(N^{1-p})$. For the remainder at order M , take $p \geq 1 + (M+1)/3$ and enough formal terms. Since p can be taken as large as desired, every prescribed finite algebraic order of the explicit approximation transfers to the exact sequence. All truncations have the same q_0 , hence the same C and the same $\gamma = 2CF'(0) = 4\kappa A_\infty$. This proves (F1).

11 Coefficient extraction and computational checks

Solving the scalar ratio matching (F30) using (F29) gives

$$h_1 = 53a^2/45, \quad h_2 = 79a/27, \quad h_3 = 143/210 - 5216a^3/42525.$$

The normalized endpoint profile is

$$\frac{\sum_k \varepsilon^k G_k(\varepsilon)}{F'(0)\varepsilon} = 1 + \frac{a}{3}\varepsilon^2 + \frac{4}{3}\varepsilon^3 + O(\varepsilon^4).$$

Consequently the logarithmic endpoint corrections in time N are

$$\frac{53a^2}{45}N^{-1/3} + \frac{88a}{27}N^{-2/3} + \left(\frac{141}{70} - \frac{5216a^3}{42525}\right)N^{-1}.$$

Substituting $N = 2n$ and $a = 2^{-1/3}z$ yields (F2), and exponentiation gives (F3).

12 The rational coefficient field

The recursion immediately puts every coefficient at time N in $\mathbb{Q}[a]$, where $a = 2^{-1/3}z$. The substitution $N = 2n$ initially gives the weaker field $\mathbb{Q}(2^{1/3})[z]$. To justify the stronger field $\mathbb{Q}[z]$ asserted in the theorem, one needs the following mod-3 grading.

Let ϕ_a be the formal power-series solution of

$$\phi_a'' = 2(x+a)\phi_a, \quad \phi_a(0) = 0, \quad \phi_a'(0) = 1.$$

Every coefficient of ϕ_a belongs to $\mathbb{Q}[a]$. At the special Airy parameter used in the proof, this is exactly $F(x)/F'(0)$. For any primitive cube root of unity ω , uniqueness of the initial-value solution gives

$$\phi_{\omega a}(\omega x) = \omega \phi_a(x), \quad \phi_{\omega a}'(\omega x) = \phi_a'(x). \quad (\text{G1})$$

Indeed, the left side of the first identity, as a function of x , obeys the same ODE as ϕ_a because $\omega^3 = 1$, and its derivative at zero is ω .

Divide every formal profile in §8 by $F'(0)$ and denote the resulting formal sum by

$$\Phi(a, \varepsilon, x) = \sum_{k \geq 0} \varepsilon^k \mathcal{G}_k(a, x), \quad \mathcal{G}_0 = \phi_a.$$

Its normalization is $\Phi(a, \varepsilon, 0) = 0$ and $\partial_x \Phi(a, \varepsilon, 0) = 1$. Denote the half-ratio of the amplitude factors by

$$R(a, \varepsilon) = 1 + \sum_{m \geq 2} s_m(a) \varepsilon^m.$$

In the formal recurrence, the rational coefficient depends only on $x\varepsilon^2$ and ε^3 ; the dilation depends only on ε^3 ; and the shifted arguments are $(x \pm \varepsilon)(1 - \varepsilon^3)^{-1/3}$. Consequently the transformation

$$(a, \varepsilon, x) \longmapsto (\omega a, \omega \varepsilon, \omega x)$$

preserves the recurrence. The transformed profile

$$\tilde{\Phi}(a, \varepsilon, x) = \omega^{-1} \Phi(\omega a, \omega \varepsilon, \omega x)$$

has the same leading solution and the same boundary and derivative normalizations, by (G1). The order-by-order polynomial recursion in §8 is unique under exactly these normalizations. Therefore

$$R(\omega a, \omega \varepsilon) = R(a, \varepsilon), \quad \Phi(\omega a, \omega \varepsilon, \omega x) = \omega \Phi(a, \varepsilon, x). \quad (\text{G2})$$

In particular,

$$s_m(\omega a) = \omega^{-m} s_m(a). \quad (\text{G3})$$

The uniquely determined coefficients $h_k(a)$ of the finite logarithmic amplitude satisfy the same grading,

$$h_k(\omega a) = \omega^{-k} h_k(a). \quad (\text{G4})$$

One way to see (G4) without manipulating a logarithm branch is to use its ratio equation. The leading ratio contributions involve $3a/\varepsilon$, multiplied by a series in ε^3 , and $\log(1 - \varepsilon^3)$. They are invariant under the transformation. The term involving $h_k \varepsilon^k$ first enters at degree $k+3$ with nonzero coefficient $-k/3$, so the unique recursive solution must have the indicated grading.

The normalized endpoint factor is

$$E(a, \varepsilon) = \frac{\Phi(a, \varepsilon, \varepsilon)}{\varepsilon}.$$

Equation (G2) shows $E(\omega a, \omega \varepsilon) = E(a, \varepsilon)$. The constant term of E is one. Thus every coefficient of both E and $\log E$ has the same mod-3 grading. Combining it with (G4), the time- N logarithmic and relative correction coefficients $b_k(a)$ obey

$$b_k(\omega a) = \omega^{-k} b_k(a). \quad (\text{G5})$$

Since b_k is a polynomial with rational coefficients, every nonzero monomial a^r in it satisfies

$$r + k \equiv 0 \pmod{3}.$$

Finally its contribution after changing variables is

$$a^r N^{-k/3} = 2^{-(r+k)/3} z^r n^{-k/3}.$$

The exponent $(r+k)/3$ is an integer, so the factor of two is rational. This proves that every forward correction coefficient at time n lies in $\mathbb{Q}[z]$, rather than merely in $\mathbb{Q}(2^{1/3})[z]$.

13 Asymptotic inversion and discrete thresholds

[**write, 2 October 2026**] The Lambert anchor, the explicit corrections (I10)–(I11) and the brackets (I18) are instances of theorems `p0:thm:lambert-core`, `p0:thm:perturbed-inversion` and `p0:thm:staircase` of the repository’s transseries volume; see Part 0, “Instances of results elsewhere in the repository”. No novelty is claimed for the inversion mechanics.

13.1 Input, normalization, and scope

Let $a_n = A082161(n)$, and let $z = a_1^{\text{Ai}} = -2.338107410459767\dots$ denote the first negative zero of the Airy function. To avoid confusing this zero with the first sequence term, this report uses z throughout. The forward theorem provides a fixed $\gamma > 0$ satisfying

$$\log a_n = \log \gamma + \log \Gamma(n+1) + n \log 4 + 3zn^{1/3} + \log n + Bn^{-1/3} + Dn^{-2/3} + L_3n^{-1} + O(n^{-4/3}), \quad (\text{I1})$$

where

$$B = \frac{53z^2}{90}, \quad D = \frac{44z}{27}, \quad L_3 = \frac{141}{140} - \frac{1304z^3}{42525}. \quad (\text{I2})$$

The error in (I1) is initially only an assertion at integer n . All conclusions about an actual sequence below use only that assertion, not unproved differentiability of its remainder.

There are three distinct inversion objects:

1. the inverse of an explicit smooth asymptotic model;
2. the inverse of a chosen interpolation of the integer sequence;
3. the discrete threshold $N(X) = \min\{n : a_n \geq X\}$.

They have different error statements. In particular, no real-valued asymptotic expansion, by itself, proves a single unqualified floor/ceiling rule at thresholds arbitrarily close to sequence values.

13.2 The factorial contributes a crucial logarithm

Stirling's formula gives

$$\log \Gamma(n+1) = n \log n - n + \frac{1}{2} \log n + \frac{1}{2} \log(2\pi) + \frac{1}{12n} + O(n^{-3}).$$

Consequently, set

$$A = 3z, \quad p = \frac{3}{2}, \quad C = \log(\gamma\sqrt{2\pi}), \quad E = L_3 + \frac{1}{12} = \frac{229}{210} - \frac{1304z^3}{42525}, \quad (\text{I3})$$

and define

$$h(n) = n \log(4n/e) + An^{1/3} + p \log n + C + Bn^{-1/3} + Dn^{-2/3} + En^{-1}. \quad (\text{I4})$$

Then (I1) is $\log a_n = h(n) + O(n^{-4/3})$. In particular, the coefficient of $\log n$ is **3/2**, not 1. Numerically,

$$\begin{aligned} A &= -7.014322231379301\dots, & B &= 3.219306132565383\dots, \\ D &= -3.810249113341843\dots, & E &= 1.482422558319535\dots \end{aligned}$$

An equally useful model retains the factorial exactly:

$$h_\Gamma(n) = \log \gamma + \log \Gamma(n+1) + n \log 4 + An^{1/3} + \log n + Bn^{-1/3} + Dn^{-2/3} + L_3/n. \quad (\text{I5})$$

Its difference from (I4) is $-1/(360n^3) + O(n^{-5})$, below the precision needed here. Its inverse differs from the inverse of (I4) by $O(n^{-3}/\log n)$.

For (I4),

$$h'(n) = \log(4n) + \frac{A}{3}n^{-2/3} + \frac{p}{n} - \frac{B}{3}n^{-4/3} - \frac{2D}{3}n^{-5/3} - En^{-2} \sim \log(4n) > 0, \quad (\text{I6})$$

and $h''(n) \sim 1/n$. Thus h has a unique increasing large-argument inverse.

13.3 Lambert W anchor and the first Airy correction

Put

$$T = \log X, \quad x = \frac{T}{W_0(4T/e)}, \quad s = \log(4x) = 1 + W_0(4T/e), \quad q = p \log x + C. \quad (\text{I7})$$

Here W_0 is the principal real Lambert W branch and X is sufficiently large. The identity

$$x \log(4x/e) = T$$

holds exactly. The leading correction needed to undo the negative Airy term is positive:

$$h^{-1}(T) = x - \frac{A}{s}x^{1/3} - \frac{q}{s} + O\left(\frac{x^{-1/3}}{s}\right). \quad (\text{I8})$$

Equivalently, the displayed first terms are

$$x - \frac{3z}{\log(4x)}x^{1/3} - \frac{3}{2} + \frac{\frac{3}{2}\log 4 - \log(\gamma\sqrt{2\pi})}{\log(4x)}. \quad (\text{I9})$$

This shows the hierarchy clearly: the Airy correction grows like $x^{1/3}/\log x$, the factorial/prefactor combination gives a constant shift $-3/2$, and the unknown amplitude first enters at order $1/\log x$. Equation (I8), including its stated error scale for suitable interpolations or the discrete bracketing interpretation below, already follows from the weaker core hypothesis

$$a_n = \gamma n! 4^n e^{3zn^{1/3}} n(1 + O(n^{-1/3})).$$

It does not require the all-orders construction.

An alternative anchor explicitly absorbs the unknown amplitude: set $y = T - C$, $x = y/W_0(4y/e)$, and $q = p \log x$. Every formula below remains valid with this replacement. The unshifted version (I7) makes the dependence on γ easier to see.

13.4 Explicit inverse through the third fractional order

Define

$$\begin{aligned}
u &= -\frac{A}{s}, \\
v &= -\frac{q}{s}, \\
w &= -\frac{B}{s} + \frac{A^2}{3s^2} - \frac{A^2}{2s^3}, \\
r &= -\frac{D}{s} + \frac{A(p+q/3)}{s^2} - \frac{Aq}{s^3}, \\
k &= -\frac{E}{s} + \frac{pq}{s^2} - \frac{AB+q^2/2}{s^3} + \frac{A^3}{3s^4} - \frac{A^3}{2s^5}.
\end{aligned} \tag{I10}$$

Then the inverse of the explicit smooth model (I4) satisfies

$$\boxed{h^{-1}(T) = x + ux^{1/3} + v + wx^{-1/3} + rx^{-2/3} + kx^{-1} + O\left(\frac{x^{-4/3}}{\log(4x)}\right)}. \tag{I11}$$

The error order in (I11) is also the uncertainty imposed by the forward remainder in (I1). It is therefore the natural stopping point using the three displayed forward logarithmic coefficients.

13.4.1 Derivation

Write $t = x^{-1/3}$ and

$$\delta = u/t + v + wt + rt^2 + kt^3, \quad n = x + \delta.$$

The leading function $f(n) = n \log(4n/e)$ has expansion

$$f(x + \delta) - f(x) = s\delta + \frac{\delta^2}{2x} - \frac{\delta^3}{6x^2} + \cdots.$$

For the perturbations, use

$$\begin{aligned}
A(x + \delta)^{1/3} &= A/t + (A/3)ut + (A/3)vt^2 + ((A/3)w - (A/9)u^2)t^3 + O(t^4), \\
p \log(x + \delta) + C &= q + put^2 + pvt^3 + O(t^4), \\
B(x + \delta)^{-1/3} &= Bt - (B/3)ut^3 + O(t^4), \\
D(x + \delta)^{-2/3} &= Dt^2 + O(t^4), \\
E(x + \delta)^{-1} &= Et^3 + O(t^5).
\end{aligned}$$

The required equations, in order, are

$$\begin{aligned}
su + A &= 0, \\
sv + q &= 0, \\
sw + u^2/2 + Au/3 + B &= 0, \\
sr + uv + Av/3 + pu + D &= 0, \\
sk + v^2/2 + (u + A/3)w - u^3/6 - Au^2/9 + pv - Bu/3 + E &= 0.
\end{aligned} \tag{I12}$$

Solving (I12) gives (I10); the substantial cancellation in k is exact. SymPy verifies every coefficient from t^{-1} through t^3 is identically zero with A, B, D, E, p, q, s left symbolic.

Although q depends on x , this calculation is a Taylor expansion about the fixed anchor x . Its contribution to derivatives is already included in $p \log(x + \delta)$; one must not separately differentiate q within (I12).

For the remainder, $q = O(s)$, $u = O(1/s)$, $v = O(1)$, and $w, r, k = O(1/s)$. Hence $\delta = O(x^{1/3}/s)$ and $\delta/x = O(x^{-2/3}/s)$. All omitted terms in the residual $h(x + \delta) - T$ are $O(x^{-4/3})$. Since the local derivative is asymptotic to s , the mean-value theorem yields (I11). No derivative bound on the unknown sequence remainder was used.

13.5 Two Newton steps are a simpler practical inverse

For computation it is usually preferable to avoid manually evaluating (I10). Starting from the Lambert anchor $x_0 = x$, take

$$x_{j+1} = x_j - \frac{h(x_j) - T}{h'(x_j)}, \quad j = 0, 1. \quad (\text{I13})$$

If $n_* = h^{-1}(T)$, then

$$\begin{aligned} x_0 - n_* &= O(x^{1/3}/s), \\ x_1 - n_* &= O(x^{-1/3}/s^3), \\ x_2 - n_* &= O(x^{-5/3}/s^7). \end{aligned} \quad (\text{I14})$$

Indeed, throughout the relevant interval, h' is comparable to s and $h'' = O(1/x)$, so one Newton step squares the error and divides it by xs . The last bound is smaller than the forward uncertainty $O(x^{-4/3}/s)$. Thus two Newton steps solve the available asymptotic inverse more accurately than its supplied coefficients justify.

The same procedure works using h_Γ from (I5), with

$$h'_\Gamma(n) = \psi(n+1) + \log 4 + (A/3)n^{-2/3} + n^{-1} - (B/3)n^{-4/3} - (2D/3)n^{-5/3} - L_3 n^{-2},$$

where ψ is the digamma function. Work with $T = \log X$ rather than forming very large X .

More generally, if a smooth model has forward logarithmic error $O(n^{-\rho})$, its inverse uncertainty is $O(x^{-\rho}/s)$. Additional Newton steps can make model inversion error smaller than any prescribed finite power. Starting from (I7), after j steps the bound is

$$x_j - n_* = O\left(x^{1-2^{j+1}/3} s^{1-2^{j+1}}\right).$$

13.6 What “smooth inverse” means for an integer sequence

The asymptotic assertion (I1) alone does not specify a canonical interpolation between integers. It is incorrect simply to differentiate its $O(n^{-4/3})$ remainder as though a differentiable real-variable asymptotic theorem had been proved.

There is, however, a precise conditional statement:

- If an eventually increasing interpolation $L(t)$ satisfies $L(n) = \log a_n$ and $L(t) = h(t) + O(t^{-4/3})$ uniformly for real t , then

$$L^{-1}(T) = h^{-1}(T) + O(x^{-4/3}/s). \quad (\text{I15})$$

The argument is value bracketing using monotonicity and the slope of h ; no derivative estimate for the remainder of L is necessary.

- Such smooth increasing interpolations **exist** under (I1). For example, put $\varepsilon_n = \log a_n - h(n)$ and choose a fixed smooth bump ϕ , supported in $(-1/3, 1/3)$, with $\phi(0) = 1$. For large real t , set

$$L(t) = h(t) + \sum_n \varepsilon_n \phi(t - n).$$

The supports are disjoint, the perturbation and its first derivative are both $O(t^{-4/3})$, and therefore $L'(t) > 0$ eventually. This supplies a smooth interpolation meeting (I15), but it is an arbitrary choice, not an intrinsic continuous extension of the combinatorial sequence.

- The common interpolation that is piecewise linear in $\log a_n$ has a weaker generic comparison. Because $h''(t) \sim 1/t$, its interpolation error relative to h is $O(1/t)$, larger than the forward remainder. Its inverse satisfies only

$$L_{\text{linear}}^{-1}(T) = h^{-1}(T) + O(1/(x \log x)). \quad (\text{I16})$$

- An arbitrary monotone interpolation can differ by order 1 in its inverse. Integer samples alone do not force the sharper real-valued estimate.

Every increasing interpolation through the exact sequence nevertheless gives the same discrete threshold after taking a ceiling, with $N(X) = \text{ceil}(L^{-1}(\log X))$ for large X . Thus interpolation distinctions do not obstruct the direct discrete bracketing below.

13.7 Discrete threshold bracketing and the rounding barrier

The forward hypothesis itself implies eventual strict monotonicity. In fact,

$$\log a_{n+1} - \log a_n = \log(4n) + O(n^{-2/3}) > 0$$

under (I1). Under just the weaker core hypothesis, the error can instead be written $O(n^{-1/3})$, still sufficient. As a_n is unbounded, finitely many earlier terms do not affect $N(X)$ for sufficiently large X .

Suppose that an actual forward bound is available:

$$|\log a_n - h(n)| \leq K n^{-4/3} \quad (n \geq n_0). \quad (\text{I17})$$

Let $r_* = h^{-1}(\log X)$ and $K_* = \max(K, 1)$. For all sufficiently large r_* , one valid, deliberately conservative choice is

$$\eta = \frac{8K_* r_*^{-4/3}}{\log(4r_*)}.$$

Then

$$\boxed{\lceil r_* - \eta \rceil \leq N(X) \leq \lceil r_* + \eta \rceil.} \quad (\text{I18})$$

To verify (I18), test the integer immediately below $\text{ceil}(r_* - \eta)$ and the integer $\text{ceil}(r_* + \eta)$. They lie within a bounded distance of r_* ; there h' is at least half $\log(4r_*)$, while the allowed forward error is at most twice $Kr_*^{-4/3}$. The displacement η leaves a strict margin on either side. Eventual monotonicity gives the result.

If the explicit approximation (I11) is used in place of r_* , enlarge η by a bound for its error, which has the same order. If the two-step Newton approximation is used, its much smaller error from (I14) can be added instead.

Important qualifications:

1. Once $2\eta < 1$, (I18) leaves either one integer or two adjacent candidates.
2. If $\text{dist}(r_*, \mathbb{Z}) > \eta$, both ceilings agree and $N(X) = \text{ceil}(r_*)$.
3. If r_* is within η of an integer, the next omitted term or exact sequence evaluation can decide which adjacent candidate is correct. The supplied asymptotic does not always decide it.
4. One may write $N(X) = \text{ceil}(r_* + O(r_*^{-4/3}/\log r_*))$ only in the existential/bracketing sense of (I18). It does **not** mean $N(X) = \text{ceil}(r_*) + o(1)$ uniformly in X .
5. An unquantified O bound proves the existence of (I18) with some constant and starting point. It does not yield a numerically certified enclosure for a particular finite X . That requires effective forward constants, a certified amplitude, and numerical error control.

Under the weaker core theorem, the corresponding bracket radius is $O(r_*^{-1/3}/\log r_*)$, still tending to zero. Therefore the core theorem alone already localizes the threshold to at most two adjacent integers asymptotically.

13.8 Unknown gamma and numerical amplitude estimates

The constant γ is a fixed positive parameter, not a free constant that can be silently dropped after leading order. Changing γ to $\tilde{\gamma}$ shifts the smooth inverse, to first order, by

$$-\frac{\log(\tilde{\gamma}/\gamma)}{\log(4n)}. \quad (\text{I19})$$

A bounded multiplicative amplitude uncertainty thus causes an inverse uncertainty $O(1/\log n)$. This is $o(1)$, but is much larger than the algebraic error in (I11), and can still change the discrete answer at near-integer thresholds.

If certified information gives

$$|\log \gamma - \log \tilde{\gamma}| \leq \varepsilon,$$

the forward error budget for a model using $\tilde{\gamma}$ becomes $\varepsilon + Kn^{-4/3}$. The threshold radius is then

$$O\left(\frac{\varepsilon + r_*^{-4/3}}{\log(4r_*)}\right).$$

An uncertified fitted decimal for γ does not supply such a bound. In particular, the supplied approximate value 166.95063854245 is suitable for illustrative computations, not for asserting rigorous threshold decisions.

13.8.1 An amplitude-free locator and its existing provenance

If γ is unknown, one can still use

$$r_0 = x - \frac{3z}{\log(4x)}x^{1/3} - \frac{3}{2}, \quad x = \frac{\log X}{W_0(4 \log X/e)}.$$

The real inverse uncertainty is $O(1/\log x) = o(1)$. In discrete form, for some fixed constant K and sufficiently large X ,

$$\left\lceil r_0 - \frac{K}{\log x} \right\rceil \leq N(X) \leq \left\lceil r_0 + \frac{K}{\log x} \right\rceil.$$

This coarse result already follows from the published Elvey Price–Fang–Wallner Θ theorem: its logarithm has an $O(1)$ remainder, which is divided by the slope $\sim \log x$ on inversion. It should therefore not be presented as a consequence unique to the amplitude proof. The amplitude convergence identifies the fixed $\log \gamma$ term; the all-orders theorem provides the finer algebraic inverse corrections. As before, existence of an unspecified constant K is not a numerical finite-threshold certificate.

14 Reproducible computation and numerical limitations

The accompanying source archive is relocatable and includes every script needed to generate finite-order coefficients, verify the algebraic identities used above, and reproduce the reported numerical evidence. It requires Python with SymPy, NumPy, SciPy, and mpmath. The exact versions used in the delivered replay are recorded in its environment report.

The main replay command is `python replay.py --max-n 20000`. It verifies the SHA-256 manifest of its inputs before executing the checks, writes every result beneath `output/`, and reports any failure. The coefficient command `python coefficients.py --order 8 --output output/coefficients.json` constructs all scalar and profile terms through the requested order. Raising the order invokes the same finite polynomial algorithm, with no lookup table or fitted coefficients.

The coefficient construction is a finite, exact procedure:

1. Represent a profile by a polynomial pair (P, Q) for $PF + QF'$ and differentiate using $F'' = 2(x + a)F$
2. Expand the rational recurrence, the two shifted arguments, and the previous-time powers to the next required degree in ε
3. Collect the known forcing in the two components F and F'

4. Solve the triangular polynomial equation (F28), choose the scalar so that $Q(0) = 0$, and integrate for P with $P(0) + Q'(0) = 0$
5. Verify the resulting residual coefficient, both boundary conditions, and the mod-three grading exactly
6. Match the logarithmic one-step ratio and expand the endpoint factor to obtain forward logarithmic and relative coefficients

The archive also checks the original triangular recurrence against the transformed recurrence with exact rational arithmetic, verifies the squared gauge identities, verifies the finite frozen quasimode, and checks the inverse cancellations with symbolic parameters. Its numerical tests use the full finite support of the forward recurrence, rather than an unquantified height cutoff.

At $n = 20000$, the normalized raw ratio is 187.01754328008607. Dividing it by the first three relative corrections gives 166.95063854245257; removing the first three logarithmic corrections instead gives 166.95152887373968. These values are consistent with the empirical amplitude 166.95208957 reported by Elvey Price, Fang, and Wallner. They are not certified enclosures, and the last displayed digits should not be interpreted as established digits of γ .

The exact symbolic checks establish the displayed cancellation identities. Floating-point checks establish only reproducibility of numerical evidence. Neither kind of computation replaces the analytic compactness, stability, or endpoint arguments above. The theorem's constant is the convergent expression (F25), not a fitted decimal. In particular, certified finite-threshold decisions require effective forward constants, a certified amplitude interval, and controlled numerical evaluation.

15 Further research questions

Certified amplitude evaluation. Can the convergent spectral-product and projection representation be supplemented with explicit tail constants to give an efficient interval enclosure of γ ? The present proof gives existence and algebraic rates, but its compactness-based spectral gap has not been made numerically effective. A constructive lower gap, certified Airy residuals, and explicit endpoint bounds would make this a concrete certification problem.

[write, 5 October 2026, batch 102] Still open. Part V adds a second convergent representation, $\gamma = \gamma_r = \sqrt{2} a_\infty C_\lambda$ with a product for C_λ and a telescoping series for a_∞ whose terms are $O(n^{-4/3})$ (Theorem 73.1), and closer uncertified evidence, 166.9520924395 at $n = 3000$; see Question V.1.

Compacted binary trees. The corresponding compacted-tree recurrence has a subtraction term. Its established leading scale has the same stretched exponential and a different power of n . Extending the present positive-kernel smoothing argument requires an appropriate positive lift or a different stability estimate. The relaxed-tree proof does not automatically establish the compacted amplitude.

[write, 2 October 2026] Answered by the same batch: Part II (source 15) proves the compacted amplitude and every fixed order, Theorem 16.1. It does not use a positive lift: it treats the subtraction as a globally bounded signed delay and cancels its scalar drift. The relaxed-tree proof indeed does not by itself establish the compacted amplitude, as said above.

Higher arity. The exact gauge and parity decomposition exploit a nearest-neighbor binary recurrence. For relaxed k -ary structures, one can ask whether a comparable slowly varying transfer

operator admits a coercive continuum limit and summable ground-mode drift. The specific gauge, gap, and boundary exponent would need to be derived anew.

[**write, 2 October 2026**] Answered for every fixed arity $k \geq 3$ by the same batch: Part III (sources 18 and 66) proves positive amplitudes and every fixed order for relaxed k -ary trees, and for compacted trees and automata, starting from the Θ theorem of Ghosh Dastidar and Wallner [3]. As this question anticipates, the gauge, gap and boundary exponent are derived anew: a fixed weighted norm matching forward and adjoint boundary harmonics replaces the exact Jacobi symmetrization. Uniformity in k remains open (Section 37).

Beyond algebraic orders. The construction proves a Poincaré expansion at every fixed order. It gives no optimal-truncation theorem, no Gevrey estimate on the coefficient growth, and no exponentially small correction. Those questions require quantitative control uniform in the expansion order, together with an analysis of higher Airy modes and possible additional asymptotic sectors.

Intrinsic interpolation and inverse certification. The combinatorial sequence has no canonical real interpolation supplied by its recurrence. The threshold brackets in §13 avoid assuming one. It would be useful to combine interval evaluation of nearby exact terms with the asymptotic locator, yielding a certified algorithm that resolves the occasional two-candidate ambiguity.

Closing note of source 62. The proposed convergence argument and its scope are distinguished from the existing result in §1. No assertion of an exhaustive global priority search or of peer-reviewed acceptance is made.

Part II

Compacted binary trees

Source. Batch-77 manuscript 15, archive `compacted-binary-trees-reproducibility.zip`, main file `article/compacted-binary-trees.tex` (1,274 lines; 21-page PDF), titled “Asymptotic amplitude and all orders for compacted binary trees”, dated 2 October 2026, empty author line. Printed in full with label prefix `air:cb::`; its appendix is Appendix A of this report. Its “companion relaxed-tree theorem” is Part I.

Abstract of the source.

The number of compacted binary trees with n internal nodes is known to have order $n!4^n \exp(3zn^{1/3})n^{3/4}$, where z is the largest negative Airy zero. We prove convergence to a positive limiting amplitude and a Poincaré expansion to every finite order in $n^{-1/3}$. The exact counting recurrence is a signed three-step perturbation of the relaxed-tree Jacobi evolution. A global $O(N^{-1})$ delay bound, cancellation of its scalar drift, and a finite-memory tracking argument establish amplitude convergence. Polynomial–Airy profiles and a zero-limit bootstrap then transfer arbitrary formal orders to the exact sequence without assuming positivity of the signed propagator or an endpoint heat-kernel estimate. The published lower bound supplies strict positivity. We also give exact positive finite-state and renewal models, explain their symmetry obstructions, derive explicit Lambert W inverse corrections, and state the precise discrete threshold interpretation. A self-contained appendix proves the shared Jacobi estimates. Numerical amplitude values are evidence, not certified enclosures.

16 Statement and scope

Let $c_{n,m}$ be defined for $0 \leq m \leq n$ by

$$c_{n,0} = 1, \quad c_{n,m} = c_{n,m-1} + (m+1)c_{n-1,m} - (m-1)c_{n-2,m-1}, \quad (16.1)$$

with invalid cells zero, and put $c_n = c_{n,n}$. The mathematical index counts internal nodes, so $c_0 = c_1 = 1$ and

$$(c_n)_{n=0}^8 = (1, 1, 3, 15, 111, 1119, 14487, 230943, 4395855).$$

This is the compacted binary-tree sequence associated with OEIS A254789 [9]. Proposition 2.11 and Definition 2.12 of Elvey Price, Fang and Wallner [1] give the recurrence and the equinumerous decorated-path model used below. Their Theorem 1.1 proves

$$c_n = \Theta\left(n!4^n e^{3zn^{1/3}} n^{3/4}\right), \quad (16.2)$$

where

$$z = -2.338107410459767\dots, \quad \text{Ai}(z) = 0, \quad a = 2^{-1/3}z, \quad F(x) = \text{Ai}(z + 2^{1/3}x).$$

Here z is the largest real Airy zero and $F(x) > 0$ for $x > 0$. The asymptotic equivalent and full expansion are conjectural in Section 3.4 of that paper; they do not follow merely from its two-sided estimate.

Theorem 16.1. *There is a constant $\gamma_c > 0$ such that, for every fixed nonnegative integer M ,*

$$c_n = \gamma_c n!4^n e^{3zn^{1/3}} n^{3/4} \left(1 + \sum_{k=1}^M \mathbf{c}_k n^{-k/3} + O_M(n^{-(M+1)/3})\right), \quad \mathbf{c}_k \in \mathbb{Q}[z]. \quad (16.3)$$

The amplitude is the convergent spectral-projection expression in (24.1). The first three logarithmic corrections are

$$\begin{aligned} \log \frac{c_n}{\gamma_c n!4^n e^{3zn^{1/3}} n^{3/4}} &= \frac{53z^2}{90} n^{-1/3} + \frac{271z}{216} n^{-2/3} \\ &+ \left(\frac{393}{1120} - \frac{1304z^3}{42525} \right) n^{-1} + O(n^{-4/3}). \end{aligned} \quad (16.4)$$

This is a Poincaré expansion: the error statement applies at each fixed finite order. It asserts neither an exponentially improved transseries nor a canonical real interpolation of the integer sequence. No elementary evaluation or rigorous numerical enclosure of γ_c is claimed. The fitted decimal 173.12670485 in [1] is consistent with the computations accompanying this article, but is not part of the theorem.

[write, 5 October 2026, batch 102] Part V (Report 101) proves this theorem a second time, in even time with three-lag signed memory, and continues (16.4): with z for its a , $b_4 = -2053z^2/9720$, $b_5 = z(2869124z^3 - 329492565)/1377810000$ and $b_6 = -(320744z^3 + 10924065)/132269760$, (89.1). With Corollary 66.3, the normalized ratio $c_n/(4nc_{n-1})$ holds through n^{-3} (the display after (76.3)). Its six-correction value at $n = 3000$ is 173.1267049959 (uncertified).

The argument is an independently checked research proof. This description is not a claim of publication or peer review. A bounded search of primary literature and the ProveIt repository found no prior resolution of this unrestricted compacted amplitude problem; that search is not a guarantee of priority or exhaustive novelty. The later k -ary work [3] and the bounded-right-height problem [6] have different stated conclusions or restrictions.

[write, 2 October 2026] The repository search (recorded in `15-compacted-src-literature-and-provenance.md` at commit `89daa64f`) was right when made. The repository's A254789 material is now this report, which the same batch delivered together with the relaxed, fixed-arity and automaton companions.

17 Exact positive path models

17.1 Four states and the signed recurrence

A path from $(0,0)$ to (n,m) stays in $0 \leq m \leq n$, using horizontal steps H and vertical steps V . An H step at ordinate m has one of the labels $1, \dots, m+1$. The forbidden pattern is HHV in which the two horizontal labels are equal and greater than one. A trailing equal pair is allowed until a V step follows it.

At a path prefix define four states:

- Z : no trailing H , so the prefix is empty or ends in V ;
- A : the last H label is 1;
- B : the last H label exceeds 1, without a trailing equal non-1 pair;
- C : the trailing two H labels are equal and exceed 1.

Write $\mathbf{F}(n, m) = (Z, A, B, C)^\top$. The exact matrices are

$$H_m^{(4)} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ m & m & m-1 & m-1 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \quad V^{(4)} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (17.1)$$

with

$$\mathbf{F}(n, m) = H_m^{(4)} \mathbf{F}(n-1, m) + V^{(4)} \mathbf{F}(n, m-1), \quad \mathbf{F}(0, 0) = (1, 0, 0, 0)^\top.$$

Invalid cells are zero. At $m=0$, states B, C are unreachable. Replacing $m-1$ by $\max(m-1, 0)$ gives globally nonnegative formal matrices without changing the reachable evolution.

To verify the transitions, a new label 1 always enters A . After a prefix in Z or A , each of the m non-1 labels enters B . After a prefix in B or C , precisely one label enters C and the other $m-1$ enter B . A V accepts exactly Z, A, B . In particular, a prefix in C is not discarded prematurely: another H replaces the trailing pair.

If $c = Z + A + B + C$, these transitions imply

$$\begin{aligned} A(n, m) &= c_{n-1, m}, & B(n, m) + C(n, m) &= m c_{n-1, m}, \\ C(n, m) &= m c_{n-2, m}, & Z(n, m) &= c_{n, m-1} - (m-1) c_{n-2, m-1} \end{aligned} \quad (17.2)$$

for the last identity when $m \geq 1$. Adding them gives (16.1). At $n = m > 0$ the last step cannot be H , so $c_n = Z(n, n)$.

17.2 A three-state local kernel

States Z and A have identical outgoing transitions and can be merged into $E = Z + A$. In the state order (E, B, C) ,

$$H_m = \begin{pmatrix} 1 & 1 & 1 \\ m & m-1 & m-1 \\ 0 & 1 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{e} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{t} = (1, 1, 1). \quad (17.3)$$

The same convention at $m = 0$ applies. For every fixed $m > 0$, the reachability matrix $(\mathbf{e}, H_m \mathbf{e}, H_m^2 \mathbf{e})$ has determinant m^2 , while the matrix with rows $\mathbf{t}, \mathbf{t}V, \mathbf{t}VH_m$ has determinant -1 . Their Hankel product has determinant $-m^2$. Thus this fixed-altitude linear suffix-counting representation has dimension three; a two-state linear representation of that same problem is impossible. This statement does not exclude coordinate-dependent or nonlocal encodings.

With physical time $N = n + m$, height $j = n - m$, and factorial normalization

$$\mathbf{f}_N(j) = \frac{\mathbf{F}_3((N+j)/2, (N-j)/2)}{((N+j)/2)!},$$

the positive nearest-neighbor kernel is

$$\mathbf{f}_N(j) = \frac{2}{N+j} H_{(N-j)/2} \mathbf{f}_{N-1}(j-1) + V \mathbf{f}_{N-1}(j+1). \quad (17.4)$$

Only valid parity and support are used; its initial vector is $\mathbf{f}_0(0) = \mathbf{e}$.

17.3 A scalar positive renewal representation

A horizontal run of length k at altitude m , completed by a V , has admissible weight

$$w_m(0) = 1, \quad w_m(1) = m + 1, \quad w_m(k) = (m^2 + m + 1)(m + 1)^{k-2} \quad (k \geq 2). \quad (17.5)$$

Indeed among the $(m + 1)^k$ labelings, exactly $m(m + 1)^{k-2}$ have a forbidden final pair; earlier pairs do not immediately precede a V . Let $a(n, m)$ count paths ending in V , with $a(0, 0) = 1$ and all other $a(n, 0) = 0$. Then

$$\begin{aligned} a(n, m+1) &= \sum_{k=0}^{n-m} w_m(k) a(n-k, m), & n \geq m+1, \\ c_{n,m} &= \sum_{k=0}^{n-m} (m+1)^k a(n-k, m), & a(n, n) = c_n. \end{aligned} \quad (17.6)$$

The completed-run generating function is

$$W_m(t) = \sum_{k \geq 0} w_m(k) t^k = \frac{1 - mt^2}{1 - (m+1)t}.$$

At the vertical-step clock, distance $h = n - m$ changes to $h' = h + k - 1$. For $b_m(h) = a(m + h, m)/(m + h)!$, the transfer weight from h to h' is

$$w_m(k) \frac{(m + h)!}{(m + h + k)!}, \quad k \geq 0, \quad h' \geq 0.$$

This kernel is scalar and positive, but has arbitrarily large upward jumps and downward jumps of size at most one.

17.4 Why the positive models are not symmetric Jacobi reductions

Several exact obstructions delimit direct symmetrization:

1. For $m > 0$, $\text{rank } H_m = 2$ and

$$H_m^2 = (m + 1, m^2, m)^\top (1, 1, 1).$$

Its characteristic polynomial is $\lambda^2(\lambda - m - 1)$, but the zero eigenspace has dimension one. Its nontrivial zero Jordan block prevents similarity to a real symmetric matrix, even under a general invertible change of basis. In particular, no positive-definite common metric makes both H_m and V self-adjoint.

2. In the height/state transfer there is a positive directed edge $B_h \rightarrow E_{h+1}$ and no reverse V edge. Positive diagonal conjugation preserves such zero patterns. The renewal kernel similarly has a $+2$ jump without a -2 jump.
3. Freeze $m = 1$, use horizontal coefficient $1/2$ and vertical coefficient 1 , and truncate heights to $0, 1, 2, 3$. The resulting 12 by 12 matrix has characteristic polynomial

$$\frac{\lambda^6}{4}(4\lambda^6 - 12\lambda^4 + 6\lambda^2 - 1).$$

The cubic in λ^2 has discriminant -432 , so this frozen model has nonreal eigenvalues and cannot be similar to a self-adjoint matrix. This is an obstruction for the stated frozen example, not every time-dependent compacted transfer.

4. A self-adjoint moment representation $w(k) = \langle v, S^k v \rangle$ would have nonnegative first Hankel minor, whereas

$$w_m(0)w_m(2) - w_m(1)^2 = -m < 0 \quad (m > 0).$$

These observations do not rule out dilations or asymptotic effective self-adjoint descriptions. They motivate the signed-delay route below, which requires neither a symmetric lift nor a new singular gap for the three-state kernel.

18 The shared Jacobi evolution

All vectors below are zero-padded elements of $\ell^2(\mathbb{N}_0)$. For $N \geq 1$ define

$$D_N(j)^2 = \frac{\Gamma(N + 2)\Gamma(N + 1)}{\Gamma(N - j + 2)\Gamma(N + j + 1)}, \quad 0 \leq j \leq N + 1, \quad D_N(0) = 1. \quad (18.1)$$

Let S_N be the symmetric Jacobi matrix on $0, \dots, N+1$ with edge weights

$$b_{N,j} = \sqrt{\frac{N-j+2}{N+j}} \quad (1 \leq j \leq N+1), \quad Q_N(j) = \sqrt{1 - \frac{j(j-1)}{N(N+1)}}. \quad (18.2)$$

Both are zero outside that finite interval. Let λ_N be the largest eigenvalue of S_N , and ψ_N its positive unit eigenvector. If E_p projects onto height parity p , put

$$\begin{aligned} g_N &= \sqrt{2} E_{N \bmod 2} \psi_N, & h_N &= \sqrt{2} E_{(N-1) \bmod 2} \psi_N, \\ P_N &= \prod_{i=1}^N \lambda_i, & M_N &= \frac{S_N Q_N}{\lambda_N}, & \Pi_N &= I - |g_N\rangle\langle g_N| \end{aligned} \quad (18.3)$$

on the occupied parity subspace. Both g_N and h_N are unit vectors.

Lemma 18.1. *There are positive constants c, κ such that, for all sufficiently large N and any y on the preceding parity,*

$$\begin{aligned} \|M_N\| &\leq 1, & \|\Pi_N M_N y\| &\leq (1 - cN^{-2/3}) \|y\|, \\ \|M_N g_{N-1} - g_N\| &= O(N^{-1}), & \|M_N^* g_N - g_{N-1}\| &= O(N^{-1}), \\ t_N := \langle g_N, M_N g_{N-1} \rangle &= 1 + O(N^{-4/3}), \\ \lambda_N &= 2 + 2aN^{-2/3} + 3N^{-1} + O(N^{-4/3}), \\ P_N &= \kappa 2^N e^{3aN^{1/3}} N^{3/2} (1 + O(N^{-1/3})). \end{aligned} \quad (18.4)$$

Moreover ψ_N is within $O(N^{-4/3})$ in ℓ^2 of the normalization of cutoff samples of

$$\begin{aligned} F(x) + N^{-2/3} \left[\frac{2(-a+2x)}{15} F(x) + \frac{x(2a+x)}{15} F'(x) \right] \\ + N^{-1} \frac{F(x) - xF'(x)}{3}, \quad x = (j+1)N^{-1/3}, \end{aligned} \quad (18.5)$$

with cutoff $\chi(j/N^{1/2})$. These normalized samples have time derivative $O(N^{-1})$, one-site shift difference $O(N^{-1/3})$, and bounded polynomial moments on the Airy scale.

A complete proof is given in Appendix A. The gap in (18.4) is a parity-restricted singular-value gap. It is not a second spectral-radius estimate on the whole space, where $-\lambda_N$ is also an eigenvalue.

The relaxed counts $r_{n,m}$ satisfy

$$r_{n,0} = 1, \quad r_{n,m} = r_{n,m-1} + (m+1)r_{n-1,m},$$

with invalid cells zero. They count the same decorated paths without the forbidden HHV restriction. Thus $0 \leq c_{n,m} \leq r_{n,m}$. The common gauge turns the relaxed recurrence into $v_N^{(r)} = S_N Q_N v_{N-1}^{(r)}$; consequently $\|v_N^{(r)} / P_N\| \leq 1$. Only this domination and the explicit Jacobi estimates, not an automatic extension of a relaxed amplitude theorem, will be used to prove Theorem 16.1.

19 A globally bounded signed delay

Set

$$e_{N,j} = \frac{c_{(N+j)/2, (N-j)/2}}{((N+j)/2)!}, \quad v_N(j) = \frac{e_{N,j}}{D_N(j)}$$

on $0 \leq j \leq N$, $j \equiv N \pmod{2}$, and zero otherwise. The scalar form of (17.4), equivalently direct substitution in (16.1), is

$$e_{N,j} = \frac{N-j+2}{N+j} e_{N-1,j-1} + e_{N-1,j+1} - \frac{2(N-j-2)}{(N+j)(N+j-2)} e_{N-3,j-1}. \quad (19.1)$$

The delayed term is used only for a valid source $(N-3, j-1)$; it is zero at $j=0$ and outside that support. In particular its coefficient at $j=N-2$ is zero. This convention excludes the spurious negative rational coefficient at the nonphysical source for $j=N$.

For all sufficiently large N , the exact gauge recurrence is

$$v_N = S_N Q_N v_{N-1} - L_N v_{N-3}, \quad (L_N f)(j) = \ell_{N,j} f(j-1), \quad (19.2)$$

where on $1 \leq j \leq N-2$

$$\ell_{N,j} = \frac{2(N-j-2)}{(N+j)(N+j-2)} \frac{D_{N-3}(j-1)}{D_N(j)}, \quad (19.3)$$

and it is zero elsewhere. Cancellation of factorials gives

$$\left(\frac{D_{N-3}(j-1)}{D_N(j)} \right)^2 = \frac{(N-j+1)(N-j)(N+j)(N+j-1)(N+j-2)(N+j-3)}{(N+1)N^2(N-1)^2(N-2)}. \quad (19.4)$$

For $N \geq 6$ this ratio is uniformly bounded on the entire physical support; the rational prefactor in (19.3) is $O(N^{-1})$. Hence

$$\|L_N\| \leq CN^{-1}. \quad (19.5)$$

This is a global operator bound, not an estimate limited to the Airy window.

For $1 \leq j \leq 2N^{1/2}$ on the physical support, expanding (19.3)–(19.4) gives

$$\ell_{N,j} = \frac{2}{N} + O\left(\frac{j+1}{N^2}\right).$$

Let U be the unilateral upward shift: $(Uf)(0) = 0$ and $(Uf)(j) = f(j-1)$. The finite profile in Lemma 18.1, its Airy moments, and (19.5) imply

$$\left\| L_N g_{N-3} - \frac{2}{N} U g_{N-3} \right\| = O(N^{-5/3}), \quad \|U g_{N-3} - g_N\| = O(N^{-1/3}). \quad (19.6)$$

For the first estimate the typical height is $O(N^{1/3})$, and the $O(N^{-4/3})$ quasimode error is multiplied by $O(N^{-1})$. Cutoff tails are exponentially small. The second estimate uses a one-site sample shift and an $O(N^{-1})$ time change. The shifted parity is exactly the parity occupied at time N .

20 Cancellation of the scalar drift

Put

$$R_N = P_N N^{-1/4}, \quad u_N = v_N / R_N, \quad (20.1)$$

and begin the following recurrences at any fixed sufficiently large N . Then

$$u_N = \eta_N M_N u_{N-1} - B_N u_{N-3}, \quad \eta_N = \left(\frac{N}{N-1} \right)^{1/4}, \quad B_N = \frac{R_{N-3}}{R_N} L_N. \quad (20.2)$$

Since $R_{N-3}/R_N = 1/8 + O(N^{-2/3})$, Lemma 18.1 and (19.6) yield

$$\begin{aligned}\eta_N &= 1 + \frac{1}{4N} + O(N^{-2}), \quad \|B_N\| = O(N^{-1}), \\ \|B_N g_{N-3} - \frac{1}{4N} g_N\| &= O(N^{-4/3}), \\ b_N := \langle g_N, B_N g_{N-3} \rangle &= \frac{1}{4N} + O(N^{-4/3}), \\ \eta_N t_N - 1 - b_N &= O(N^{-4/3}).\end{aligned}\tag{20.3}$$

The last line cancels the nonsummable order- N^{-1} drift. It explains the additional factor $N^{-1/4}$ relative to the relaxed spectral normalization.

Lemma 20.1 (Finite-memory bounds). *Suppose nonnegative x_N satisfy, eventually,*

$$x_N \leq (1 - cN^{-2/3})x_{N-1} + CN^{-1}x_{N-3} + CN^{-r}.\tag{20.4}$$

For every fixed real r , $x_N = O(N^{2/3-r})$. A sum of power forcings has the sum of the corresponding bounds. If a polynomially bounded sequence δ_N satisfies

$$|\delta_N| \leq CN^{-1}(|\delta_{N-1}| + |\delta_{N-2}|) + O(N^{-r}),\tag{20.5}$$

then $\delta_N = O(N^{-r})$.

Proof. For the first statement use the barrier $KN^{2/3-r}$. Its difference from the two homogeneous terms on the right of (20.4) is

$$KcN^{-r} + O(KN^{2/3-r-1}).$$

The error is smaller by $N^{-1/3}$ regardless of the sign of $2/3 - r$. Choose a sufficiently large starting time and then K large enough to dominate both the forcing and the finite initial triple. Induction proves the estimate. Adding barriers handles sums of powers. For the second statement, repeatedly substitute the polynomial bound in (20.5); each substitution reduces its exponent by one until the forcing dominates. Equivalently the barrier KN^{-r} gives an eventual induction, since C/N tends to zero. $\square$

21 Convergence of the scalar amplitude

The positive gauge preserves $0 \leq v_N \leq v_N^{(r)}$ entrywise. The relaxed contraction therefore gives the initial bound

$$\|u_N\| \leq N^{1/4}.\tag{21.1}$$

Decompose

$$u_N = A_N g_N + w_N, \quad w_N \perp g_N.$$

Applying Lemma 18.1 to the transverse part of (20.2), and bounding the delayed whole vector using (21.1), gives

$$\|w_N\| \leq (1 - c'N^{-2/3})\|w_{N-1}\| + O(N^{-3/4}).$$

The factor $\eta_N = 1 + O(N^{-1})$ is absorbed in the gap. Lemma 20.1 then gives

$$\|w_N\| = O(N^{-1/12}).\tag{21.2}$$

No cancellation or positivity of the delayed evolution is needed here.

Projecting the recurrence gives

$$A_N = \eta_N t_N A_{N-1} - b_N A_{N-3} + \xi_N, \quad |\xi_N| \leq CN^{-1}(\|w_{N-1}\| + \|w_{N-3}\|). \quad (21.3)$$

The one-step cross term uses $\|M_N^* g_N - g_{N-1}\| = O(N^{-1})$ and orthogonality; the delayed cross term uses only $\|B_N\| = O(N^{-1})$. Set $\delta_N = A_N - A_{N-1}$. Then

$$\delta_N = b_N(\delta_{N-1} + \delta_{N-2}) + (\eta_N t_N - 1 - b_N)A_{N-1} + \xi_N. \quad (21.4)$$

Equations (20.3), (21.1) and (21.2) give (20.5) with $r = 13/12$. Thus $\delta_N = O(N^{-13/12})$ is summable, and A_N has a finite limit A_∞ . Since u_N and g_N are nonnegative, $A_\infty \geq 0$.

Now A_N is bounded and $w_N \rightarrow 0$, so u_N is bounded. Repeating the transverse and scalar estimates improves them to

$$A_N = A_\infty + O(N^{-1/3}), \quad \|w_N\| = O(N^{-1/3}), \quad \|u_N\| = O(1). \quad (21.5)$$

There is not yet an endpoint conclusion: the normalized ground endpoint itself is only of order $N^{-1/2}$.

22 Formal profiles at every finite order

Let $\varepsilon = N^{-1/3}$ and $x = (j+1)\varepsilon$. Seek an approximate solution to the ungauged recurrence in the form

$$Z_{N,j} = H_N \Phi_N(x), \quad \Phi_N(x) = \sum_{k=0}^K \varepsilon^k G_k(x), \quad G_0 = F, \quad (22.1)$$

with $G_k(0) = G'_k(0) = 0$ for $k \geq 1$, and formal ratio

$$\frac{H_N}{H_{N-1}} \sim 2 \left(1 + \sum_{m \geq 2} s_m \varepsilon^m \right). \quad (22.2)$$

The one-step arguments are $(x \pm \varepsilon)(1 - \varepsilon^3)^{-1/3}$, and the delayed argument is $(x - \varepsilon)(1 - 3\varepsilon^3)^{-1/3}$. The horizontal and delayed coefficients in (19.1) become

$$\begin{aligned} \mathcal{A}(x, \varepsilon) &= \frac{1 - x\varepsilon^2 + 3\varepsilon^3}{1 + x\varepsilon^2 - \varepsilon^3}, \\ \beta(x, \varepsilon) &= \frac{2\varepsilon^3(1 - x\varepsilon^2 - \varepsilon^3)}{(1 + x\varepsilon^2 - \varepsilon^3)(1 + x\varepsilon^2 - 3\varepsilon^3)}. \end{aligned} \quad (22.3)$$

After division by H_{N-1} , the delayed term has the extra factor

$$\frac{H_{N-3}}{H_{N-1}} = \left(\frac{H_{N-1}}{H_{N-2}} \frac{H_{N-2}}{H_{N-3}} \right)^{-1},$$

whose leading constant is $1/4$. As β starts at degree three, neither the new profile nor the new scalar in those previous ratios changes the leading Airy inversion at the current degree.

At formal degree m , the equation is

$$\mathcal{L}G_{m-2} - 2s_m F + R_m(x)F + S_m(x)F' = 0, \quad \mathcal{L} = \partial_x^2 - 2(x + a), \quad (22.4)$$

where $R_m, S_m \in \mathbb{Q}[a][x]$ are already known; in particular $s_2 = a$. For a polynomial pair P, Q ,

$$\mathcal{L}(PF + QF') = (P'' + 4(x + a)Q' + 2Q)F + (2P' + Q'')F'.$$

Thus solving $\mathcal{L}G = RF + SF'$ reduces to

$$-\frac{1}{2}Q''' + 4(x + a)Q' + 2Q = R - \frac{1}{2}S'. \quad (22.5)$$

On polynomials of degree at most d , its matrix is triangular with nonzero diagonal $4j + 2$, $0 \leq j \leq d$. It is therefore invertible over $\mathbb{Q}[a]$. Changing s_m shifts the constant term of Q by s_m , so it uniquely enforces $Q(0) = 0$. The integration constant of P then enforces $P(0) + Q'(0) = 0$. Since $F(0) = 0$ and $F'(0) \neq 0$, these are exactly the value and derivative normalizations of G . This proves existence and uniqueness at every finite formal order.

The initial scalar coefficients are

$$\begin{aligned} s_2 &= a, & s_3 &= \frac{13}{12}, & s_4 &= \frac{29a^2}{270}, \\ s_5 &= -\frac{11a}{324}, & s_6 &= \frac{2054835 - 140528a^3}{1360800}. \end{aligned} \quad (22.6)$$

A genuine finite logarithmic amplitude may consequently be chosen as

$$\log H_N = N \log 2 + 3aN^{1/3} + \frac{13}{12} \log N + \sum_{k=1}^J h_k N^{-k/3}. \quad (22.7)$$

At degree $k + 3$ in the one-step ratio, the new h_k has coefficient $-k/3$, so any finite list of ratio coefficients can be matched uniquely. For every truncation the leading multiplicative constant in H_N is one. Initially,

$$h_1 = \frac{53a^2}{45}, \quad h_2 = \frac{235a}{108}, \quad h_3 = -\frac{641}{1680} - \frac{5216a^3}{42525}. \quad (22.8)$$

23 Global residuals and the zero-limit bootstrap

23.1 Approximate solutions in the normalized norm

Fix any $p > 1$. Take sufficiently many terms of (22.1) and (22.7), multiply by the cutoff $\chi(j/N^{1/2})$, restrict to occupied parity, and define

$$z_N = D_N^{-1} Z_N / R_N.$$

Then

$$\|z_N\| = O(1), \quad r_N := z_N - \eta_N M_N z_{N-1} + B_N z_{N-3}, \quad \|r_N\| = O(N^{-p}), \quad (23.1)$$

and

$$\langle g_N, z_N \rangle \longrightarrow q_0 = \kappa^{-1} \sqrt{I/2} > 0, \quad I = \int_0^\infty F(x)^2 dx. \quad (23.2)$$

Here is the required global justification. On $j \leq 2N^{1/2}$ the factorial product for D_N has $\log D_N^{-1} = O(j^2/N)$, so D_N^{-1} is uniformly bounded there. Taylor remainders in (22.3), the time dilations and the finite amplitude ratios are bounded by an arbitrarily prescribed power of $N^{-1/3}$ times a fixed polynomial–Airy envelope. All such envelopes are square-integrable. The coefficient denominators

stay uniformly away from zero. Cutoff derivative errors, and differences between one- and three-step cutoff supports, are exponentially small in $N^{1/4}$, up to powers of N . The bounded gauge preserves this property. At the left boundary all fictitious samples vanish exactly, because every $G_k(0) = 0$; the upper moving boundary lies outside the cutoff for large N . These facts give the norm residual after summation.

For the normalization, $H_N/R_N \sim \kappa^{-1}N^{-1/6}$, and a parity-restricted Airy sample has norm asymptotic to $N^{1/6}\sqrt{I/2}$. Its direction tends to g_N , proving (23.2). Only the leading asymptotic of P_N has been used: the original approximate recurrence is divided by the exact R_N , so an all-orders expansion of the relaxed spectral product is unnecessary.

23.2 A bootstrap with vanishing limiting projection

Set

$$C_* = A_\infty/q_0, \quad E_N = u_N - C_*z_N = \alpha_N g_N + W_N, \quad W_N \perp g_N. \quad (23.3)$$

Equations (21.5) and (23.2) imply $\alpha_N \rightarrow 0$. The error recurrence is

$$E_N = \eta_N M_N E_{N-1} - B_N E_{N-3} - C_* r_N.$$

The same projection estimates as before imply

$$\begin{aligned} \|W_N\| &\leq (1 - cN^{-2/3}) \|W_{N-1}\| + CN^{-1} \|W_{N-3}\| \\ &\quad + CN^{-1}(|\alpha_{N-1}| + |\alpha_{N-3}|) + CN^{-p}, \\ |\delta_N| &\leq CN^{-1}(|\delta_{N-1}| + |\delta_{N-2}|) + CN^{-4/3}|\alpha_{N-1}| \\ &\quad + CN^{-1}(\|W_{N-1}\| + \|W_{N-3}\|) + CN^{-p}, \end{aligned} \quad (23.4)$$

where $\delta_N = \alpha_N - \alpha_{N-1}$. If $\alpha_N = O(N^{-q})$, $q \geq 0$, Lemma 20.1 gives

$$\|W_N\| = O(N^{-q-1/3} + N^{2/3-p}), \quad \delta_N = O(N^{-q-4/3} + N^{-p}). \quad (23.5)$$

The latter is summable. Crucially, summing the differences to infinity uses the known zero limit and yields

$$\alpha_N = O(N^{-q-1/3} + N^{1-p}). \quad (23.6)$$

Begin with $q = 0$, since u_N and z_N are bounded, and iterate finitely many times. The version of Lemma 20.1 for sums of powers handles the residual floor throughout. Thus

$$\|E_N\| = O(N^{1-p}). \quad (23.7)$$

Neither positivity nor contractivity of the signed delayed propagator is assumed. The argument uses the gap for the unperturbed Jacobi evolution, smallness of the delay, and the zero limiting scalar projection.

24 The endpoint and the common positive amplitude

Point evaluation has norm one on ℓ^2 , so $|E_N(0)| \leq \|E_N\| = O(N^{1-p})$. For even N , the leading normalized approximate endpoint is of order $N^{-1/2}$, because

$$H_N/R_N \sim \kappa^{-1}N^{-1/6}, \quad \Phi_N(\varepsilon) \sim F'(0)\varepsilon.$$

We may choose p arbitrarily large. For the relative remainder in (16.3), it is enough to take $p \geq 3/2 + (M+1)/3$ and retain enough formal terms. Thus (23.7) supplies every fixed finite relative endpoint order; no endpoint smoothing estimate for the signed kernel is needed.

If $A_\infty = 0$, then $C_* = 0$. Equation (23.7) with $p > 3/2$ would imply $u_N(0) = o(N^{-1/2})$. On the other hand, the known lower bound (16.2), divided by R_{2n} , implies $u_{2n}(0) \geq cN^{-1/2}$ for large even $N = 2n$. This contradiction proves $A_\infty > 0$.

Since $D_N(0) = 1$ and $c_n = n!e_{2n,0}$, conversion from N to $2n$ gives

$$\boxed{\gamma_c = 2^{3/4}C_*F'(0) = 2^{7/4}\kappa A_\infty > 0, \quad A_\infty = \lim_{N \rightarrow \infty} \left\langle g_N, v_N / (P_N N^{-1/4}) \right\rangle.} \quad (24.1)$$

We used the Airy identity $F'(0)/\sqrt{I} = \sqrt{2}$, proved in the appendix. Every truncation has the same q_0 and the same leading normalization of H_N . Hence C_* and γ_c are independent of the expansion order. This completes the analytic part of Theorem 16.1.

25 Explicit coefficients and their rational field

The normalized endpoint factor begins

$$\frac{\Phi_N(\varepsilon)}{F'(0)\varepsilon} = 1 + \frac{a}{3}\varepsilon^2 + \frac{13}{12}\varepsilon^3 + O(\varepsilon^4).$$

Combining this with (22.8), then setting $N = 2n$, gives (16.4). Exponentiation gives

$$\begin{aligned} \mathfrak{c}_1 &= \frac{53z^2}{90}, \\ \mathfrak{c}_2 &= \frac{z(2809z^3 + 20325)}{16200}, \\ \mathfrak{c}_3 &= \frac{2084278z^6 + 43365690z^3 + 21487275}{61236000}. \end{aligned} \quad (25.1)$$

These coefficients were independently checked by a direct polynomial linear-system solve, separate from the triangular recursion.

The formal construction first gives coefficients in $\mathbb{Q}[a]$. To obtain $\mathbb{Q}[z]$ after the change of time scale, let ω be a primitive cube root of unity. For general complex a , define ϕ_a by the entire initial-value problem

$$\phi_a''(x) = 2(x+a)\phi_a(x), \quad \phi_a(0) = 0, \quad \phi_a'(0) = 1.$$

At the physical parameter $a = 2^{-1/3}z$ it equals $F/F'(0)$. At other parameters it need not be a decaying Airy function. Uniqueness of this initial-value problem gives

$$\phi_{\omega a}(\omega x) = \omega \phi_a(x).$$

The full formal recurrence, including its delayed dilation and both previous ratios, is invariant under $(a, \varepsilon, x) \mapsto (\omega a, \omega \varepsilon, \omega x)$. Uniqueness of (22.4)–(22.5) therefore implies

$$s_m(\omega a) = \omega^{-m} s_m(a).$$

The finite logarithm and normalized endpoint inherit the same grading: every monomial a^r at order k has $r+k$ divisible by three. Therefore

$$a^r N^{-k/3} = 2^{-(r+k)/3} z^r n^{-k/3}$$

has a rational power of two as its multiplier. All final relative coefficients lie in $\mathbb{Q}[z]$, as asserted.

26 Inverse asymptotics and discrete thresholds

[write, 2 October 2026] The anchor (26.3), the corrections (26.5) and the bracket (26.9) are instances of theorems `p0:thm:lambert-core`, `p0:thm:perturbed-inversion` and `p0:thm:staircase` of the repository's transseries volume (Part 0). No novelty is claimed for the inversion mechanics.

26.1 The smooth logarithmic model

Define

$$\begin{aligned} A &= 3z, & p &= \frac{5}{4}, & C &= \log(\gamma_c \sqrt{2\pi}), \\ B &= \frac{53z^2}{90}, & D &= \frac{271z}{216}, & E &= \frac{1459}{3360} - \frac{1304z^3}{42525}. \end{aligned} \quad (26.1)$$

Here the p in this section is the fixed exponent $5/4$, rather than the freely chosen residual order above. Set

$$h(t) = t \log(4t/e) + At^{1/3} + p \log t + C + Bt^{-1/3} + Dt^{-2/3} + Et^{-1}. \quad (26.2)$$

Stirling's formula and Theorem 16.1 give $\log c_n = h(n) + O(n^{-4/3})$. The factorial contributes both $\frac{1}{2} \log n$ and $1/(12n)$; omitting either changes the inverse. As $t \rightarrow \infty$, $h'(t) \sim \log(4t) > 0$ and $h''(t) \sim 1/t$.

26.2 The Lambert anchor and explicit corrections

For $T = \log Y$ put

$$x = \frac{T}{W_0(4T/e)}, \quad s = \log(4x), \quad q = p \log x + C. \quad (26.3)$$

Then $x \log(4x/e) = T$. With

$$\begin{aligned} u &= -\frac{A}{s}, & v &= -\frac{q}{s}, \\ w &= -\frac{B}{s} + \frac{A^2}{3s^2} - \frac{A^2}{2s^3}, \\ r &= -\frac{D}{s} + \frac{A(p+q/3)}{s^2} - \frac{Aq}{s^3}, \\ k &= -\frac{E}{s} + \frac{pq}{s^2} - \frac{AB+q^2/2}{s^3} + \frac{A^3}{3s^4} - \frac{A^3}{2s^5}, \end{aligned} \quad (26.4)$$

the inverse of the explicit smooth model satisfies

$$h^{-1}(T) = x + ux^{1/3} + v + wx^{-1/3} + rx^{-2/3} + kx^{-1} + O(x^{-4/3}/s). \quad (26.5)$$

In particular,

$$h^{-1}(T) = x - \frac{3z}{s}x^{1/3} - \frac{5}{4} + \frac{(5/4)\log 4 - C}{s} + O(x^{-1/3}/s). \quad (26.6)$$

The Airy displacement is positive because $z < 0$. The prefactor/factorial shift is $-5/4$, and the amplitude first appears at order $1/\log x$.

For verification, write $t = x^{-1/3}$ and $\delta = u/t + v + wt + rt^2 + kt^3$. The successive coefficients of $h(x + \delta) - T$, from t^{-1} through t^3 , are

$$\begin{aligned} su + A, & \quad sv + q, \\ sw + u^2/2 + Au/3 + B, \\ sr + uv + Av/3 + pu + D, \\ sk + v^2/2 + (u + A/3)w - u^3/6 - Au^2/9 + pv - Bu/3 + E. \end{aligned} \tag{26.7}$$

All vanish under (26.4). Since $q = O(s)$, $u = O(s^{-1})$, $v = O(1)$ and $w, r, k = O(s^{-1})$, the remaining residual is $O(x^{-4/3})$. Division by the local derivative $\sim s$ proves (26.5).

26.3 Newton inversion and higher finite orders

Starting at $x_0 = x$, use

$$x_{j+1} = x_j - \frac{h(x_j) - T}{h'(x_j)}.$$

One Newton step squares the error and multiplies it by $O((xs)^{-1})$. The initial error is $O(x^{1/3}/s)$, so in general

$$x_j - h^{-1}(T) = O\left(x^{1-2^{j+1}/3} s^{1-2^{j+1}}\right).$$

In particular two steps give $O(x^{-5/3}s^{-7})$, smaller than the forward uncertainty $O(x^{-4/3}/s)$. One may retain $\log \Gamma(t + 1)$ instead of truncating Stirling, with the corresponding changes to (26.2). Higher forward orders produce higher smooth models. A logarithmic forward error $O(n^{-\rho})$ becomes an inverse uncertainty $O(x^{-\rho}/\log x)$; enough Newton steps make numerical inversion error smaller than any prescribed finite forward order.

These statements invert explicit smooth models. They never differentiate an unknown remainder of the integer sequence, and do not define an intrinsic real interpolation of the combinatorial counts.

26.4 The exact discrete interpretation

Define

$$N_c(Y) = \min\{n : c_n \geq Y\}.$$

The forward expansion implies eventual strict monotonicity, since

$$\log c_{n+1} - \log c_n = \log(4n) + O(n^{-2/3}) > 0.$$

Suppose $|\log c_n - h(n)| \leq Kn^{-4/3}$ for $n \geq n_0$, and let $r_* = h^{-1}(\log Y)$. For all sufficiently large Y , put

$$\eta = \frac{8 \max(K, 1) r_*^{-4/3}}{\log(4r_*)}. \tag{26.8}$$

Then

$$\lceil r_* - \eta \rceil \leq N_c(Y) \leq \lceil r_* + \eta \rceil. \tag{26.9}$$

Indeed evaluate the integer immediately below the lower ceiling and the integer at the upper ceiling. Near r_* the derivative of the model is at least $\frac{1}{2} \log(4r_*)$, and the forward error at most $2Kr_*^{-4/3}$. The stated displacement gives a strict margin in both directions. Eventual monotonicity completes the argument; any finite initial segment is irrelevant for sufficiently large Y .

When $2\eta < 1$, at most two adjacent integers remain. If $\text{dist}(r_*, \mathbb{Z}) > \eta$, both ceilings agree and give the exact threshold. Replacing r_* by (26.5) or a Newton iterate requires enlarging η by the model-inversion error. A bare ceiling of a truncated asymptotic is not uniformly valid near integer boundaries.

An unspecified asymptotic K proves an eventual theorem, not a certified finite- Y algorithm. Certification also needs effective forward constants, a rigorous amplitude enclosure, and controlled numerical evaluation errors. The fitted decimal for γ_c provides none of these by itself.

26.5 What already follows from the published bounds

The two-sided estimate (16.2) already implies, for some $K' > 0$ and large Y ,

$$\left[x - \frac{3z}{s}x^{1/3} - \frac{5}{4} - \frac{K'}{\log x} \right] \leq N_c(Y) \leq \left[x - \frac{3z}{s}x^{1/3} - \frac{5}{4} + \frac{K'}{\log x} \right]. \quad (26.10)$$

The logarithms of the unknown multiplicative constants are $O(1)$, which becomes $O(1/\log x)$ after inversion. Thus the coarse localization is not new to the amplitude theorem. The new result identifies the fixed amplitude term and permits the finer algebraic corrections and arbitrarily narrow eventual brackets.

Replacing γ_c by $\tilde{\gamma}$ changes the smooth inverse, to first order, by

$$-\frac{\log(\tilde{\gamma}/\gamma_c)}{\log(4n)}.$$

A fixed bounded amplitude error tends to zero in the inverse, yet still exceeds the algebraic precision in (26.5) and can change a threshold near an integer.

27 Cancellation in the compacted to relaxed ratio

The independently checked companion relaxed-tree theorem (Part I, (F1)–(F2); source 15 cites the companion article here) has positive amplitude γ_r (the γ of Part I) and the logarithmic expansion

$$\begin{aligned} \log \frac{r_n}{\gamma_r n! 4^n e^{3zn^{1/3}} n} &= \frac{53z^2}{90} n^{-1/3} + \frac{44z}{27} n^{-2/3} \\ &+ \left(\frac{141}{140} - \frac{1304z^3}{42525} \right) n^{-1} + O(n^{-4/3}). \end{aligned} \quad (27.1)$$

Using that companion theorem, subtraction from (16.4) gives the algebraic corollary

$$\log \frac{c_n}{r_n} = \log \frac{\gamma_c}{\gamma_r} - \frac{1}{4} \log n - \frac{3z}{8} n^{-2/3} - \frac{21}{32} n^{-1} + O(n^{-4/3}), \quad (27.2)$$

and hence

$$\frac{c_n}{r_n} = \frac{\gamma_c}{\gamma_r} n^{-1/4} \left(1 - \frac{3z}{8} n^{-2/3} - \frac{21}{32} n^{-1} + O(n^{-4/3}) \right). \quad (27.3)$$

The $n^{-1/3}$ correction cancels exactly. Since $z < 0$, the first surviving correction is positive. No certified numerical value of the amplitude ratio is supplied.

For the corresponding explicit smooth logarithmic models, let $n_r = h_r^{-1}(T)$ and $n_c = h_c^{-1}(T)$, using their true amplitudes. Subtracting the models and applying the mean-value theorem yields

$$n_c - n_r = \frac{\frac{1}{4} \log n_r - \log(\gamma_c/\gamma_r)}{\log(4n_r)} + O(n_r^{-2/3}/\log n_r). \quad (27.4)$$

This real-model inverse gap tends to $1/4$. Integer threshold differences are subject to rounding and need not converge. The exact thresholds satisfy

$$N_r(Y) \leq N_c(Y) \leq N_r(Y) + 1 \quad (27.5)$$

for sufficiently large Y : use $c_n \leq r_n$ and eventual $c_{n+1} > r_n$. The latter follows already from the published two-sided estimates, since c_{n+1}/r_n grows at least as a positive multiple of $n^{3/4}$. Thus (27.5) is also a consequence of the known bounds, rather than a new amplitude-level conclusion.

28 Reproducibility and numerical limitations

The accompanying reproducibility archive contains the exact finite-state checks, coefficient construction, independent coefficient calculation, gauge and delay checks, inverse algebra, and exact forward recurrence. The computations require Python with SymPy, mpmath, NumPy and SciPy. Run `python replay.py` from the archive root. It verifies its SHA-256 manifest, runs all seven checks under `output/mathematical-replay/`, compares independent coefficients and the exact-state fixture, and checks that its inputs remain unchanged. The option `-manifest-only` checks input integrity without running the mathematical checks. The README records the environment and dependencies; complete approved relaxed proof sources and the coefficient-field supplement are included under `dependencies/relaxed/`. The article source can be rebuilt with `bash build_pdf.sh` and a TeX distribution providing pdfLaTeX and the packages listed in its preamble. In the archive this writes only beneath `output/article/`; beside a standalone article it writes to the local build directory and replaces the adjacent PDF.

[write, 2 October 2026] This paragraph describes source 15's delivered archive (commit 096ee7b87). Its `dependencies/relaxed/` files are earlier versions of Part I's text, of its review (shipped as `62-relaxed-verif-mathematical-review.md`) and of its Section 12; they are not shipped. The build script and the manuscript are not shipped either (this report is built as its README says), and the delivered README wrote the build command as `basharticle/build_pdf.sh`. The README of this report lists the shipped files.

The coefficient algorithm has no fitted input:

1. Represent each profile by the polynomial pair (P, Q) for $PF + QF'$.
2. Expand the rational coefficients, time dilations and previous ratios to the required degree, reducing derivatives with $F'' = 2(x + a)F$.
3. Solve the triangular polynomial equation (22.5), choose the scalar to enforce $Q(0) = 0$, and choose the remaining integration constant to enforce $P(0) + Q'(0) = 0$.
4. Match the finite logarithmic amplitude and expand the endpoint factor; then convert $N = 2n$ and $a = 2^{-1/3}z$.

5. Verify the residual coefficient, both boundary conditions and the mod-three grading exactly. An independent linear-system implementation checks the displayed initial coefficients.

Exact computation finds agreement among the signed, four-state and three-state recurrences on all 5151 cells $0 \leq m \leq n \leq 100$. The scalar renewal gives the same diagonal and all V -ending entries. Direct, nonmemoized enumeration of 709506 accepted labeled prefixes with at most seven horizontal steps agrees state by state with the four-state kernel; 13317 attempted vertical moves are rejected. These are finite checks of models whose equality was proved above.

The forward computation uses exact integer compacted counts through $n = 2500$. Removing the first three logarithmic corrections at that point gives an amplitude estimate of approximately 173.121045803661366, consistent with the published fit 173.12670485. Neither number is a rigorous enclosure, and the final displayed digits are not established digits of γ_c .

Symbolic checks verify finite algebraic identities; numerical checks verify the reproducibility of numerical evidence. Neither replaces the compactness, stability, or endpoint arguments. The amplitude in the theorem is (24.1), not a decimal fit.

29 Further questions

Certified amplitude evaluation. An effective version of the Airy gap and explicit residual and tail constants could turn (24.1) into an interval algorithm. The present argument establishes convergence and algebraic rates without furnishing all constants needed for a finite numerical certificate.

Positive effective operators. The exact three-state and renewal representations remain useful for probabilistic or nonselfadjoint analysis. The obstructions above specify what a direct symmetrization cannot do; they do not preclude a controlled asymptotic elimination of fast internal modes.

[write, 5 October 2026, batch 102] A further use of them is proposed in Part V, Question V.5: as positive evolutions in which Report 100's intrinsic positivity argument (Lemma 72.1) might replace the published lower bound. Report 100 had asked for such an encoding (Question V.6).

Higher arity and beyond algebraic orders. Higher-arity compacted recurrences require their own transfer analysis. The present fixed-order construction gives no bound uniform in the expansion order, no Gevrey estimate, and no exponentially small correction. Those questions require additional information on coefficient growth and higher Airy modes.

[write, 2 October 2026] For every fixed arity $k \geq 3$ the compacted recurrence is treated in Part III (source 18), as a finite-memory perturbation with a summable delay; the binary case, where the delay is not summable, is the one treated here. The questions of uniformity in the order, Gevrey bounds and exponentially small corrections remain open for every k .

Certified inverse thresholds. Combining interval evaluation of the amplitude with effective forward error bounds and exact neighboring counts would resolve the occasional two-integer ambiguity in (26.9).

Part III

Fixed arity $k \geq 3$

Source. Three manuscripts. (1) Batch-77 manuscript 18, archive `fixed-arity-airy-package.zip`, main file `report/fixed-arity-airy-expansions.tex` (515 lines; 12-page PDF), titled “Airy expansions for fixed arity trees and finite language automata”, author line “Research report”, dated 2 October 2026: Sections 30–37, label prefix `air:ka:.` (2) Batch-77 manuscript 66, archive `ternary-airy-amplitudes-result.zip`: the ternary section, Sections 38–46, label prefix `air:t3:.` (3) Batch-77 manuscript 29, archive `larger-alphabet-automata-report.zip`, handed over from cluster P4: the comparison section, Sections 47–52, label prefix `air:ka:al:.` All three are printed in full.

[**write, 2 October 2026**] Source 18’s leading proof extends source 66’s ternary proof. Its detailed proof note (shipped as `18-arity-fixed-arity-proof.md`) says it “extends the independently approved ternary fixed-weight proof” and names as its base a producer workspace path, `/workspace/shared/ternary-airy-amplitude-research/amplitude-proof-candidate.md`, which is not shipped under that name. The file is source 66’s `amplitude-proof-candidate.md`, shipped as `66-ternary-amplitude-proof-candidate.md`; source 18’s independent analytic audit pins it by SHA-256 (`18cdbc4e...`). Source 18’s all-orders proof note (`18-arity-all-orders-proof.md`) says that the structure of its error bootstrap is that of source 66’s `all-orders-reduction.md` (shipped as `66-ternary-all-orders-reduction.md`). The ternary section below prints source 66 as that base.

Abstract of the source.

For every fixed integer arity $k \geq 3$, we obtain positive asymptotic amplitudes and expansions to every fixed finite order for relaxed trees, compacted trees, and minimal deterministic finite automata accepting finite languages. The first two relative corrections are explicit and common to all three models. We determine the first model-dependent logarithmic ratio corrections, prove eventual decrease of the normalized ratios, and give an inverse-count expansion with the necessary interpolation qualification. The method combines exact boundary harmonics, a fixed weighted singular Airy limit, polynomial–Airy recursion, and a finite-memory perturbation argument. The leading amplitudes are positive but are not evaluated explicitly. No assertion is uniform as k grows, and no convergence or exponentially accurate completion of the infinite formal expansion is claimed.

30 Results and indexing

Fix $k \geq 3$, write $q = k - 1$, and let $a = a_1 < 0$ be the largest zero of A_i . Set

$$B = (2/q)^{1/3}, \quad \lambda = a/B, \quad \gamma = 3(kq/2)^{1/3}a, \quad \eta = (2k - 1)/3, \quad (30.1)$$

and define the common relaxed scale

$$\mathcal{G}_{k,n} = (n!)^q \left(\frac{k^k}{q^q} \right)^n e^{\gamma n^{1/3}} n^\eta. \quad (30.2)$$

Let $R_n = r(qn, n)$ and $C_n = c(qn, n)$ count relaxed and compacted k -ary trees, respectively. Let $B_n = b(qn, n)$ use the source DFA normalization: n transient states, together with one rejecting sink. The auxiliary source value specified below is part of this normalization.

For ternary relaxed trees, A082162 has offset 1 and begins 1, 7, 139, 5711, $\dots$; our additional convention is $R_0 = 1$. For arity four, A102102 has offset 0 and begins 1, 1, 15, 1000, 189035, $\dots$. The array A128249 uses alphabet size equal to its column index plus one, so arity k corresponds to column $k - 1$; its sequence is read antidiagonally. These identifiers describe relaxed counts, not the distinct minimal DFA counts.

Theorem 30.1 (All finite orders). *There are constants $A_k^R, A_k^C, A_k^B > 0$ and real coefficients $c_{k,r}^X$, $X \in \{R, C, B\}$, such that for each fixed $M \geq 0$,*

$$R_n = A_k^R \mathcal{G}_{k,n} \left[1 + \sum_{r=1}^M c_{k,r}^R n^{-r/3} + O_{k,M}(n^{-(M+1)/3}) \right], \quad (30.3)$$

$$C_n = A_k^C \mathcal{G}_{k,n} \left[1 + \sum_{r=1}^M c_{k,r}^C n^{-r/3} + O_{k,M}(n^{-(M+1)/3}) \right], \quad (30.4)$$

$$B_n = A_k^B 2^n \mathcal{G}_{k,n} \left[1 + \sum_{r=1}^M c_{k,r}^B n^{-r/3} + O_{k,M}(n^{-(M+1)/3}) \right]. \quad (30.5)$$

Each model has one and the same amplitude for every M . All three share their first two corrections. With

$$h_1 = \frac{\lambda^2(3q^2 + 27q + 23)}{45q}, \quad (30.6)$$

these are

$$c_{k,1}^X = k^{-1/3} h_1, \quad (30.7)$$

$$c_{k,2}^X = k^{-2/3} \left[\frac{h_1^2}{2} + \frac{\lambda(4q^3 + 321q^2 + 429q + 126)}{270q} \right]. \quad (30.8)$$

The published theorem of Dastidar and Wallner [3] supplies the relaxed Θ scale, including its positive lower bound. The equivalences, finite-order expansions, and comparisons here are additional arguments; they are not attributed to that published theorem. Finite-order expansion means a separate controlled remainder for every fixed M , not convergence of the resulting infinite series.

[write, 2 October 2026] At $k = 2$ the formulas (30.7)–(30.8) reduce exactly to the relaxed binary coefficients c_1, c_2 of Part I, (F3) (checked symbolically when this report was written; source 18’s proof note `18-arity-all-orders-proof.md` observes it for $c_{k,1}$). The theorem itself is claimed only for $k \geq 3$, and at $k = 2$ it fails as stated for the compacted and automaton models: their powers are $n^{3/4}$ (Part II) and $n^{7/8}$ (Part IV), not $n^{(2k-1)/3} = n$, and their second corrections differ from the relaxed one. See “Where the method changes” in Part 0.

Theorem 30.2 (First model-dependent defects). *The normalized limits*

$$\rho_C = \frac{A_k^C}{A_k^R} = \lim \frac{C_n}{R_n} > 0, \quad \rho_B = \frac{2A_k^B}{A_k^R} = \lim \frac{B_n}{2^{n-1}R_n} > 0$$

obey

$$\log \frac{C_n}{\rho_C R_n} = \frac{(q/k)^k}{k-2} n^{-(k-2)} + O_k(n^{-(k-2)-1/3}), \quad (30.9)$$

$$\log \frac{B_n}{\rho_B 2^{n-1} R_n} = \frac{(q/k)^k}{2(k-2)} n^{-(k-2)} + O_k(n^{-(k-2)-1/3}). \quad (30.10)$$

Thus all coefficients with $r < 3(k-2)$ agree with the relaxed ones. At $r = 3(k-2)$ the compacted coefficient increases by $(q/k)^k/(k-2)$ and the DFA coefficient by half that amount. Both normalized ratios approach their limits from above and are eventually strictly decreasing.

For $k = 3$, the common corrections are

$$c_1 = \frac{89a^2}{90 \cdot 3^{1/3}}, \quad c_2 = 3^{-2/3} \left(\frac{7921a^4}{16200} + \frac{115a}{27} \right),$$

and the logarithmic ratio defects begin with $8/(27n)$ for compacted trees and $4/(27n)$ for DFAs.

[**write, 2 October 2026**] These are the c_1, c_2 of source 66's Theorem 38.2 (symbolic difference zero). Source 66 also gives the relaxed c_3 at $k = 3$, which this section does not.

31 Exact arrays and boundary harmonics

The source recurrences, for $m \geq 1$ and $x \geq qm$, are

$$r(x, m) = r(x, m-1) + (m+1)r(x-1, m), \quad (31.1)$$

$$c(x, m) = c(x, m-1) + (m+1)c(x-1, m) - (m-1)c(x-k, m-1), \quad (31.2)$$

$$b(x, m) = 2b(x, m-1) + (m+1)b(x-1, m) - mb(x-k, m-1). \quad (31.3)$$

Their ordinary base rows are $r(x, 0) = c(x, 0) = b(x, 0) = 1$ for $x \geq 0$. Values below the physical diagonal are zero, apart from the DFA auxiliary $b(-1, 0) = 1$. Other missing values are zero. In particular,

$$b(q, 1) = 1, \quad b(x, 1) = 2^{x-q+1} - 1 \quad (x \geq q).$$

The last identity follows from $b(x, 1) = 2b(x-1, 1) + 1$ after its first site. The compacted first row agrees with the relaxed row because $m-1 = 0$ there.

For the relaxed model introduce

$$i = x + m, \quad j = x - qm, \quad x = (qi + j)/k, \quad d_i(j) = \frac{q^{2x}}{x!} r(x, m).$$

Then $d_0(0) = 1$ and, on $J_i = \{0 \leq j \leq i : j \equiv i \pmod{k}\}$,

$$(T_i u)(j) = U_i(j)u(j-1) + u(j+q), \quad U_i(j) = \frac{q^2(i-j+k)}{qi+j} = q - \frac{qk(j-q)}{qi+j}. \quad (31.4)$$

Missing inputs are zero. The endpoint conversion is exact:

$$R_n = \frac{(qn)!}{q^{2qn}} d_{kn}(0). \quad (31.5)$$

Let T denote the infinite critical map with up coefficient q and down coefficient 1. Set $h_j = j + 1$ and define l_j by

$$\sum_{j \geq 0} l_j z^j = \frac{qk/2}{q - kz + z^k}. \quad (31.6)$$

Then $Th = kh$ and $T^*l = kl$, including the bottom equations, with $l_j = 0$ at negative sites. The exact fixed norm is

$$\|u\|_\omega^2 = \sum_j \omega_j |u_j|^2, \quad \omega_j = \frac{l_j}{j+1}, \quad \pi_j = l_j(j+1).$$

The weighted adjoint also satisfies $T^\dagger h = kh$.

Lemma 31.1 (Positive fixed weight). *For fixed k , the weight is positive and bounded above and below. More precisely,*

$$l_j = j + \frac{q+2}{3} + E_j, \quad \omega_j = 1 + \frac{q-1}{3(j+1)} + \frac{E_j}{j+1},$$

where every fixed-order finite difference of E_j decays exponentially. Hence $|\Delta^m(\omega_j - 1)| \leq C_{k,m}(j+1)^{-m-1}$.

Proof. Factor

$$q - kz + z^k = (1 - z) \left(q - \sum_{m=1}^q z^m \right) = (1 - z)^2 \sum_{r=0}^{q-1} (q - r) z^r.$$

The middle factor has no zero in $|z| < 1$; equality in the triangle inequality on the unit circle forces $z = 1$. That zero is simple, so all zeros of the last polynomial lie outside the unit disk. Its value and derivative at 1 are $qk/2$ and $qk(q-1)/6$, which give the displayed expansion. Positivity follows from the renewal generating function

$$\frac{k/2}{(1 - z)(1 - q^{-1} \sum_{m=1}^q z^m)}.$$

Indeed if the increments are uniform on $\{1, \dots, q\}$ and τ_j is their first passage time above j , then $l_j = (k/2)\mathbb{E}\tau_j = \mathbb{E}S_{\tau_j}$. Here $\tau_j \leq j + 1$, so the stopped-sum identity is a finite sum of independent-increment expectations. Thus $j + 1 \leq l_j \leq j + q$; in particular $1 \leq \omega_j \leq q$. $\square$

32 Positive forms and the singular Airy limit

For $g_j = u_j/h_j$, exact conditional variance of the h -Doob kernel gives

$$k^2 \|u\|_\omega^2 - \|Tu\|_\omega^2 = \sum_{\substack{j \geq 1 \\ \text{output phase}}} q \omega_j j(j+k) |g_{j-1} - g_{j+q}|^2. \quad (32.1)$$

On an input phase r , put $H_m = r + km + 1$. Direct telescoping gives

$$\sum_{m \geq 0} H_m H_{m+1} \left| \frac{u_{m+1}}{H_{m+1}} - \frac{u_m}{H_m} \right|^2 = \sum_{m \geq 0} |u_{m+1} - u_m|^2 + \frac{k|u_0|^2}{r+1}. \quad (32.2)$$

Both formulas include the terminal edge to zero for a finite-support critical vector.

For the physical transfer define

$$\alpha_i = 1 + \frac{q^2 - 1}{qi + 1}, \quad Q_i(j, t) = \frac{T_i(j, t)h_t}{k\alpha_i h_j}, \quad r_i(j) = \sum_t Q_i(j, t), \quad c_i(t) = \pi_t^{-1} \sum_j \pi_j Q_i(j, t).$$

Active up edges have $j \geq 1$, and $U_i(j) \leq q\alpha_i$. Thus $r_i, c_i \leq 1$. Write $v_i(j) = q - U_i(j)$. Every physical input has its up output, so

$$1 - c_i(t) = \frac{\alpha_i - 1 + v_i(t+1)l_{t+1}/(kl_t)}{\alpha_i}. \quad (32.3)$$

At the extra bottom sites $0 \leq t \leq q-1$, this is exactly

$$1 - c_i(t) = \frac{kt(qi + q)}{\alpha_i(qi + 1)(qi + t + 1)}.$$

For all physical $t \geq 1$ it is at least $c_k t/i$. The negative values of v_i at some bottom sites are therefore accounted for, not discarded.

With $p = Q_i(j, j-1)$ and $b = Q_i(j, j+q)$, assigning zero probability to missing inputs, the positive identity is

$$\begin{aligned} \frac{k^2 \alpha_i^2 \|u\|_\omega^2 - \|T_i u\|_\omega^2}{k^2 \alpha_i^2} &= \sum_t \pi_t (1 - c_i(t)) |g_t|^2 \\ &\quad + \sum_j \pi_j p b |g_{j-1} - g_{j+q}|^2 \\ &\quad + \sum_j \pi_j (1 - r_i(j)) (p |g_{j-1}|^2 + b |g_{j+q}|^2). \end{aligned} \quad (32.4)$$

This follows from the two-point variance identity. All three terms are nonnegative, and replacing α_i^2 by 1 changes the form only by $O_k(i^{-1}) \|u\|^2$.

Proposition 32.1 (Compact singular Airy limit). *For each fixed m , the m th largest singular value in the fixed weighted phase norms satisfies*

$$\sigma_{i,m} = k \left[1 + \frac{a_m}{B} i^{-2/3} + o_k(i^{-2/3}) \right],$$

where a_m are the Airy zeros in decreasing order. The leading right singular vector converges, under mesh interpolation, to the positive normalized $\text{Ai}(a_1 + Bx)$.

Proof. Put $\varepsilon = i^{-1/3}$ and divide the unshifted defect by $2k^2\varepsilon^2$. For $x = \varepsilon(j+1)$, take $u_j = \sqrt{k}\varepsilon^{3/2}h_j G(x)$ and $F = xG$. Riemann sums in (32.4) give

$$\|u\|^2 \longrightarrow \int |F|^2, \quad \frac{k^2 \|u\|^2 - \|T_i u\|^2}{2k^2 \varepsilon^2} \longrightarrow \int \left(\frac{q}{2} |F'|^2 + x |F|^2 \right).$$

The variance supplies $q/2$ times the radial energy; column and row losses each supply half the potential because $v_i(j) = kx\varepsilon^2 + O_k(\varepsilon^3)$ away from zero.

Bounded forms give the tail bound $\sum_{\varepsilon j \geq R} \omega_j |u_j|^2 \leq C_k/R$. To control the bottom, first localize with a cutoff supported in $x \leq 2R$ and equal to one on $x \leq R$. This costs $O_{k,R}(\varepsilon^2) \|u\|^2$ in the unscaled defect. On its support the variance coefficients are comparable to the critical ones. Now (32.2) applies, including its zero terminal edge, since the localized vector is far below the physical top. It bounds $\sum |\Delta u|^2 + |u_r|^2$ by $C_{k,R}\varepsilon^2$. The interpolation of $u/\sqrt{k\varepsilon}$, extended to zero at $x = 0$, has a bounded local H^1 norm and zero trace. Rellich compactness, the negligible mass near zero from this bound, and the tail estimate give strong L^2 compactness.

For the liminf retain the nonnegative terms on $\delta \leq x \leq R$, pass by weak lower semicontinuity and strong L^2 , then let $\delta \downarrow 0$ and $R \uparrow \infty$. The limiting radial domain is the Dirichlet form domain: the identity $\int x^2 |(F/x)'|^2 = \int |F'|^2$ holds by cutoff approximation, with Hardy's inequality controlling zero. Compactly supported smooth functions give recovery sequences. Min-max therefore gives convergence of each fixed low eigenvalue to that of $-(q/2)D^2 + x$ on the half-line with zero boundary. Those eigenvalues are $-a_m/B$. Simplicity gives ground-state convergence. $\square$

33 Evolving profiles and a positive amplitude

Let $f(x) = \text{Ai}(a + Bx)$ and

$$p(x) = -\frac{q+3}{6q}x^2 - \frac{(q-1)\lambda}{3q}x, \quad \beta = \frac{7k-6}{6}.$$

Fix a sufficiently large starting time I , so all scalar factors and profiles below are defined and positive where required; the earlier evolution is absorbed into finite initial data. Choose a smooth cutoff equal to one up to $x = \log i$ and zero after $2 \log i$, and let

$$F_i(j) = \chi_i(x)[f(x) + \varepsilon p(x)f(x)], \quad N_i = \|F_i\|_\omega, \quad \psi_i = F_i/N_i,$$

$$t_i = k(1 + \lambda i^{-2/3} + \beta i^{-1}), \quad s_i = t_i N_i / N_{i-1}.$$

The exact previous-time dilation is $(i-1)^{-1/3} = \varepsilon(1 + \varepsilon^3/3 + O(\varepsilon^6))$. Writing $g = pf$, the forward expansion is

$$\begin{aligned} T_i F_{i-1} &= kf + k\varepsilon g + k\varepsilon^2((q/2)f'' - xf) \\ &\quad + k\varepsilon^3 \left[\frac{q(q-1)}{6} f''' + \frac{4}{3} x f' + kf + \frac{q}{2} g'' - xg \right] + O_k(\varepsilon^4 W(x)). \end{aligned}$$

Here W is a square-integrable polynomial Airy envelope allowing $|f'|$ as well as $|f|$; it need not vanish at zero. Substitution of $f'' = (2/q)(x + \lambda)f$ and p cancels the f' coefficient and gives βf . Thus

$$\|T_i \psi_{i-1}/s_i - \psi_i\| = O_k(i^{-4/3}), \quad \|T_i^\dagger \psi_i/s_i - \psi_{i-1}\| = O_k(i^{-1}). \quad (33.1)$$

For clarity, the adjoint estimate retains the exact cancellation $T^\dagger h = Th$. At $t \geq q$ the difference $T^\dagger - T$ has Taylor moments $M_0 = O(h^{-3})$, $M_1, M_2 = O(h^{-2})$, and $hM_0 + M_1 = 0$. Subtracting $f'(0)\varepsilon h$ leaves a profile and derivatives bounded by $C_k \varepsilon^3(h^3 + h^2)$ and its derivative analogues near zero. The finitely many bottom sites use the same exact linear cancellation directly. In the bulk, Airy bounds apply. This proves the second estimate in (33.1); separately estimating the two weight errors would not suffice.

Euler–Maclaurin on each phase, using the exact algebraic-plus-exponential decomposition of ω , yields

$$N_{r,\varepsilon}^2 = C_{-1}\varepsilon^{-1} + C_0 + C_1\varepsilon + O_k(\varepsilon^2), \quad C_{-1} = \frac{1}{k} \int f^2 > 0,$$

with the three written constants independent of phase. The squared profile and its first derivative vanish at zero; the algebraic-weight quotient also vanishes there. Hence phase dependence first enters at absolute order ε^2 . In particular,

$$N_i/N_{i-1} = 1 + O_k(i^{-1}), \quad N_i \asymp_k i^{1/6}, \quad \psi_i(0) \asymp_k i^{-1/2}.$$

Normalize by $A_i = T_i/s_i$ and decompose $y_i = a_i\psi_i + z_i$, $z_i \perp \psi_i$. The residuals give off-diagonal blocks of sizes $O(i^{-1})$ and $O(i^{-4/3})$, and a central entry $1 + O(i^{-4/3})$. Proposition 32.1 gives the stable block bound $1 - c_k i^{-2/3}$: on $\psi_{i-1}^\perp$ the squared Rayleigh quotient is at most $\sigma_{i,2}^2 + (\sigma_{i,1}^2 - \sigma_{i,2}^2)o(1)$.

The quantity $|a_i| + K i^{-1/3} \|z_i\|$ is bounded for sufficiently large K . Stable convolution then gives $\|z_i\| = O(i^{-2/3})$ and $a_i = a_\infty + O(i^{-1/3})$. At residue-zero endpoints, $z_i(0)/\psi_i(0) = O(i^{-1/6})$. Finally,

$$\prod_{m=I+1}^i t_m = \text{const} \cdot k^i e^{3\lambda i^{1/3}} i^\beta (1 + O_k(i^{-1/3})), \quad F_i(0) \sim B \text{Ai}'(a) i^{-1/3}.$$

Norm factors telescope. Conversion (31.5) and Stirling give the scale (30.2). The published positive lower bound for this exact relaxed recurrence forces its finite limiting amplitude to be positive.

34 Recursion and transfer to all finite orders

Set $\tau = (1 - \varepsilon^3)^{-1/3}$ and

$$U(\varepsilon, x) = \frac{q^2[1 - x\varepsilon^2 + (k+1)\varepsilon^3]}{q + x\varepsilon^2 - \varepsilon^3}.$$

For $\Phi = \sum_{m \geq 0} \varepsilon^m \phi_m$ and scalar step series $\sigma(\varepsilon) = k + \sum_{r \geq 2} \sigma_r \varepsilon^r$, the exact formal evolving equation is

$$U\Phi(\varepsilon\tau, (x - \varepsilon)\tau) + \Phi(\varepsilon\tau, (x + q\varepsilon)\tau) = \sigma(\varepsilon)\Phi(\varepsilon, x). \quad (34.1)$$

Take $\phi_0 = f$, $\sigma_2 = k\lambda$, and require every $\phi_m(0) = 0$. At order ε^{m+2} the only new unknowns are ϕ_m and σ_{m+2} , with equation

$$kL_q\phi_m = \sigma_{m+2}f + P_m f + Q_m f', \quad L_q = (q/2)D^2 - x - \lambda.$$

All known forcing coefficients are polynomials. Writing $\phi_m = A_m f + D_m f'$, elimination reduces polynomial inversion to

$$\left[-\frac{q}{4}D^3 + 2(x + \lambda)D + 1 \right] D_m = P - Q'/2, \quad A'_m = Q/q - D''_m/2. \quad (34.2)$$

The operator has diagonal $2d+1$ in polynomial degree d . Adding a constant to its right side changes only D_m 's constant term, so the new scalar uniquely enforces $D_m(0) = 0$. Choose $A_m(0) = 0$ to fix the amplitude gauge. This proves an obstruction-free recursion to every finite order.

For scalar matching write

$$H_i = k^i e^{3\lambda i^{1/3}} i^\beta \exp\left(\sum_{r=1}^J h_r i^{-r/3}\right)$$

with leading constant one. The new h_r first enters $\log(H_i/H_{i-1})$ at order ε^{r+3} with coefficient $-r/3$; all later coefficients can therefore be matched. The first two additional scalar stages are

$$\sigma_4 = -\frac{\lambda^2 k(6q^2 - 81q + 46)}{270q},$$

$$\sigma_5 = -\frac{\lambda k(8q^3 - 327q^2 + 657q - 92)}{810q}.$$

They give (30.6) and

$$h_2 = \frac{\lambda(4q^3 + 309q^2 + 531q - 46)}{270q}.$$

The endpoint profile begins

$$\Phi_i(0) = \varepsilon f'(0)(1 + e_2 \varepsilon^2 + \dots), \quad e_2 = \frac{\lambda(6q^2 - 51q + 86)}{135q}.$$

There is no relative ε endpoint term. Stirling starts at ε^3 , so $c_1 = k^{-1/3}h_1$ and $c_2 = k^{-2/3}(h_1^2/2 + h_2 + e_2)$, as stated.

We give the analytic transfer, since formal matching alone is insufficient. For each $p > 1$, take sufficiently many complete stages and the logarithmic Airy cutoff. All denominators are bounded away from zero on its support; every fixed Taylor remainder is a power of ε times a square-integrable polynomial Airy envelope. Squared mesh sums cost $O(\varepsilon^{-1})$, exactly canceled by $N_i^2 \asymp \varepsilon^{-1}$. Every bottom input is exactly zero and every top cutoff has a macroscopic margin. Cutoff errors decay faster than every power. Hence

$$\|T_i(H_{i-1}\Phi_{i-1}) - H_i\Phi_i\| \leq C_{k,p}H_iN_i i^{-p}.$$

Choose a fixed starting time L , let $P_i = \prod_{m=L}^i t_m$, $S_i = P_i N_i / N_{L-1}$, and $P_i / [k^i e^{3\lambda i^{1/3}} i^\beta] \rightarrow \kappa > 0$. Then

$$Z_i = \frac{\kappa}{N_{L-1}} \frac{H_i \Phi_i}{S_i}$$

has central projection tending to one for every truncation, and normalized defect $O(i^{-p})$. This fixes a common amplitude.

Subtract $a_\infty Z_i$ from the exact normalized solution and write the error as $\alpha_i \psi_i + w_i$, with $\alpha_i \rightarrow 0$. The leading blocks give

$$|\alpha_i - \alpha_{i-1}| \leq C i^{-4/3} |\alpha_{i-1}| + C i^{-1} \|w_{i-1}\| + C i^{-p},$$

$$\|w_i\| \leq (1 - c i^{-2/3}) \|w_{i-1}\| + C i^{-4/3} |\alpha_{i-1}| + C i^{-p}.$$

If $\alpha_i = O(i^{-\nu})$, stable convolution gives $\|w_i\| = O(i^{-\nu-2/3} + i^{2/3-p})$. Backward summation from the zero central limit then improves this to $\alpha_i = O(i^{-\nu-1/3} + i^{1-p})$. Starting at $\nu = 0$ and iterating finitely many times proves total norm error $O(i^{1-p})$. Endpoint evaluation loses at most $i^{1/2}$, so arbitrary p gives every finite asymptotic order with the same amplitude.

35 Compacted trees and automata as finite memory perturbations

For compacted trees use $a(x, m) = c(x, m)$, and for DFAs use $a(x, m) = b(x, m)/2^m$. Transform a by the same factorial and exponential weights. Through time $i = k$, the diagonal entries are

$d_i(i) = q^{2i}/i!$; before k these are the only entries. At time k the extra bottom entry is

$$d_k(0) = \frac{q^{2q}}{q!} \quad \text{for compacted trees,} \quad d_k(0) = \frac{q^{2q}}{2q!} \quad \text{for DFAs.}$$

The second value must not be replaced by the relaxed seed. For $i \geq k+1$ the exact recurrence is

$$d_i = T_i d_{i-1} - D_i d_{i-k-1}, \quad (D_i u)(j) = \frac{\delta_m q^{2k}}{(X)_k} u(j-1), \quad (35.1)$$

where $X = (qi + j)/k$, $m = (i - j)/k$, and $\delta_m = m - 1$ for compacted trees or $m/2$ for normalized DFAs. Define $\delta_0 = 0$. Missing delayed inputs are zero. Every denominator is nonzero in this time range; the auxiliary negative source has already been handled in the seed and is never assigned a factorial transform.

35.1 A positive comparison despite the signed recurrence

Write ℓ_m for the horizontal-run length at level m before its rise. The relaxed run weight is $V_m(\ell) = (m+1)^\ell$. The exact compacted completed-run weight is

$$V_m(\ell) \left(1 - \frac{m}{(m+1)^k} \mathbf{1}_{\ell \geq k} \right).$$

For normalized DFAs the first allowed run has weight $V_0(\ell)/2$ for every $\ell \geq q$, and later runs have weight

$$V_m(\ell) \left(1 - \frac{1}{2(m+1)^q} \mathbf{1}_{\ell \geq k} \right), \quad m \geq 1.$$

To verify these factors, remove the last k horizontals of a marked completed run before a rise. Its removed relaxed weight is $(m+1)^k$; the negative source coefficient at the next level is m or $(m+1)/2$. Removal preserves every preceding barrier constraint and gives exactly the signed source term. Expanding the product over marked runs reproduces the recurrence. For the shortest initial DFA run, $\ell = q < k$, the auxiliary seed produces the same half factor as the ordinary subtraction does when $\ell \geq k$. At the final diagonal endpoint there is no open terminal horizontal run.

Consequently,

$$\begin{aligned} 0 < P_C &:= \prod_{m \geq 1} \left(1 - \frac{m}{(m+1)^k} \right) \leq \frac{C_n}{R_n} \leq 1, \\ 0 < P_B &:= \prod_{m \geq 1} \left(1 - \frac{1}{2(m+1)^q} \right) \leq \frac{B_n}{2^{n-1} R_n} \leq 1. \end{aligned} \quad (35.2)$$

These products are positive because the defects are summable for fixed $k \geq 3$. They will force strictly positive limiting amplitudes.

[write, 2 October 2026] For automata, this comparison, with the finite- n constant $P_{k,n} = \prod_{j=2}^n (1 - 1/(2j^{k-1})) \geq P_B$ in place of P_B , is Theorem 47.1 of the comparison section (source 29), its earliest form; at $k = 3$ it is (44.1) of the ternary section. Here it is a step of the proof, and the run weights are verified combinatorially; source 29 verifies them through the completed-run generating series.

35.2 Global control and the delayed bootstrap

For all sufficiently large i , every physical $X - r$, $0 \leq r < k$, is bounded below by a positive k -dependent multiple of i , while $|\delta_m| = O_k(i)$. Bounded weight ratios and the injective height shift give

$$\|D_i\| = O_k(i^{1-k}) = O_k(i^{-q})$$

globally, not merely in the Airy window. After division by the relaxed S_i , the delayed scalar product over $k + 1$ steps stays bounded. The central/stable inequalities therefore acquire terms

$$C_k i^{-q} (|a_{i-k-1}| + \|z_{i-k-1}\|).$$

For $Q_i = |a_i| + K i^{-1/3} \|z_i\|$, these are bounded by $C_k i^{1/3-q} Q_{i-k-1}$. The running maximum of Q is controlled by the convergent product of $1 + C i^{-4/3} + C i^{1/3-q}$. Thus a_i is bounded and initially $z_i = O(i^{1/3})$. Reinsertion gives a delayed source no larger than $O(i^{-5/3})$, and hence the same $z_i = O(i^{-2/3})$ and convergent central projection as before. The bounds (35.2) force positive amplitudes.

At arbitrary order the delayed normalized error inequalities add $C i^{-q} \|w_{i-k-1}\|$ and $C i^{-q} |\alpha_{i-k-1}|$. For each power supersolution, its relative one-step change is $O(i^{-1})$, and the delayed coefficient is $O(i^{-q})$; both are absorbed by the stable decrement of order $i^{-2/3}$. Thus the same bootstrap gives

$$\|w_i\| = O(i^{-\nu-2/3} + i^{2/3-p}), \quad \alpha_i = O(i^{-\nu-1/3} + i^{1-p}).$$

This proves the all-finite-orders transfer for signed solutions without invoking positivity of their propagators or a spectral theorem for a companion matrix.

35.3 The delayed formal equation and first ratio terms

Let $\eta = (1 - l\varepsilon^3)^{-1/3}$ and

$$\mathcal{D}_k(\varepsilon, x) = \prod_{r=0}^{k-1} [q + x\varepsilon^2 - (1 + kr)\varepsilon^3].$$

The exact scaled delay coefficients in the Airy window are

$$\begin{aligned} \beta^B(\varepsilon, x) &= \frac{q^{2k} k^{k-1}}{2} \frac{\varepsilon^{3q} (1 - x\varepsilon^2 + \varepsilon^3)}{\mathcal{D}_k(\varepsilon, x)}, \\ \beta^C(\varepsilon, x) &= q^{2k} k^{k-1} \frac{\varepsilon^{3q} (1 - x\varepsilon^2 - q\varepsilon^3)}{\mathcal{D}_k(\varepsilon, x)}. \end{aligned}$$

Dividing by H_{i-1} gives the delayed formal equation

$$\begin{aligned} &U\Phi(\varepsilon\tau_1, (x - \varepsilon)\tau_1) + \Phi(\varepsilon\tau_1, (x + q\varepsilon)\tau_1) \\ &- \beta^X(\varepsilon, x) \left(\prod_{l=1}^k \sigma(\varepsilon\tau_l)^{-1} \right) \Phi(\varepsilon\tau_{k+1}, (x - \varepsilon)\tau_{k+1}) = \sigma(\varepsilon)\Phi(\varepsilon, x). \end{aligned} \quad (35.3)$$

There are k inverse scalar factors here, versus $k + 1$ after division by H_i . The delay begins at ε^{3q} , so at each new stage it adds only known lower-order forcing. The polynomial inverse (34.2) remains unchanged. Rational denominators stay bounded away from zero on the cutoff; there are only finitely many additional fixed shifts. The previous arbitrary-order remainder proof therefore applies in full, with the same exact bottom zeros and common normalization.

At first altered order the DFA subtraction is $q^k/(2k)\varepsilon^{3q}f$, while the compacted value is twice this. It is absorbed solely into the scalar, giving

$$\sigma_{3q}^B - \sigma_{3q}^R = -\frac{q^k}{2k}, \quad \sigma_{3q}^C - \sigma_{3q}^R = -\frac{q^k}{k}.$$

Earlier profiles agree, and the new profile at this first altered stage agrees too. Since h_{3q-3} enters the scalar logarithm with coefficient $-(q-1)$, the first DFA change in it is $q^k/[2k^2(q-1)]$, with twice this for compacted trees. At $i = kn$ this proves (30.9)–(30.10).

The positive first correction implies approach from above. For strict decrease, use enough finite terms to leave a remainder $o(n^{-(k-1)})$ in the logarithmic ratio. Subtract its values at $n+1$ and n . The leading difference is $-(k-2)An^{-(k-1)}$ with $A > 0$; all higher retained powers and the difference of the two remainders are smaller. This avoids differentiating an uncontrolled asymptotic remainder.

35.4 Cancellation between the two ratios

Let $u_n = C_n/R_n$ and $v_n = B_n/(2^{n-1}R_n)$, with positive limits $u_\infty = \rho_C$ and $v_\infty = \rho_B$. A stronger cancellation holds:

$$\log \frac{u_n}{u_\infty} - 2 \log \frac{v_n}{v_\infty} = -\frac{q^q}{k^k} n^{-q} + O_k(n^{-q-1/3}). \quad (35.4)$$

Every preceding fractional power vanishes. In particular the coefficient is $-4/27$ for $k = 3$ and $-27/256$ for $k = 4$.

Here is the order argument that controls nonlinear feedback. Put $d = 3q \geq 6$ and compare each formal solution to the relaxed one in the same boundary gauge. The first scalar change has order ε^d and is pure f forcing, so its profile correction is zero. Thus the individual profile changes begin at order ε^{d-1} , and the inverse scalar-product changes begin at order ε^d . Every nonlinear response product has order at least $2d-1 > d+3$. Through order $d+3$ the response is therefore genuinely linear.

The exact coefficient identity is

$$\beta^C - 2\beta^B = -\frac{q^{2k}}{(X)_k} = -q^k k^k \varepsilon^{d+3} + O(\varepsilon^{d+5}).$$

After the k inverse scalar factors, the combined forcing is $+q^k \varepsilon^{d+3} f$. The polynomial inverse therefore gives

$$\sigma^C - 2\sigma^B + \sigma^R = q^k \varepsilon^{d+3} + O(\varepsilon^{d+4}),$$

with zero combined profiles through index $d+1$. Taking scalar logarithms introduces nonlinear terms only at order $2d$, still too late. Scalar matching then gives the combined coefficient $h_d^C - 2h_d^B + h_d^R = -q^{k-1}/k$. The endpoint logarithms contribute nothing through order ε^d , since all profiles vanish at zero and their combined correction starts at index $d+2$. Evaluation at $i = kn$ gives (35.4); the all-finite-orders theorem supplies its remainder with the same amplitudes.

36 Inverse counting scales and integer thresholds

[write, 2 October 2026] The Lambert anchor and the threshold rule are instances of theorems `p0:thm:lambert-core`, `p0:thm:perturbed-inversion` and `p0:thm:staircase` of the repository's transseries volume (Part 0). No novelty is claimed for the inversion mechanics.

This section applies separately to any of the three models. Write its expansion as

$$T_n = C(n!)^q A^n e^{\gamma n^{1/3}} n^\eta \left(1 + \sum_{r=1}^M c_r n^{-r/3} + O(n^{-(M+1)/3}) \right),$$

where $A = k^k/q^q$ for relaxed or compacted trees and $A = 2k^k/q^q$ for DFAs. The amplitude C is the corresponding positive model constant. Set

$$\nu = \frac{q}{2} + \eta = \frac{7k-5}{6}, \quad d = \log C + \frac{q}{2} \log(2\pi), \quad L = \log y.$$

Let $x(y)$ denote the inverse of the *log-linearly interpolated exact counts*: between consecutive integers interpolate $\log T_n$, then invert that eventually increasing function. With the principal Lambert function W_0 , define

$$w = W_0 \left(\frac{A^{1/q} L}{qe} \right), \quad n_0 = \frac{L}{qw}, \quad D = q(w+1).$$

Then

$$x(y) = n_0 - \frac{\gamma n_0^{1/3} + \nu \log n_0 + d}{D} + o\left(\frac{1}{\log n_0}\right). \quad (36.1)$$

An equivalent form displaying the constant displacement is

$$x(y) = n_0 - \frac{\gamma n_0^{1/3}}{D} - \frac{\nu}{q} - \frac{d - (\nu/q) \log A}{D} + o\left(\frac{1}{\log n_0}\right).$$

Indeed n_0 solves $qn_0 \log n_0 + (\log A - q)n_0 = L$, and the derivative of this leading expression is D . Inserting the displayed correction leaves logarithmic residual $O(n_0^{-1/3}) = o(1)$, including the omitted first finite-order term; the actual chord slopes are asymptotic to $q \log n_0$. Division by the slope proves (36.1).

For arbitrary finite inverse accuracy, take a sufficiently long smooth logarithmic truncation $H_M(t)$ of the proven expansion, interpolate its integer values linearly, and invert that interpolation. At nearby integer sites the logarithmic error is $O(n^{-(M+1)/3})$, which interpolation preserves. The inverse error is therefore

$$O\left(\frac{n_0^{-(M+1)/3}}{\log n_0}\right).$$

This is an effective all-finite-orders inversion procedure once the amplitude is specified; it does not numerically determine the amplitude.

A smooth inverse alone omits an oscillatory lattice correction. If $H(z) = \log y$ for a sufficiently accurate smooth logarithmic truncation and $\theta = \{z\}$, the first smooth-to-chord inverse correction is

$$-\frac{q\theta(1-\theta)}{2zH'(z)} + o\left(\frac{1}{z \log z}\right).$$

This follows from the chord excess $H''(z)\theta(1-\theta)/2$, with $H''(z) \sim q/z$. Finally, for large y , the exact threshold

$$N(y) = \min\{n : T_n \geq y\} = \lceil x(y) \rceil.$$

It need not equal the ceiling of a finite approximation to $x(y)$ when that approximation is too close to an integer.

37 Checks scope and further research

The independent verification chain separates analytic proofs from finite calculations. The exact fixed-weight and phase identities pass 2,095 rational checks for $k = 3, \dots, 11$. The source seeds, completed-run sums, and signed transforms pass 2,403 rational checks for $k = 3, \dots, 8$ and $n \leq 7$. A separate signed-model replay passes 11,016 independent checks. The first two relaxed coefficients were independently recomputed by a separate symbolic direct-linear-system method through order ε^5 . These checks guard algebra and indexing; compactness, tracking, and error transfer have separate proofs.

The only external asymptotic growth input needed for positivity is the relaxed lower bound in Dastidar–Wallner [3, Section 3.4, equation (9)]. It occurs before the paper’s subsequent high-path upper-bound argument. In particular, the present signed-delay route does not use a pointwise bound $U_i \leq q$, which is false at some active bottom edges, and needs no bridge-tail repair. The valid bound used here is $U_i(j) \leq q\alpha_i$ for active $j \geq 1$.

[**write, 2 October 2026**] That the published proof uses a false bound $U_i \leq q$ is source 18’s statement (and source 66’s, Section 44.1); this report has not checked it against the paper. See Part 0, “A correction of the literature that this report has not verified”.

The following remain distinct questions rather than consequences of this report.

1. **Explicit amplitude evaluation.** Construct a certified convergent numerical method with rigorous tail bounds for A_k^R, A_k^C, A_k^B , or obtain a structural representation that evaluates them. Positivity alone does not provide their values.
2. **Large-order control.** Determine the growth of the recursively generated coefficients, possible Gevrey bounds, optimal truncation, and exponentially small remainders. All finite orders do not by themselves imply an exponentially accurate or transseries completion.
3. **Growing arity.** Study whether a simultaneous k, n regime exists. The fixed-weight roots, spectral constants, and remainder bounds here may deteriorate with k ; no uniformity has been proved.
4. **Monotonicity at every size.** Are C_n/R_n and $B_n/(2^{n-1}R_n)$ nonincreasing for every n and fixed $k \geq 3$? The theorem proves only eventual decrease. The positive completed-run formula alone does not couple the bridge measures at different terminal sizes.
5. **Further model comparisons.** Beyond the cancellation in (35.4), determine additional universal combinations and their higher corrections. At sufficiently high orders the response becomes nonlinear, so further cancellation cannot be assumed from the first-order defect alone.
6. **Sharper endpoint errors.** Boundary-adapted resolvent or pointwise estimates may reduce the $i^{1/2}$ loss in norm-to-endpoint extraction. This could improve efficient rigorous numerical evaluation even though arbitrary finite orders are already available.

The ternary case (source 66)

Source. Batch-77 manuscript 66, archive `ternary-airy-amplitudes-result.zip`, main file `report/ternary-airy-amplitudes.tex` (656 lines; 14-page PDF), titled “Airy amplitudes for ternary trees and finite language automata”, author line “Research report”, dated 2 October 2026. Printed in full as the ternary section of Part III, label prefix `air:t3:`.

Abstract of source 66.

We prove positive limiting amplitudes for relaxed ternary trees and for minimal deterministic finite automata accepting finite languages over a three-letter alphabet. For relaxed trees we also construct an asymptotic expansion to every fixed finite order. The proof uses a fixed weighted Hilbert norm that matches the forward and adjoint boundary harmonics, a compact singular Airy limit, and evolving polynomial–Airy profiles. A corrected uniform bridge-tail estimate transfers the relaxed amplitude to the automata. The leading scales were established or motivated in earlier work; the results here strengthen the relaxed Theta estimate to an equivalent and transfer an equivalent to the ternary automata. No all-orders automaton expansion or general-alphabet theorem is asserted.

[**write, 2 October 2026**] Source 66 is the case $k = 3$ of source 18 and the base proof that source 18 generalizes (see the note at the head of this Part). Its Theorems 38.1 and 38.2 are the case $k = 3$ of Theorems 30.1 and 30.2: R_n and B_n are Part III’s counts at $k = 3$, $C_R = A_3^R$, $C_B = A_3^B$, $\rho = \rho_B$, and P_3 is the product P_B of (35.2) at $q = 2$, by Euler’s sine product. They are kept in source 66’s normalization because the proof that follows is that base proof; nothing in them is a separate result. Source 66’s own additions are the third coefficient c_3 and the scalar coefficients $\mathfrak{s}_3, \dots, \mathfrak{s}_6$; the explicit ternary weight of Lemma 39.1, with $1 \leq \omega \leq 3/2$ (Lemma 31.1 gives $1 \leq \omega \leq q$); and the bridge-tail route to the automaton amplitude of Section 44, a second route beside source 18’s signed-delay route. Its statements “no general-alphabet theorem” and “no all-orders automaton expansion” are about source 66 alone: source 18 proves both for every $k \geq 3$.

38 Statements and normalizations

Let R_n count relaxed ternary trees with n internal nodes, including $R_0 = 1$. This is the sequence A082162, beginning

$$1, 1, 7, 139, 5711, 408354, 45605881, \dots$$

Let B_n count minimal DFAs accepting finite languages on a fixed three-letter alphabet, with n transient states and one rejecting sink. Thus $B_0 = 1$ and

$$B_1 = 1, \quad B_2 = 14, \quad B_3 = 532, \quad B_4 = 42644.$$

These are different counting sequences. In particular, A082162 is not the minimal finite-language DFA sequence.

Write $a = a_1 < 0$ for the largest real zero of Ai, and put

$$\gamma = 3 \, 3^{1/3} a.$$

The published relaxed-tree theorem of Dastidar and Wallner [3] gives the Theta scale used below, including its positive lower bound. Our amplitude and finite-order conclusions are additional arguments, not statements attributed to that paper.

Theorem 38.1 (Positive amplitudes; the case $k = 3$). *There are constants $C_R, C_B > 0$ such that*

$$R_n \sim C_R (n!)^2 \left(\frac{27}{4} \right)^n e^{\gamma n^{1/3}} n^{5/3}, \quad (38.1)$$

$$B_n \sim C_B (n!)^2 \left(\frac{27}{2} \right)^n e^{\gamma n^{1/3}} n^{5/3}. \quad (38.2)$$

Moreover,

$$\frac{B_n}{2^{n-1}R_n} \longrightarrow \rho \in [P_3, 1], \quad P_3 = \frac{2\sqrt{2}}{\pi} \sin \frac{\pi}{\sqrt{2}} > 0, \quad C_B = \frac{\rho C_R}{2}.$$

Theorem 38.2 (Every fixed finite order for relaxed trees; the case $k = 3$). *For every integer $M \geq 0$ there are computable constants $c_1, \dots, c_M$ such that*

$$R_n = C_R(n!)^2 \left(\frac{27}{4}\right)^n e^{\gamma n^{1/3}} n^{5/3} \left(1 + \sum_{r=1}^M c_r n^{-r/3} + O(n^{-(M+1)/3})\right).$$

The same positive C_R works for every M . The assertion is an asymptotic expansion to each fixed finite order, not convergence of an infinite series.

The first coefficients, in exactly this normalization, are

$$\begin{aligned} c_1 &= 3^{-1/3} \frac{89a^2}{90}, \\ c_2 &= 3^{-2/3} \left(\frac{7921a^4}{16200} + \frac{115a}{27} \right), \\ c_3 &= \frac{704969a^6}{13122000} + \frac{115931a^3}{85050} + \frac{1514}{945}. \end{aligned}$$

The recursion and remainder argument, rather than finite symbolic checks alone, justify these coefficients. Neither theorem determines the leading constants in closed form.

[**write, 2 October 2026**] c_1 and c_2 are the $k = 3$ values displayed after Theorem 30.2; c_3 is source 66's own. A numerical check made at intake (exact R_n for $n \leq 640$; not shipped) is consistent with c_3 : with it the successive differences of the corrected amplitude shrink at the rate $n^{-4/3}$ predicts, without it at the rate n^{-1} .

39 Exact transfer and the boundary weight

The relaxed array has support $x \geq 2m \geq 0$ and satisfies

$$r_{x,m} = r_{x,m-1} + (m+1)r_{x-1,m}, \quad r_{x,0} = 1.$$

Set $i = x + m$, $j = x - 2m$, and use the exact normalization

$$d_i(j) = \frac{4^x}{x!} r_{x,m}.$$

This gauge multiplies every path to a fixed endpoint by the same positive factor, so normalized bridge measures are unchanged. The resulting array satisfies

$$d_i(j) = U_i(j)d_{i-1}(j-1) + d_{i-1}(j+2), \tag{39.1}$$

$$U_i(j) = \frac{4(i-j+3)}{2i+j} = 2 - v_i(j), \quad v_i(j) = \frac{6(j-2)}{2i+j}, \tag{39.2}$$

$$R_n = \frac{(2n)!}{16^n} d_{3n}(0). \tag{39.3}$$

Here $d_0(0) = 1$ and all missing coordinates are zero. At time i the physical space is

$$J_i = \{j : 0 \leq j \leq i, j \equiv i \pmod{3}\}.$$

The rectangular transfer T_i maps the space on J_{i-1} to that on J_i .

The critical infinite-phase transfer is

$$(Tu)(j) = 2u(j-1) + u(j+2), \quad j \geq 0.$$

It is not diagonally symmetrizable: its two directed jumps have unequal lengths. The following fixed weight is the useful replacement for a self-adjoint transfer.

Lemma 39.1 (Compatible forward and adjoint harmonics). *Define*

$$h(j) = j + 1, \quad \ell(j) = j + \frac{4}{3} + \frac{1}{6} \left(-\frac{1}{2}\right)^j, \quad \omega(j) = \frac{\ell(j)}{h(j)}, \quad \pi(j) = \ell(j)h(j).$$

Then $1 \leq \omega(j) \leq 3/2$ and

$$Th = 3h, \quad T^*\ell = 3\ell,$$

where $\ell(-1) = \ell(-2) = 0$. In the fixed phase norms $\|u\|_\omega^2 = \sum_j \omega(j)|u(j)|^2$, one has $\|T\| \leq 3$. If $g(j) = u(j)/h(j)$ and the input is finitely supported, then

$$9\|u\|_\omega^2 - \|Tu\|_\omega^2 = \sum_{\substack{j \geq 1 \\ j \text{ in output phase}}} 2\omega(j)j(j+3)|g(j-1) - g(j+2)|^2. \quad (39.4)$$

Proof. Substitute the formulas into the two recurrences. In particular the adjoint bottom equations are $2\ell(1) = 3\ell(0)$ and $2\ell(2) = 3\ell(1)$; the geometric term is essential there. The h -conjugate of $T/3$ has row probabilities

$$p_j = \frac{2j}{3(j+1)}, \quad q_j = \frac{j+3}{3(j+1)}.$$

At $j = 0$, only the second term remains and equals one. The adjoint identity makes π invariant between successive residue phases. Jensen's inequality proves the contraction. Its exact two-point variance identity gives (39.4), including the last edge from the input support to its zero extension. $\square$

The weighted adjoint $T^\dagger = \omega^{-1}T^*\omega$ also sends h to $3h$. Thus both sides have the same boundary harmonic profile. The unweighted singular profiles, by contrast, have residue-dependent extrapolation lengths; ignoring that mismatch does not prove an amplitude theorem.

[write, 2 October 2026] This weight is Lemma 31.1 at $k = 3$: there $\sum_j l_j z^j = 3/(2 - 3z + z^3)$, whose coefficients are $l_j = j + \frac{4}{3} + \frac{1}{6}(-\frac{1}{2})^j$. It is the “matched forward and adjoint boundary profiles” that source 29 asks for in Section 51.

40 The singular Airy limit

Put $\alpha_i = 1 + 3/(2i + 1)$ and conjugate the finite transfer by h :

$$Q_i(j, k) = \frac{T_i(j, k)h(k)}{3\alpha_i h(j)}, \quad r_i(j) = \sum_k Q_i(j, k), \quad c_i(k) = \frac{1}{\pi(k)} \sum_j \pi(j) Q_i(j, k).$$

Since $U_i(j) \leq 2\alpha_i$ on active up edges, both r_i and c_i are at most one. The input column loss is exactly

$$1 - c_i(k) = \frac{\alpha_i - 1}{\alpha_i} + \frac{v_i(k+1)\ell(k+1)}{3\alpha_i \ell(k)}. \quad (40.1)$$

It vanishes at $k = 0$ and is at least ck/i for $k \geq 1$, with an absolute $c > 0$. One may take $c = 4/9$ for all physical input sites. There is no missing upper column correction: both possible nonnegative output rows are present.

For each row write $p = Q_i(j, j-1)$, $q = Q_i(j, j+2)$, assigning zero to a missing input. The following decomposition is exact:

$$\begin{aligned} 9\alpha_i^2 \|u\|_\omega^2 - \|T_i u\|_\omega^2 &= 9\alpha_i^2 \left\{ \sum_k \pi(k)(1 - c_i(k))|g(k)|^2 \right. \\ &\quad + \sum_j \pi(j)pq|g(j-1) - g(j+2)|^2 \\ &\quad \left. + \sum_j \pi(j)(1 - r_i(j))[p|g(j-1)|^2 + q|g(j+2)|^2] \right\}. \end{aligned} \quad (40.2)$$

It follows by summing the column masses and applying

$$p|a|^2 + q|b|^2 - |pa + qb|^2 = pq|a - b|^2 + (1 - p - q)(p|a|^2 + q|b|^2).$$

Every term in (40.2) is nonnegative.

Lemma 40.1 (Compact Airy limit). *For each fixed $m \geq 1$, the m th largest singular value of T_i in the fixed weighted norms satisfies*

$$\sigma_{i,m} = 3 \left(1 + a_m i^{-2/3} + o(i^{-2/3}) \right),$$

where $a_1 > a_2 > \dots$ are the negative Airy zeros. The normalized first right singular vector tends, under mesh- $3i^{-1/3}$ interpolation, to the positive normalized $\text{Ai}(a_1 + x)$ on $\mathbb{R}_+$.

Proof. Let $\varepsilon = i^{-1/3}$ and consider the self-adjoint singular forms

$$\mathcal{E}_i(u) = \frac{9 \|u\|_\omega^2 - \|T_i u\|_\omega^2}{18\varepsilon^2}.$$

Replacing 9 by $9\alpha_i^2$ changes this form by $O(\varepsilon) \|u\|_\omega^2$. In particular the forms are bounded below uniformly. We give the compactness and limiting constants explicitly.

On an input phase r , let $H_k = r + 3k + 1$ and write $u_k = u(r + 3k)$. Direct telescoping gives

$$\sum_{k \geq 0} H_k H_{k+1} \left| \frac{u_{k+1}}{H_{k+1}} - \frac{u_k}{H_k} \right|^2 = \sum_{k \geq 0} |u_{k+1} - u_k|^2 + \frac{3|u_0|^2}{r+1}. \quad (40.3)$$

The input is extended by zero at infinity. If F_i is linear between the nodes

$$x_k = \varepsilon H_k, \quad F_i(x_k) = \frac{u_k}{\sqrt{3\varepsilon}}, \quad F_i(0) = 0,$$

then the right side of (40.3), divided by $9\varepsilon^2$, is exactly $\int |F'_i|^2$.

The moving variance sum omits the critical top edge to zero. Therefore localize first: multiply u_k by a cutoff equal to one for $x_k \leq R$ and zero for $x_k \geq 2R$. For fixed R and large i , these edges are far below the physical top and U_i is bounded below. Writing $\mathcal{Q}$ for the radial sum in (40.3), product differences and $H_{k+1}/H_k \leq 4$ give

$$\mathcal{Q}(\chi_R u) \leq C \mathcal{Q}(u; x \leq 2R + 3\varepsilon) + C\varepsilon^2 R^{-2} \|u\|^2.$$

Hence bounded norms and bounded $\mathcal{E}_i$ imply uniform local H^1 bounds for the zero-anchored interpolants. In particular

$$\int_0^\delta |F_i|^2 \leq \frac{\delta^2}{2} \int_0^\delta |F'_i|^2,$$

and the corresponding nodal mass is $O(\delta^2 + \varepsilon^2)$. The lower endpoint cannot carry concentrated mass.

The column loss (40.1) also yields

$$\sum_{\varepsilon k \geq R} \omega(k) |u(k)|^2 \leq C/R \quad (40.4)$$

for bounded form energy. Local compactness, the lower-end estimate and (40.4) give global strong L^2 compactness. Weighted nodal norms, unweighted nodal norms and interpolated norms have the same limit: away from zero $\omega \rightarrow 1$, and the two tails are uniformly negligible. The same holds for inner products.

On any compact interval $0 < \delta < x < R$, the variance term in (40.2) converges to $\int x^2 |(F/x)'|^2$. Each loss term contributes one half of $\int x |F|^2$. Indeed $v_i(j) = 3x\varepsilon^2 + O(\varepsilon^3)$ there, and both normalized row and column losses equal $x\varepsilon^2 + O(\varepsilon^3)$. Local strong L^2 convergence and the gradient bound justify translated sampling. Discarding the nonnegative terms outside $[\delta, R]$, then sending $\delta \downarrow 0$ and $R \uparrow \infty$, gives the liminf form

$$\mathcal{E}(F) = \int_0^\infty (|F'|^2 + x|F|^2) dx, \quad F \in H_0^1(\mathbb{R}_+).$$

Here $\int x^2 |(F/x)'|^2 = \int |F'|^2$ by Hardy's inequality and integration by parts. The zero trace was supplied by (40.3); no Neumann or Robin boundary has been introduced.

For recovery sequences sample smooth compactly supported functions inside $(0, \infty)$ as $u_k = \sqrt{3\varepsilon} F(x_k)$. Direct Riemann sums give the norm and form limits. Such functions form a core of the displayed form. Finite-dimensional recovery spaces give the min-max upper bounds. For the lower bounds, compactness of the first m discrete eigenvectors preserves their orthonormality and gives an m -dimensional limiting subspace; apply the liminf to its linear combinations. The limiting operator is $-\partial_x^2 + x$ with Dirichlet boundary, with eigenvalues $-a_m$. Thus

$$\frac{9 - \sigma_{i,m}^2}{18\varepsilon^2} \longrightarrow -a_m.$$

Simplicity of the first limit eigenvalue gives ground-vector convergence. The first discrete vector can be chosen positive because $T_i^\dagger T_i$ is irreducible on its input residue chain. $\square$

Remark 40.2. The lemma concerns one occupied residue channel. On the direct sum of all three channels, the first three singular values have the same leading Airy level. It would be incorrect to use a_2 as the second level on that full space.

41 An evolving Airy profile

Let $f(x) = \text{Ai}(a+x)$ and

$$g(x) = \left(-\frac{5x^2}{12} - \frac{ax}{6} \right) f(x).$$

Choose a fixed smooth cutoff shape, equal to one up to one and zero past two, and apply it at $x/\log i$. With $\varepsilon = i^{-1/3}$ define

$$\begin{aligned} F_i(j) &= \chi_i(\varepsilon(j+1)) [f(\varepsilon(j+1)) + \varepsilon g(\varepsilon(j+1))], \\ N_i &= \|F_i\|_\omega, \quad \psi_i = F_i/N_i, \\ t_i &= 3 \left(1 + ai^{-2/3} + \frac{5}{2i} \right), \quad s_i = t_i \frac{N_i}{N_{i-1}}. \end{aligned}$$

All estimates from this point begin at one fixed sufficiently large time I . Earlier evolution is absorbed into its finite initial vector.

Lemma 41.1 (Two-sided evolving residual). *Uniformly through all three phases,*

$$\|T_i \psi_{i-1}/s_i - \psi_i\|_\omega = O(i^{-4/3}), \quad (41.1)$$

$$\|T_i^\dagger \psi_i/s_i - \psi_{i-1}\|_\omega = O(i^{-1}). \quad (41.2)$$

Also $N_i \asymp i^{1/6}$, $N_i/N_{i-1} = 1 + O(i^{-1})$, and, when $i \equiv 0 \pmod{3}$, $\psi_i(0) \asymp i^{-1/2}$.

Proof. The profile is evolving, not a frozen eigenvector. In particular $(i-1)^{-1/3} = \varepsilon(1 + \varepsilon^3/3 + O(\varepsilon^6))$ must be retained. At $x = \varepsilon(j+1)$, Taylor expansion of the uncut $T_i F_{i-1}$ gives

$$3f + 3\varepsilon g + 3\varepsilon^2(f'' - xf) + \varepsilon^3\{f''' + 4xf' + 9f + 3(g'' - xg)\} + O(\varepsilon^4).$$

The $4xf'$ includes xf' from the time dilation. Using $f'' = (x+a)f$,

$$f''' + 4xf' + 9f = 10f + (5x+a)f',$$

while

$$3\{g'' - (x+a)g\} = -\frac{5}{2}f - (5x+a)f'.$$

The result is $t_i(f + \varepsilon g) + O(\varepsilon^4)$. The remainder has a square-integrable polynomial–Airy envelope, uniformly on the moving support. A mesh sum costs $O(\varepsilon^{-1})$ in squared norm, canceled by $N_i^2 \asymp \varepsilon^{-1}$. The lower missing input corresponds exactly to $x = 0$, where $f = g = 0$. This proves (41.1) after normalization.

We spell out the adjoint boundary estimate, where an unweighted argument would fail. The exact weighted adjoint is

$$(T_i^\dagger u)(t) = \frac{U_i(t+1)\omega(t+1)u(t+1) + \omega(t-2)u(t-2)}{\omega(t)}.$$

Let $B = \omega^{-1}T^*\omega$ be the critical weighted adjoint, and let T^* here denote its unweighted counterpart. For $t \geq 2$ put

$$a_t = \frac{2(\omega(t+1) - \omega(t))}{\omega(t)}.$$

Both B and T^* send h to $3h$ at those sites. Hence, exactly,

$$[(B - T^*)F](t) = a_t h(t+1) \left[\frac{F(t+1)}{h(t+1)} - \frac{F(t-2)}{h(t-2)} \right]. \quad (41.3)$$

Its coefficient is bounded by $C((t+1)^{-1} + 2^{-t})$. Write $F(t)/h(t) = \varepsilon \Phi_\varepsilon(\varepsilon(t+1))$, where $\Phi_\varepsilon(x) = (f(x) + \varepsilon g(x))/x$. For $0 \leq x \leq 1$, $|\Phi'_\varepsilon(x)| \leq C(x + \varepsilon)$ because $f(0) = f''(0) = 0$. Thus (41.3) is $O(\varepsilon^3)$ per site near the bottom. Above the Airy scale, the same estimate has an integrable polynomial–Airy envelope. The operator $T^* - T$ is a third finite difference and has the same order.

At the exceptional sites check directly:

$$(BF)(0) = \frac{3}{2}F(1), \quad (BF)(1) = 2F(2), \quad (TF)(0) = F(2), \quad (TF)(1) = 2F(0) + F(3).$$

The linear terms cancel, and the quadratic term of f vanishes. The remaining terms are $O(\varepsilon^3)$. The difference of the adjoint and forward inhomogeneous perturbations is

$$v_i(t)F(t-1) - v_i(t+1)\frac{\omega(t+1)}{\omega(t)}F(t+1).$$

A first-difference estimate gives normalized norm $O(\varepsilon^3)$; at fixed t it is $O(\varepsilon^4)$. Replacing F_i by F_{i-1} has normalized cost $O(\varepsilon^3)$ by time dilation. These facts prove (41.2).

All cutoff discrepancies occur at $x \geq \log i - O(\varepsilon)$, where the Airy envelope is bounded by a polynomial times $e^{-c(\log i)^{3/2}}$. They are smaller than every fixed power of i .

Finally, Euler–Maclaurin on a residue phase r gives

$$N_{r,\varepsilon}^2 = C_{-1}\varepsilon^{-1} + C_0 + C_1\varepsilon + O(\varepsilon^2), \quad C_{-1} = \frac{1}{3} \int_0^\infty f^2 > 0, \quad (41.4)$$

with the first three constants independent of r . To see the phase order, set $H_\varepsilon = (f + \varepsilon g)^2$. Then $H_\varepsilon(0) = H'_\varepsilon(0) = 0$; ordinary phase-sum endpoint corrections begin at $O(\varepsilon^2)$. The algebraic weight contributes $\varepsilon H_\varepsilon(x)/(3x)$, whose quotient vanishes at zero, and its first phase-dependent correction is again $O(\varepsilon^2)$. The geometric part of ω is $O(\varepsilon^2)$ in the sum. Formula (41.4) proves the adjacent-time ratio, including the change of phase. The endpoint estimate follows from $f(\varepsilon) = \text{Ai}'(a)\varepsilon + O(\varepsilon^3)$. $\square$

42 Tracking and the relaxed amplitude

Normalize the evolution by $A_i = T_i/s_i$ and decompose an exact solution as

$$y_i = A_i y_{i-1} = b_i \psi_i + z_i, \quad z_i \perp \psi_i.$$

The two residuals give the block estimates

$$b_i = (1 + O(i^{-4/3}))b_{i-1} + \beta_i z_{i-1}, \quad \|\beta_i\| = O(i^{-1}), \quad (42.1)$$

$$z_i = v_i b_{i-1} + D_i z_{i-1}, \quad \|v_i\| = O(i^{-4/3}). \quad (42.2)$$

The vector symbol v_i in this section is unrelated to the scalar up-weight loss in (39.2).

Lemma 42.1 (Compressed contraction). *For some $c > 0$ and all sufficiently large i , $\|D_i\| \leq 1 - ci^{-2/3}$.*

Proof. The explicit input profile converges to the same Airy ground function as the first right singular vector, so its squared overlap is $1 - o(1)$. If $x \perp \psi_{i-1}$, singular decomposition gives

$$\|T_i x\|^2 \leq [\sigma_{i,2}^2 + (\sigma_{i,1}^2 - \sigma_{i,2}^2)o(1)] \|x\|^2.$$

By Lemma 40.1 and $s_i = 3(1 + ai^{-2/3} + O(i^{-1}))$, division by s_i^2 leaves a strict loss of order $i^{-2/3}$. Output projection cannot increase the norm. This argument uses singular values of one channel and does not assume self-adjointness of T_i or a frozen eigenvector equation for ψ_i . $\square$

Let $q_i = |b_i|$ and $Z_i = \|z_i\|$. The quantity $q_i + Ki^{-1/3}Z_i$ is bounded for sufficiently large fixed K . Indeed the negative term $-Kci^{-1}Z_{i-1}$ absorbs the $O(i^{-1})Z_{i-1}$ cross term, while the remaining multiplicative errors are summable $O(i^{-4/3})$. Iterating the stable block gives

$$Z_i = O(i^{-2/3}), \quad b_i = b_\infty + O(i^{-1/3}).$$

For the convolution estimate, sources older than $i/2$ have stretched-exponential suppression; the remaining kernel is bounded by $\exp[-c(i-m)i^{-2/3}]$ and has total mass $O(i^{2/3})$.

Endpoint evaluation is bounded in the fixed norm. Since $\psi_i(0) \asymp i^{-1/2}$ at residue-zero times,

$$\frac{z_i(0)}{\psi_i(0)} = O(i^{-1/6}) \longrightarrow 0.$$

The scalar norm factors telescope, while

$$\prod_{m=I}^i t_m = K_I 3^i e^{3ai^{1/3}} i^{5/2} (1 + O(i^{-1/3})), \quad F_i(0) \sim \text{Ai}'(a) i^{-1/3}.$$

Consequently

$$d_{3n}(0) \sim C_d 27^n e^{\gamma n^{1/3}} n^{13/6}. \quad (42.3)$$

The constant is finite. It is strictly positive because the published relaxed Theta lower bound in [3] rules out a zero limit for this exact solution. Thus the proof does not assume positivity of a nonnormal spectral projection. Equation (39.3) and Stirling prove (38.1). The same argument applies to every fixed-time initial vector, giving a finite limiting endpoint ratio to the relaxed reference solution; that limit need not be known to be positive.

43 Polynomial Airy recursion to every finite order

Put $\tau = (1 - \varepsilon^3)^{-1/3}$ and seek formal series

$$\Phi(\varepsilon, x) = \sum_{m \geq 0} \varepsilon^m \phi_m(x), \quad \mathfrak{s}(\varepsilon) = 3 + \sum_{m \geq 2} \mathfrak{s}_m \varepsilon^m, \quad \phi_0 = f, \quad \mathfrak{s}_2 = 3a.$$

The exact formal evolving equation is

$$\mathcal{U}(\varepsilon, x) \Phi(\varepsilon \tau, (x - \varepsilon) \tau) + \Phi(\varepsilon \tau, (x + 2\varepsilon) \tau) = \mathfrak{s}(\varepsilon) \Phi(\varepsilon, x), \quad (43.1)$$

where

$$\mathcal{U}(\varepsilon, x) = 2 - \frac{6(x\varepsilon^2 - 3\varepsilon^3)}{2 + x\varepsilon^2 - \varepsilon^3}.$$

Require $\phi_m(0) = 0$ at every order. At order ε^{m+2} the only new profile is ϕ_m , and the only new scalar is $\mathfrak{s}_{m+2}$. Zero drift cancels the intervening order. After reducing derivatives by $f'' = (x+a)f$, the equation is

$$3(\partial_x^2 - x - a)\phi_m = \mathfrak{s}_{m+2}f + P_m f + Q_m f', \quad (43.2)$$

with known polynomials P_m, Q_m .

For a polynomial–Airy function $Af + Bf'$,

$$(\partial_x^2 - x - a)(Af + Bf') = [A'' + 2(x+a)B' + B]f + [2A' + B'']f'.$$

After dividing (43.2) by three, polynomial forcing $Pf + Qf'$ is solved by

$$\mathcal{K}B := -\frac{1}{2}B''' + 2(x+a)B' + B = P - \frac{1}{2}Q', \quad A' = \frac{1}{2}(Q - B''). \quad (43.3)$$

On polynomials of degree at most d , $\mathcal{K}$ has diagonal entries $1, 3, \dots, 2d+1$ and lower-degree remaining terms. It is invertible. Adding a constant to the right side adds precisely that constant to B . The new scalar therefore uniquely enforces $B(0) = 0$, which is the boundary condition $\phi_m(0) = 0$. Fix $A(0) = 0$ to remove the remaining amplitude gauge. This constructs every finite stage with no Fredholm obstruction or logarithmic profile.

For example, the first scalar coefficients in this gauge are

$$\mathfrak{s}_3 = \frac{15}{2}, \quad \mathfrak{s}_4 = \frac{23a^2}{45}, \quad \mathfrak{s}_5 = \frac{11a}{270}, \quad \mathfrak{s}_6 = -\frac{1714a^3}{14175} - \frac{5293}{1890}.$$

Two independent exact symbolic recursions verify these values. The infinite-stage existence statement follows from (43.3), not from those finite computations.

43.1 Uniform remainders and a common amplitude

Fix $p > 1$, take an integer $K \geq \max(4, \lceil 3p \rceil)$, construct profiles through $K-2$ and scalars through K , and allow one further order if desired. Let

$$G_i = 3^i e^{3ai^{1/3}} i^{5/2}, \quad H_i = G_i \exp \left(\sum_{r=1}^{K-3} h_r i^{-r/3} \right)$$

with leading constant exactly one. Match $\log(H_i/H_{i-1})$ to $\log \mathfrak{s}(i^{-1/3})$. The first two nonconstant orders already agree. The new h_r first appears at order ε^{r+3} with coefficient $-r/3$, so every further scalar equation is uniquely soluble.

Apply the same cutoff to the finite profile sum Φ_i . Each fixed derivative of each polynomial–Airy correction has a bound $C(1+x)^D e^{-cx^{3/2}}$ for $x \geq 1$ and is bounded near zero. On $x \leq 2 \log i$, the rational denominators in (43.1) stay uniformly away from zero. Taylor remainders are powers of ε times a common integrable polynomial–Airy envelope; spatial displacements are $O(\varepsilon + \varepsilon^3 x)$. Thus weighted mesh sums divide by $N_i^2 \asymp \varepsilon^{-1}$ without logarithmic loss. Bottom values vanish exactly, the physical top lies beyond the cutoff, and cutoff discrepancies are superpolynomially small. These estimates prove

$$\|T_i[H_{i-1}\Phi_{i-1}] - H_i\Phi_i\|_\omega \leq C_p H_i N_i i^{-p}. \quad (43.4)$$

Constants and starting indices may depend on the requested finite order.

Fix the same lower index I in every construction and write

$$P_i = \prod_{m=I}^i t_m \sim \kappa G_i, \quad S_i = \prod_{m=I}^i s_m = P_i \frac{N_i}{N_{I-1}}.$$

Then

$$Z_i = \frac{\kappa}{N_{I-1}} \frac{H_i \Phi_i}{S_i}$$

has projection onto ψ_i tending to one, independently of truncation. Indeed $H_i/G_i \rightarrow 1$ and $\Phi_i/N_i - \psi_i \rightarrow 0$. Dividing (43.4) by S_i yields $\|Z_i - A_i Z_{i-1}\| = O(i^{-p})$. The reciprocal constants in this normalization are essential; one cannot independently absorb a different constant into each finite truncation.

43.2 The zero-limit bootstrap

Let the exact normalized solution have limiting projection C , and set $E_i = y_i - CZ_i = \eta_i \psi_i + w_i$, where $\eta_i \rightarrow 0$. The already proved block estimates imply

$$\begin{aligned} |\eta_i - \eta_{i-1}| &\leq C i^{-4/3} |\eta_{i-1}| + C i^{-1} \|w_{i-1}\| + C i^{-p}, \\ \|w_i\| &\leq (1 - c i^{-2/3}) \|w_{i-1}\| + C i^{-4/3} |\eta_{i-1}| + C i^{-p}. \end{aligned}$$

If $\eta_i = O(i^{-q})$, the contraction convolution gives

$$\|w_i\| = O(i^{-q-2/3} + i^{2/3-p}).$$

Summing the central increment *backward*, using $\eta_\infty = 0$ and $p > 1$, gives

$$\eta_i = O(i^{-q-1/3} + i^{1-p}).$$

Starting at bounded η and iterating finitely many times yields $\|E_i\| = O(i^{1-p})$. Endpoint extraction costs at most $i^{1/2}$, so the relative endpoint error is $O(i^{3/2-p})$. For the M -term theorem choose $p > 3/2 + (M+1)/3$ and sufficiently many formal terms. The sufficiently large starting index of these estimates may depend on M , while the scalar normalization retains its fixed lower limit I . This transfers every finite order with the same amplitude.

Finally, every ϕ_m vanishes at zero, so $\Phi_i(0)$ has an ordinary expansion in ε after factoring out $\text{Ai}'(a)\varepsilon$. Independent calculations give

$$h_1 = \frac{89a^2}{90}, \quad h_2 = \frac{571a}{135}, \quad \frac{\Phi_i(0)}{\text{Ai}'(a)\varepsilon} = 1 + \frac{4a}{135}\varepsilon^2 - \frac{103}{810}\varepsilon^3 + O(\varepsilon^4).$$

Combining the scalar factor, endpoint series, $i = 3n$, and Stirling proves Theorem 38.2, including its displayed coefficients.

44 Positive defects and the automaton amplitude

The ternary DFA recurrence is

$$b_{x,m} = 2b_{x,m-1} + (m+1)b_{x-1,m} - mb_{x-3,m-1}, \quad x \geq 2m.$$

The base row is $b_{x,0} = 1$ for $x \geq 0$. The necessary auxiliary boundary value is $b_{-1,0} = 1$; all other out-of-domain values vanish. Equivalently the first row is $b_{x,1} = 2^{x-1} - 1$ for $x \geq 2$. This boundary convention reproduces the source's tabulated initial counts.

We use a positive completed-run representation rather than the signed recurrence. At level $m \geq 1$, put

$$V_m(\ell) = (m+1)^\ell, \quad W_m(\ell) = \begin{cases} 2(m+1)^\ell, & 0 \leq \ell < 3, \\ 2(m+1)^\ell - (m+1)^{\ell-2}, & \ell \geq 3. \end{cases}$$

Start both completed-run arrays at $a(x,1) = A(x,1) = 1$ for $x \geq 2$, with triangular support. Convolution by V_m gives $a(x, m+1)$; convolution by W_m gives $A(x, m+1)$. The endpoints are $R_n = a(2n, n)$ and $B_n = A(2n, n)$. For verification,

$$\sum_{\ell \geq 0} W_m(\ell) t^\ell = \frac{2 - (m+1)t^3}{1 - (m+1)t},$$

and appending the final horizontal run recovers the signed recurrence. In particular all weights in this representation are nonnegative.

Pointwise,

$$2 \left(1 - \frac{1}{2(m+1)^2} \right) V_m(\ell) \leq W_m(\ell) \leq 2V_m(\ell).$$

Induction gives

$$\prod_{j=2}^n \left(1 - \frac{1}{2j^2} \right) \leq \frac{B_n}{2^{n-1}R_n} \leq 1. \quad (44.1)$$

Euler's sine product evaluates the infinite lower product as P_3 in Theorem 38.1.

[**write, 2 October 2026**] This is the case $k = 3$ of Theorem 47.1 (source 29), whose proof of the run weights goes through the completed-run generating series as here.

Under the relaxed bridge measure to $(2n, n)$, let L_m be the horizontal-run length at vertical level m before rising to $m+1$. Then

$$\rho_n := \frac{B_n}{2^{n-1}R_n} = \mathbb{E}_n \prod_{m=1}^{n-1} \left(1 - \frac{\mathbf{1}_{\{L_m \geq 3\}}}{2(m+1)^2} \right). \quad (44.2)$$

The factor 2^{n-1} is constant over the colored relaxed paths, so coloring does not change this bridge probability.

44.1 A corrected uniform high-path bound

[**write, 2 October 2026**] That the published lemma asserts $U_i(j) \leq 2$ is source 66's statement, not checked against the paper by this report (Part 0). The repair below, with the monotonicity of u_r and the bridge comparison, is a second route to the automaton amplitude at $k = 3$. Source 18 needs none of it: its signed-delay route (Section 35.1 and the following subsections) uses only $U_i(j) \leq q\alpha_i$.

The proof of the high-path lemma in [3] (published Lemma 15, page 15:12; Lemma 14 in arXiv v1) contains the pointwise assertion $U_i(j) \leq 2$. That assertion is false near the boundary, for example $U_4(1) = 8/3$. The following repair proves the required conclusion.

At active up destinations $j \geq 1$,

$$U_i(j) \leq U_i(1) = 2 \left(1 + \frac{3}{2i+1} \right) \leq 2 \left(1 + \frac{3}{2i} \right).$$

The product of the extra factors along any path of length t is at most $Ct^{3/2}$. Thus

$$d_{3x}(3y) \leq Cx^{3/2}2^{2x+y} \binom{3x}{x-y}. \quad (44.3)$$

For continuations to $(T, 0)$, write $p(r, s; T)$ for their total weight and $u_r(s) = p(r, s; T)/(s+1)$. Its backward coefficients are

$$A_r(s) = \frac{4(r-s+3)(s+2)}{(2r+s+3)(s+1)}, \quad B_r(s) = \frac{(s-1)_+}{s+1}.$$

For $s \geq 1$ their row sum is

$$V_r(s) = 3 - 6 \frac{s-1}{s+1} \frac{s+2}{2r+s+3},$$

which decreases because both nonnegative fractions increase. At $s = 0$ the sole coefficient is at least four, whereas $V_r(1) = 3$, so the boundary comparison also holds. Adjacent same-phase starts $s, s+3$ have next supports $\{s-2, s+1\}$ and $\{s+1, s+4\}$. If the next normalized continuation decreases, all values on the first support are at least the common middle value and all on the second are at most it. The row-sum comparison preserves monotonicity. The terminal sequence is one at zero and zero at other physical sites, so reverse induction proves that u_r decreases on each phase. Consequently

$$p(3x, 3y; T) \leq (3y+1)p(3x, 0; T),$$

and comparison through height zero yields

$$\mathbb{P}_T\{\text{path visits } (3x, 3y)\} \leq (3y+1) \frac{d_{3x}(3y)}{d_{3x}(0)}. \quad (44.4)$$

The relaxed lower bound gives $d_{3x}(0) \geq c27^x e^{-Cx^{1/3}} x^{13/6}$. Furthermore,

$$\frac{2^{2x+y}}{27^x} \binom{3x}{x-y} = \mathbb{P}\{\text{Bin}(3x, 1/3) = x-y\} \leq e^{-2y^2/(3x)}.$$

The last inequality follows from the binomial lower-tail Chernoff bound. Combining (44.3) and (44.4), summing over $y > x^{3/4}$ and $y \leq x$, and then over $x > I$, bounds the forbidden bridge mass by

$$\sum_{x>I} Cx^{4/3} \exp\left(-\frac{2}{3}\sqrt{x} + Cx^{1/3}\right) \longrightarrow 0. \quad (44.5)$$

This estimate is uniform in the terminal size. The correction to the false critical-weight domination is only polynomial and is harmless here.

44.2 From fixed height to fixed time

In (44.2), first retain only factors with $m < M$, obtaining $\rho_n^{[M]}$. Pointwise product comparison gives, uniformly in n ,

$$|\rho_n - \rho_n^{[M]}| \leq \frac{1}{2} \sum_{j>M} j^{-2} \leq \frac{1}{2M}.$$

For fixed M , additionally ignore a retained factor whose terminating rise occurs after path time L , obtaining $\rho_n^{[M,L]}$. This depends only on a fixed finite prefix. Summing the reweighted prefixes gives a finite initial vector at time L ; afterward the transfer is exactly relaxed. The tracking theorem therefore proves convergence of $\rho_n^{[M,L]}$ as $n \rightarrow \infty$.

If a retained rise occurs at time $i > L$, then its level is $m + 1 \leq M$ and its transformed height is $j = i - 3(m + 1) \geq i - 3M$. At the preceding block time $p = 3\lfloor i/3 \rfloor$, the height is at least $p - 3M$, because a forward step raises height by at most one. For fixed M and sufficiently large p , this exceeds $3(p/3)^{3/4}$. The uniform tail bound (44.5) therefore gives

$$\sup_n |\rho_n^{[M]} - \rho_n^{[M,L]}| \rightarrow 0 \quad (L \rightarrow \infty).$$

Thus $\rho_n^{[M]}$ converges as a uniform limit of convergent sequences; then the $1/(2M)$ estimate proves convergence of ρ_n . Its limit lies in $[P_3, 1]$ by (44.1). Combining with the relaxed equivalent proves the DFA part of Theorem 38.1, including $C_B = \rho C_R/2$.

This transfer is qualitative in its truncation levels. It does not yield a finite-order DFA expansion.

45 Inverse asymptotics and integer thresholds

[write, 2 October 2026] An instance of theorems `p0:thm:lambert-core`, `p0:thm:perturbed-inversion` and `p0:thm:staircase` of the repository's transseries volume (Part 0); no novelty is claimed for the inversion mechanics.

Let T_n denote either R_n or B_n , and set respectively

$$(A, C) = (27/4, C_R) \quad \text{or} \quad (27/2, C_B).$$

For $y \rightarrow \infty$, define

$$L = \log y, \quad w = W_0 \left(\frac{\sqrt{A} L}{2e} \right), \quad n_0 = \frac{L}{2w}, \quad D = 2(w + 1),$$

where W_0 is the principal real Lambert branch. Let $x(y)$ be the inverse of the linear interpolation of $\log T_n$ between consecutive integers, with target value $\log y$.

Corollary 45.1 (Inverse equivalent and corrections). *For both counting sequences,*

$$\begin{aligned} x(y) &= n_0 - \frac{\gamma n_0^{1/3} + (8/3) \log n_0 + \log(2\pi C)}{D} + o(1/\log n_0) \\ &= n_0 - \frac{\gamma n_0^{1/3}}{D} - \frac{4}{3} - \frac{\log(2\pi C) - (4/3) \log A}{D} + o(1/\log n_0). \end{aligned} \tag{45.1}$$

The least integer threshold $\nu(y) = \min\{n : T_n \geq y\}$ equals $\lceil x(y) \rceil$ eventually. In particular (45.1) gives its value with an $O(1)$ final error. Rounding the displayed approximation itself is not an exact uniform formula near integer thresholds.

Proof. Stirling and the leading equivalent give

$$\log T_n = H_0(n) + g(n) + o(1),$$

where

$$H_0(t) = 2t \log t + (\log A - 2)t, \quad g(t) = \gamma t^{1/3} + \frac{8}{3} \log t + \log(2\pi C).$$

The coefficient $8/3$ includes the logarithm from the two factorials. Consecutive log increments are $2 \log n + \log A + o(1) > 0$, so both inverses are eventually well defined. Linear interpolation of the smooth model adds $O(1/n)$, and interpolation of the sequence's $o(1)$ error remains uniformly $o(1)$.

The equation $H_0(n_0) = L$ has the stated exact Lambert solution, with $H'_0(n_0) = D$. Put $\delta = -g(n_0)/D = O(n_0^{1/3}/\log n_0)$. Taylor's formula gives

$$H_0(n_0 + \delta) + g(n_0 + \delta) - L = O(\delta^2/n_0) + O(n_0^{-2/3}|\delta|) = o(1).$$

The actual interpolant has slopes asymptotic to $2 \log n_0$, proving the error in (45.1). The second expression follows from $(8/3) \log n_0 = (4/3)D - (4/3) \log A$. Exact rounding applies to the actual inverse by monotonicity, not automatically to its approximation. $\square$

In particular $x(y) \sim L/(2 \log L)$. Since $\gamma < 0$, the stretched correction increases the inverse by

$$-\frac{\gamma n_0^{1/3}}{D} \sim \frac{-\gamma}{2^{4/3}} \frac{L^{1/3}}{(\log L)^{4/3}}.$$

For relaxed trees, Theorem 38.2 also gives constructive inverses to every finite order, provided the interpolation convention is retained. Taking logarithms and Stirling yields a smooth truncation

$$H_M(t) = H_0(t) + g(t) + \sum_{r=1}^M \lambda_r t^{-r/3},$$

with error $O(n^{-(M+1)/3})$ at integers. Let $\mathcal{I}$ denote linear interpolation and define x_M by $(\mathcal{I}H_M)(x_M) = L$. Then

$$x(y) - x_M = O\left(\frac{n_0^{-(M+1)/3}}{\log n_0}\right).$$

This statement requires no derivative bounds on the enumeration remainder. A smooth-model inverse instead solves $H_M(z_M) = L$ and can be obtained by Newton iteration from n_0 ; its error squares with factor $O((n_0 \log n_0)^{-1})$ at each step. These two inverses differ at high order. In fact, with $\theta = \{z_M\}$,

$$x_M - z_M = -\frac{\theta(1-\theta)}{z_M H'_M(z_M)} + O\left(\frac{z_M^{-5/3}}{\log z_M}\right).$$

The fractional-part term comes from the chord error of the convex leading logarithm. A pure smooth inverse series must not silently be identified with the actual log-linear inverse beyond this order. No analogous all-orders DFA conclusion is needed or asserted here.

46 Verification and scope

The accompanying reproducibility package contains two independent exact polynomial–Airy recursion implementations, rational checks of the critical and moving boundary identities, exact completed-run and backward-monotonicity checks, and diagnostic numerical residual and inverse tests. The diagnostic floating checks verify implementations and signs; the proofs above supply the infinite-range conclusions.

Independent reviews cover the singular compactness and quasimode argument, the every-fixed-order transfer, the DFA amplitude transfer with the repaired tail, and the inverse conventions. The report keeps the inherited relaxed Theta theorem separate from its new proof steps. It includes no full third-party paper and makes no claim of an exhaustive novelty search. All results here are for the ternary case. The leading amplitudes are positive but are not numerically or algebraically evaluated.

Orders of growth by the completed-run comparison (source 29)

Source. Batch-77 manuscript 29, archive `larger-alphabet-automata-report.zip`, main file `article/larger-alphabet-automata.tex` (344 lines; 6-page PDF), titled “Finite languages over larger alphabets: Summable exclusion and sharp order estimates”, dated 2 October 2026, empty author line. Handed over by cluster P4 and printed in full as the comparison section of Part III, label prefix `air:ka:al:`.

Abstract of source 29.

For each fixed alphabet size $k \geq 3$, we transfer the known sharp order of relaxed k -ary trees to minimal deterministic automata recognizing finite languages. The proof is a positive completed-run representation: exclusion reduces each run weight by at most $1/[2(m+1)^{k-1}]$, whose sum converges. We obtain an explicit uniform comparison constant, the factorial, exponential, Airy and polynomial scales, a uniform finite-level approximation, and a logarithmic inverse enclosure. We do not prove a limiting amplitude or an all-orders expansion. A boundary calculation explains why the binary one-step singular-profile argument does not extend without modification.

[**write, 2 October 2026**] Source 29 is the earliest form of the positive completed-run comparison that sources 18 and 66 use as a proof step ((35.2), (44.1)); Theorem 47.1 is where this report states it. Its Θ conclusion (47.4) is superseded by Theorem 30.1, which proves a positive amplitude and every fixed order. What remains source 29’s own: the explicit finite- n constant $P_{k,n}$, uniform in $k \geq 3$ and $n \geq 1$; the explicit lower bound $P_k \geq (2\sqrt{2}/\pi) \sin(\pi/\sqrt{2})$; the uniform finite-level approximation of Proposition 49.1; the Θ -only inverse enclosure of Theorem 50.1, with the injection $B_{n+1} \geq kB_n$; the count by total states; and the boundary obstruction of Section 51, which the ternary weight (Lemma 39.1) and Lemma 31.1 resolve. Source 66’s transfer audit lists source 29’s research proof among its audited inputs.

47 Statement and state convention

Fix an integer $k \geq 3$, and write $q = k - 1$. Let B_n count finite languages over a fixed ordered k -letter alphabet whose minimal complete deterministic automaton has exactly n transient states and one rejecting sink, up to state isomorphism. Thus $B_0 = 1$ counts the empty language, and $B_1 = 1$ counts $\{\varepsilon\}$. Let R_n count relaxed k -ary trees with n internal nodes, in the convention of Ghosh Dastidar–Wallner [3].

Set

$$C_k = \frac{2k^k}{q^q}, \quad \beta_k = 3z \left(\frac{kq}{2} \right)^{1/3}, \quad \alpha_k = \frac{2k-1}{3}, \quad (47.1)$$

where $z = -2.338107410\dots$ is the largest negative zero of Ai.

Theorem 47.1. *For every $n \geq 1$,*

$$P_{k,n} \leq \frac{B_n}{2^{n-1}R_n} \leq 1, \quad P_{k,n} = \prod_{j=2}^n \left(1 - \frac{1}{2j^{k-1}} \right). \quad (47.2)$$

Furthermore,

$$P_k := \prod_{j=2}^{\infty} \left(1 - \frac{1}{2j^{k-1}} \right) \geq \frac{2\sqrt{2}}{\pi} \sin \frac{\pi}{\sqrt{2}} > 0. \quad (47.3)$$

Consequently, for each fixed $k \geq 3$,

$$B_n = \Theta \left((n!)^{k-1} C_k^n \exp(\beta_k n^{1/3}) n^{\alpha_k} \right). \quad (47.4)$$

The only asymptotic input is the published relaxed-tree theorem [3, Theorem 1]. The comparison (47.2) is exact and elementary. Its lower constant is uniform in both $k \geq 3$ and $n \geq 1$; the unspecified constants in (47.4) may depend on k . For ternary alphabets the scales are

$$(n!)^2 (27/2)^n \exp(3 \cdot 3^{1/3} z n^{1/3}) n^{5/3}.$$

Remark 47.2 (Total states). The exponent $\alpha_k = (2k-1)/3$ uses *transient* states. If $N = n+1$ is the total number of states and $D_N = B_{N-1}$, then

$$D_N = \Theta \left((N!)^{k-1} C_k^N \exp(\beta_k N^{1/3}) N^{(2-k)/3} \right).$$

Indeed $(N-1)! = N!/N$, and the remaining shifted factors change only by a bounded factor tending to a positive constant.

48 Exact recurrence and positive completed runs

The arrays in [3, Propositions 8 and 12] satisfy, on $x \geq qm \geq q$,

$$r_{x,m} = r_{x,m-1} + (m+1)r_{x-1,m}, \quad (48.1)$$

$$b_{x,m} = 2b_{x,m-1} + (m+1)b_{x-1,m} - mb_{x-k,m-1}. \quad (48.2)$$

Their base row is $r_{x,0} = b_{x,0} = 1$ for $x \geq 0$, and they vanish when $x < qm$ for $m \geq 1$. Retain the auxiliary value $b_{-1,0} = 1$; all other negative-coordinate values are zero. Then

$$R_n = r_{qn,n}, \quad B_n = b_{qn,n}.$$

The auxiliary boundary is essential: omitting it would give $B_1 = 2$, in conflict with the combinatorial interpretation and Table 3 of [3]. Equivalently, prescribe $b_{x,1} = 2^{x-k+2} - 1$ for $x \geq k-1$ and use (48.2) above that row. This makes explicit the first-state convention implicit in the terse published recurrence and consistent with the binary case [2].

[write, 2 October 2026] The binary recurrence with this auxiliary value is Part IV's (53.1).

For $m \geq 1$ and $\ell \geq 0$ define

$$V_m(\ell) = (m+1)^\ell, \quad W_m(\ell) = 2(m+1)^\ell - \mathbf{1}_{\{\ell \geq k\}}(m+1)^{\ell-k+1}. \quad (48.3)$$

Let $a(x, 1) = A(x, 1) = 1$ for $x \geq q$. Both arrays vanish for $x < qm$. For $x \geq q(m+1)$ set

$$a(x, m+1) = \sum_{\ell=0}^{x-qm} V_m(\ell) a(x-\ell, m), \quad (48.4)$$

$$A(x, m+1) = \sum_{\ell=0}^{x-qm} W_m(\ell) A(x-\ell, m). \quad (48.5)$$

Proposition 48.1. *All these weights are nonnegative. For $x \geq qm$ and $m \geq 1$,*

$$r_{x,m} = \sum_{\ell=0}^{x-qm} (m+1)^\ell a(x-\ell, m), \quad (48.6)$$

$$b_{x,m} = \sum_{\ell=0}^{x-qm} (m+1)^\ell A(x-\ell, m). \quad (48.7)$$

In particular $R_n = a(qn, n)$ and $B_n = A(qn, n)$ for $n \geq 1$.

Proof. The completed-run generating series is

$$\sum_{\ell \geq 0} W_m(\ell) t^\ell = \frac{2 - (m+1)t^k}{1 - (m+1)t}.$$

Thus (48.5) and (48.7) give $A(x, m+1) = 2b_{x,m} - (m+1)b_{x-k,m}$, while adding the trailing horizontal steps gives $b_{x,m+1} = A(x, m+1) + (m+2)b_{x-1,m+1}$. Their combination is (48.2). The first row in (48.7) is $\sum_{\ell=0}^{x-q} 2^\ell = 2^{x-k+2} - 1$. Uniqueness of the triangular recurrence proves the identity. Replacing the numerator by 1 gives (48.1) and (48.6). At $x = qn, m = n$ only the term $\ell = 0$ survives. $\square$

49 Summable exclusion and its consequences

Proof of Theorem 47.1. For every $m \geq 1$ and $\ell \geq 0$,

$$2 \left(1 - \frac{1}{2(m+1)^{k-1}} \right) V_m(\ell) \leq W_m(\ell) \leq 2V_m(\ell).$$

Induction in the nonnegative convolutions (48.4)–(48.5) therefore proves

$$2^{m-1} P_{k,m} a(x, m) \leq A(x, m) \leq 2^{m-1} a(x, m).$$

The common trailing-run convolution preserves these inequalities for r, b . Taking the endpoint proves (47.2). Since $k \geq 3$, the sum of the deficits $1/(2j^{k-1})$ converges and all factors are positive; hence $P_k > 0$. The factors increase with k . Euler's sine product gives

$$P_3 = \frac{\prod_{j=1}^{\infty} (1 - 1/(2j^2))}{1 - 1/2} = \frac{2\sqrt{2}}{\pi} \sin \frac{\pi}{\sqrt{2}}.$$

Finally, the published theorem [3] is

$$R_n = \Theta\left((n!)^{k-1} \left(\frac{k^k}{(k-1)^{k-1}}\right)^n \exp(\beta_k n^{1/3}) n^{(2k-1)/3}\right).$$

Combining it with (47.2) proves (47.4). $\square$

The same comparison holds when $k = 2$, but its lower product then tends to zero like $n^{-1/2}$. It cannot recover the binary-DFA power $7/8$. The summability threshold at $k = 3$ is the reason this larger-alphabet leading-order extension is simpler than the binary amplitude problem.

[**write, 2 October 2026**] The binary amplitude problem is solved in Part IV (source 48): $b_n \sim \gamma n! 8^n e^{3zn^{1/3}} n^{7/8}$ with $\gamma > 0$, by cancelling a non-summable drift rather than by a summable comparison.

Proposition 49.1 (Uniform finite-level approximation). *For an integer $M \geq 1$, retain W_m at levels $m < M$, replace it by $2V_m$ for $m \geq M$, and keep the first row unchanged. Write $B_n^{[M]}$ for the resulting endpoint count. For every $n \geq 1$ and $k \geq 3$,*

$$0 \leq 1 - \frac{B_n}{B_n^{[M]}} \leq \frac{1}{2} \sum_{j>M} j^{-(k-1)} \leq \frac{1}{2(k-2)M^{k-2}}. \quad (49.1)$$

Proof. The exact ratio lies between 1 and $\prod_{j=M+1}^{\max(M,n)} (1 - 1/(2j^{k-1}))$, by the same convolution comparison. Use $1 - \prod_j (1 - u_j) \leq \sum_j u_j$ and the integral bound for the decreasing tail. $\square$

If $B_n^{[M]}/(2^{n-1}R_n)$ converges for every fixed M , then (49.1) proves convergence of $B_n/(2^{n-1}R_n)$ by a uniform approximation argument. This reduces, but does not solve, the ratio-limit problem.

[**write, 2 October 2026**] The ratio-limit problem is solved in Part III: for $k = 3$ by exactly this route in Section 44.2 (source 66), and for every $k \geq 3$, with the first defect and eventual monotonicity, by Theorem 30.2 (source 18).

50 A logarithmic inverse enclosure

[**write, 2 October 2026**] An instance of theorems `p0:thm:lambert-core` and `p0:thm:staircase` of the repository's transseries volume (Part 0); no novelty is claimed for the inversion mechanics. Theorem 30.1 now gives the amplitude term that this enclosure leaves open, and Section 36 the corresponding finer inverse.

This section uses only (47.4); it does not posit an unknown amplitude. Put

$$\lambda_k = C_k^{1/q}/e, \quad \eta_k = \alpha_k + q/2 = \frac{7k-5}{6},$$

and define for large positive t

$$F_k(t) = qt \log(\lambda_k t) + \beta_k t^{1/3} + \eta_k \log t. \quad (50.1)$$

Stirling's formula and (47.4) give the precise bounded-error statement

$$\log B_n = F_k(n) + O_k(1). \quad (50.2)$$

In particular the coefficient of $\log n$ is η_k , not α_k .

For $y \rightarrow \infty$, let W be the positive real Lambert function and set

$$t = \frac{y}{qW(\lambda_k y/q)}, \quad L = 1 + \log(\lambda_k t), \quad (50.3)$$

$$\nu_k(y) = t - \frac{\beta_k t^{1/3} + \eta_k \log t}{qL}. \quad (50.4)$$

Theorem 50.1. *There are constants $K_k, y_k > 0$ such that, for $y \geq y_k$,*

$$\left\lceil \nu_k(y) - \frac{K_k}{\log \nu_k(y)} \right\rceil \leq I_k(e^y) \leq \left\lceil \nu_k(y) + \frac{K_k}{\log \nu_k(y)} \right\rceil, \quad (50.5)$$

where $I_k(X) = \min\{n \geq 1 : B_n \geq X\}$. Moreover

$$n = \nu_k(\log B_n) + O_k(1/\log n).$$

The constants here are existence bounds, not effective numerical certificates.

Proof. Write $f(t) = qt \log(\lambda_k t)$ and $g(t) = \beta_k t^{1/3} + \eta_k \log t$. Formula (50.3) gives $f(t) = y$ and $f'(t) = qL$. With $\delta = -g(t)/(qL) = O(t^{1/3}/\log t)$, Taylor's formula yields

$$F_k(t + \delta) - y = O(\delta^2/t) + O((t^{-2/3} + t^{-1})|\delta|) = o(1).$$

Since $F'_k(t) \sim q \log t > 0$, (50.2) implies the asserted $O(1/\log n)$ inversion at sequence values. Choose a constant bounding the error in (50.2). Increasing K_k if necessary makes every sufficiently large integer below $\nu_k(y) - K_k/\log \nu_k(y)$ have $\log B_n < y$, and every sufficiently large integer at or above the upper endpoint have $\log B_n \geq y$. Taking ceilings gives (50.5).

The threshold is well defined: for every nonempty finite language L , each language aL , with a a letter, has one more transient state in its minimal complete automaton. The old initial residual L is reached after a ; all its residuals remain reachable, and the new initial residual has strictly larger maximum word length. The k constructions are disjoint and injective, so $B_{n+1} \geq kB_n$ for $n \geq 1$. $\square$

The enclosure respects integer rounding. It does not assert a smooth interpolation canonically determined by the sequence, nor an everywhere unambiguous rounded estimate without the displayed error interval.

51 What remains open and a boundary obstruction

Neither the relaxed theorem nor the comparison proves that the quotient in (47.4) has a limit. A positive limiting amplitude and higher-order endpoint corrections for this family remain unproved here. A relaxed amplitude theorem alone would still require control of the finite-level perturbed ratios in Proposition 49.1.

[**write, 2 October 2026**] A positive limiting amplitude, and the endpoint corrections to every fixed order, are proved for every fixed $k \geq 3$ in Theorem 30.1 (source 18; for $k = 3$ the amplitude also in Theorem 38.1, source 66). The obstruction below is resolved there by the fixed weight $\omega = \ell/h$, which matches the forward and adjoint boundary harmonics on every residue phase (Lemmas 39.1 and 31.1); the paragraph after Lemma 39.1 names the residue-dependent extrapolation lengths computed here.

The critically tilted relaxed transfer has two shifts:

$$(T_i f)(j) = U(i, j)f(j-1) + f(j+k-1), \quad U(i, j) = \frac{(k-1)^2(i-j+k)}{(k-1)i+j}.$$

For $k \geq 3$, its directed zero pattern has a forward edge of length one without its reverse. Positive diagonal conjugation cannot make it symmetric. Although $T_i^* T_i$ is Jacobi on residue classes modulo k , the following exact frozen calculation prevents an automatic binary singular-profile proof.

Let $T = q \text{ up} + \text{down}_q$ on the half-line, and let S be the unilateral lower shift in residue coordinate $\ell \geq 0$. From input residue r to output residue $r+1 \pmod k$, the matrix is

$$T_r = qI + S^* \quad (0 \leq r < q), \quad T_q = I + qS.$$

At the bulk squared edge k^2 , a formal linear solution $\ell + c$ of the right and left Jacobi boundary equations has constants

input residue	c_{right}	c_{left}	physical extrapolated zero
$0 \leq r < q$	q/k	1	$r - q$
$r = q$	1	$1/k$	-1

For example the first right equation is $q^2 c + q(c+1) = k^2 c$, giving $c = q/k$; the first left equation has bottom diagonal $q^2 + 1$ and gives $c = 1$. In physical coordinates the right and left zeros agree within one step, but change between successive residues. The boundary offsets cycle as $q, q-1, \dots, 1, 1$, so only the binary case has a constant offset. Thus an $O(i^{-1/3})$ boundary-scale mismatch may occur between successive singular profiles, and an $O(i^{-1})$ tracking estimate requires an additional cancellation argument. The finite operator's Airy asymptotics are not asserted by this frozen calculation.

A whole k -step analysis or matched forward and adjoint boundary profiles could address this issue. Also $U(i, j)$ can exceed q near the boundary, so global contraction requires its own normalization. These analytic questions are separate from the positive proof of Theorem 47.1.

52 Verification and literature scope

The attached standard-library Python verifier checks signed and positive recurrences, all-cell comparisons, truncation bounds, and published terms for $2 \leq k \leq 8$ through 16 transient states. Independent small-automaton checks and mathematical review are recorded separately. In particular,

$$\begin{aligned} k = 3 : & \quad 1, 1, 14, 532, 42644, 6011320, 1330452032, \dots, \\ k = 4 : & \quad 1, 1, 30, 3900, 1460700, 1220162880, \dots \end{aligned}$$

A331120 is the binary minimal-DFA sequence. A082162 counts relaxed ternary trees, not minimal ternary DFAs. No larger-arity minimal-DFA OEIS identifier was verified in the bounded search.

The 2024 primary paper’s conclusion leaves compacted and minimal-DFA asymptotics to a projected long version. The current author bibliography, focused later-literature searches, and authorized Provel semantic searches were checked on 2 October 2026; no matching later result was located. Wallner’s 2026 decompression paper [7] concerns a different statistic on chains. These are bounded overlap checks, not a universal novelty guarantee. The attached receipts specify the queries and limitations.

[**write, 2 October 2026**] The repository searches were right when made (the receipt refers to commit `4b874cea0`). The repository’s material on these automata is now this report, which the same batch delivered. The raw search receipt `sources/overlap.json` is not shipped; see the README.

Part IV

Binary finite-language automata

Source. Batch-77 manuscript 48, archive `oeis-dfa-finite-language-report.zip`, main file `article/dfa-finite-languages.tex` (967 lines; 16-page PDF), titled “Positive amplitude and all orders for finite binary languages”, dated 2 October 2026, empty author line, placed by commit `aa7345800` (cluster P2). Label prefix `air:dfa:.` Its passages that are word for word Part II’s are replaced by pointers, and its appendix is Appendix A; every passage in which the automaton differs is printed.

Abstract of the source.

A331120 counts finite binary languages by the number of transient states in their minimal deterministic automata. The known estimate has order $n!8^n \exp(3zn^{1/3})n^{7/8}$, where z is the largest negative Airy zero. We strengthen it to a positive limiting amplitude and a relative expansion at every finite order in $n^{-1/3}$. An exact nonnegative renewal model establishes positivity and domination, while a half-strength signed delay of a contractive Jacobi evolution identifies the polynomial correction. A finite-memory projection argument and an arbitrary-order residual bootstrap yield endpoint accuracy without assuming that the signed propagator is positive. We give three explicit logarithmic corrections, a constructive coefficient recursion, a selected smooth inverse, and the proper distinction between inverse interpolation and discrete thresholds. The shared spectral estimates are proved in a self-contained appendix. The amplitude exists and is positive; its decimal digits are not certified.

53 Statement and mathematical scope

Let $b_{n,m}$ be the array

$$\begin{aligned} b_{n,m} &= 2b_{n,m-1} + (m+1)b_{n-1,m} - mb_{n-2,m-1} & (n \geq m \geq 1), \\ b_{n,0} &= 1 & (n \geq -1), \quad b_{n,m} = 0 \quad (m > n). \end{aligned} \tag{53.1}$$

The auxiliary value $b_{-1,0} = 1$ is retained. Put $b_n = b_{n,n}$. This is A331120, beginning

1, 1, 6, 60, 900, 18480, 487560, 15824880, 612504240, . . .

It counts minimal complete deterministic automata over a fixed binary alphabet, up to isomorphism of states, with n transient states and one rejecting sink. Equivalently it counts the corresponding finite languages by minimal-state complexity. Proposition 5 of [2] proves the exact recurrence. The same paper's Theorem 1 proves

$$b_n = \Theta\left(n!8^n e^{3zn^{1/3}} n^{7/8}\right), \quad (53.2)$$

where $z = -2.338107410459767\dots$ is the largest real zero of Ai . The Airy exponential and exponent $7/8$ are therefore established inputs. The additional result here is a convergent positive amplitude and its complete finite algebraic correction hierarchy.

Set $a = 2^{-1/3}z$ and $F(x) = \text{Ai}(z + 2^{1/3}x)$, so $F'' = 2(x + a)F$, $F(0) = 0$, and $F'(0) > 0$.

Theorem 53.1. *There is a constant $\gamma > 0$ such that, for every fixed integer $M \geq 0$,*

$$b_n = \gamma n!8^n e^{3zn^{1/3}} n^{7/8} \left(1 + \sum_{k=1}^M \mathbf{c}_k n^{-k/3} + O_M(n^{-(M+1)/3})\right). \quad (53.3)$$

All $\mathbf{c}_k$ are constructible elements of $\mathbb{Q}[z]$. In logarithmic form the first three corrections are

$$\begin{aligned} \log \frac{b_n}{\gamma n!8^n e^{3zn^{1/3}} n^{7/8}} &= \frac{53z^2}{90} n^{-1/3} + \frac{623z}{432} n^{-2/3} \\ &+ \left(\frac{3497}{4480} - \frac{1304z^3}{42525}\right) n^{-1} + O(n^{-4/3}). \end{aligned} \quad (53.4)$$

This is a Poincaré expansion to every fixed finite order. It is not a claim of an exponentially improved transseries or a canonical analytic continuation of an integer sequence. Strict positivity of γ uses the published lower bound in (53.2) only after the convergence and all-orders error estimates have been proved.

[**write, 5 October 2026, batch 102**] Part V (Report 103) proves this theorem a second time, in even time, taking positivity from the comparison $b_n \geq 2^{n-1}c_n$ and the compacted lower bound only. It continues (53.4) by $d_4 = -5429z^2/19440$, (101.1), gives $\mathbf{c}_4$ (c_4 in (101.2)) and the normalized ratio $b_n/(8nb_{n-1})$ through $n^{-7/3}$, (101.3).

54 Exact positivity and the initial boundary

Normalize uniformly by $c_{n,m} = b_{n,m}/2^m$. Then

$$c_{n,m} = c_{n,m-1} + (m+1)c_{n-1,m} - \frac{m}{2}c_{n-2,m-1}. \quad (54.1)$$

In particular $c_{n,0} = 1$, $c_{-1,0} = 1$, and $c_{n,1} = (2^n - 1)/2$. A normalization by 2^{m-1} only for $m \geq 1$ requires a separate boundary row; (54.1) avoids that exception.

[**write, 5 October 2026, batch 102**] Report 103 in Part V uses the same uniform normalization, $B(n, m) = b(n, m)/2^m$ (this Part's $c_{n,m}$), and goes further: it states that the transformed arrays printed on pages 11:6–11:7 of [2], normalized by 2^{m-1} for $m \geq 1$, give $\frac{5}{4}$ instead of $\frac{3}{2}$ at one entry when read uniformly (Section 93). That statement about the printed paper has not been checked here (Question V.4); Report 103 adds that it is not a claim that the published theorem is false.

Proposition 54.1 (Positive renewal representation). *Set $A(n, 1) = 1$ for every $n \geq 1$ and $A(n, m) = 0$ for $n < m$. For $m \geq 1$ put*

$$W_m(k) = \begin{cases} 2, & k = 0, \\ 2(m+1), & k = 1, \\ (2(m+1)^2 - (m+1))(m+1)^{k-2}, & k \geq 2. \end{cases} \quad (54.2)$$

For $n \geq m+1$, define

$$A(n, m+1) = \sum_{k=0}^{n-m} W_m(k) A(n-k, m).$$

Then

$$b_{n,m} = \sum_{k=0}^{n-m} (m+1)^k A(n-k, m) \quad (m \geq 1). \quad (54.3)$$

All weights are nonnegative, and $b_n = A(n, n)$ for $n \geq 1$.

Proof. The completed-run generating series is

$$\sum_{k \geq 0} W_m(k) t^k = \frac{2 - (m+1)t^2}{1 - (m+1)t}.$$

Consequently completing a run followed by a vertical step gives $A(n, m) = 2b_{n,m-1} - mb_{n-2,m-1}$ for $m \geq 2$. Adding possible trailing horizontal steps in (54.3) recovers (53.1). For $m = 1$, the initial renewal values give $b_{n,1} = \sum_{k=0}^{n-1} 2^k = 2^n - 1$, which agrees with (53.1), including its initial value $b_{-1,0}$. The triangular cutoff ensures that only $k = 0$ survives on $n = m$. Uniqueness of the triangular recurrence proves the assertion. $\square$

Combinatorially A counts prefixes ending in a vertical step. At height m , a horizontal run has $m+1$ choices per edge and two vertical colors; the subtracted completed-run decorations are exactly the forbidden repeated transition signature. The exceptional first row includes the mandatory initial horizontal edge and the requirement that the first non-sink state be accepting. The algebraic proof above makes the initial convention exact.

Let $r_{n,m}$ denote the relaxed array $r_{n,0} = 1$, $r_{n,m} = r_{n,m-1} + (m+1)r_{n-1,m}$, with invalid entries zero. Since $c_{n,m} \geq 0$, induction in (54.1) yields

$$0 \leq c_{n,m} \leq r_{n,m}. \quad (54.4)$$

This is a domination of exact counts; no positivity of the signed linear propagator will be assumed.

55 The shared Jacobi evolution

[write, 2 October 2026] Sections 55–60 and Appendix A of source 48 follow the compacted-tree argument of Part II line by line; only the renewal model, the delay coefficient (with its $m = 1$ term), the drift $1/(8N)$, the seeds and the coefficients differ. Source 48 does not say so. Its Section 3 up to the paragraph below is, word for word, the beginning of Part II's Section 18: the gauge (18.1), the Jacobi matrix S_N with edge weights $b_{N,j}$ and the contraction Q_N (18.2), the vectors g_N, h_N and the operators P_N, M_N, Π_N (18.3), Lemma 18.1 with (18.4) and the finite profile (18.5), and the remark that the gap is parity-restricted. It is not reprinted, and the

references below to these displays and to Lemma 18.1 are to Part II. Source 48 adds only the conventions $D_0(0) = 1$, $P_0 = 1$ and $v_0^{(r)} = \delta_0$. Its appendix “Self-contained Jacobi estimates” is Appendix A with “DFA theorem” in place of “compacted theorem”, and is not reprinted either.

The relaxed counts $r_{n,m}$ satisfy

$$r_{n,0} = 1, \quad r_{n,m} = r_{n,m-1} + (m+1)r_{n-1,m},$$

with invalid cells zero. By (54.4), $0 \leq c_{n,m} \leq r_{n,m}$. The common gauge turns the relaxed recurrence into $v_N^{(r)} = S_N Q_N v_{N-1}^{(r)}$; consequently $\|v_N^{(r)} / P_N\| \leq 1$. Only this domination and the explicit Jacobi estimates, not an automatic extension of a relaxed amplitude theorem, will be used to prove Theorem 53.1.

56 A globally bounded signed delay

Set

$$e_{N,j} = \frac{c_{(N+j)/2, (N-j)/2}}{((N+j)/2)!}, \quad v_N(j) = \frac{e_{N,j}}{D_N(j)}$$

on $0 \leq j \leq N$, $j \equiv N \pmod{2}$, and zero otherwise. For $N \geq 3$, direct substitution in (54.1) gives

$$e_{N,j} = \frac{N-j+2}{N+j} e_{N-1,j-1} + e_{N-1,j+1} - \frac{N-j}{(N+j)(N+j-2)} e_{N-3,j-1}. \quad (56.1)$$

The physical recurrence starts at $N = 3$, using the exact initial vectors

$$e_0(0) = 1, \quad e_1(1) = 1, \quad e_2(0) = e_2(2) = \frac{1}{2},$$

with all other entries zero. At $N = 2, j = 0$ the fictitious $c_{-1,0} = 1$ contributes to (54.1); omitting it would change the seed. For $N \geq 3$ the delay is used only on $1 \leq j \leq N-2$ with the correct parity. It is zero at $j = 0$ and absent on $m = 0$. It is *nonzero* at $m = 1, j = N-2$, unlike the compacted-tree delay.

[**write, 2 October 2026**] In Part II the delay coefficient $2(N-j-2)/((N+j)(N+j-2))$ vanishes at $j = N-2$ (Section 19); here it is $(N-j)/((N+j)(N+j-2))$, nonzero there.

For all sufficiently large N , the exact gauge recurrence is

$$v_N = S_N Q_N v_{N-1} - L_N v_{N-3}, \quad (L_N f)(j) = \ell_{N,j} f(j-1), \quad (56.2)$$

where on $1 \leq j \leq N-2$

$$\ell_{N,j} = \frac{N-j}{(N+j)(N+j-2)} \frac{D_{N-3}(j-1)}{D_N(j)}, \quad (56.3)$$

and it is zero elsewhere. Cancellation of factorials gives

$$\left(\frac{D_{N-3}(j-1)}{D_N(j)} \right)^2 = \frac{(N-j+1)(N-j)(N+j)(N+j-1)(N+j-2)(N+j-3)}{(N+1)N^2(N-1)^2(N-2)}. \quad (56.4)$$

For $N \geq 6$ this ratio is uniformly bounded on the entire physical support; the rational prefactor in (56.3) is $O(N^{-1})$. Hence

$$\|L_N\| \leq C N^{-1}. \quad (56.5)$$

This is a global operator bound, not an estimate limited to the Airy window.

For $1 \leq j \leq 2N^{1/2}$ on the physical support, expanding (56.3)–(56.4) gives

$$\ell_{N,j} = \frac{1}{N} + O\left(\frac{j+1}{N^2}\right).$$

Let U be the unilateral upward shift: $(Uf)(0) = 0$ and $(Uf)(j) = f(j-1)$. The finite profile in Lemma 18.1, its Airy moments, and (56.5) imply

$$\left\|L_N g_{N-3} - \frac{1}{N} U g_{N-3}\right\| = O(N^{-5/3}), \quad \|U g_{N-3} - g_N\| = O(N^{-1/3}). \quad (56.6)$$

For the first estimate the typical height is $O(N^{1/3})$, and the $O(N^{-4/3})$ quasimode error is multiplied by $O(N^{-1})$. Cutoff tails are exponentially small. The second estimate uses a one-site sample shift and an $O(N^{-1})$ time change. The shifted parity is exactly the parity occupied at time N .

57 Cancellation of the scalar drift

Put

$$R_N = P_N N^{-1/8}, \quad u_N = v_N / R_N, \quad (57.1)$$

and begin the following recurrences at any fixed sufficiently large N . Then

$$u_N = \eta_N M_N u_{N-1} - B_N u_{N-3}, \quad \eta_N = \left(\frac{N}{N-1}\right)^{1/8}, \quad B_N = \frac{R_{N-3}}{R_N} L_N. \quad (57.2)$$

Since $R_{N-3}/R_N = 1/8 + O(N^{-2/3})$, Lemma 18.1 and (56.6) yield

$$\begin{aligned} \eta_N &= 1 + \frac{1}{8N} + O(N^{-2}), \quad \|B_N\| = O(N^{-1}), \\ \|B_N g_{N-3} - \frac{1}{8N} g_N\| &= O(N^{-4/3}), \\ b_N := \langle g_N, B_N g_{N-3} \rangle &= \frac{1}{8N} + O(N^{-4/3}), \\ \eta_N t_N - 1 - b_N &= O(N^{-4/3}). \end{aligned} \quad (57.3)$$

The last line cancels the nonsummable order- N^{-1} drift. It explains the additional factor $N^{-1/8}$ relative to the relaxed spectral normalization.

[**write, 2 October 2026**] Source 48 states here the finite-memory Lemma 20.1, with (20.4), (20.5) and its proof, word for word as in Part II; it is not reprinted, and the references to it below are to Part II.

58 Convergence of the scalar amplitude

The positive gauge preserves $0 \leq v_N \leq v_N^{(r)}$ entrywise. The relaxed contraction therefore gives the initial bound

$$\|u_N\| \leq N^{1/8}. \quad (58.1)$$

Decompose

$$u_N = A_N g_N + w_N, \quad w_N \perp g_N.$$

Applying Lemma 18.1 to the transverse part of (57.2), and bounding the delayed whole vector using (58.1), gives

$$\|w_N\| \leq (1 - c' N^{-2/3}) \|w_{N-1}\| + O(N^{-7/8}).$$

The factor $\eta_N = 1 + O(N^{-1})$ is absorbed in the gap. Lemma 20.1 then gives

$$\|w_N\| = O(N^{-5/24}). \quad (58.2)$$

No cancellation or positivity of the delayed evolution is needed here.

Projecting the recurrence gives

$$A_N = \eta_N t_N A_{N-1} - b_N A_{N-3} + \xi_N, \quad |\xi_N| \leq C N^{-1} (\|w_{N-1}\| + \|w_{N-3}\|). \quad (58.3)$$

The one-step cross term uses $\|M_N^* g_N - g_{N-1}\| = O(N^{-1})$ and orthogonality; the delayed cross term uses only $\|B_N\| = O(N^{-1})$. Set $\delta_N = A_N - A_{N-1}$. Then

$$\delta_N = b_N (\delta_{N-1} + \delta_{N-2}) + (\eta_N t_N - 1 - b_N) A_{N-1} + \xi_N. \quad (58.4)$$

Equations (57.3), (58.1) and (58.2) give (20.5) with $r = 29/24$. Thus $\delta_N = O(N^{-29/24})$ is summable, and A_N has a finite limit A_∞ . Since u_N and g_N are nonnegative, $A_\infty \geq 0$.

Now A_N is bounded and $w_N \rightarrow 0$, so u_N is bounded. Repeating the transverse and scalar estimates improves them to

$$A_N = A_\infty + O(N^{-1/3}), \quad \|w_N\| = O(N^{-1/3}), \quad \|u_N\| = O(1). \quad (58.5)$$

There is not yet an endpoint conclusion: the normalized ground endpoint itself is only of order $N^{-1/2}$.

59 Formal profiles at every finite order

Let $\varepsilon = N^{-1/3}$ and $x = (j+1)\varepsilon$. Seek an approximate solution to the ungauged recurrence in the form

$$Z_{N,j} = H_N \Phi_N(x), \quad \Phi_N(x) = \sum_{k=0}^K \varepsilon^k G_k(x), \quad G_0 = F, \quad (59.1)$$

with $G_k(0) = G'_k(0) = 0$ for $k \geq 1$, and formal ratio

$$\frac{H_N}{H_{N-1}} \sim 2 \left(1 + \sum_{m \geq 2} s_m \varepsilon^m \right). \quad (59.2)$$

The one-step arguments are $(x \pm \varepsilon)(1 - \varepsilon^3)^{-1/3}$, and the delayed argument is $(x - \varepsilon)(1 - 3\varepsilon^3)^{-1/3}$. The horizontal and delayed coefficients in (56.1) become

$$\begin{aligned} \mathcal{A}(x, \varepsilon) &= \frac{1 - x\varepsilon^2 + 3\varepsilon^3}{1 + x\varepsilon^2 - \varepsilon^3}, \\ \beta(x, \varepsilon) &= \frac{\varepsilon^3(1 - x\varepsilon^2 + \varepsilon^3)}{(1 + x\varepsilon^2 - \varepsilon^3)(1 + x\varepsilon^2 - 3\varepsilon^3)}. \end{aligned} \quad (59.3)$$

After division by H_{N-1} , the delayed term has the extra factor

$$\frac{H_{N-3}}{H_{N-1}} = \left(\frac{H_{N-1}}{H_{N-2}} \frac{H_{N-2}}{H_{N-3}} \right)^{-1},$$

whose leading constant is $1/4$. As β starts at degree three, neither the new profile nor the new scalar in those previous ratios changes the leading Airy inversion at the current degree.

[**write, 2 October 2026**] Source 48 continues with the paragraph of Part II that gives the formal equation at degree m , (22.4), in which $R_m, S_m \in \mathbb{Q}[a][x]$ are already known and $s_2 = a$, and its triangular polynomial reduction (22.5), with existence and uniqueness under $Q(0) = 0$ and $P(0) + Q'(0) = 0$. The paragraph is word for word Part II's and is not reprinted; here the delayed coefficient is this Part's (59.3).

The initial scalar coefficients are

$$\begin{aligned} s_2 &= a, & s_3 &= \frac{29}{24}, & s_4 &= \frac{29a^2}{270}, \\ s_5 &= -\frac{103a}{648}, & s_6 &= \frac{5341815 - 562112a^3}{5443200}. \end{aligned} \quad (59.4)$$

A genuine finite logarithmic amplitude may consequently be chosen as

$$\log H_N = N \log 2 + 3aN^{1/3} + \frac{29}{24} \log N + \sum_{k=1}^J h_k N^{-k/3}. \quad (59.5)$$

At degree $k+3$ in the one-step ratio, the new h_k has coefficient $-k/3$, so any finite list of ratio coefficients can be matched uniquely. For every truncation the leading multiplicative constant in H_N is one. Initially,

$$h_1 = \frac{53a^2}{45}, \quad h_2 = \frac{551a}{216}, \quad h_3 = \frac{2371}{6720} - \frac{5216a^3}{42525}. \quad (59.6)$$

60 Global residuals and the zero-limit bootstrap

60.1 Approximate solutions in the normalized norm

Fix any $p > 1$. Take sufficiently many terms of (59.1) and (59.5), multiply by the cutoff $\chi(j/N^{1/2})$, restrict to occupied parity, and define

$$z_N = D_N^{-1} Z_N / R_N.$$

Then

$$\|z_N\| = O(1), \quad r_N := z_N - \eta_N M_N z_{N-1} + B_N z_{N-3}, \quad \|r_N\| = O(N^{-p}), \quad (60.1)$$

and

$$\langle g_N, z_N \rangle \longrightarrow q_0 = \kappa^{-1} \sqrt{I/2} > 0, \quad I = \int_0^\infty F(x)^2 dx. \quad (60.2)$$

Here is the required global justification. On $j \leq 2N^{1/2}$ the factorial product for D_N has $\log D_N^{-1} = O(j^2/N)$, so D_N^{-1} is uniformly bounded there. Taylor remainders in (59.3), the time dilations and

the finite amplitude ratios are bounded by an arbitrarily prescribed power of $N^{-1/3}$ times a fixed polynomial–Airy envelope. All such envelopes are square-integrable. The coefficient denominators stay uniformly away from zero. Cutoff derivative errors, and differences between one- and three-step cutoff supports, are exponentially small in $N^{1/8}$, up to powers of N . The bounded gauge preserves this property. At the left boundary all fictitious samples vanish exactly, because every $G_k(0) = 0$; the upper moving boundary lies outside the cutoff for large N . These facts give the norm residual after summation.

[**write, 2 October 2026**] Part II has $N^{1/4}$ here. The cutoff $\chi(j/N^{1/2})$ is the same, so both statements hold; the one with $N^{1/4}$ is the stronger.

For the normalization, $H_N/R_N \sim \kappa^{-1}N^{-1/6}$, and a parity-restricted Airy sample has norm asymptotic to $N^{1/6}\sqrt{I/2}$. Its direction tends to g_N , proving (60.2). Only the leading asymptotic of P_N has been used: the original approximate recurrence is divided by the exact R_N , so an all-orders expansion of the relaxed spectral product is unnecessary.

60.2 A bootstrap with vanishing limiting projection

[**write, 2 October 2026**] Source 48’s Section 8.2 is, word for word, Part II’s subsection of the same title (Section 23.2), equations (23.3)–(23.7), read with this Part’s η_N , B_N , r_N , z_N and Lemma 20.1. It is not reprinted. It sets

$$C_* = A_\infty/q_0, \quad E_N = u_N - C_*z_N = \alpha_N g_N + W_N, \quad W_N \perp g_N, \quad (60.3)$$

and concludes

$$\|E_N\| = O(N^{1-p}), \quad (60.4)$$

assuming neither positivity nor contractivity of the signed delayed propagator.

61 The endpoint and the common positive amplitude

Point evaluation has norm one on ℓ^2 , so $|E_N(0)| \leq \|E_N\| = O(N^{1-p})$. For even N , the leading normalized approximate endpoint is of order $N^{-1/2}$, because

$$H_N/R_N \sim \kappa^{-1}N^{-1/6}, \quad \Phi_N(\varepsilon) \sim F'(0)\varepsilon.$$

We may choose p arbitrarily large. For the relative remainder in (53.3), it is enough to take $p \geq 3/2 + (M+1)/3$ and retain enough formal terms. Thus (60.4) supplies every fixed finite relative endpoint order; no endpoint smoothing estimate for the signed kernel is needed.

If $A_\infty = 0$, then $C_* = 0$. Equation (60.4) with $p > 3/2$ would imply $u_N(0) = o(N^{-1/2})$. On the other hand, the known lower bound (53.2), divided by $2^n n! R_{2n}$, implies $u_{2n}(0) \geq cN^{-1/2}$ for large even $N = 2n$. This contradiction proves $A_\infty > 0$.

Since $D_N(0) = 1$ and $b_n = 2^n n! e_{2n,0}$, conversion from N to $2n$ gives

$$\begin{aligned} \gamma &= 2^{7/8} C_* F'(0) = 2^{15/8} \kappa A_\infty > 0, \\ A_\infty &= \lim_{N \rightarrow \infty} \langle g_N, v_N / (P_N N^{-1/8}) \rangle. \end{aligned} \quad (61.1)$$

We used the Airy identity $F'(0)/\sqrt{I} = \sqrt{2}$, proved in the appendix. Every truncation has the same q_0 and the same leading normalization of H_N . Hence C_* and γ are independent of the expansion order. This completes the analytic part of Theorem 53.1.

62 Explicit coefficients and their rational field

The normalized endpoint factor begins

$$\frac{\Phi_N(\varepsilon)}{F'(0)\varepsilon} = 1 + \frac{a}{3}\varepsilon^2 + \frac{29}{24}\varepsilon^3 + O(\varepsilon^4).$$

Combining this with (59.6), then setting $N = 2n$, gives (53.4). Exponentiation gives

$$\begin{aligned} \mathfrak{c}_1 &= \frac{53z^2}{90}, \\ \mathfrak{c}_2 &= \frac{z(5618z^3 + 46725)}{32400}, \\ \mathfrak{c}_3 &= \frac{8337112z^6 + 200508660z^3 + 191198475}{244944000}. \end{aligned} \tag{62.1}$$

These coefficients were independently checked by a direct polynomial linear-system solve, separate from the triangular recursion.

[**write, 5 October 2026, batch 102**] The next coefficients are in Part V (Report 103, which writes a for z , d_j for the logarithmic and c_j for the relative coefficients): $d_4 = -5429z^2/19440$ and $\mathfrak{c}_4 = z^2(220933468z^6 + 10228873860z^3 + 53801804475)/44089920000$, (101.1)–(101.2); its scalar coefficients σ_m are $2s_m$ in the variable $2^{2/3}z$ (Remark 66.2), and it adds $\sigma_7 = -767\kappa^2/23328$, (98.9).

[**write, 2 October 2026**] The rest of source 48’s Section 10, the proof that every coefficient lies in $\mathbb{Q}[z]$ by the grading under $(a, \varepsilon, x) \mapsto (\omega a, \omega \varepsilon, \omega x)$, is word for word the second half of Part II’s Section 25, its invariance statement covering “the full formal recurrence, including its delayed dilation and both previous ratios”. It is not reprinted.

63 Selected smooth inverse and discrete thresholds

[**write, 2 October 2026**] The anchor $X = Y/W((8/e)Y)$ and the corrections are instances of theorems `p0:thm:lambert-core` and `p0:thm:perturbed-inversion` of the repository’s transseries volume, and the threshold remark is the separation step of `p0:thm:staircase` (Part 0). No novelty is claimed for the inversion mechanics.

Write ℓ_k for the logarithmic correction coefficients in Theorem 53.1. For each finite M define a positive-real model by exponentiating

$$\mathcal{F}_M(x) = \log \gamma + \log \Gamma(x+1) + x \log 8 + 3zx^{1/3} + \frac{7}{8} \log x + \sum_{k=1}^M \ell_k x^{-k/3}. \tag{63.1}$$

Its derivative is $\log(8x) + O(x^{-2/3})$, positive for sufficiently large x . Therefore $\mathcal{F}_M$ has a unique eventual inverse $x_M(Y)$. No values of b_n at nonintegers are asserted.

Put

$$Y = \log y, \quad w = W((8/e)Y), \quad X = Y/w.$$

The principal real Lambert function is used. Then X solves $X(\log X + \log 8 - 1) = Y$. Stirling expansion gives

$$\mathcal{F}_M(x) = x(\log x + \log 8 - 1) + 3zx^{1/3} + \frac{11}{8} \log x + \log(\gamma\sqrt{2\pi}) + O(x^{-1/3}).$$

Consequently

$$x_M(Y) = X - \frac{3zX^{1/3} + \frac{11}{8} \log X + \log(\gamma\sqrt{2\pi})}{1 + w} + O\left(\frac{X^{-1/3}}{\log X}\right). \quad (63.2)$$

Indeed the correction is $O(X^{1/3}/\log X)$; the change in the stretched term's derivative contributes $O(X^{-1/3}/(\log X)^2)$ to the index, and the quadratic contribution from the factorial core is $O(X^{-1/3}/(\log X)^3)$. The first omitted forward coefficient is at most the remainder displayed in (63.2).

[write, 5 October 2026, batch 102] Report 103 in Part V makes the $X^{-1/3}$ term explicit: with $L = \log(8X) = 1 + w$ and $d_1 = 53z^2/90$, it is $X^{-1/3}[-d_1/L + 3z^2/L^2 - 9z^2/(2L^3)]$, with remainder $O(X^{-2/3}/L)$, (102.11).

Higher corrections can be generated by reversion of (63.1), or by Newton iteration

$$x^{(r+1)} = x^{(r)} - \frac{\mathcal{F}_M(x^{(r)}) - Y}{\mathcal{F}'_M(x^{(r)})}.$$

On $x \asymp X$, $\mathcal{F}''_M(x) = O(1/x)$ and $\mathcal{F}'_M(x) \asymp \log X$, so each sufficiently accurate Newton step squares the index error up to the factor $O((X \log X)^{-1})$. Finite forward orders and finite Newton depth therefore give any desired finite algebraic inverse accuracy. They remain distinct errors: solving a fixed model more accurately does not improve its agreement with b_n . Precisely, Theorem 53.1 and the mean-value theorem imply

$$x_M(\log b_n) - n = O_M\left(\frac{n^{-(M+1)/3}}{\log n}\right). \quad (63.3)$$

For arbitrary y , the first integer index with $b_n \geq y$ is a different object. The asymptotics show eventual strict increase, since $b_{n+1}/b_n \sim 8n$, but a threshold still requires checking the adjacent integers or a certified bracket. An unspecified big- O constant is not a certified finite threshold algorithm.

64 Verification and numerical limitations

The accompanying code separates analytic proof from finite evidence. The exact script checks 5,050 triangular cells through $n = 100$ against the positive renewal construction, verifies 5,049 normalized recurrence instances, includes the nonzero $m = 1$ delay, and reproduces all 20 displayed OEIS terms. The symbolic coefficient generator implements the triangular polynomial inversion. A separate direct-linear-system computation checks its coefficients without importing the generator. The analytic estimates above, rather than these finite checks, establish the theorem.

For orientation only, direct floating evaluation of the normalized amplitude gives approximately 178.317, 164.945, 150.038, 136.080, and 131.985 at $n = 10, 20, 40, 80, 100$, respectively. These slowly

varying values are *not* enclosures of γ , do not give certified digits, and are not used in the proof. Inverse numerical tests, when a trial amplitude is chosen, verify only the specified model with that trial parameter.

[write, 5 October 2026, batch 102] Report 103 in Part V carries the uncorrected values to 102.077 ($n = 1000$) and 94.052 ($n = 3000$), and, after dividing out four relative corrections, finds 76.4392, 76.4385, 76.4383 at $n = 1000, 2000, 3000$: evidence for $\gamma \approx 76.438$, uncertified.

The package contains this standalone article, its source and build script, exact and symbolic verification programs and JSON outputs, a separate mathematical review, bounded literature and overlap receipts, a clean replay script, and a file manifest. The PDF build and all pages were visually checked separately from mathematical review.

[write, 2 October 2026] This paragraph describes source 48’s delivered archive (commit 096ee7b87); its manuscript, PDF, build script and checksum manifest are not shipped. The README of this report lists the shipped files.

65 Literature boundary and further questions

The earlier binary-DFA result [2] provides the two-sided order, not the positive limiting amplitude. The later general-arity paper [3] gives a broader exact recurrence and a relaxed-tree asymptotic theorem. Its general-arity setting does not supply the present endpoint amplitude. Ordinary minimal automata for arbitrary languages have different asymptotics and should not be confused with this finite-language class. The official entry [9], the cited papers, the author’s current project bibliography, and focused 2025–2026 searches were checked on 2 October 2026. No later amplitude theorem was located; this bounded search is not a guarantee of universal novelty.

[write, 2 October 2026] The repository searches recorded in `48-dfa-literature-status.md` (at commit `a68b926cf`) were right when made. The same batch delivered the relaxed, compacted and fixed-arity companions, printed as Parts I–III of this report; source 48 cites neither its binary companions nor their shared text.

Several extensions deserve separate investigation.

1. Obtain certified decimal enclosures of γ by effective error bounds or positive interval propagation. Existence alone does not do this.
2. For a normalized exclusion coefficient $(\alpha m + \beta)$, the same formal drift is $\alpha/(4N)$, suggesting power $1 - \alpha/4$. The relaxed, compacted, and present DFA cases have parameters $(0, 0)$, $(1, -1)$, and $(1/2, 0)$, respectively. Nonnegative renewal weights and upper domination can hold on a larger parameter set, but a matching lower bound or another positivity proof is still required. No general parameter theorem is claimed here.
3. Extend the method to the non-nearest-neighbor general-arity recurrences, controlling the non-selfadjoint modes rather than assuming that the binary Jacobi argument automatically applies.

[**write, 2 October 2026**] For every fixed alphabet size $k \geq 3$ this is done in Part III (source 18), with a fixed weighted norm and a summable delay instead of the binary Jacobi argument, as this item anticipates.

4. Analyze exponentially small corrections beyond the present finite-order Poincaré expansion.

Part V

Second routes: an even-time Jacobi product and signed memory

Source. Bundle Reports 100, 101 and 103 of batch 102, delivered in the arrival commit 60f54ea06 and placed in this directory by commit 6ab1f1979 (both 5 October 2026), all dated 2 October 2026: A082161_Positive_Airy_Amplitude_and_All_Orders_Source.zip (relaxed trees, A082161), A254789_Positive_Airy_Amplitude_and_All_Orders_Source.zip (compacted trees, A254789) and A331120_Minimal_Binary_Automata_All_Orders_Source.zip (minimal binary automata, A331120). The author line of Reports 100 and 101, on the title page and in the PDF metadata, is “Research prepared for private review”; that of Report 103 is “Research prepared for Vladimir”, and Report 103’s bibliography calls Report 100 a “companion research report prepared for private review”. No tool or person is named. Printed in full except for two restatements of Report 100 (Section 66.4); label prefixes `air:jp:rb:`, `air:jp:cb:`, `air:jp:dfa:`, and `air:jp:` for this Part’s own material.

66 Reading Part V: sources, second routes and notation

This section and Section 105 were written when the three reports were added, on 5 October 2026; so were the passages marked [**write, 5 October 2026, batch 102**]. Everything else in Sections 67–104 is the reports’ text.

The three reports treat the three binary sequences of Parts I, II and IV and prove the same theorems: a positive limiting amplitude and a Poincaré expansion to every fixed order in $n^{-1/3}$, with Lambert- W inverses. They were written without knowledge of those Parts and reach the theorems by a second route. Time is eliminated to even steps, $n = N/2$, so the relaxed evolution becomes a product of positive semidefinite two-step Jacobi matrices, and boundary values are recovered with a killed lazy-walk kernel. For compacted trees and automata the subtraction becomes signed memory with three lags, whose scalar drift cancels as in Parts II and IV. Report 100 is the base: Reports 101 and 103 cite its Sections 2–5 for every Jacobi estimate and ship its archive unchanged.

66.1 Sources and provenance

Report	Wrapper / main file	Lines	pp.	Pin	Labels	Sections
100	Relaxed_Tree_Amplitude_Source / relaxed_tree_amplitude.tex	992	24	4b874cea0 ^a	air:jp:rb:	67–79
101	compacted_tree_amplitude / compacted_tree_amplitude.tex	693	19	none ^b	air:jp:cb:	80–91
103	minimal_dfa_amplitude / minimal_dfa_amplitude.tex	927	23	none ^b	air:jp:dfa:	92–104

^aA repository collision check recorded in Report 100’s provenance file (shipped as `data/102-100-relaxed-PROVENANCE.json`), not a continuation pin; `4b874cea0` is an ancestor of this report’s first placement commit `34f1acd4b`.

^bEach embeds Report 100’s delivered PDF and source archive, byte for byte (SHA-256 `72e79141...` for the archive, the digest Report 103 prints); no repository path is named.

The bundle numbers 100, 101 and 103 are report numbers of the delivering pipeline, unrelated to batch numbers; the other ten reports of batch 102 went to other reports of the collection. Each report calls itself research, disclaims publication, repository modification and external review, and ships an exact checking suite. The intake reran every suite on copies: all pass (Report 100’s `verify.py` and its 53 negative tests, Report 101’s compacted verifier with its 10 negative cases and its release ledger, Report 103’s verifier and negative tests), and its outputs match the delivered logs except for the recorded Python version and run time. The README of this report lists the shipped files, their delivered names and the rerun route.

The sibling report. Bundle Reports 104 and 102, of the same delivery series, prove the positive amplitude and all finite orders for tree-child networks of every fixed combining degree and for A213863, with the same lemma family: positive Jacobi products, the killed half-line kernel, the forced-product bootstrap, and the restart comparison of Lemma 72.1 for positivity. They are printed as Parts III and IV of the sibling report `../a213863-tree-child-networks` (its Sections “A strictly positive limiting amplitude” of Part III and “Convergence and strict positivity of the amplitude” of Part IV). Report 102 says that its all-order proof “does not require a companion relaxed-tree article”; that article is presumably Report 100 (same series and wording), though Report 102 names neither title nor digest. The two reports stay separate; neither is an input to the other.

66.2 What is new, and what is a second route

In the table, “second route” means the same theorem proved again by a different or parallel argument; “same” means equal printed values; “new” means not printed elsewhere in this report. Equalities of coefficients were recomputed symbolically at the write (difference zero); they are also forced by uniqueness of the coefficients of an asymptotic expansion.

Part V	Earlier Parts	Status
Theorems 67.1, 75.1 (relaxed amplitude, all orders)	Part I, (F1), (F24)–(F25)	second route
Lemma 72.1 (positivity from the exact recurrence alone)	positivity from the Elvey Price–Fang–Wallner lower bound in every Part	new ; Remark 66.1

Part V	Earlier Parts	Status
(69.5), endpoint error $O(n^{-2/3})$	$g_N(0)$ with error $O(N^{-1/3})$, Section 6	sharper
Proposition 69.1, Sections 69–71	Sections 2–5, Appendix A	second route (two-step matrices)
Theorem 73.1: product (73.2), series (73.3)	(F22)–(F23), (F25)	product: even-time counterpart; series: new in form
Propositions 74.1, 87.1, 98.1 (formal profiles)	Section 8, Section 22, Part IV’s formal section	second route; $\sigma_m = 2s_m$ (Remark 66.2)
Lemmas 75.2, 86.1, 100.1 (forced bootstraps)	zero-limit projection bootstraps of Parts I, II, IV	second route
b_1, b_2, b_3 of (76.1)	(F2)	same
b_4, b_5, b_6 of (76.1); ratio (76.2) through n^{-3}	—	new
Theorems 80.1, 80.2 (compactd)	Theorem 16.1	second route
Lemma 83.1 (gauged memory, $\tilde{B}_n \phi_{n-2} \approx (4/n)\phi_n$)	(19.5), (20.3), Lemma 20.1	second route (three lags)
compactd b_1, b_2, b_3 and d_1, d_2, d_3	(16.4), (25.1)	same
compactd b_4, b_5, b_6 ((89.1); formal in (76.3)); compactd ratio through n^{-3}	—	new , proved (Corollary 66.3)
product (85.3) for K_c	(24.1)	form new
Theorem 92.1 (automata)	Theorem 53.1	second route; positivity from the compactd lower bound
$d_1, d_2, d_3, c_1, c_2, c_3, \sigma_3, \dots, \sigma_6$ of Report 103	(53.4), (62.1), (59.4)	same ($\sigma_m = 2s_m$)
d_4, c_4, σ_7 ; ratio (101.3) through $n^{-7/3}$; the $X^{-1/3}$ term of (102.11)	(63.2) stops at $O(X^{-1/3}/\log X)$	new
inverse theorems, threshold rules, Newton counts (Sections 77, 90, 102)	inverse sections of Parts I, II, IV	same theorems; instances of the transseries volume (Part 0)
166.9520924395, 166.95208957422; 173.1267049959, 173.1267048501473; 76.438	166.9506, 166.9515; 173.12105; 131.985 at $n = 100$	evidence, uncertified

66.3 Notation and reading conventions

No letter of the three reports was renamed. Their macros were mapped to this report’s (Report 100’s $\mathbb{N}$ is printed $\mathbb{N}_0$, as it meant; Report 103’s norm, which it defines as the Euclidean norm, is printed with the subscript 2). The reports agree with one another in most letters and differ from the earlier Parts in many; the table gives the meaning in Part V and the reading to avoid.

Symbol	Meaning in Part V	Do not read it as
a, a_1	The largest Airy zero, $-2.338107\dots$: Parts I, II, IV’s z . Report 100 writes a_1 until its Section 74, then $a = a_1$; $a_2, a_3, \dots$ are the further zeros.	Parts I, II, IV’s $a = 2^{-1/3}z$, which is not the zero.
$\kappa, \varkappa$	$2^{2/3}a$ in the formal sections (κ in Report 100, $\varkappa$ in Reports 101 and 103): twice Parts I, II, IV’s a . Reports 101 and 103 also use $\kappa_n = \lambda_n - 1/n$, resp. $\lambda_n - 1/(2n)$.	Part I’s κ , the spectral-product constant (F22).

Symbol	Meaning in Part V	Do not read it as
n, N, ε, t	n is even time, half of the earlier Parts' time N ; $\varepsilon = n^{-1/3}$. In the formal sections $N = 2n$ is one-step time and $t = N^{-1/3} = 2^{-1/3}\varepsilon$, the earlier Parts' ε . In Report 100's Lemma 72.1, N is a restart time.	The earlier Parts' $\varepsilon = N^{-1/3}$ when ε appears.
x_k, X, x	$x_k = X_k = (k + \frac{1}{2})\varepsilon$, a half-grid in even time; in the formal sections $x = (m + 1)t$, the earlier Parts' $x_j = (j + 1)\varepsilon$.	
$S_n, J_n, \lambda_n, \phi_n$	S_n : diagonal symmetrizer. $J_n = S_n^{-1}A_nS_n$: two-step Jacobi matrix, positive semidefinite, top eigenvalue $\lambda_n \approx 4$, Perron vector ϕ_n .	The earlier Parts' S_N , the Jacobi matrix itself, with $\lambda_N \approx 2$ and Perron vector g_N .
$R_n, R_n^g, R(n, m)$	$R_n = S_{n-1}/S_n$, a diagonal gauge (R_n^g in Report 103). $R(n, m)$: the relaxed triangle (Reports 100, 103).	Parts II, IV's $R_N = P_N N^{-1/4}$, $P_N N^{-1/8}$; Part III's counts R_n .
$C_n, C(n, m)$	C_n : the bidiagonal factor of $J_n = C_n C_n^T$; $C(n, m)$: the compacted triangle.	Part III's counts C_n .
b	Report 100: $b_n(k)$ off-diagonal of J_n , b_n the perpendicular norm (Section 72), b_j logarithmic coefficients, $b_j^{(c)}$ formal compacted ones. Report 101: $b_n(k)$ an elimination coefficient, b_j logarithmic coefficients, b_n a perpendicular norm in Lemma 86.1. Report 103: $b_n, b(n, m)$ automaton counts.	The earlier Parts' Jacobi weights $b_{N,j}$ and scalar b_N .
d_j, c_j	Report 101: d_j <i>multiplicative</i> coefficients. Report 103: d_j <i>logarithmic</i> , c_j <i>multiplicative</i> .	Part I's c_k and Parts II, IV's $\mathbf{c}_k$ are multiplicative.
$c_n, B(n, m)$	c_n : compacted counts (Reports 100, 101); c_n^{tree} in Report 103. $B(n, m) = b(n, m)/2^m$ (Report 103) is Part IV's $c_{n,m}$.	Part IV's $c_{n,m}$ is not a compacted count.
$a_n, A_n, A_n^*, a_\infty, A_\infty$	$a_n = \langle \phi_n, v_n \rangle$ is a scalar projection (Reports 100, 101), except in Report 100's Section 77, where a_n is r_n or c_n . A_n : the two-step operator. A_n^*, A_∞ (Report 103): projection and its limit.	Part I's counting sequence a_n and its projection constant A_∞ , which differs from a_∞ here.
Γ_n, Γ	$\Gamma_n = \prod_{j \leq n} \kappa_j$ (Reports 101, 103); $\Gamma(x + 1)$ is Euler's gamma function in the inverse sections.	
$B_n, F_n, \mathcal{B}_n, \mathcal{F}_n, M_n, N_n$	Memory matrices of lags two and three (Report 101: B_n, F_n ; Report 103: $\mathcal{B}_n, \mathcal{F}_n$) and their normalized forms M_n, N_n .	Parts II, IV's delay B_N and M_N .
$W, B, \mathcal{B}, D$	Formal coefficients: W is the earlier Parts' $\mathcal{A}$; Report 100's B and Report 101's $\mathcal{B}$ are Part II's β ; Report 103's D is Part IV's β . $W_n(k)$ is a lag-three band.	Part IV's run weights $W_m(k)$.
$s(t), \sigma_m, \ell_j$	$s(t) = H_N/H_{N-1} = 2 + \kappa t^2 + \sum_{m \geq 3} \sigma_m t^m$ and carrier coefficients ℓ_j .	$\sigma_m = 2s_m$ and $\ell_j = h_j$ of the earlier Parts (Remark 66.2), not $\sigma_m = s_m$.
$\gamma_r, \gamma_c, \gamma_D$	The amplitudes: Part I's γ , Part II's γ_c , Part IV's γ .	Part III's γ , an exponent coefficient.
C_λ, K_c, K_D, P	Normalizing constants of Λ_n and Γ_n ; $P = J_\infty/4$, the killed lazy walk.	Part I's κ and the earlier Parts' spectral product P_N .
α, K, L	$\alpha = 1$ or $3/4$ (Report 100's inverse), α_n a projection in the forced lemmas; K a Newton step count or a barrier constant; L a smoothing length or $\log(4x_0)$, $\log(8X)$.	

66.4 How Part V was assembled

1. *Order and numbering.* Report 100 is printed first, because the other two cite it; then Report 101 and Report 103. Sections are numbered on from this report's Section 65: Report 100's Section k is Section $k + 66$, Report 101's is $k + 79$, Report 103's is $k + 91$; theorems keep their position in their section and equations are numbered within sections. Appendix A and every earlier number are unchanged.
2. *Two restatements replaced by pointers.* Report 101's Section 3 (Section 82) and Report 103's Section 3 (Section 94) restate Report 100's definitions and summarize its proofs. Their definitions and summaries are replaced by notes that point to Report 100 and name every change of letter; the statements they cite, Lemma 82.1 and Proposition 94.1, are kept, with each line traced to Report 100. Six unreferenced display labels of those passages are therefore not printed.
3. *Parallel text printed twice.* Reports 101 and 103 run the same signed-memory argument (exact elimination, gauged memory bounds, tracking with drift cancellation, forced memory lemma, formal algorithm, residuals, transfer) with different coefficients, drift $1/(4n)$ against $1/(8n)$, and different exponents. They are paraphrases of each other, not copies (the intake measured 6–7% shared 8-grams), and are both printed. Report 101's inverse theorem and threshold corollary are Report 100's at $\alpha = 3/4$; they are kept, with a note.
4. *Citations.* The bibliographies are merged into the report's: `epflw` and `maple` are new entries; Report 100's `kary` is `gdw`; Report 103's `trees` is `epfw`; Report 100's `oeisr`, `oeisc` are `oeis`; the companion entry `relaxed` of Reports 101 and 103 is replaced by references to Report 100's sections here.
5. *Omitted.* Only title blocks, tables of contents, running heads (Report 103's reads "Report 103") and the separate bibliographies. Each report opens with its abstract as delivered.

66.5 Consequences drawn at the write

The three statements below are the write's. Their proofs are complete and use only results printed or cited in this report; Corollary 66.3 rests, through Report 101 and Part II, on the published compacted lower bound. (Dated note, 5 October 2026: the second sentence first read "Their proofs are complete and use only results printed in this report." It is changed because the scope of the claim changes. Remark 66.1 uses no external analytic input, but Corollary 66.3 needs $\gamma_c > 0$, which Report 101 and Part II take from the published lower bound (80.4) of [1], and the identification of Report 101's printed $b_1, \dots, b_6$ with the coefficients of its theorem rests on Report 100's exact certificate suite. The proof of the corollary now states both dependencies; see the independent check recorded at the end of Section 105.)

Remark 66.1 (Positivity in Part I without the published lower bound). The positive amplitude of Part I follows without the lower half of the Elvey Price–Fang–Wallner Θ theorem: Part I's projection constant satisfies $A_\infty = \gamma_r/(4\kappa) > 0$, and $\gamma = \gamma_r$. Consequently both halves of the relaxed Θ bound (67.3) follow from Report 100 alone.

Proof [write]. Report 100's proof of Theorem 67.1 (Sections 68–73) uses no published lower bound: (67.3) is invoked only in its Section 77, and positivity comes from Lemma 72.1. With $a_1 = z$ it gives $r_n = \gamma_r n! 4^n e^{3zn^{1/3}} n(1 + O(n^{-1/3}))$ with $\gamma_r > 0$. Part I proves (F24) before it uses any lower bound:

for even N , $d_{N,0} = 2\kappa A_\infty 2^N e^{3aN^{1/3}} N + O(2^N e^{3aN^{1/3}} N^{2/3})$, with $a = 2^{-1/3}z$ and $d_{2n,0} = r_{n,n}/n! = a_n/n!$ (Part 0, normalization dictionary). At $N = 2n$ one has $2^N e^{3aN^{1/3}} N = 2 \cdot 4^n e^{3zn^{1/3}} n$, so

$$a_n = 4\kappa A_\infty n! 4^n e^{3zn^{1/3}} n + O(n! 4^n e^{3zn^{1/3}} n^{2/3}).$$

Since $a_n = r_n$, dividing both formulas by $n! 4^n e^{3zn^{1/3}} n$ and letting $n \rightarrow \infty$ gives $4\kappa A_\infty = \gamma_r > 0$. As $\kappa > 0$ by (F22), $A_\infty > 0$, which is what Part I infers from the lower bound, and (F25) gives $\gamma = 4\kappa A_\infty = \gamma_r$. The asymptotic formula of Theorem 67.1 with $\gamma_r \in (0, \infty)$ implies both bounds of (67.3) for r_n . $\square$

Remark 66.2 (The two normalizations of the scalar series). For each of the three models, the formal problem of Report 100 (relaxed or compacted), Report 101 or Report 103 is the formal problem of Part I, II or IV in other letters. Hence the profiles agree, $f_j = G_j$; the carrier coefficients agree, $\ell_j = h_j$; and $\sigma_m = 2s_m$ for every $m \geq 2$, where σ_m is written in $\kappa = \varkappa = 2^{2/3}z$ and s_m in the earlier Parts' $a = 2^{-1/3}z = \kappa/2$.

Proof [write]. Both sides use $t = N^{-1/3}$ (the earlier Parts' ε) and the height variable $x = (m+1)t$ (there $x = (j+1)\varepsilon$, j the height). The carrier $A(x) = \text{Ai}(z + 2^{1/3}x)$ is the earlier Parts' F , and $v = 2x + \kappa = 2(x+a)$, so $A'' = vA$ is $F'' = 2(x+a)F$. The downward coefficient W of (74.5), (87.4) and (98.4) is Part I's downward coefficient in its formal section and $\mathcal{A}$ of (22.3), (59.3); Report 100's B and Report 101's $\mathcal{B}$ are Part II's β , and Report 103's D is Part IV's β ; the shifted arguments $(x \pm t)(1-t^3)^{-1/3}$, $(x-t)(1-3t^3)^{-1/3}$ and the time dilations $t_j = t(1-jt^3)^{-1/3}$ are the same; and in both the delayed term is divided by $H_{N-1}/H_{N-3} = s(t_1)s(t_2)$. The gauges agree: $f_0 = A = F = G_0$ and $f_j(0) = f'_j(0) = 0 = G_j(0) = G'_j(0)$ for $j \geq 1$. Finally $s(t) = H_N/H_{N-1}$ on both sides, written $2 + \kappa t^2 + \sum \sigma_m t^m$ here and $2(1 + \sum s_m \varepsilon^m)$ there. Each side constructs its formal solution among series $\sum_j t^j f_j$ with $f_j = P_j A + Q_j A'$, P_j, Q_j polynomials, and s a formal power series, and proves that the solution with these gauges is unique in that class (Propositions 74.1, 87.1, 98.1; Section 8, Section 22 and Part IV's formal section, which takes this proof from Part II's paragraph and cites it). The polynomial–Airy restriction is what excludes a Bi-type homogeneous solution. At the first order where two solutions differ, the difference g of the profiles satisfies $g'' - vg = cA$ for a constant c , with $g(0) = g'(0) = 0$; over all smooth functions uniqueness fails, since $g = \frac{c}{2}A' + dA + eB$, with B a Bi-type solution of $B'' = vB$, meets both gauges for every c , whereas within the class the gauge $Q(0) = 0$ forces $c = 0$ and then $g = 0$. So the two solutions coincide, which gives $f_j = G_j$ and $\sigma_m = 2s_m$ ($\sigma_2 = \kappa = 2a = 2s_2$). The carriers (75.3), (88.2), (99.2) and (F30), (22.7), (59.5) have the same leading factors ($\frac{3}{2}\kappa = 3a$, exponents $\frac{4}{3}, \frac{13}{12}, \frac{29}{24}$) and leading constant one, and both are fixed uniquely by matching s , so $\ell_j = h_j$. As checks: $\sigma_3, \dots, \sigma_6$ of Report 100 (relaxed and compacted) and of Report 103 are twice (F29), (22.6) and (59.4) (difference zero), and Report 100's $\ell_1 = 53\kappa^2/180$ equals $53a^2/45$, the value of h_1 in (22.8) and (59.6) (it is common to the three models). $\square$

Corollary 66.3 (Report 100's compacted formal expansions are genuine). *With a the Airy zero and $b_j^{(c)}$ as in (76.3),*

$$\log \frac{c_n}{n! 4^n e^{3an^{1/3}} n^{3/4}} = \log \gamma_c + \sum_{j=1}^6 b_j^{(c)} n^{-j/3} + O(n^{-7/3}),$$

and the formal compacted ratio printed after (76.3) holds with the remainder $O(n^{-10/3})$ in place of "...".

Proof [write]. Report 101’s Theorem 80.2 gives the genuine expansion (89.1) to every depth J , with one γ_c , and its printed $b_1, \dots, b_6$ are Report 100’s $b_1^{(c)}, \dots, b_6^{(c)}$ term by term. This is the first claim, with $J = 6$. For the ratio take $J = 9$ at n and $n - 1$ and subtract. The factorials give $n!/(n(n-1)!) = 1$, γ_c cancels, and

$$\log \frac{c_n}{4nc_{n-1}} = 3a(n^{1/3} - (n-1)^{1/3}) + \frac{3}{4} \log \frac{n}{n-1} + \sum_{j=1}^9 b_j(n^{-j/3} - (n-1)^{-j/3}) + O(n^{-10/3}).$$

Expanding $(n-1)^{-s} = n^{-s}(1 - 1/n)^{-s}$ in powers of $1/n$ shows that b_j first enters at $n^{-j/3-1}$, so b_7, b_8, b_9 contribute $O(n^{-10/3})$. Exponentiating, the coefficients through n^{-3} are polynomials in $a, b_1, \dots, b_6$; with the printed values they are those of the display (symbolic recomputation at the write, difference zero). That the coefficients of Theorem 80.2 are the printed $b_1, \dots, b_6$ is Report 101’s computation by the algorithm of Section 87, certified by the exact formal suite of Report 100 (Section 76); it is not derived in the text. Unlike Remark 66.1, this corollary also inherits the published compacted lower bound (80.4), through which both Report 101 and Part II obtain $\gamma_c > 0$. Part II’s Theorem 16.1 gives the same conclusion independently: by Remark 66.2 its formal solution and carrier are those from which Report 100 computes the $b_j^{(c)}$. $\square$

Report 100: relaxed binary trees

Source. Bundle Report 100, archive A082161_Positive_Airy_Amplitude_and_All_Orders_Source.zip, main file Relaxed_Tree_Amplitude_Source/relaxed_tree_amplitude.tex (992 lines; 24-page PDF), titled “A positive Airy amplitude for relaxed binary trees: A Jacobi product proof with all-order expansions and inverse asymptotics”, dated 2 October 2026, author line “Research prepared for private review”. Printed in full, label prefix `air:jp:rb::`; its Sections 1–13 are Sections 67–79.

Abstract of Report 100.

Let r_n be the number of relaxed binary trees of size n , OEIS A082161, and let a_1 be the largest zero of the Airy function. We prove that a constant $\gamma_r \in (0, \infty)$ exists such that

$$r_n = \gamma_r n! 4^n \exp(3a_1 n^{1/3}) n(1 + O(n^{-1/3})).$$

The proof converts the exact counting recurrence into a product of positive, contractive operators obtained from finite Jacobi matrices. A quantitative Dirichlet Airy limit gives a spectral gap and a slowly varying Perron direction. Summable errors imply convergence of its scalar projection; a separate positivity argument prevents a zero limit. A killed lazy-walk kernel transfers the Hilbert-space estimate to the boundary coordinate that counts trees. We also prove a full Poincaré expansion to every fixed algebraic order, with one compatible positive amplitude, and give a convergent amplitude representation, an exact polynomial coefficient algorithm, and rigorous eventual inverse statements. Numerical checks and formal identities are distinguished from analytic theorems. The compacted-tree analogue, OEIS A254789, is included only for comparison and formal computation: its positive amplitude is not proved here. No worldwide priority claim or certified numerical value of either amplitude is made.

67 Result and relation to earlier work

Write $R(n, m) = 0$ for $n < m$ or $m < 0$, and set $R(n, 0) = 1$ for $n \geq 0$. For $n \geq m \geq 1$, define

$$R(n, m) = (m+1)R(n-1, m) + R(n, m-1), \quad r_n = R(n, n). \quad (67.1)$$

Thus $r_0, r_1, \dots = 1, 1, 3, 16, 127, 1363, 18628, \dots$. This is the relaxed binary-tree counting triangle of [1]; it fixes every startup and indexing convention used below. Let

$$a_1 > a_2 > \dots$$

be the negative zeros of Ai , where $\text{Ai}''(z) = z \text{Ai}(z)$ and Ai decays at positive infinity. In particular $a_1 = -2.338107410459767\dots$

Theorem 67.1 (Positive amplitude). *There is a constant $\gamma_r \in (0, \infty)$ for which*

$$r_n = \gamma_r n! 4^n e^{3a_1 n^{1/3}} n \left(1 + O(n^{-1/3})\right). \quad (67.2)$$

The constant has the convergent finite-matrix representation in Theorem 73.1.

The 2020 version of Elvey Price, Fang, and Wallner’s paper, published in 2021, proves the two bounds

$$r_n = \Theta\left(n! 4^n e^{3a_1 n^{1/3}} n\right), \quad c_n = \Theta\left(n! 4^n e^{3a_1 n^{1/3}} n^{3/4}\right), \quad (67.3)$$

where c_n counts compacted binary trees [1, Theorem 1.1]. That paper explicitly leaves even existence of the two amplitudes unresolved and gives the numerical estimates

$$\gamma_r \approx 166.95208957, \quad \gamma_c \approx 173.12670485.$$

These numbers are the earlier authors’ estimates, not newly certified values. Their Sections 3.3–3.4 already propose Airy expansions and use high-order numerical extrapolation. Accordingly the formal ansatz below is not presented as a new general idea.

The April 2026 revision of Elvey Price, Fang, Louf, and Wallner [4, Section 10.1] again distinguishes the tree problem from its asymptotic-equality results: the associated walk can leave the boundary and its asymptotic initial condition is unknown. Its general theorem therefore cannot be invoked without checking additional hypotheses. The present proof instead establishes convergence and positivity directly for the relaxed-tree transfer product.

Theorem 67.1 is supported here by an analytic argument, not by fitted values or finite computation. The exact formal coefficient identities in Section 74 are a separate algebraic result. They are transferred to genuine all-order asymptotics of r_n in Theorem 75.1, using uniform residual bounds and a forced-product bootstrap. The compacted recurrence has a subtraction term and does not inherit the positive product used in the proof. We do not claim an amplitude theorem for c_n , an exponentially improved transseries, a closed form for γ_r , or exhaustive literature priority.

[write, 5 October 2026, batch 102] Here a_1 is Part I’s z , and Theorems 67.1 and 75.1 are a second route to Part I’s (F1): γ_r is Part I’s γ (both are the limit of $r_n/(n! 4^n e^{3a_1 n^{1/3}} n)$) and b_1, b_2, b_3 of (76.1) are (F2). The route differs: an even-time, positive semidefinite two-step Jacobi product and a lazy-walk sine kernel, where Part I uses the one-step matrix S_N and a Catalan kernel. What Report 100 adds is listed in Section 66.2; the chief item is Lemma 72.1, which proves $\gamma_r > 0$ without the Elvey Price–Fang–Wallner lower bound that every earlier Part uses. The compacted amplitude that this paragraph does not claim is proved by Report 101 in this Part (Theorems 80.1 and 80.2) and by Part II.

68 The exact finite Jacobi product

[write, 5 October 2026, batch 102] Sections 68–71 are a third write-up of the estimates that Part I, Sections 2–5, and Appendix A prove, now in even time: n here is half of the earlier Parts' time N , $\varepsilon = n^{-1/3}$, S_n is a diagonal symmetrizer (there S_N is the Jacobi matrix itself), J_n is the two-step Jacobi matrix with top eigenvalue $\lambda_n \approx 4$ (there $\lambda_N \approx 2$), and R_n is a diagonal gauge (there $R_N = P_N N^{-1/4}$). Reports 101 and 103 cite these four sections. The reading conventions of the whole Part are in Section 66.3.

68.1 Rotated coordinates and the two step recurrence

For integers $t \geq h \geq 0$ of the same parity set

$$d(t, h) = \frac{R((t+h)/2, (t-h)/2)}{((t+h)/2)!}.$$

Set $d(t, h) = 0$ outside this wedge or off parity, and $d(0, 0) = 1$. Substitution in (67.1) gives

$$d(t, h) = \frac{t-h+2}{t+h} d(t-1, h-1) + d(t-1, h+1), \quad d(t, -1) = 0, \quad (68.1)$$

for the noninitial allowed entries. In particular

$$r_n = n! d(2n, 0).$$

Define the finite vector $e_n(k) = d(2n, 2k)$ for $0 \leq k \leq n$, and extend it by zero to $\ell^2(\mathbb{N}_0)$. Eliminating the intermediate odd time in (68.1) gives

$$(A_n u)(0) = u(0) + u(1), \quad (68.2)$$

$$(A_n u)(k) = L_n(k)u(k-1) + D_n(k)u(k) + u(k+1), \quad (68.3)$$

where, for $1 \leq k \leq n$,

$$L_n(k) = \frac{(n-k+1)^2}{(n+k)(n+k-1)}, \quad D_n(k) = \frac{2n-2k+1}{n+k}. \quad (68.4)$$

The last unused coordinate of u is zero. Thus $e_n = A_n e_{n-1}$, after zero extension. The exceptional first row (68.2) must be retained; using the bulk formula at $k = 0$ gives the wrong boundary phase later.

68.2 Symmetrization and positive semidefiniteness

Set $S_n(0) = 1$ and

$$S_n(k) = \frac{n!}{(n-k)!} \sqrt{\frac{(n-1)!n!}{(n+k-1)!(n+k)!}}, \quad 1 \leq k \leq n. \quad (68.5)$$

We use the same letter for this positive diagonal matrix. Since $S_n(k)/S_n(k-1) = \sqrt{L_n(k)}$, the conjugate

$$J_n = S_n^{-1} A_n S_n$$

is symmetric. Its diagonal is $D_n(k)$ for $k \geq 1$, its first diagonal entry is 1, and its positive off-diagonal between $k - 1$ and k is

$$b_n(k) = \frac{n - k + 1}{\sqrt{(n + k)(n + k - 1)}}. \quad (68.6)$$

In fact J_n is positive semidefinite. Let C_n be the $(n + 1) \times n$ bidiagonal matrix with

$$C_n(k, k) = \sqrt{\frac{n - k}{n + k}}, \quad 0 \leq k < n, \quad (68.7)$$

$$C_n(k, k - 1) = \sqrt{\frac{n - k + 1}{n + k}}, \quad 1 \leq k \leq n. \quad (68.8)$$

A direct multiplication gives $J_n = C_n C_n^\top$, including both endpoints. Positivity of all off-diagonals and the finite Perron theorem give a simple largest eigenvalue $\lambda_n > 0$ and a strictly positive unit eigenvector ϕ_n on $0, \dots, n$. All eigenvalues lie in $[0, \lambda_n]$. For initialization put $\phi_0 = \delta_0$.

68.3 The change of diagonal gauge

For $n \geq 2$, define the zero-extended diagonal matrix

$$R_n(k) = \frac{S_{n-1}(k)}{S_n(k)} = \frac{n - k}{n} \sqrt{\frac{(n + k - 1)(n + k)}{n(n - 1)}}, \quad 0 \leq k < n, \quad (68.9)$$

and $R_n(k) = 0$ for $k \geq n$. At $n = 1$ put $R_1(0) = 1$ and all other entries zero. The notation R_n for the diagonal gauge is distinct from the two-variable counting triangle $R(n, m)$.

Lemma 68.1 (Gauge contraction). *For all $n \geq 1$ and $0 \leq k \leq n$,*

$$0 \leq R_n \leq I, \quad 0 \leq 1 - R_n(k) \leq C \frac{k^2}{n^2}. \quad (68.10)$$

Proof. The case $n = 1$ is immediate. For $n \geq 2$, $R_n(0) = 1$ and $R_n(1)^2 = 1 - n^{-2}$. The logarithmic derivative of the continuous expression in (68.9), for $k < n$, is

$$-\frac{1}{n - k} + \frac{1}{2(n + k - 1)} + \frac{1}{2(n + k)},$$

which is nonpositive for $k \geq 1/2$. Thus $R_n(k) \leq 1$ at each integer k . For $1 \leq k \leq n/2$, expand the square-root factors with uniform Taylor remainder. The terms linear in k/n cancel and $|1 - R_n(k)| \leq C(k^2 + k)/n^2 \leq 2Ck^2/n^2$. For $k > n/2$, use $1 - R_n(k) \leq 1 \leq 4k^2/n^2$. The case $k = 0$ has zero defect. $\square$

Let $\Lambda_0 = 1$, $\Lambda_n = \prod_{j=1}^n \lambda_j$, and

$$v_n = \Lambda_n^{-1} S_n^{-1} e_n, \quad H_n = J_n / \lambda_n, \quad T_n = H_n R_n. \quad (68.11)$$

The exact normalized evolution is

$$v_n = T_n v_{n-1}, \quad v_0 = \delta_0. \quad (68.12)$$

Every T_n is entrywise nonnegative and has operator norm at most one. Hence $\|v_n\|_2 \leq 1$. Symmetry is used for H_n , not asserted for T_n ; the matrices at different times need not commute.

69 A quantitative frozen Airy theorem

Proposition 69.1 (Frozen spectral estimates). *Let $\lambda_n^{(2)}$ denote the second largest eigenvalue of J_n . For all sufficiently large n ,*

$$\lambda_n = 4 + 4a_1 n^{-2/3} + 6n^{-1} + O(n^{-4/3}), \quad (69.1)$$

$$0 \leq \lambda_n^{(2)}/\lambda_n \leq 1 - cn^{-2/3}, \quad (69.2)$$

$$\|k^p \phi_n\|_2 \leq C_p n^{p/3} \quad (p \geq 0 \text{ fixed}), \quad (69.3)$$

$$\|\phi_n - \phi_{n-1}\|_2 \leq Cn^{-1}, \quad (69.4)$$

$$\phi_n(0) = \sqrt{2} n^{-1/2} \left(1 + O(n^{-2/3})\right). \quad (69.5)$$

Changing dimensions are always interpreted by zero extension.

The next three sections prove all parts, including the endpoint estimate. Put

$$\varepsilon = n^{-1/3}, \quad x_k = \varepsilon(k + 1/2).$$

The half-grid shift reflects the exceptional first row: the constant-coefficient limiting equation has ghost value $u(-1) = -u(0)$, so its Dirichlet point is $k = -1/2$.

69.1 The continuum operator and its normalization

Consider the Dirichlet operator

$$\mathcal{H} = -\frac{d^2}{dx^2} + 8x \quad \text{on } L^2(0, \infty).$$

Its closed quadratic form is $\int_0^\infty (|f'|^2 + 8x|f|^2) dx$ with zero trace at 0. The growing potential makes its form embedding compact: restriction to a finite interval is compact by the one-dimensional Rellich theorem and the potential bounds the remaining mass by $O(1/R)$. Solving the differential equation with the decaying Airy solution and the Dirichlet condition gives its eigenvalues and first unit eigenfunction as

$$E_j = -4a_j, \quad f(x) = \frac{\sqrt{2} \operatorname{Ai}(2x + a_1)}{\operatorname{Ai}'(a_1)}. \quad (69.6)$$

The normalization follows without an undetermined constant: the derivative of $\operatorname{Ai}'(z)^2 - z \operatorname{Ai}(z)^2$ is $-\operatorname{Ai}(z)^2$. Integrating from the zero a_1 to infinity gives $\int_{a_1}^\infty \operatorname{Ai}(z)^2 dz = \operatorname{Ai}'(a_1)^2$. Thus $\int_0^\infty f^2 = 1$, $f'(0) = 2\sqrt{2}$, $f''(0) = 0$, and $f'''(0) = 4a_1 f'(0)$.

69.2 Variational identification and the spectral gap

Set $b_n(0) = b_n(n+1) = 0$, take $D_n(0) = 1$, and put

$$V_n(k) = 4 - D_n(k) - b_n(k) - b_n(k+1).$$

The exact quadratic form is

$$\langle u, (4I - J_n)u \rangle = \sum_{k=0}^{n-1} b_n(k+1) |u(k+1) - u(k)|^2 + \sum_{k=0}^n V_n(k) |u(k)|^2. \quad (69.7)$$

For $k \geq 1$, each term in $(2 - D_n(k)) + (1 - b_n(k)) + (1 - b_n(k + 1))$ is nonnegative, and

$$V_n(k) \geq \frac{4k - 1}{n + k} \geq \frac{3k}{2n}. \quad (69.8)$$

At zero, $V_n(0) = 3 - b_n(1) \geq 2$. On every fixed scaled interval the edge weights tend uniformly to 1, and away from zero $V_n(k)/\varepsilon^2 \rightarrow 8x_k$ uniformly.

We give both min–max bounds to avoid identifying the Perron eigenvalue from a quasimode alone. Sampling a fixed finite-dimensional space of smooth compactly supported Dirichlet trial functions as $\sqrt{\varepsilon} g(x_k)$ gives convergence of its norm and scaled form to their continuum values. Smooth Dirichlet functions are dense in the continuum form domain, so the finite-dimensional min–max principle gives the upper bounds for each fixed low eigenvalue of $(4I - J_n)/\varepsilon^2$.

For the lower bounds, suppose $\|u\|_2 = 1$ and the form in (69.7) is at most $C\varepsilon^2$. Equation (69.8) implies

$$\sum k|u(k)|^2 \leq C/\varepsilon, \quad \sum_{x_k \geq R} |u(k)|^2 \leq C/R.$$

Interpolate $u(k)/\sqrt{\varepsilon}$ linearly between the nodes x_k , and linearly from value 0 at $x = 0$ to its first node. On each fixed interval the resulting functions have bounded H^1 seminorm. The first half-cell has derivative energy exactly

$$\frac{2|u(0)|^2}{\varepsilon^2},$$

controlled by the boundary term since $V_n(0) \geq 2$. This proves the zero-trace condition with the correct coefficient. Local Rellich compactness and the uniform mass bound yield strong global L^2 compactness. More explicitly, on a fixed interval the linear interpolant differs from the piecewise-constant embedding by at most $C\varepsilon$ times its H^1 seminorm in L^2 ; the first half-cell vanishes in the limit, and both tails have mass $O(1/R)$. Thus norms and, by polarization, inner products are preserved.

Weak lower semicontinuity gives the gradient-energy lower bound, including the first segment. On fixed intervals bounded away from zero, the uniform potential convergence gives the lower bound for $\int 8x|g|^2$; the remaining potential is nonnegative. Increasing these intervals proves the full continuum form bound. Apply this compactness argument simultaneously to an orthonormal basis of each fixed-dimensional low-energy eigenspace. Preservation of inner products preserves its dimension, and continuum min–max gives the lower eigenvalue bound. Consequently, for every fixed j ,

$$\frac{4 - \lambda_n^{(j)}}{\varepsilon^2} \rightarrow E_j = -4a_j. \quad (69.9)$$

Since $E_2 > E_1$, this proves (69.2) and isolates the top eigenvalue on the ε^2 scale.

69.3 The quantitative quasimode and the coefficient six

Sample $\tilde{q}_n(k) = \sqrt{\varepsilon} f(x_k)$ for $0 \leq k \leq n$. Midpoint quadrature and Airy decay give $\|\tilde{q}_n\|_2 = 1 + O(\varepsilon^2)$; let q_n be its unit normalization. We claim

$$\left\| (J_n - [4 + 4a_1\varepsilon^2 + 6\varepsilon^3])q_n \right\|_2 \leq C\varepsilon^4. \quad (69.10)$$

For $k \geq 1$, with $x = x_k$, the exact entries yield

$$D_n(k) = 2 - 4x\varepsilon^2 + 3\varepsilon^3 + O(\varepsilon^4(1 + x^2)), \quad (69.11)$$

$$b_n(k) = 1 - 2x\varepsilon^2 + \frac{5}{2}\varepsilon^3 + O(\varepsilon^4(1 + x^2)), \quad (69.12)$$

$$b_n(k+1) = 1 - 2x\varepsilon^2 + \frac{1}{2}\varepsilon^3 + O(\varepsilon^4(1 + x^2)). \quad (69.13)$$

Each remainder is uniform for $1 \leq k \leq n/2$: write its denominator as products of factors $1 + x\varepsilon^2 - c\varepsilon^3$, bounded away from zero in this range, and use Taylor's formula with a uniform remainder. Also $|b_n(k+1) - b_n(k)| \leq C\varepsilon^3$. Pair the two expansions of $f(x \pm \varepsilon)$ before estimating them. The first, second, and third derivative terms are respectively

$$\varepsilon(b_n(k+1) - b_n(k))f', \quad \frac{\varepsilon^2}{2}(b_n(k+1) + b_n(k))f'', \quad \frac{\varepsilon^3}{6}(b_n(k+1) - b_n(k))f'''. \quad (69.14)$$

The third is $O(\varepsilon^6|f'''|)$; the fourth-order remainder is at most $C\varepsilon^4 \sup_{|y-x| \leq \varepsilon} |f^{(4)}(y)|$. It follows that the absolute bulk error after subtracting

$$4f + \varepsilon^2(f'' - 8xf) + 6\varepsilon^3f$$

is bounded by $C\varepsilon^4 E(x)$, where one may take an envelope dominating

$$(1 + x^2)|f| + (1 + x)|f''| + |f'| + |f'''| + \sup_{|y-x| \leq \varepsilon, y \geq 0} |f^{(4)}(y)|.$$

The differential equation and Airy decay bound this by $E(x) = C(1 + x)^{10}e^{-cx^{3/2}}$, after enlarging C near zero. Therefore $\varepsilon \sum_k E(x_k)^2 \leq C$ and the ℓ^2 bulk residual of $\sqrt{\varepsilon}f$ is $O(\varepsilon^4)$. For $k > n/2$, $x_k \geq n^{2/3}/2$ and all terms have exponentially small norm. The upper truncation is covered by the same estimate.

At the exceptional first row, use $D_n(0) = 1$, $b_n(1) = \sqrt{n/(n+1)}$, and expand $f(3\varepsilon/2) - 3f(\varepsilon/2)$. The terms of degree one cancel; those of degree three cancel the $4a_1\varepsilon^2 f(\varepsilon/2)$ term because $f'''(0) = 4a_1 f'(0)$. What remains is $O(\varepsilon^4)$ before the sampling factor $\sqrt{\varepsilon}$, hence $O(\varepsilon^{9/2})$. This proves (69.10) globally.

For a finite symmetric matrix, the norm of a unit-vector residual bounds the distance of its proposed eigenvalue from the spectrum. By (69.9), the eigenvalue within $C\varepsilon^4$ is the top one. This proves (69.1). Expanding q_n in an orthonormal eigenbasis and using the gap $c\varepsilon^2$ gives, with positive signs,

$$\|\phi_n - q_n\|_2 \leq C\varepsilon^2. \quad (69.14)$$

This completes the frozen eigenvalue analysis; localization and boundary evaluation require further estimates.

70 Localization and drift of the Perron vector

Let J_∞ have diagonal 1 at zero, diagonal 2 elsewhere, and off-diagonals 1. Entrywise $0 \leq J_n \leq J_\infty$. Because ϕ_n is nonnegative, this comparison is valid in its quadratic form and gives

$$\sum_{k \geq 0} |\phi_n(k+1) - \phi_n(k)|^2 + 2\phi_n(0)^2 = \langle \phi_n, (4I - J_\infty)\phi_n \rangle \leq 4 - \lambda_n \leq C\varepsilon^2. \quad (70.1)$$

The elementary discrete Sobolev inequality

$$\|u\|_\infty^2 \leq 2 \|u\|_2 \|u(\cdot + 1) - u\|_2$$

follows by telescoping $u(k)^2$ to zero at infinity and applying Cauchy–Schwarz. It implies $\|\phi_n\|_\infty \leq C\sqrt{\varepsilon}$.

Fix a small $c_0 > 0$ and let $K = \lceil A/\varepsilon \rceil$ for a sufficiently large constant A . For $k \geq K$ use the trial sequence

$$w(k) = \exp[-c_0\varepsilon(k - K)].$$

The quotient of $J_n w$ by w equals

$$D_n(k) + b_n(k)e^{c_0\varepsilon} + b_n(k+1)e^{-c_0\varepsilon},$$

with the absent final term omitted at $k = n$. The row sum is at most $4 - ck/n$ by (69.8). The even perturbation from the exponentials is $O(\varepsilon^2)$ and the odd one is $O(\varepsilon/n)$, since $|b_n(k+1) - b_n(k)| \leq C/n$. Thus, for large fixed A ,

$$(J_n w)(k) \leq (\lambda_n - \varepsilon^2)w(k), \quad k > K.$$

At any strict maximum of $\phi_n(k)/w(k)$ in $K < k \leq n$, positivity of the neighboring entries and the preceding strict supersolution inequality contradict $J_n \phi_n = \lambda_n \phi_n$. This finite maximum principle includes $k = n$, where the missing neighbor is omitted. Comparison at K proves

$$\phi_n(k) \leq C\sqrt{\varepsilon} e^{-c_0\varepsilon(k-K)} \quad (k \geq K). \quad (70.2)$$

Together with the supremum bound for $k < K$, this yields

$$\sum k^{2p} \phi_n(k)^2 \leq C_p \varepsilon^{-2p}$$

for every fixed $p \geq 0$, proving (69.3).

The exact coefficient formulas also give, on the common domain of J_n and J_{n-1} , differences bounded by $C(k+1)/n^2$. The new edge is $O(1/n)$, and at its old incident coordinate $k = n - 1$ this has the same bound. The old terminal diagonal causes no difficulty; the new diagonal multiplies the zero extension. Since a tridiagonal matrix has only three contributions per coordinate, the first moment bound therefore gives

$$\|(J_n - J_{n-1})\phi_{n-1}\|_2 \leq Cn^{-2} \|(k+1)\phi_{n-1}\|_2 \leq Cn^{-5/3}. \quad (70.3)$$

The spectral limit (69.9) puts λ_{n-1} in the isolated neighborhood of the largest eigenvalue of J_n ; every other eigenvalue has distance at least $cn^{-2/3}$. Apply (70.3) to the residual $(J_n - \lambda_{n-1})\phi_{n-1}$, expand in the eigenbasis of J_n , and use the positive signs of both unit Perron vectors. This proves $\|\phi_n - \phi_{n-1}\|_2 \leq C/n$, including the change of dimension. Finally Lemma 68.1 and the second moment estimate imply

$$\|(I - R_n)\phi_n\|_2 + \|(I - R_n)\phi_{n-1}\|_2 \leq Cn^{-4/3}. \quad (70.4)$$

71 A killed lazy walk and boundary smoothing

Put $P = J_\infty/4$. Away from zero its row is that of a lazy nearest-neighbor walk with weights $1/4, 1/2, 1/4$. The first row has total mass $1/2$, so the walk can be killed at the boundary. The

half-integer sine transform diagonalizes P : the functions $\sin((k+1/2)\theta)$, $0 < \theta < \pi$, have normalized eigenvalue $\cos^2(\theta/2)$ and the orthogonality factor is $2/\pi$. Consequently

$$P^m(0,0) = \frac{2}{\pi} \int_0^\pi \cos^{2m}(\theta/2) \sin^2(\theta/2) d\theta \leq C(m+1)^{-3/2}, \quad (71.1)$$

$$\|\delta_0^\top P^l\|_2 = \sqrt{P^{2l}(0,0)} \leq C(l+1)^{-3/4}. \quad (71.2)$$

For the inequality in (71.1), on a fixed neighborhood of zero use $\cos^{2m}(\theta/2) \leq e^{-cm\theta^2}$ and $\sin^2(\theta/2) \leq C\theta^2$; away from zero the integral decays geometrically. The equality in (71.2) uses symmetry of P . Positivity, $R_j \leq I$, and $J_j \leq 4P$ entrywise imply

$$0 \leq T_n \cdots T_{n-l+1} \leq \left(\prod_{j=n-l+1}^n 4/\lambda_j \right) P^l$$

entrywise. If $0 \leq l \leq n^{2/3}$, then $j \asymp n$ on the interval and (69.1) bounds the scalar product uniformly: its logarithm is $O(ln^{-2/3}) = O(1)$. Thus

$$\|\delta_0^\top T_n \cdots T_{n-l+1}\|_2 \leq C(l+1)^{-3/4}, \quad l \leq n^{2/3}. \quad (71.3)$$

No commutation is used; entrywise domination passes through products of nonnegative matrices. The same argument gives

$$\|\delta_0^\top H_n^l\|_2 \leq C(l+1)^{-3/4}, \quad 0 \leq l \leq n^{2/3}. \quad (71.4)$$

We can now complete Proposition 69.1. Let $z = \phi_n - q_n$. Equations (69.10), (69.14), and (69.1) give

$$\|z\|_2 \leq C\varepsilon^2, \quad z = H_n z + \rho, \quad \|\rho\|_2 \leq C\varepsilon^4.$$

Iterate $L = \lfloor n^{2/3} \rfloor$ times and use (71.4):

$$|z(0)| \leq C(L+1)^{-3/4} \varepsilon^2 + C\varepsilon^4 \sum_{l=0}^{L-1} (l+1)^{-3/4} = O(\varepsilon^{7/2}).$$

Since $f(0) = f''(0) = 0$ and $f'(0) = 2\sqrt{2}$, the normalized sample obeys

$$q_n(0) = \sqrt{2} \varepsilon^{3/2} (1 + O(\varepsilon^2)).$$

The preceding error is a relative $O(\varepsilon^2)$, proving (69.5). A plain ℓ^2 quasimode estimate would not suffice for this relative endpoint conclusion, because the boundary value itself is small.

[write, 5 October 2026, batch 102] The relative endpoint error $O(n^{-2/3})$ of (69.5) is sharper than the $O(N^{-1/3})$ of Part I's endpoint $g_N(0) = 2N^{-1/2}(1 + O(N^{-1/3}))$ (Section 6, after (F21)). The leading constants agree: at $N = 2n$, $2N^{-1/2} = \sqrt{2}n^{-1/2}$. The two statements concern different unit vectors (the Perron vector of the two-step J_n here, that of the one-step S_N there), so the sharpening is of the endpoint law, not of the earlier lemma as stated. Only the weaker rate is needed for the amplitude.

72 Tracking the Perron direction and proving positivity

Write

$$a_n = \langle \phi_n, v_n \rangle, \quad u_n = (I - \phi_n \phi_n^\top) v_n, \quad b_n = \|u_n\|_2.$$

We first prove convergence and only then nonvanishing of the scalar limit. Since $\|v_n\|_2 \leq 1$, the gap, drift, and gauge estimates give

$$b_n \leq (1 - cn^{-2/3})b_{n-1} + C/n. \quad (72.1)$$

For completeness, apply the orthogonal projection off ϕ_n to $H_n R_n v_{n-1}$. On that perpendicular space H_n contracts by at most $1 - cn^{-2/3}$. The old perpendicular part contributes at most b_{n-1} before this contraction, since R_n is contractive; the old parallel direction differs from ϕ_n after applying R_n by $O(n^{-1})$, by (69.4) and (70.4). This proves (72.1).

Comparison with $Kn^{-1/3}$ gives

$$b_n = O(n^{-1/3}). \quad (72.2)$$

Indeed, for large K , the contraction loss cK/n dominates the forcing C/n , while $K(n-1)^{-1/3} - Kn^{-1/3} = O(Kn^{-4/3})$. Enlarge K for the finite initial range. Since $H_n \phi_n = \phi_n$ and R_n is symmetric, the scalar increment is exactly

$$\begin{aligned} a_n - a_{n-1} &= a_{n-1} [\langle \phi_n, R_n \phi_{n-1} \rangle - 1] \\ &\quad + \langle R_n \phi_n - \phi_{n-1}, u_{n-1} \rangle. \end{aligned} \quad (72.3)$$

The first bracket is $O(n^{-4/3})$: its part with $R_n = I$ equals $-\|\phi_n - \phi_{n-1}\|_2^2/2 = O(n^{-2})$, and the remaining part is controlled by (70.4). The second term is $O(n^{-1}b_{n-1}) = O(n^{-4/3})$. Hence the increments are absolutely summable and

$$a_n \longrightarrow a_\infty \geq 0, \quad |a_n - a_\infty| + \|v_n - a_\infty \phi_n\|_2 \leq Cn^{-1/3}. \quad (72.4)$$

Lemma 72.1 (The limiting projection is strictly positive). *The scalar a_∞ in (72.4) satisfies $a_\infty > 0$.*

Proof. Start the same normalized evolution at a sufficiently large time N with $w_N = \phi_N$. It has norm at most one and zero initial perpendicular error. All constants above are independent of N ; the scalar barrier gives $b_n(w) \leq Cn^{-1/3}$ for $n \geq N$. Summing the analogue of (72.3) yields

$$|a_\infty(w) - 1| \leq C \sum_{n>N} n^{-4/3} \leq C' N^{-1/3}.$$

Choose N with this bound at most $1/2$. The actual vector v_N has strictly positive coordinates on its entire support $0, \dots, N$, by induction in the exact recurrence. So does ϕ_N , and the finite minimum of their positive coordinate ratios gives $\delta > 0$ with $v_N \geq \delta \phi_N$. Subsequent positive evolution preserves this inequality: $v_n \geq \delta w_n$. Taking inner products with $\phi_n \geq 0$ and then limits gives $a_\infty \geq \delta a_\infty(w) > 0$. $\square$

This argument requires neither a uniform-in- N lower bound for δ nor an assumption that the original vector starts near the Perron direction. Boundedness and positivity of each finite vector alone would not have proved a positive limiting projection.

[write, 5 October 2026, batch 102] Lemma 72.1 is the one genuinely new step of Report 100 relative to this report. (The same restart comparison proves positivity in bundle Reports 104 and 102 for tree-child networks, Parts III and IV of the sibling report `../a213863-tree-child-networks`.) Every earlier Part obtains strict positivity of its amplitude from a published lower bound: Part I from the lower half of the Elvey Price–Fang–Wallner Θ theorem (Section 7, after (F24)), Parts II and IV from the compacted and automaton lower bounds, Part III from the Ghosh Dastidar–Wallner relaxed lower bound. The proof above uses only the positivity of the exact recurrence, the contraction $\|T_n\| \leq 1$ and the tracking estimates, whose constants do not depend on the restart time N . It was checked at the intake: v_N and ϕ_N are strictly positive on $0, \dots, N$, every T_n is entrywise nonnegative, and $\|w_n\|_2 \leq 1$ bounds the restarted scalar projections, so $|a_\infty(w) - 1| \leq C'N^{-1/3}$ holds with C' independent of N . Consequences: Report 100 re-derives both halves of the relaxed bound (67.3) with no external analytic input, and Part I’s positivity step can rest on this lemma instead (Remark 66.1). Whether the argument also replaces the published lower bounds in the signed-memory cases is Question V.5 of Section 105.

To recover the boundary coordinate, let $z_n = v_n - a_\infty \phi_n$. Its exact defect equation is

$$z_n = T_n z_{n-1} + a_\infty \eta_n, \quad \eta_n = T_n \phi_{n-1} - \phi_n, \quad \|\eta_n\|_2 \leq C/n.$$

Iterate over the final $L = \lfloor n^{2/3} \rfloor$ steps. By (71.3) and (72.4), the starting error and the forcing satisfy

$$\begin{aligned} |z_n(0)| &\leq CL^{-3/4}n^{-1/3} + \frac{C}{n} \sum_{l=0}^{L-1} (l+1)^{-3/4} \\ &= O(n^{-5/6}). \end{aligned}$$

Combining with (69.5) and Lemma 72.1 gives

$$v_n(0) = \sqrt{2} a_\infty n^{-1/2} \left(1 + O(n^{-1/3})\right). \quad (72.5)$$

73 The amplitude and its convergent representation

From (69.1),

$$\log(\lambda_n/4) = a_1 n^{-2/3} + \frac{3}{2} n^{-1} + O(n^{-4/3}).$$

The differences between $n^{-2/3}$ and $3(n^{1/3} - (n-1)^{1/3})$, and between n^{-1} and $\log(n/(n-1))$, are summable with tails smaller than $O(n^{-1/3})$. Thus

$$C_\lambda = \lim_{n \rightarrow \infty} \frac{\Lambda_n}{4^n e^{3a_1 n^{1/3}} n^{3/2}} \in (0, \infty), \quad (73.1)$$

with relative error $O(n^{-1/3})$.

Theorem 73.1 (A convergent amplitude formula). *With $J_n, R_n, \lambda_n, \phi_n, v_n$ defined by the finite exact recurrences above,*

$$C_\lambda = \frac{\lambda_1}{4} e^{-3a_1} \prod_{j=2}^{\infty} \left\{ \frac{\lambda_j}{4} e^{-3a_1(j^{1/3} - (j-1)^{1/3})} \left(\frac{j-1}{j} \right)^{3/2} \right\}, \quad (73.2)$$

$$a_\infty = 1 + \sum_{n=1}^{\infty} \langle R_n \phi_n - \phi_{n-1}, v_{n-1} \rangle, \quad (73.3)$$

$$\gamma_r = \sqrt{2} a_\infty C_\lambda \in (0, \infty). \quad (73.4)$$

The product factors are $1 + O(j^{-4/3})$ and the series summands are $O(n^{-4/3})$. Both normalizations are canonical once the positive unit Perron eigenvectors and the initial vector $v_0 = \delta_0$ are specified.

Proof. Equation (73.2) is the telescoping normalization of (73.1). Equation (73.3) is the telescoping identity $a_n - a_{n-1} = \langle R_n \phi_n - \phi_{n-1}, v_{n-1} \rangle$, with $a_0 = 1$; its summability was proved by (72.3), rather than by a norm bound of order n^{-1} alone. The series sum is positive by Lemma 72.1. Since $S_n(0) = 1$, the original count is

$$r_n = n! \Lambda_n v_n(0).$$

Multiplying (73.1) and (72.5) proves (67.2) and (73.4). $\square$

This is a convergent characterization, not a closed form. The proof's O -constants are not made explicit, so the representation alone does not certify decimal digits or a numerical cutoff. Its finite truncations may be evaluated and compared with exact recurrence data; such comparisons remain numerical checks until explicit tail bounds and interval arithmetic are supplied.

[write, 5 October 2026, batch 102] The product (73.2) is the even-time counterpart of Part I's spectral product (F22)–(F23): a convergent product for the normalization of $\Lambda_n = \prod_{j \leq n} \lambda_j$, where (F22) normalizes the product P_N of the one-step eigenvalues. The two constants $\bar{C}_\lambda$ and κ are different numbers, and so are a_∞ and Part I's A_∞ ; only the products $\gamma_r = \sqrt{2} a_\infty C_\lambda$ and $\gamma = 4\kappa A_\infty$ (F25) coincide. The telescoping series (73.3) for the projection constant is new in form; Part I defines A_∞ as a limit.

74 An exact polynomial Airy coefficient algorithm

We first construct one compatible formal expansion to every order. Section 75 will transfer it to the actual relaxed-tree solution. For comparison, define the compacted triangle by $C(n, 0) = 1$, zero outside its wedge, and

$$C(n, m) = (m+1)C(n-1, m) + C(n, m-1) - (m-1)C(n-2, m-1). \quad (74.1)$$

The last term at $n = m = 1$ is zero without evaluating a negative-index entry. Its diagonal is $c_n = C(n, n)$, beginning 1, 1, 3, 15, 111, 1119, 14487, ... Both the recurrence and its nonnegative counting interpretation are from [1]. The rotated compacted recurrence is

$$\begin{aligned} u_{N,m} = & \frac{N-m+2}{N+m} u_{N-1,m-1} + u_{N-1,m+1} \\ & - \frac{2(N-m-2)}{(N+m)(N+m-2)} u_{N-3,m-1}, \end{aligned} \quad (74.2)$$

with $u_{N,-1} = 0$, and with startup entries supplied by the triangle. Only large N is used in the formal calculation. At $m = 0$ the ghost term is absent exactly.

Put

$$t = N^{-1/3}, \quad x = (m+1)t, \quad a = a_1, \quad \kappa = 2^{2/3}a, \quad A(x) = \text{Ai}(a + 2^{1/3}x), \quad v(x) = 2x + \kappa.$$

Here $A'' = vA$ and primes denote differentiation in x . Seek

$$F(t, x) = \sum_{j \geq 0} t^j f_j(x), \quad f_j = P_j A + Q_j A', \quad s(t) = 2 + \kappa t^2 + \sum_{k \geq 3} \sigma_k t^k, \quad (74.3)$$

with $P_0 = 1, Q_0 = 0$, and with boundary and scalar gauge

$$Q_j(0) = 0, \quad P_j(0) + Q'_j(0) = 0 \quad (j \geq 1). \quad (74.4)$$

Equivalently $f_j(0) = f'_j(0) = 0$ for $j \geq 1$. The derivative gauge removes the freedom to move a scalar correction between F and the carrier.

Let $t_q = t(1 - qt^3)^{-1/3}$ and

$$W(t, x) = \frac{1 - xt^2 + 3t^3}{1 + xt^2 - t^3}, \quad B(t, x) = \frac{2t^3(1 - xt^2 - t^3)}{(1 + xt^2 - t^3)(1 + xt^2 - 3t^3)}.$$

The exact formal relaxed equation is

$$\begin{aligned} s(t)F(t, x) &= W(t, x)F(t_1, (x - t)(1 - t^3)^{-1/3}) \\ &\quad + F(t_1, (x + t)(1 - t^3)^{-1/3}). \end{aligned} \quad (74.5)$$

For compacted trees one subtracts

$$\frac{B(t, x)}{s(t_1)s(t_2)}F(t_3, (x - t)(1 - 3t^3)^{-1/3}) \quad (74.6)$$

from its right-hand side. These identities follow by substituting $u_{N,m} = h_N F(t, x)$ and $h_N/h_{N-1} = s(t)$ in the corresponding lattice recurrences.

Proposition 74.1 (Formal existence and uniqueness at every order). *For each of the two equations there are unique polynomials $P_j, Q_j \in \mathbb{Q}[\kappa, x]$ and $\sigma_j \in \mathbb{Q}[\kappa]$ satisfying (74.3)–(74.4). They can be generated by a triangular recursion requiring only rational polynomial arithmetic.*

Proof. The coefficients of t^0, t^1 cancel identically and the t^2 equation is $A'' - (2x + \kappa)A = 0$. Suppose all coefficients through t^{k-1} have been removed. Because $A'' = vA$, all derivatives of polynomial pairs multiplying A, A' remain such pairs. Write the known coefficient of t^k in right side minus left side as $R_0A + R_1A'$. The new unknown $f_{k-2} = PA + QA'$ and scalar σ_k enter as

$$(D^2 - v)(PA + QA') - \sigma_k A + R_0A + R_1A' = 0.$$

Indeed

$$(D^2 - v)(PA + QA') = (P'' + 2vQ' + 2Q)A + (2P' + Q'')A'.$$

Eliminating P leaves

$$\mathcal{T}Q = 2R_0 - R'_1 - 2\sigma_k, \quad \mathcal{T}Q = Q''' - 4(2x + \kappa)Q' - 4Q.$$

On x^d the leading coefficient of $\mathcal{T}$ is $-4(2d + 1)$, and its other terms have smaller degree. Hence it is invertible on every finite polynomial degree space over $\mathbb{Q}[\kappa]$. Put

$$\begin{aligned} \tilde{q} &= \mathcal{T}^{-1}(2R_0 - R'_1), & \sigma_k &= -2\tilde{q}(0), \\ Q &= \tilde{q} + \sigma_k/2, & P(x) &= \int_0^x \frac{-R_1(u) - Q''(u)}{2} du - Q'(0). \end{aligned} \quad (74.7)$$

Since $\mathcal{T}$ sends a constant c to $-4c$, these formulas satisfy both differential identities, $Q(0) = 0$, and $P(0) + Q'(0) = 0$. The same identities and gauges imply uniqueness. For compacted trees, B starts at order t^3 , so the new pair and new scalar do not enter (74.6) at the current order t^k . The same induction applies. This proves an infinite *formal* series without asserting its convergence. $\square$

A direct descending implementation writes $\tilde{q} = \sum q_d x^d$ and $2R_0 - R'_1 = \sum v_d x^d$, pads higher q 's with zero, and uses

$$q_d = \frac{-v_d - 4\kappa(d+1)q_{d+1} + (d+3)(d+2)(d+1)q_{d+3}}{4(2d+1)}. \quad (74.8)$$

The first values are

$$f_1 = -\frac{2}{3}x^2 A, \quad \sigma_3 = \begin{cases} 8/3 & \text{relaxed,} \\ 13/6 & \text{compacted,} \end{cases} \quad \sigma_4 = \frac{29\kappa^2}{540}.$$

For relaxed trees,

$$f_2 = \frac{2(\kappa + 15x^4 + 111x)}{135}A + \frac{x(-4\kappa + 51x)}{270}A',$$

$$\sigma_5 = -\frac{23\kappa}{81}, \quad \sigma_6 = -\frac{8783\kappa^3 - 595080}{340200}.$$

For compacted trees,

$$f_2 = \frac{8\kappa + 120x^4 + 753x}{540}A + \frac{x(-4\kappa + 51x)}{270}A',$$

$$\sigma_5 = -\frac{11\kappa}{324}, \quad \sigma_6 = -\frac{17566\kappa^3 - 2054835}{680400}.$$

The source archive records the full exact polynomials through f_7 and the scalars through σ_9 . Finite symbolic verification is a check on these explicit values, not a replacement for the indefinite recursion proof above.

75 From formal profiles to all algebraic orders

Theorem 75.1 (All-order relaxed-tree expansion). *There is one compatible sequence of real coefficients $b_1, b_2, \dots$, computable by Proposition 74.1, such that for every fixed integer $J \geq 0$,*

$$r_n = \gamma_r n! 4^n e^{3an^{1/3}} n \exp\left(\sum_{j=1}^J b_j n^{-j/3}\right) \left(1 + O(n^{-(J+1)/3})\right). \quad (75.1)$$

The positive constant γ_r is the same at every depth and is the constant of Theorem 67.1. Equivalently the exponential has a full Poincaré expansion in $n^{-1/3}$. The constants implicit in the remainder may depend on J .

We prove this by finite truncations with genuine uniform residual bounds. Neither convergence of the infinite series nor positivity of the truncated higher-order profiles is needed.

75.1 A compatible scalar carrier

Define ℓ_j by the formal identity

$$\log(s(t)/2) = \frac{3\kappa}{2t} [1 - (1 - t^3)^{1/3}] - \frac{4}{3} \log(1 - t^3)$$

$$+ \sum_{j \geq 1} \ell_j t^j [1 - (1 - t^3)^{-j/3}]. \quad (75.2)$$

The terms of degrees two and three agree because $\sigma_2 = \kappa$ and $\sigma_3 = 8/3$. At degree $j + 3$, the new ℓ_j has coefficient $-j/3$, so it is uniquely determined. In particular $\ell_1 = 53\kappa^2/180$. For each fixed depth $M \geq 1$, put

$$h_N^{[M]} = 2^N \exp\left(\frac{3\kappa}{2} N^{1/3}\right) N^{4/3} \exp\left(\sum_{j=1}^M \ell_j N^{-j/3}\right), \quad (75.3)$$

$$F^{[M]}(t, x) = \sum_{j=0}^M t^j (P_j A + Q_j A'). \quad (75.4)$$

The exact ratio $h_N^{[M]}/h_{N-1}^{[M]}$ agrees with $s(t)$ through at least degree $M + 3$. Generating the needed ℓ_j may use formal pairs beyond the retained profile depth, which Proposition 74.1 already supplies. Crucially the carrier's leading multiplicative constant is 1 at every depth. Independently initializing products of truncated ratios would introduce depth-dependent constants and is not an equivalent construction.

Use even-time variables

$$\varepsilon = n^{-1/3}, \quad X = (k + 1/2)\varepsilon, \quad N = 2n, \quad m = 2k, \quad t = 2^{-1/3}\varepsilon, \quad x = 2^{2/3}X.$$

Define the uncut approximation

$$E_n^{[M]}(k) = \frac{2^{-4/3}}{\text{Ai}'(a)} h_{2n}^{[M]} F^{[M]}(t, x). \quad (75.5)$$

Its leading term, writing $\mathcal{B}_n = 4^n e^{3an^{1/3}} n^{4/3}$, is

$$\frac{\mathcal{B}_n}{\text{Ai}'(a)} \text{Ai}(a + 2X).$$

More explicitly its scalar correction is $\exp(\sum_{j=1}^M 2^{-j/3} \ell_j \varepsilon^j)$ and its profile corrections have polynomial coefficients

$$\overline{P}_j(X) = 2^{-j/3} P_j(2^{2/3}X), \quad \overline{Q}_j(X) = 2^{(1-j)/3} Q_j(2^{2/3}X),$$

with $\kappa = 2^{2/3}a$ substituted. Since $X_0 = \varepsilon/2$ and $\text{Ai}(a) = 0$,

$$E_n^{[M]}(0) = 4^n e^{3an^{1/3}} n (1 + O(\varepsilon)), \quad (75.6)$$

with leading coefficient exactly 1, independent of M . Its power coefficients stabilize under deeper truncation.

75.2 Uniform residuals and the exact ghost boundary

Fix M and any constant $C_0 > 0$. On $0 \leq m \leq C_0 \sqrt{N}$, the one-step residual of $h_N^{[M]} F^{[M]}$, divided by $h_N^{[M]}$, is bounded by

$$C t^{M+3} (1+x)^D e^{-cx^{3/2}}, \quad (75.7)$$

for constants depending on M, C_0 . Here are the uniform remainder details. In this region $x \leq C t^{-1/2}$, so $|xt^2| \leq C t^{3/2}$; all denominators $1 + xt^2 - c_0 t^3$ stay away from zero. Finite rational expansions with exact geometric remainders are bounded by a selected power of t times a polynomial in x .

The shifted profile argument differs from x by $\pm t + O(xt^3 + t^4) = O(t)$. Every fixed derivative of a polynomial–Airy pair is bounded by $C(1+x)^D e^{-cx^{3/2}}$, uniformly also on the Taylor interval after adjusting constants. Taylor’s integral remainder has the same form. The finite-carrier ratio and the powers $(1-t^3)^r$ are analytic at $t=0$ after the known factors are removed. Products preserve the envelope bounds. Formal cancellation through degree $M+2$ proves (75.7).

The statement includes $m=0$: the ghost evaluation $m=-1$ has $x=0$, and every $f_j(0)=0$ exactly before expansion. Thus no asymptotic boundary condition has been substituted for the true one. Eliminating the odd step gives the even residual $E_n^{[M]} - A_n E_{n-1}^{[M]}$ as a bounded combination of the residual at N and two at $N-1$. All coefficients are uniformly bounded on this region, and the forbidden-site term is absent at zero. The same order therefore holds for the even-time residual.

Take a fixed smooth χ with $0 \leq \chi \leq 1$, $\chi=1$ on $[0, 1/2]$, and $\chi=0$ on $[1, \infty)$, and define

$$\tilde{E}_n^{[M]}(k) = \chi(k/\sqrt{n}) E_n^{[M]}(k).$$

The transition region has $X \asymp n^{1/6}$, so all profiles and their neighbors there are polynomial multiples of $e^{-cn^{1/4}}$. The cutoff error is smaller than every power of n after division by the carrier, including the change from the cutoff at $n-1$ to that at n . Applying A_n may enlarge the residual support to $k \leq \sqrt{n}+1$; the estimates are applied on this padded support. Since the X -mesh is ε ,

$$\sum_{k \geq 0} (1+X_k)^{2D} e^{-2cX_k^{3/2}} = O(\varepsilon^{-1}).$$

Consequently

$$\left\| \tilde{E}_n^{[M]} - A_n \tilde{E}_{n-1}^{[M]} \right\|_2 = O(\mathcal{B}_n \varepsilon^{M+5/2}).$$

The gauge is harmless on the padded support: the exact product for S_n gives $-\log S_n(k) = O(k^2/n + k/n) = O(1)$ for $k \leq \sqrt{n} + O(1)$, so S_n^{-1} is uniformly bounded there. Also $\mathcal{B}_n/\Lambda_n = O(\varepsilon^{1/2})$ by (73.1). We have proved the genuine normalized residual estimate

$$\left\| \Lambda_n^{-1} S_n^{-1} (\tilde{E}_n^{[M]} - A_n \tilde{E}_{n-1}^{[M]}) \right\|_2 = O(\varepsilon^{M+3}) = O(n^{-(M+3)/3}). \quad (75.8)$$

The normalized approximants $U_n^{[M]} = \Lambda_n^{-1} S_n^{-1} \tilde{E}_n^{[M]}$ satisfy

$$\left\| U_n^{[M]} - \frac{1}{\sqrt{2}C_\lambda} \phi_n \right\|_2 = O(n^{-1/3}). \quad (75.9)$$

To see the constant, the leading profile after normalization is $n^{-1/6} \text{Ai}(a+2X)/(C_\lambda \text{Ai}'(a)) = (\sqrt{2}C_\lambda)^{-1} \tilde{q}_n(k)$. The carrier corrections and the finitely many profile corrections have weighted relative ℓ^2 norm $O(\varepsilon)$. The exact gauge yields $S_n^{-1} - 1 = O(\varepsilon(1+X^2))$ in that weighted norm; its remote tail is exponentially small. The normalized Perron vector differs from the leading Airy sample by $O(\varepsilon^2)$, and (73.1) has relative error $O(\varepsilon)$. This proves (75.9), uniformly in n for each fixed depth.

75.3 A forced product bootstrap

Lemma 75.2 (Transfer with a vanishing limiting projection). *Suppose finite-support signed vectors obey*

$$z_n = T_n z_{n-1} + \rho_n, \quad \|\rho_n\|_2 = O(n^{-p}), \quad p > 4/3.$$

If $\|z_n\|_2 = O(n^{-1/3})$ and $\langle \phi_n, z_n \rangle \rightarrow 0$, then

$$\|z_n\|_2 = O(n^{-(p-1)}), \quad z_n(0) = O(n^{-1/2-(p-1)}). \quad (75.10)$$

Proof. Let $\alpha_n = \langle \phi_n, z_n \rangle$ and $\beta_n = \|(I - \phi_n \phi_n^\top) z_n\|_2$. The same projection calculation as (72.1) and (72.3), with the additional forcing, gives

$$\begin{aligned}\beta_n &\leq (1 - cn^{-2/3})\beta_{n-1} + Cn^{-1}|\alpha_{n-1}| + Cn^{-p}, \\ |\alpha_n - \alpha_{n-1}| &\leq Cn^{-4/3}|\alpha_{n-1}| + Cn^{-1}\beta_{n-1} + Cn^{-p}.\end{aligned}$$

If $|\alpha_n| + \beta_n = O(n^{-q})$, the first stable scalar recurrence gives

$$\beta_n = O(n^{-q-1/3} + n^{-p+2/3}).$$

The scalar estimate used here is elementary: a sufficiently large multiple of $n^{-r+2/3}$ is a supersolution of $x_n \leq (1 - cn^{-2/3})x_{n-1} + O(n^{-r})$ for every fixed r ; the finite initial part decays at least as $e^{-c'n^{1/3}}$. Because $\alpha_n \rightarrow 0$, sum the second inequality from $n+1$ to infinity, using the improved β estimate. All terms are summable, initially with $q = 1/3$ and thereafter with larger q . This gives

$$\alpha_n = O(n^{-q-1/3} + n^{-p+1}).$$

Thus a bootstrap improves q to $\min(q + 1/3, p - 1)$. Finitely many repetitions reach $p - 1$, proving the norm estimate. Finally iterate the forced equation over $L = \lfloor n^{2/3} \rfloor$ steps and apply (71.3):

$$|z_n(0)| \leq CL^{-3/4}n^{-(p-1)} + Cn^{-p} \sum_{l=0}^{L-1} (l+1)^{-3/4} = O(n^{-p+1/2}) + O(n^{-p+1/6}).$$

This is (75.10). The proof uses no sign assumption on z_n . $\square$

Proof of Theorem 75.1. Set $z_n = v_n - \gamma_r U_n^{[M]}$. By (73.4), (72.4), and (75.9), $\|z_n\|_2 = O(n^{-1/3})$ and its limiting Perron projection is zero. Equation (75.8) supplies the forcing with $p = (M + 3)/3$; take $M \geq 2$ to ensure $p > 4/3$. Lemma 75.2 gives a boundary error $O(n^{-1/2-M/3})$. Multiplication by Λ_n , and division by the diagonal carrier $4^n e^{3an^{1/3}} n$, changes this to the relative error $O(n^{-M/3})$. The endpoint series of $E_n^{[M]}$ has leading coefficient one and compatible coefficients at all depths. Given J , choose $M \geq \max(2, J + 1)$, expand its endpoint through order J , and absorb its tail and the transfer error into $O(n^{-(J+1)/3})$. Formal logarithm converts the multiplicative coefficients to the b_j in (75.1). The amplitude is the already fixed γ_r , so it is not reselected at each depth. $\square$

Theorem 75.1 is an all-order algebraic, or Poincaré, expansion in a single dominant Airy sector. It does not identify contributions smaller than every power. A full transseries would require additional sectors, their amplitudes, and corresponding remainder estimates; these have not been established here.

76 Explicit diagonal coefficients and numerical checks

Formal integration of $\log s$ by (75.2) and endpoint evaluation $F(t, t)$ give the coefficients b_j of Theorem 75.1. The first six relaxed log coefficients are

$$\begin{aligned}b_1 &= \frac{53a^2}{90}, & b_2 &= \frac{44a}{27}, \\ b_3 &= \frac{171315 - 5216a^3}{170100}, & b_4 &= -\frac{422a^2}{1215}, \\ b_5 &= \frac{a(717281a^3 - 298772235)}{344452500}, & b_6 &= \frac{2250428a^3 - 20962395}{41334300}.\end{aligned}\tag{76.1}$$

In particular, the multiplicative expansion starts with

$$1 + \frac{53a^2}{90}n^{-1/3} + \left[\frac{1}{2} \left(\frac{53a^2}{90} \right)^2 + \frac{44a}{27} \right] n^{-2/3} + O(n^{-1}).$$

The all-order theorem justifies termwise shifts to any fixed required depth: take a deeper forward expansion than the desired ratio error before dividing. Thus the normalized relaxed ratio has

$$\begin{aligned} \frac{r_n}{4nr_{n-1}} &= 1 + an^{-2/3} + n^{-1} + \frac{41a^2}{135}n^{-4/3} + \frac{20a}{81}n^{-5/3} \\ &\quad + \frac{176a^3 - 1215}{170100}n^{-2} - \frac{427a^2}{3645}n^{-7/3} \\ &\quad - \frac{a(1035598a^3 + 3591495)}{103335750}n^{-8/3} + \frac{7774a^3}{295245}n^{-3} + O(n^{-10/3}). \end{aligned} \quad (76.2)$$

The coefficients through the familiar leading terms agree with the formal analysis in [1, Section 3.4]. The further displayed exact values are reproducible derivations relative to the inspected sources, not a claim that no one has computed them previously.

[write, 5 October 2026, batch 102] With $a = z$, the coefficients b_1, b_2, b_3 are Part I's (F2) (171315/170100 = 141/140 and 5216/170100 = 1304/42525). The coefficients b_4, b_5, b_6 and the ratio expansion (76.2) are new to this report: Part I prints three logarithmic corrections and no ratio expansion. The conversion from (76.1) to (76.2), and from b_1, b_2, b_3 to Part I's (F3), was recomputed symbolically at the write (difference zero). Through n^{-3} the ratio involves only a and $b_1, \dots, b_6$, because b_j first enters at $n^{-j/3-1}$; its $O(n^{-10/3})$ remainder uses Theorem 75.1 at depth $J = 9$.

76.1 Compacted formal comparison

For compacted trees the formal diagonal exponent is $\alpha = 3/4$. If one writes

$$\log \frac{c_n}{n!4^n e^{3an^{1/3}} n^{3/4}} \hat{=} \log \gamma_c + \sum_{j \geq 1} b_j^{(c)} n^{-j/3},$$

the hat explicitly denotes a formal ansatz, not an asymptotic theorem for the actual counts. The same algebra gives

$$\begin{aligned} b_1^{(c)} &= \frac{53a^2}{90}, & b_2^{(c)} &= \frac{271a}{216}, \\ b_3^{(c)} &= \frac{477495 - 41728a^3}{1360800}, & b_4^{(c)} &= -\frac{2053a^2}{9720}, \\ b_5^{(c)} &= \frac{a(2869124a^3 - 329492565)}{1377810000}, & b_6^{(c)} &= -\frac{320744a^3 + 10924065}{132269760}. \end{aligned} \quad (76.3)$$

The formal ratio is

$$\begin{aligned} \frac{c_n}{4nc_{n-1}} &\hat{=} 1 + an^{-2/3} + \frac{3}{4}n^{-1} + \frac{41a^2}{135}n^{-4/3} + \frac{20a}{81}n^{-5/3} \\ &\quad + \frac{704a^3 + 207765}{680400}n^{-2} - \frac{454a^2}{3645}n^{-7/3} \\ &\quad - \frac{a(8284784a^3 + 153117585)}{826686000}n^{-8/3} + \frac{11327392a^3 + 50486895}{330674400}n^{-3} + \dots \end{aligned} \quad (76.4)$$

The common $n^{-5/3}$ coefficient is not a copying error: it equals $a(\alpha + 1/3) - (2/3)b_2$, and the changes in α and b_2 cancel. No convergence of the compacted normalization or analytic remainder for these coefficients follows from the formal recursion alone.

[write, 5 October 2026, batch 102] The analytic input that this subsection lacks now exists twice: Report 101 (Theorem 80.2, with the same six coefficients in (89.1)) and Part II (Theorem 16.1, whose (F2)-type corrections are $b_1^{(c)}, b_2^{(c)}, b_3^{(c)}$). By Corollary 66.3 both hatted expansions above hold as genuine asymptotic expansions, the ratio with remainder $O(n^{-10/3})$.

76.2 What the finite calculations do and do not establish

Exact rational small-index tests check the original meander recurrence against the two-step operator, the diagonal normalization $r_n/n!$, the symmetrizer, the bidiagonal positive-semidefinite factorization, and the gauge contraction. These tests detect algebraic and indexing errors but do not establish asymptotic estimates.

Exact symbolic calculations check every formal recurrence coefficient through t^9 , the boundary and derivative gauge, agreement of the triangular solver with an independently set-up generic linear system, and an independent endpoint-ratio conversion. The latter computes the ratio directly from $s_N s_{N-1} F_N(0)/F_{N-2}(0)$, rather than reusing the integration of $\log s$. These certify finitely many polynomial identities. Proposition 74.1 proves their all-order continuation, and Section 75 supplies the analytic transfer only for relaxed trees.

The integer dynamic program computes both counting triangles through $n = 3000$ using exact integers and only a bounded number of rows. The archived checkpoint JSON stores high-precision derived diagnostics, not the exact integer count array itself. The script can regenerate the integer counts before evaluation. At $n = 3000$, the uncorrected normalized quantities are approximately 205.00390917 for relaxed trees and 213.44032023 for compacted trees. Removing the six displayed log corrections gives approximately 166.9520924395 and 173.1267049959. These are diagnostic values, with no rigorous numerical error certificate for the limiting amplitudes.

n	Relaxed ratio residual	Compacted formal ratio residual
500	$1.274 \cdot 10^{-10}$	$2.344 \cdot 10^{-10}$
1000	$1.177 \cdot 10^{-11}$	$2.146 \cdot 10^{-11}$
2000	$1.106 \cdot 10^{-12}$	$1.995 \cdot 10^{-12}$
3000	$2.790 \cdot 10^{-13}$	$5.000 \cdot 10^{-13}$

Here the residual is the absolute difference between the exact-count ratio evaluated at high precision and the displayed series truncated through n^{-3} . The observed decay corroborates the calculations; it is not a proof of the compacted expansion. Polynomial extrapolations using checkpoints from 1600 through 3000 give stable-looking values near 166.9520895742200 and 173.1267048501473. None of the extra digits is certified, and we do not use them anywhere in the proofs.

Frozen tridiagonal spectral diagnostics through $n = 100000$ likewise agree with the scales in Proposition 69.1. For example, the scaled gap $(\lambda_n - \lambda_n^{(2)})n^{2/3}$ changes from about 6.1823 at $n = 100$ to 6.9897 at $n = 100000$, and $\sqrt{n}\phi_n(0)$ changes from 1.3665 to 1.4137, approaching $\sqrt{2}$. These are floating-point checks, separate from the min-max and smoothing arguments.

[write, 5 October 2026, batch 102] Earlier evidence in this report: Part I gives 166.9506 and 166.9515 at $n = 20000$ after three corrections, Part II gives 173.12105 at $n = 2500$, and the intake extrapolated 166.9508 and 173.1263 from exact counts to $n = 900$. Report 100's six-correction values at $n = 3000$ are closer to the published fits 166.95208957 and 173.12670485. The intake recomputed the ratio residuals of the table at $n = 500$ and $n = 1000$ from exact counts and found the same values, and the relaxed six-correction quotient 166.95227 at $n = 500$ and 166.95213 at $n = 1000$. None of these numbers is an enclosure.

77 Rigorous inverse asymptotics

Let a_n denote r_n with $\alpha = 1$ or c_n with $\alpha = 3/4$. This section first uses only the already established Θ bounds (67.3), so its first theorem applies to both sequences. The distinction between recovering the index of an exact sequence value and inverting an arbitrary threshold is essential.

77.1 An eventual inverse requiring no amplitude value

For $Y > 0$ put

$$x_0(Y) = \frac{Y}{W_0(4Y/e)}, \quad L_0(Y) = \log(4x_0(Y)), \quad (77.1)$$

$$z_\alpha(Y) = x_0(Y) - \frac{3ax_0(Y)^{1/3} + (\alpha + 1/2) \log x_0(Y) + \frac{1}{2} \log(2\pi)}{L_0(Y)}, \quad (77.2)$$

where W_0 is the positive real branch of Lambert's W function. All formulas are used only for large positive Y .

Theorem 77.1 (Exact-value inverse from two-sided bounds). *For either counting sequence,*

$$z_\alpha(\log a_n) = n + O(1/\log n). \quad (77.3)$$

In particular, rounding $z_\alpha(\log a_n)$ to the nearest integer returns n for all sufficiently large n .

Proof. Stirling's formula and (67.3) imply

$$\log a_n = G(n) + O(1),$$

where

$$G(x) = x(\log(4x) - 1) + 3ax^{1/3} + (\alpha + 1/2) \log x + \frac{1}{2} \log(2\pi).$$

Let $G_0(x) = x(\log(4x) - 1)$ and $g(x) = G(x) - G_0(x)$. The Lambert identity gives $G_0(x_0) = Y$ and $G'_0(x_0) = L_0$. Set $h = -g(x_0)/L_0$, so $z_\alpha = x_0 + h$ and $h = O(x_0^{1/3}/\log x_0)$. Taylor expansion gives

$$G(z_\alpha) - Y = g'(x_0)h + O(h^2/x_0) = O(x_0^{-1/3}/\log x_0).$$

The omitted smaller terms are absorbed in this bound since $g'(x) = ax^{-2/3} + (\alpha + 1/2)/x$. At $Y = \log a_n$, the mean value theorem and $G'(x) \sim \log x$ give (77.3); both x_0 and z_α are asymptotic to n . The error tends to zero, proving the rounding assertion. $\square$

The theorem is eventual, and the unspecified constants in the Θ bounds do not supply an effective cutoff. In the numerical checks, at $n = 3000$ the error $z_\alpha(\log a_n) - n$ is approximately +0.55818 for relaxed trees and +0.56223 for compacted trees. Nearest-integer recovery therefore still fails at that moderate index. The theorem is not a certified finite-range algorithm.

77.2 Arbitrary thresholds and the required exact comparison

First establish monotonicity separately. For relaxed trees, (67.1) gives

$$r_n = R(n, n-1) \geq nr_{n-1} \quad (n \geq 2).$$

For compacted trees, nonnegativity of the counting triangle and (74.1) give

$$c_n \geq nc_{n-1} - (n-2)c_{n-2} \geq 2c_{n-1} \quad (n \geq 2),$$

inductively, beginning with $c_0 = c_1 = 1$ and $c_2 = 3$. Both sequences are strictly increasing from index 1 onward. Direct differentiation of (77.2) gives $z'_\alpha(Y) \sim 1/\log x_0 > 0$.

Corollary 77.2 (Two candidates for a threshold inverse). *For a sufficiently large real threshold y , let*

$$k(y) = \max\{n : a_n \leq y\}, \quad j_0 = \text{round}(z_\alpha(\log y)).$$

Then $j_0 \in \{k(y), k(y) + 1\}$ and

$$k(y) = \begin{cases} j_0, & a_{j_0} \leq y, \\ j_0 - 1, & a_{j_0} > y. \end{cases}$$

Thus one exact or rigorously certified comparison selects the floor inverse. The least index $j(y)$ with $a_{j(y)} \geq y$ is obtained by the analogous single comparison: return j_0 if $a_{j_0} \geq y$, and $j_0 + 1$ otherwise. Equality belongs to the first branch.

Proof. With $k = k(y)$, monotonicity and Theorem 77.1 put $z_\alpha(\log y)$ between $k + o(1)$ and $k + 1 + o(1)$. For large k , nearest-integer rounding therefore lies in $\{k, k + 1\}$. Monotonicity of a_n proves the floor comparison rule, including equality. For the least-index rule, the same argument puts j_0 in $\{j(y) - 1, j(y)\}$; at equality the exact-value rounding theorem applies. The stated single comparison then selects the least index. $\square$

This rule also has an unspecified eventual cutoff. An input arbitrarily close to a sequence value needs an exact or certified comparison; unverified floating-point evaluation of an asymptotic series cannot settle which side of the threshold it lies on.

77.3 Arbitrary-order relaxed inverses

The all-order forward theorem now gives much sharper inverse remainders for relaxed trees, with the canonically defined γ_r . For $J \geq 0$, set

$$\mathcal{F}_J(x) = \log \Gamma(x+1) + x \log 4 + 3ax^{1/3} + \log x + \log \gamma_r + \sum_{j=1}^J b_j x^{-j/3}. \quad (77.4)$$

For large x , $\mathcal{F}'_J(x) \sim \log x > 0$. Let $x_J(Y)$ be its unique sufficiently large real root of $\mathcal{F}_J(x) = Y$.

Corollary 77.3 (Sharpened relaxed inverse). *For every fixed $J \geq 0$,*

$$x_J(\log r_n) = n + O\left(\frac{n^{-(J+1)/3}}{\log n}\right). \quad (77.5)$$

Proof. Taking logarithms in (75.1) gives $\log r_n = \mathcal{F}_J(n) + O(n^{-(J+1)/3})$. The mean value theorem and $\mathcal{F}'_J(x) \sim \log x$ give (77.5). $\square$

This is an unconditional mathematical statement with a positive, convergently represented constant; using an uncertified decimal approximation for that constant does not give a certified numerical inverse. One may compute the root by Newton iteration from the Lambert start (77.1), continuing until the numerical error is smaller than the desired inverse scale. A single truncated Newton step is not automatically accurate to arbitrary order. For compacted trees the same implication holds *conditionally* if its corresponding forward expansion and positive amplitude are established; only Theorem 77.1 and Corollary 77.2 are unconditional for it here.

[**write, 5 October 2026, batch 102**] The condition is now met: the compacted forward expansion with a positive amplitude is Report 101's Theorem 80.2 and Part II's Theorem 16.1, and Report 101 states the unconditional compacted inverse as Corollary 90.3. The inverse statements of this section are instances of the transseries volume's theorems named in Part 0 ("Instances of results elsewhere in the repository"); no novelty is claimed for the inversion mechanics. Theorem 77.1 and Corollary 77.2 are second routes to Part I's amplitude-free locator (Section 13.8.1) and threshold brackets (Section 13.7), and the Newton count K below is an explicit specialization of the j -step bound of Section 13.5.

77.4 A quantified finite Newton algorithm

For each prescribed inverse order, one can replace the implicit root by an explicitly bounded number of Newton steps. With $Y \rightarrow \infty$, start at x_0 in (77.1), put $L = \log(4x_0)$, and iterate

$$x_{k+1} = x_k - \frac{\mathcal{F}_J(x_k) - Y}{\mathcal{F}'_J(x_k)}, \quad K = \lceil \log_2(J+4) \rceil - 1. \quad (77.6)$$

The derivative is

$$\mathcal{F}'_J(x) = \psi(x+1) + \log 4 + ax^{-2/3} + x^{-1} - \sum_{j=1}^J \frac{j}{3} b_j x^{-j/3-1},$$

where ψ is the digamma function. For fixed J , the K th iterate satisfies

$$x_K(\log r_n) = n + O\left(n^{-(J+1)/3}/\log n\right). \quad (77.7)$$

To prove the iteration count, let x_* be the root of $\mathcal{F}_J(x) = Y$. The Lambert equation and the size of the Airy term give $x_* - x_0 = O(x_0^{1/3}/L)$. On $x_0 \pm o(x_0)$, $\mathcal{F}'_J \sim L$ and $\mathcal{F}''_J = O(1/x_0)$. Taylor's theorem bounds the exact Newton root errors $e_k = x_k - x_*$ by

$$|e_{k+1}| \leq C|e_k|^2/(x_0 L).$$

The iterates stay in that neighborhood, and induction gives

$$e_k = O\left(\frac{x_0^{1-2^{k+1}/3}}{L^{2^{k+1}-1}}\right).$$

The first two errors are $O(x_0^{-1/3}/L^3)$ and $O(x_0^{-5/3}/L^7)$. Since $2^{K+1} \geq J + 4$, this root error is no larger than the model error in Corollary 77.3, proving (77.7).

There is also a version using only a finite Stirling polynomial in place of $\log \Gamma$. Substitute

$$(x + \tfrac{1}{2}) \log x - x + \tfrac{1}{2} \log(2\pi) + \sum_{r=1}^R \frac{B_{2r}}{2r(2r-1)x^{2r-1}}, \quad R = \max\left(0, \left\lceil \frac{J-2}{6} \right\rceil\right),$$

where B_{2r} are Bernoulli numbers, and differentiate that expression consistently in every step. Its remainder is $O(x^{-2R-1})$ and $2R+1 \geq (J+1)/3$, so its root differs from x_* by at most the desired inverse error. The same K steps therefore suffice. Each fixed order requires finitely many arithmetic operations, logarithms, real fractional powers, and one principal Lambert- W evaluation, together with the constants a, γ_r, b_j and Bernoulli numbers. This is an asymptotic algorithm with the true constants and sufficiently accurate evaluations; uncertified amplitude digits or unbounded floating-point errors do not provide a certified numerical implementation.

78 Further questions and proof boundaries

[write, 5 October 2026, batch 102] These questions, with those of Reports 101 and 103, are re-scoped in Section 105, the Part's register of open and answered questions.

Effective amplitude enclosures. The canonical product and series provide a route to rigorous numerical bounds, but effective constants for the spectral, drift, projection, and boundary estimates are still needed. A useful next goal is an interval computation of the finite part together with an explicit tail enclosure. It would turn the quoted numerical amplitude into certified digits and make inverse cutoffs effective.

Compacted trees. The subtraction in (74.2) obstructs the direct positive Jacobi construction. A positive multistate encoding that records the forbidden local configurations, or a stable replacement that preserves the scalar amplitude, could address this. Upper and lower positive comparison recurrences alone can leave different constants and do not prove an amplitude limit. No compacted amplitude theorem is claimed by this article.

[write, 5 October 2026, batch 102] Answered without a positive encoding: Report 101 (Theorems 80.1 and 80.2) and Part II prove the compacted amplitude and every fixed order by treating the subtraction as signed memory whose scalar drift cancels. Part II also has exact positive four-state, three-state and renewal encodings, with obstructions to a symmetric Jacobi reduction (the section after Section 16). A proof of the compacted amplitude through a positive encoding remains open (Question V.6).

Exponentially small Airy sectors. The gap suggests relaxation on a stretched-exponential scale related to other Airy eigenvalues. To construct a genuine transseries one would need evolving excited-mode amplitudes, control of mode coupling, and remainder bounds beyond every algebraic order. The all-order Poincaré result does not determine these data, and formal subtraction of its infinite series is not justified.

Coefficient growth and summability. The triangular recursion proves existence at every fixed order. It does not bound growth as the order tends to infinity. Determining a Gevrey class, optimal truncation scale, or a summability mechanism would require estimates uniform in the truncation order, which the present envelope argument intentionally does not supply.

Higher arity and related recurrences. The relaxed k -ary results in [3] retain a Θ formulation. The present strategy suggests looking for an exact finite positive representation, a limiting operator with a simple ground state and gap, summable gauge losses, and an endpoint kernel estimate. Each ingredient must be proved for the new recurrence; universality of the Airy shape alone is insufficient.

[write, 5 October 2026, batch 102] Answered for every fixed arity $k \geq 3$ by Part III, for relaxed trees, compacted trees and automata, with a fixed weighted norm in place of an exact symmetrization; uniformity in k remains open there.

79 Reproducibility and provenance

The companion source archive contains this standalone L^AT_EX source, the compiled PDF, a build script, and a separate `checks/` directory. The latter includes exact rational and symbolic tests, finite baseline coefficient data, and optional floating-point diagnostics. Its README specifies the replay commands and dependencies. Checks fail with explicit exceptions rather than relying on Python assertions, and corrupted-input regression tests run in both normal and optimized Python modes. The exact checks and numerical diagnostics are clearly labeled; no numerical output is promoted to an analytic theorem.

The report builds with pdfL^AT_EX and standard AMS, Latin Modern, geometry, microtype, hyperref, and fancyhdr packages. The exact Python checks use Python 3 and SymPy. Optional numerical diagnostics use mpmath, NumPy, and SciPy as specified in their README. The provided build and replay commands do not fetch external source files or modify a research repository. A fresh archive extraction is rebuilt and replayed as part of release validation. The delivered PDF is rendered and inspected page by page, including displayed equations, tables, references, and page boundaries.

Primary provenance is explicit: the counting recurrences, initial Θ theorem, earlier Airy ansatz, and original amplitude estimates are from [1]; the April 2026 statement of the remaining boundary difficulty is from [4]. The exact sequence identifiers are linked in [9]. The original authors’ Maple supplement is linked in [5]; it is not redistributed in this archive. The proof in Sections 68–75 is presented self-contained rather than as an invocation of an unverified general transfer theorem.

The accompanying research inspected the primary 2020 and April 2026 papers and related literature, and found no duplicate relaxed-amplitude result in the relevant indices and project summaries of the inspected repository snapshot. That bounded search is not an exhaustive worldwide priority search. No publication, repository edit, external submission, or certification by an independent journal is implied.

[write, 5 October 2026, batch 102] The “inspected repository snapshot” is commit 4b874cea0 (Report 100’s provenance record, shipped as `data/102-100-relaxed-PROVENANCE.json`). The sentence was true there and is stale now: Part I of this report, placed in commit 34f1acd4b on 2 October 2026 from an archive that arrived the same day, proves the same relaxed amplitude theorem, and Part II the compacted one. The “companion source archive” is Report 100’s

delivered archive, which survives in commit 60f54ea06; the README of this report lists the files shipped from it and their names here.

Report 101: compacted binary trees

Source. Bundle Report 101, archive A254789_Positive_Airy_Amplitude_and_All_Orders_Source.zip, main file compacted_tree_amplitude/compacted_tree_amplitude.tex (693 lines; 19-page PDF), titled “A positive Airy amplitude for compacted binary trees: Signed memory stability with all order expansions and inverse asymptotics”, dated 2 October 2026, author line “Research prepared for private review”. Printed in full except for the restatement of Report 100 in its Section 3, label prefix `air:jp:cb:`; its Sections 1–12 are Sections 80–91.

Abstract of Report 101.

Let c_n count compacted binary trees of size n , OEIS A254789, and let a_1 be the largest zero of the Airy function. We prove

$$c_n = \gamma_c n! 4^n e^{3a_1 n^{1/3}} n^{3/4} (1 + O(n^{-1/3})), \quad 0 < \gamma_c < \infty.$$

The exact normalized recurrence contains negative memory and does not inherit the positive contraction of relaxed trees. We eliminate odd times, control the resulting two memory operators globally after a diagonal gauge, and normalize by the relaxed Jacobi eigenvalue minus $1/n$. Counting domination first gives an $O(n^{1/4})$ bound. An Airy spectral gap and a cancellation of the scalar memory drift then give an absolutely convergent amplitude. Boundary smoothing and the published two-sided bound make that amplitude strictly positive. A signed forced-memory lemma and compatible polynomial-Airy profiles prove a full Poincaré expansion to every fixed algebraic order. We give exact coefficient and inverse algorithms, a convergent amplitude representation, and reproducible finite certificates. The required relaxed Jacobi lemmas are stated precisely and are proved in the companion article, supplied unchanged with this report. Numerical amplitude digits are not certified; no exponentially small transseries sector or exhaustive literature-priority claim is made.

80 The counting problem and the result

Set $C(n, 0) = 1$, and $C(n, m) = 0$ outside $0 \leq m \leq n$. For $n \geq m \geq 1$,

$$C(n, m) = (m + 1)C(n - 1, m) + C(n, m - 1) - (m - 1)C(n - 2, m - 1). \quad (80.1)$$

A term with zero prefactor is taken to be zero, without evaluating an invalid entry. The diagonal $c_n = C(n, n)$ begins

$$1, 1, 3, 15, 111, 1119, 14487, 230943, 4395855, 97608831, \dots$$

This is the compacted binary-tree triangle of [1, Proposition 2.11]. These are ordered binary compressed trees; the left and right child positions are distinct. Let $a = a_1$ be the largest real zero of Ai , with $\text{Ai}''(z) = z \text{Ai}(z)$ and decay at positive infinity.

Theorem 80.1 (Positive amplitude). *There exists $\gamma_c \in (0, \infty)$ such that*

$$c_n = \gamma_c n! 4^n e^{3a n^{1/3}} n^{3/4} (1 + O(n^{-1/3})). \quad (80.2)$$

The constant admits the absolutely convergent representation in Section 85.

Theorem 80.2 (All algebraic orders). *There is one compatible sequence $d_1, d_2, \dots \in \mathbb{Q}[a]$ such that, for every fixed integer $J \geq 0$,*

$$c_n = \gamma_c n! 4^n e^{3an^{1/3}} n^{3/4} \left(1 + \sum_{j=1}^J d_j n^{-j/3} + O(n^{-(J+1)/3}) \right). \quad (80.3)$$

The same positive γ_c is used at every depth. The coefficients are computed by the exact polynomial recursion in Section 87.

The previously established result is

$$c_n = \Theta\left(n! 4^n e^{3an^{1/3}} n^{3/4}\right), \quad (80.4)$$

proved in [1, Theorem 1.1]. That paper explicitly leaves amplitude existence unresolved; its numerical estimate $\gamma_c \approx 173.12670485$ is not a certified value. It already develops a formal Airy ansatz and high-order extrapolation, so the formal ansatz itself is not claimed as new. The April 2026 revision of [4, Section 10.1] explains why its equality theorem does not automatically settle this tree problem: the walk can leave the boundary and its asymptotic initial condition is unknown.

Here the missing amplitude step is a signed-memory stability argument. It uses both the published bound (80.4) and the independently proved relaxed-tree Jacobi estimates of the companion article, Report 100 (Sections 67–79 here). The latter is a mathematical dependency, not an appeal to an unproved compacted analogue. Its unchanged PDF and source archive accompany this article. Its statements that compacted trees remain unresolved describe that earlier article’s scope; the present companion supplies the additional argument. No numerical fit is used in either theorem above.

[**write, 5 October 2026, batch 102**] Theorems 80.1 and 80.2 are a second route to Part II’s Theorem 16.1 (a here is Part II’s z , γ_c the same constant), by essentially the same mechanism in even time: counting domination gives an $O(n^{1/4})$ start, the transverse error decays, and the scalar memory drift $1/(4n)$ cancels into a delayed difference. Part II works with the one-step delay and the normalization $R_N = P_N N^{-1/4}$ (Sections 19–22); here the memory has three lags and the normalization is $\kappa_n = \lambda_n - 1/n$. Both routes take positivity from the published lower bound (80.4). New in Report 101 relative to Part II: the genuine coefficients b_4, b_5, b_6 of (89.1), the ratio expansion of Section 89, and the product formula (85.3). The “unchanged PDF and source archive” of the companion are byte copies of Report 100’s PDF and archive inside Report 101’s delivery; they are not shipped here (Report 100 is printed above, and its archive survives in commit 60f54ea06).

81 The exact signed memory recurrence

For $N \geq M \geq 0$ of the same parity put

$$E(N, M) = \frac{C((N+M)/2, (N-M)/2)}{((N+M)/2)!},$$

and put $E = 0$ outside this wedge or off parity. The recurrence is

$$\begin{aligned} E(N, M) &= \frac{N-M+2}{N+M} E(N-1, M-1) + E(N-1, M+1) \\ &\quad - \frac{2(N-M-2)}{(N+M)(N+M-2)} E(N-3, M-1), \end{aligned} \quad (81.1)$$

$$E(N, 0) = E(N-1, 1). \quad (81.2)$$

The first equation is for $M > 0$ away from its finitely many startup singularities. Initialize those entries from (80.1); $E(0, 0) = 1$. Define

$$u_n(k) = E(2n, 2k), \quad 0 \leq k \leq n, \quad o_n(k) = E(2n-1, 2k+1), \quad 0 \leq k < n.$$

All vectors below are zero-extended to $\ell^2(\mathbb{N}_0)$, and $c_n = n!u_n(0)$. Direct substitution gives

$$\begin{aligned} u_n(k) &= a_n(k)o_n(k-1) + o_n(k) - b_n(k)o_{n-1}(k-1), \\ o_n(k) &= p_n(k)u_{n-1}(k) + u_{n-1}(k+1) - q_n(k)u_{n-2}(k), \end{aligned} \quad (81.3)$$

with $u_n(0) = o_n(0)$ and

$$\begin{aligned} a_n(k) &= \frac{n-k+1}{n+k}, & b_n(k) &= \frac{n-k-1}{(n+k)(n+k-1)}, \\ p_n(k) &= \frac{n-k}{n+k}, & q_n(k) &= \frac{n-k-2}{(n+k)(n+k-1)}. \end{aligned}$$

Eliminating both odd-time vectors, for $n \geq 4$,

$$u_n = A_n u_{n-1} - B_n u_{n-2} + F_n u_{n-3}. \quad (81.4)$$

The forward operator is the relaxed-tree operator

$$\begin{aligned} (A_n w)(0) &= w(0) + w(1), \\ (A_n w)(k) &= \frac{(n-k+1)^2}{(n+k)(n+k-1)} w(k-1) + \frac{2n-2k+1}{n+k} w(k) + w(k+1), \end{aligned} \quad (81.5)$$

for $1 \leq k \leq n$. The memory operators are

$$(B_n w)(0) = \frac{n-2}{n(n-1)} w(0), \quad (81.6)$$

$$(B_n w)(k) = P_n(k)w(k-1) + Q_n(k)w(k),$$

$$P_n(k) = \frac{(n-k-1)(2n-2k+1)}{(n+k)(n+k-1)(n+k-2)},$$

$$Q_n(k) = \frac{2n-2k-3}{(n+k)(n+k-1)},$$

$$(F_n w)(k) = W_n(k)w(k-1), \quad (81.7)$$

$$W_n(k) = \frac{(n-k-1)(n-k-2)}{(n+k)(n+k-1)(n+k-2)(n+k-3)},$$

with $(F_n w)(0) = 0$. The input supports are $0, \dots, n-2$ for B_n and $0, \dots, n-3$ for F_n .

For clarity, the five algebraic identities producing these coefficients are

$$\begin{aligned} L_n(k) &= a_n(k)p_n(k-1), & D_n(k) &= a_n(k) + p_n(k), \\ P_n(k) &= a_n(k)q_n(k-1) + b_n(k)p_{n-1}(k-1), \\ Q_n(k) &= q_n(k) + b_n(k), & W_n(k) &= b_n(k)q_{n-1}(k-1). \end{aligned}$$

At row zero the exact equation is

$$u_n(0) = u_{n-1}(0) + u_{n-1}(1) - \frac{n-2}{n(n-1)} u_{n-2}(0).$$

Apparent negative coefficients beyond the stated input supports multiply zero and are not entries of the actual matrices. All actual entries of B_n, F_n are nonnegative; $P_n(n-1) = 0$ and $W_n(n-2) = 0$. The sign before B_n in (81.4) is nevertheless negative. We never infer positivity or contraction of the full compacted evolution from the entrywise signs of its individual matrices.

82 The relaxed Jacobi estimates used here

[write, 5 October 2026, batch 102] Report 101 opens this section by restating, word for word in substance, Report 100's definitions of Section 68: the symmetrizer S_n (68.5), the symmetric positive semidefinite $J_n = S_n^{-1}A_nS_n = C_nC_n^\top$ with diagonal 1 at zero and $(2n - 2k + 1)/(n + k)$ elsewhere, its top eigenvalue λ_n and positive unit eigenvector ϕ_n , $H_n = J_n/\lambda_n$, the gauge R_n (68.9) (with $R_1(0) = 1$), $T_n = H_nR_n$ and $\Lambda_n = \prod_{j \leq n} \lambda_j$. It is not reprinted. The one change of letter: Report 101 writes $j_n(k)$ for the off-diagonal entry of J_n that Report 100 calls $b_n(k)$ (68.6), because $b_n(k)$ is an elimination coefficient in Section 81. Lemma 82.1 is kept, because the rest of Report 101 cites it display by display.

Lemma 82.1 (Companion Jacobi estimates). *For all sufficiently large n , with constants independent of n ,*

$$\lambda_n = 4 + 4an^{-2/3} + 6n^{-1} + O(n^{-4/3}), \quad (82.1)$$

$$\|(I - \phi_n\phi_n^\top)H_n\| \leq 1 - cn^{-2/3}, \quad (82.2)$$

$$0 \leq R_n \leq I, \quad 1 - R_n(k) \leq Ck^2/n^2, \quad (82.3)$$

$$\|(k+1)^p\phi_n\| \leq C_p n^{p/3}, \quad p \geq 0 \text{ fixed}, \quad (82.4)$$

$$\|\phi_n - \phi_{n-1}\| \leq C/n, \quad (82.5)$$

$$\|(I - R_n)\phi_n\| + \|(I - R_n)\phi_{n-1}\| \leq Cn^{-4/3}, \quad (82.6)$$

$$\|Z\phi_n - \phi_n\| \leq Cn^{-1/3}, \quad (82.7)$$

$$\phi_n(0) = \sqrt{2}n^{-1/2}(1 + O(n^{-2/3})), \quad (82.8)$$

where $(Zw)(k) = w(k-1)$ and $w(-1) = 0$. Moreover, for $0 \leq l \leq \lfloor n^{2/3} \rfloor$,

$$\|\delta_0^\top T_n \cdots T_{n-l+1}\| \leq C(l+1)^{-3/4}. \quad (82.9)$$

For the actual relaxed counting vector d_n , its normalization $v_n^r = \Lambda_n^{-1}S_n^{-1}d_n$ satisfies $\|v_n^r\| \leq 1$.

Here $d_n(k) = R(n+k, n-k)/(n+k)!$, where R obeys (80.1) without its subtraction term and with the same boundary conditions. These estimates are proved in Report 100's Sections 2–5 (Sections 68–71 here), included as a dependency. The half-line energy bound (70.1),

$$\sum_{k \geq 0} |\phi_n(k+1) - \phi_n(k)|^2 + 2\phi_n(0)^2 \leq Cn^{-2/3},$$

implies (82.7), including both zero-extension endpoints. We apply (82.9) only to the relaxed propagator, never to the signed compacted product.

[write, 5 October 2026, batch 102] Between the first and the last sentence of this paragraph Report 101 summarizes how Report 100 proves the lemma: min–max convergence of $n^{2/3}(4I - J_n)$ to the Dirichlet operator $-\partial_x^2 + 8x$ with ground state $\sqrt{2}\text{Ai}(2x+a)/\text{Ai}'(a)$ on the half grid $x_k = (k + \frac{1}{2})n^{-1/3}$, the quasimode residual that gives $6/n$, exponential barriers for the moments, the gap applied to $(J_n - J_{n-1})\phi_{n-1}$ for the drift, and the killed lazy-walk kernel P with $P^m(0,0) \leq C(m+1)^{-3/2}$ for (82.9); it adds that the companion “retains the full variational, residual, boundary, and localization proofs”. That summary repeats Sections 69–71 and is not reprinted; the derivation of (82.7), which Report 100 does not state, is kept. Each line of the

lemma is a line of Report 100: (82.1) is (69.1); (82.2) follows from (69.2) because J_n is positive semidefinite; (82.3) is Lemma 68.1; (82.5) and (82.8) are (69.4) and (69.5), and (82.4) follows from (69.3) since $(k+1)^p \leq 2^p(k^p+1)$; (82.6) is (70.4); (82.9) is (71.3); and $\|v_n^r\| \leq 1$ is the contraction stated after (68.12). Report 101 ships the companion's PDF and source archive unchanged (byte copies of Report 100's archive); they are not shipped here, since Report 100 is printed in full above.

83 Global bounds for the two gauged memory operators

Put $\tilde{B}_n = S_n^{-1}B_nS_{n-2}$ and $\tilde{F}_n = S_n^{-1}F_nS_{n-3}$.

Lemma 83.1. *On their entire finite input supports,*

$$\|\tilde{B}_n\| \leq C/n, \quad \|\tilde{F}_n\| \leq C/n^2, \quad (83.1)$$

$$\|\tilde{B}_n\phi_{n-2} - (4/n)\phi_n\| \leq Cn^{-4/3}. \quad (83.2)$$

Proof. The diagonal gauge ratio is $S_{n-2}(k)/S_n(k) = R_n(k)R_{n-1}(k) \leq 1$. On the lower band,

$$\frac{S_{n-2}(k-1)}{S_n(k)} = \frac{R_n(k-1)R_{n-1}(k-1)}{j_n(k)}.$$

Hence the corresponding gauged entry is bounded by

$$\frac{P_n(k)}{j_n(k)} = \frac{n-k-1}{n-k+1} \frac{2n-2k+1}{(n+k-2)\sqrt{(n+k)(n+k-1)}} \leq \frac{C}{n}.$$

The diagonal bound follows from Q_n and its exceptional first row. For $\tilde{F}_n$ there is one further factor $R_{n-2}(k-1) \leq 1$, and

$$\frac{W_n(k)}{j_n(k)} = \frac{(n-k-1)(n-k-2)}{n-k+1} \frac{1}{(n+k-2)(n+k-3)\sqrt{(n+k)(n+k-1)}} \leq \frac{C}{n^2}.$$

These estimates include k close to n : the small Jacobi edge is canceled by the displayed numerators. Summing the norms of the one or two diagonal bands proves (83.1).

For the finer assertion, on inputs supported at $0, \dots, n-2$ write

$$\tilde{B}_n = \frac{2}{n}(I+Z) - \frac{1}{n}\Pi_0 + V_n, \quad (83.3)$$

where Π_0 projects onto coordinate zero. Each band of V_n has row- k entry bounded by $C(k+1)/n^2$. On $k \leq n/2$, the explicit rational and square-root formulas give $P_n(k)/j_n(k) = 2/n + O((k+1)/n^2)$ and $Q_n(k) = 2/n + O((k+1)/n^2)$, uniformly. The gauge products differ from one by $O(k^2/n^2)$, adding only $O(k^2/n^3)$. At row zero the true entry is $1/n + O(n^{-2})$, explaining $-\Pi_0/n$. For $k > n/2$, the crude band bounds already imply the claimed $C(k+1)/n^2$ error, including the zero upper-edge entries.

For $\psi = \phi_{n-2}$, the moment bound gives $\|V_n\psi\| \leq Cn^{-5/3}$. Also $\|Z\psi - \psi\| \leq Cn^{-1/3}$ and $\|\Pi_0\psi\| \leq \|Z\psi - \psi\|$. Applying (83.3) gives $\tilde{B}_n\psi = (4/n)\psi + O_{\ell^2}(n^{-4/3})$. Finally $\|\phi_n - \psi\| \leq C/n$ by two applications of (82.5), proving (83.2). $\square$

84 Normalization and convergence of the scalar amplitude

Set

$$\kappa_n = \lambda_n - 1/n, \quad \Gamma_n = \prod_{j=1}^n \kappa_j, \quad v_n = \Gamma_n^{-1} S_n^{-1} u_n. \quad (84.1)$$

Every κ_n is positive: $\lambda_1 = 3/2$, and for $n \geq 2$ the Rayleigh quotient at δ_0 gives $\lambda_n \geq 1 > 1/n$. Thus

$$\kappa_n = 4 + 4an^{-2/3} + 5/n + O(n^{-4/3}).$$

Define

$$M_n = \frac{\tilde{B}_n}{\kappa_n \kappa_{n-1}}, \quad N_n = \frac{\tilde{F}_n}{\kappa_n \kappa_{n-1} \kappa_{n-2}}, \quad \rho_n = \frac{\lambda_n}{\kappa_n}.$$

Equation (81.4) becomes exactly

$$v_n = \rho_n T_n v_{n-1} - M_n v_{n-2} + N_n v_{n-3}. \quad (84.2)$$

The estimates already proved imply

$$\begin{aligned} \rho_n &= 1 + \frac{1}{4n} + O(n^{-5/3}), \\ \|M_n\| &\leq C/n, \quad \|N_n\| \leq C/n^2, \\ \|M_n \phi_{n-2} - (1/(4n)) \phi_n\| &\leq Cn^{-4/3}. \end{aligned} \quad (84.3)$$

84.1 A comparison bound that starts the bootstrap

Compacted decorated paths form a subset of relaxed decorated paths. Equivalently, [1, Lemma 5.1 and its proof] gives $0 \leq u_n \leq d_n$ coordinatewise. The positive diagonal gauge preserves this order, so

$$0 \leq v_n \leq (\Lambda_n / \Gamma_n) v_n^r.$$

Moreover,

$$\log \frac{\Lambda_n}{\Gamma_n} = \sum_{j=1}^n -\log \left(1 - \frac{1}{j\lambda_j} \right) = \frac{1}{4} \log n + O(1),$$

because $1/(j\lambda_j) = 1/(4j) + O(j^{-5/3})$. Since $\|v_n^r\| \leq 1$,

$$\|v_n\| \leq Cn^{1/4}. \quad (84.4)$$

This is a bound on the actual counting solution. It assumes no positive or contractive compacted operator on arbitrary initial data.

84.2 Perpendicular tracking

Write

$$a_n = \langle \phi_n, v_n \rangle, \quad w_n = v_n - a_n \phi_n, \quad \beta_n = \|w_n\|, \quad Q_n^\perp = I - \phi_n \phi_n^\top.$$

The counting solution has $a_n \geq 0$. Gauge contraction and Perron drift imply

$$\|Q_n^\perp R_n \phi_{n-1}\| \leq C/n, \quad \|Q_n^\perp R_n w_{n-1}\| \leq \beta_{n-1}.$$

Since H_n commutes with $Q_n^\perp$, the gap absorbs $\rho_n = 1 + O(1/n)$ and gives, for all large n ,

$$\beta_n \leq (1 - c'n^{-2/3})\beta_{n-1} + \frac{C}{n}|a_{n-1}| + \frac{C}{n}\|v_{n-2}\| + \frac{C}{n^2}\|v_{n-3}\|. \quad (84.5)$$

By (84.4) the forcing is $O(n^{-3/4})$, hence

$$\beta_n = O(n^{-1/12}). \quad (84.6)$$

We use the elementary scalar comparison: if $x_n \leq (1 - cn^{-2/3})x_{n-1} + Cn^{-r}$, with $r > 2/3$, then $x_n = O(n^{2/3-r})$. A sufficiently large multiple of that power is a supersolution for all large n , because its change between adjacent indices is smaller than its damping term; enlarge it to cover the finite prefix.

84.3 The scalar memory cancellation

Projection of (84.2), using $H_n\phi_n = \phi_n$, gives

$$a_n = \rho_n \langle R_n\phi_n, v_{n-1} \rangle - \langle \phi_n, M_nv_{n-2} \rangle + \langle \phi_n, N_nv_{n-3} \rangle. \quad (84.7)$$

Before cancellation, $\|R_n\phi_n - \phi_{n-1}\| \leq C/n$ and (84.4) imply

$$\Delta a_n := a_n - a_{n-1} = O(n^{-3/4}). \quad (84.8)$$

For the sharper estimate,

$$\langle \phi_n, R_n\phi_{n-1} \rangle - 1 = O(n^{-4/3}),$$

since its R_n -free part is $-\|\phi_n - \phi_{n-1}\|^2/2 = O(n^{-2})$ and the rest is bounded by (82.6). Therefore

$$\begin{aligned} \rho_n \langle \phi_n, R_n\phi_{n-1} \rangle - 1 &= \frac{1}{4n} + O(n^{-4/3}), \\ \langle \phi_n, M_nv_{n-2} \rangle &= \frac{1}{4n} + O(n^{-4/3}). \end{aligned}$$

The perpendicular part of the first term in (84.7) is

$$\rho_n \langle R_n\phi_n - \phi_{n-1}, w_{n-1} \rangle = O(\beta_{n-1}/n).$$

The remaining two perpendicular terms use the global norms of M_n, N_n . Thus the exact scalar equation has the form

$$\Delta a_n = \frac{1}{4n}\Delta a_{n-1} + \eta_n, \quad (84.9)$$

where

$$\begin{aligned} |\eta_n| &\leq Cn^{-4/3}(|a_{n-1}| + |a_{n-2}|) + Cn^{-2}|a_{n-3}| \\ &\quad + \frac{C}{n}(\beta_{n-1} + \beta_{n-2}) + \frac{C}{n^2}\beta_{n-3}. \end{aligned} \quad (84.10)$$

Insert $a_n = O(n^{1/4})$, (84.6), and (84.8). The largest term is $O(n^{-13/12})$, while the delayed difference contributes $O(n^{-7/4})$. Hence $\sum_n |\Delta a_n| < \infty$ and $a_n \rightarrow a_\infty \geq 0$.

Now a_n is bounded and $\beta_n \rightarrow 0$, so v_n is bounded. A second application of (84.5) gives $\beta_n = O(n^{-1/3})$. The crude scalar estimate improves to $\Delta a_n = O(1/n)$; equations (84.9)–(84.10) then give $\Delta a_n = O(n^{-4/3})$. Summing the tail proves

$$|a_n - a_\infty| + \|v_n - a_\infty\phi_n\| \leq Cn^{-1/3}. \quad (84.11)$$

The two bootstraps are sequential, not circular. The cancellation in (84.9) removes the nonsummable $1/(4n)$ scalar drift; this is the decisive effect of the negative memory.

85 Boundary smoothing and the positive amplitude

Put $z_n = v_n - a_\infty \phi_n$. The exact rewrite around the relaxed propagator is

$$\begin{aligned} z_n &= T_n z_{n-1} + g_n, \\ g_n &= (\rho_n - 1)T_n v_{n-1} - M_n v_{n-2} + N_n v_{n-3} + a_\infty(T_n \phi_{n-1} - \phi_n). \end{aligned}$$

Boundedness of v_n and the previous estimates imply $\|g_n\| \leq C/n$. The forcing may have either sign. Iterate over $L = \lfloor n^{2/3} \rfloor$ steps and use the row-norm estimate (82.9):

$$\begin{aligned} |z_n(0)| &\leq C(L+1)^{-3/4} n^{-1/3} + \frac{C}{n} \sum_{l=0}^{L-1} (l+1)^{-3/4} \\ &= O(n^{-5/6}). \end{aligned}$$

Thus

$$v_n(0) = \sqrt{2} a_\infty n^{-1/2} + O(n^{-5/6}). \quad (85.1)$$

Since $\log(\kappa_n/4) = an^{-2/3} + \frac{5}{4}n^{-1} + O(n^{-4/3})$, summation gives

$$\Gamma_n = K_c 4^n e^{3an^{1/3}} n^{5/4} (1 + O(n^{-1/3})), \quad K_c \in (0, \infty). \quad (85.2)$$

One exact convergent representation is

$$K_c = \frac{\kappa_1}{4e^{3a}} \prod_{j=2}^{\infty} \frac{\kappa_j}{4e^{3a(j^{1/3} - (j-1)^{1/3})} (j/(j-1))^{5/4}}. \quad (85.3)$$

Its logarithmic factors are $O(j^{-4/3})$, and their tail is $O(n^{-1/3})$.

Divide the published positive lower bound (80.4) by $n!\Gamma_n$. It yields $v_n(0) \geq cn^{-1/2}$ eventually. If $a_\infty = 0$, (85.1) instead gives $O(n^{-5/6})$, a contradiction. Therefore $a_\infty > 0$. Because $S_n(0) = 1$ and $c_n = n!\Gamma_n v_n(0)$, Theorem 80.1 follows with

$$\gamma_c = \sqrt{2} K_c a_\infty, \quad a_\infty = a_3 + \sum_{n=4}^{\infty} (a_n - a_{n-1}). \quad (85.4)$$

The series is absolutely convergent and all its entries are specified by the exact normalized recurrence and the finite Perron eigensystems. These formulas characterize the constant without an amplitude-limit hypothesis. They do not give certified decimal digits without explicit numerical tail bounds.

[write, 5 October 2026, batch 102] This is the one external analytic input of Report 101, as of Part II, whose endpoint section uses the same lower bound (16.2). Report 100's intrinsic positivity argument, Lemma 72.1, does not transfer verbatim, because the signed three-lag evolution (84.2) is not entrywise nonnegative; see Question V.5. The product (85.3) and the series in (85.4) are the compacted counterparts of Report 100's Theorem 73.1; Part II expresses γ_c through the one-step product instead, (24.1).

86 A forced signed memory lemma

The next result turns arbitrarily accurate approximate solutions into genuine asymptotics. It applies to signed errors and does not use the counting comparison again.

Lemma 86.1 (Forced memory transfer). *Suppose finite-support vectors z_n satisfy*

$$z_n = \rho_n T_n z_{n-1} - M_n z_{n-2} + N_n z_{n-3} + f_n, \quad (86.1)$$

where $\|f_n\| = O(n^{-p})$, $p > 4/3$, and $\|z_n\| = O(n^{-1/3})$. Then

$$\|z_n\| = O(n^{-(p-1)}), \quad z_n(0) = O(n^{-p+1/2}). \quad (86.2)$$

Proof. Put $\alpha_n = \langle \phi_n, z_n \rangle$ and $b_n = \|(I - \phi_n \phi_n^\top) z_n\|$. The initial norm hypothesis implies $\alpha_n \rightarrow 0$. Suppose presently $\|z_n\| = O(n^{-q})$, initially $q = 1/3$. Perpendicular projection as in (84.5) gives

$$b_n \leq (1 - cn^{-2/3})b_{n-1} + O(n^{-q-1}) + O(n^{-p}),$$

so the scalar comparison implies

$$b_n = O(n^{-q-1/3} + n^{-p+2/3}). \quad (86.3)$$

The uncanceled scalar projection gives $\Delta\alpha_n = O(n^{-q-1} + n^{-p})$. The refined scalar identity is

$$\Delta\alpha_n = \frac{1}{4n} \Delta\alpha_{n-1} + \eta_n + \langle \phi_n, f_n \rangle,$$

with (84.10) using α, b in place of a, β . The delayed difference is $O(n^{-q-2} + n^{-p-1})$; the other errors are bounded by (86.3). Therefore

$$\Delta\alpha_n = O(n^{-q-4/3} + n^{-p}).$$

These terms are summable because $q \geq 1/3$ and $p > 4/3$. Sum from $n+1$ to infinity using $\alpha_\infty = 0$ to get

$$\alpha_n = O(n^{-q-1/3} + n^{-p+1}).$$

Together with (86.3), this improves q to $\min(q + 1/3, p - 1)$. Finitely many iterations reach exactly $p - 1$, proving the norm assertion.

Rewrite (86.1) as $z_n = T_n z_{n-1} + h_n$, where

$$h_n = (\rho_n - 1)T_n z_{n-1} - M_n z_{n-2} + N_n z_{n-3} + f_n.$$

The improved norm gives $\|h_n\| = O(n^{-p})$. Smoothing over $L = \lfloor n^{2/3} \rfloor$ steps gives

$$|z_n(0)| \leq CL^{-3/4} n^{-(p-1)} + Cn^{-p} \sum_{l=0}^{L-1} (l+1)^{-3/4} = O(n^{-p+1/2}) + O(n^{-p+1/6}),$$

as claimed. No sign is required of z_n, f_n , or h_n . □

87 An exact polynomial Airy coefficient algorithm

Use $\varkappa = 2^{2/3}a$, distinct from the even-time normalization κ_n . Set

$$t = N^{-1/3}, \quad x = (M+1)t, \quad A(x) = \text{Ai}(a + 2^{1/3}x), \quad v(x) = 2x + \varkappa.$$

Then $A'' = vA$, $A(0) = 0$, and $A'(0) \neq 0$; primes here differentiate with respect to x . Seek formal series

$$s(t) = 2 + \varkappa t^2 + \sum_{j \geq 3} \sigma_j t^j, \quad (87.1)$$

$$F(t, x) = \sum_{j \geq 0} t^j f_j(x), \quad f_j = P_j A + Q_j A', \quad (87.2)$$

where $P_j, Q_j \in \mathbb{Q}[\varkappa, x]$, $P_0 = 1, Q_0 = 0$, with gauges

$$Q_j(0) = 0, \quad P_j(0) + Q'_j(0) = 0 \quad (j \geq 1). \quad (87.3)$$

These ensure $f_j(0) = f'_j(0) = 0$ for $j \geq 1$, while $f_0 = A$ fixes the normalization. Let $t_j = t(1 - jt^3)^{-1/3}$. The exact formal equation is

$$\begin{aligned} s(t)F(t, x) &= W(t, x)F(t_1, (x - t)(1 - t^3)^{-1/3}) \\ &\quad + F(t_1, (x + t)(1 - t^3)^{-1/3}) \\ &\quad - \frac{\mathcal{B}(t, x)}{s(t_1)s(t_2)}F(t_3, (x - t)(1 - 3t^3)^{-1/3}), \end{aligned} \quad (87.4)$$

where

$$W = \frac{1 - xt^2 + 3t^3}{1 + xt^2 - t^3}, \quad \mathcal{B} = \frac{2t^3(1 - xt^2 - t^3)}{(1 + xt^2 - t^3)(1 + xt^2 - 3t^3)}.$$

It is obtained from (81.1) by dividing through the preceding scalar carrier. In particular the two inverse factors in the last term are essential.

Proposition 87.1 (Existence and uniqueness at every order). *There is a unique compatible formal solution (87.1)–(87.4) satisfying (87.3), and all its coefficients belong to the displayed rational polynomial rings. Its first scalar correction is $\sigma_3 = 13/6$.*

Proof. The powers t^0, t^1 cancel identically and t^2 gives $A'' - (2x + \varkappa)A = 0$. Suppose the coefficients through degree t^{k-1} have vanished, for $k \geq 3$. Differentiation preserves polynomial pairs times A, A' . Let the known coefficient at t^k be $R_0 A + R_1 A'$. The new unknown profile $PA + QA'$ and coefficient σ_k enter as

$$(\partial_x^2 - v)(PA + QA') - \sigma_k A + R_0 A + R_1 A' = 0.$$

The delayed term starts at t^3 , so it contains neither new unknown at this order. Equating the two polynomial coefficients gives

$$\begin{aligned} P'' + 2vQ' + 2Q - \sigma_k + R_0 &= 0, \\ 2P' + Q'' + R_1 &= 0. \end{aligned}$$

Elimination of P yields

$$\mathcal{T}Q = 2R_0 - R'_1 - 2\sigma_k, \quad \mathcal{T}Q = Q''' - 4vQ' - 4Q.$$

On polynomials of degree at most d , $\mathcal{T}$ is triangular with diagonal entries $-4(2j+1)$ for $j = 0, \dots, d$. It is invertible and preserves $\mathbb{Q}[\varkappa, x]$. Set

$$\begin{aligned}\tilde{q} &= \mathcal{T}^{-1}(2R_0 - R'_1), & \sigma_k &= -2\tilde{q}(0), & Q &= \tilde{q} + \sigma_k/2, \\ P(x) &= \int_0^x \frac{-R_1(u) - Q''(u)}{2} du - Q'(0).\end{aligned}$$

These formulas give both differential identities and both gauges, and conversely those identities and gauges determine the same solution. The recursion extends indefinitely. Direct evaluation of its first stage gives $\sigma_3 = 13/6$. This is a finite-dimensional exact construction at each order, not a claim that the infinite series converges. $\square$

88 Compatible profiles with arbitrarily small genuine residuals

88.1 A scalar carrier with one common constant

Define coefficients $\ell_j \in \mathbb{Q}[\varkappa]$ by the formal identity

$$\begin{aligned}\log(s(t)/2) &= \frac{3\varkappa}{2t}(1 - (1 - t^3)^{1/3}) - \frac{13}{12}\log(1 - t^3) \\ &\quad + \sum_{j \geq 1} \ell_j t^j (1 - (1 - t^3)^{-j/3}).\end{aligned}\tag{88.1}$$

The t^2, t^3 coefficients match because they are $\varkappa/2$ and $13/12$. At degree $j+3$, the new ℓ_j has coefficient $-j/3 \neq 0$, so it is uniquely determined. Fix the leading multiplicative constant to one at every depth. For integer $D \geq 2$ let

$$H_N^{[D]} = 2^N \exp\left(\frac{3\varkappa}{2} N^{1/3}\right) N^{13/12} \exp\left(\sum_{j=1}^D \ell_j N^{-j/3}\right),\tag{88.2}$$

$$F^{[D]}(t, x) = \sum_{j=0}^D t^j f_j(x).\tag{88.3}$$

The exact ratio $H_N^{[D]}/H_{N-1}^{[D]}$ matches $s(t)$ through degree $D+3$. Computing these carrier coefficients may use more formal profiles than retained in $F^{[D]}$; Proposition 87.1 supplies them. Arbitrary products of truncated ratios with independently chosen initial constants must not replace (88.2), since that could change the amplitude with D .

In even time put $\varepsilon = n^{-1/3}$, $X = (k + 1/2)\varepsilon$, so $t = 2^{-1/3}\varepsilon$ and $x = 2^{2/3}X$. Define the uncut approximation

$$U_n^{[D]}(k) = \frac{2^{-13/12}}{\text{Ai}'(a)} H_{2n}^{[D]} F^{[D]}(t, x).\tag{88.4}$$

The derivative in this prefactor is the usual Airy-argument derivative. Its leading term is

$$\frac{4^n e^{3an^{1/3}} n^{13/12}}{\text{Ai}'(a)} \text{Ai}(a + 2X).$$

At $k = 0$, $\text{Ai}(a + \varepsilon)/\text{Ai}'(a) = \varepsilon + O(\varepsilon^3)$, hence

$$U_n^{[D]}(0) = 4^n e^{3an^{1/3}} n^{3/4} (1 + O(\varepsilon)).\tag{88.5}$$

Taylor expansion produces a compatible relative power series through any fixed degree not exceeding D . A newly included profile $\varepsilon^j f_j(x)$ vanishes at $x = 0$, so at $X_0 = \varepsilon/2$ it contributes only $O(\varepsilon^{j+1})$ before division by the leading ε . A newly included scalar term starts at relative degree j . Thus deeper truncations leave earlier endpoint coefficients unchanged and preserve leading coefficient one.

88.2 A uniform envelope for the delayed recurrence

For every fixed D, C_0 , the exact one-step residual of $H_N^{[D]} F^{[D]}$, divided by $H_N^{[D]}$, is bounded uniformly for $0 \leq M \leq C_0 \sqrt{N}$ by

$$Ct^{D+3}(1+x)^A e^{-cx^{3/2}}, \quad (88.6)$$

for constants $C, A, c > 0$ depending on the fixed depth. Here $x \leq Ct^{-1/2}$, so $|xt^2| \leq Ct^{3/2}$ and the denominators $1 + xt^2 - jt^3$ are uniformly separated from zero. Finite rational Taylor expansions therefore have remainders bounded by prescribed powers of t times polynomials in x . The finitely many exact carrier quotients $H_{N-j}^{[D]}/H_N^{[D]}$, $j \leq 3$, are bounded and have Taylor expansions with bounded remainders, directly from (88.2). The shifted arguments differ from x by $\pm t + O(xt^3 + t^4) = O(t)$.

Every fixed derivative of a polynomial-Airy profile is again a polynomial pair times A, A' , bounded by $C(1+x)^A e^{-cx^{3/2}}$. The same envelope, with changed constants, holds over the Taylor intervals. Multiplication preserves such envelopes. Formal cancellation removes powers through t^{D+2} , and the matching carrier ratios leave (88.6). The negative memory coefficient begins at t^3 and loses no order.

At $M = 0$, both terms at the ghost height $M - 1 = -1$ vanish exactly because every retained $f_j(0) = 0$. Thus the estimate holds for the actual exceptional boundary $E(N, 0) = E(N - 1, 1)$. No approximate boundary condition has been substituted.

88.3 Exact residual elimination and cutoff at every lag

Define odd-time approximants from the same expression $H_N^{[D]} F^{[D]}$ with the same prefactor $2^{-13/12}/\text{Ai}'(a)$ as in (88.4). Let $r_{e,n}, r_{o,n}$ denote their even and odd residuals in (81.3). Let K_n map o_n to u_n with diagonal 1 and lower diagonal $a_n(k)$, and let G_n map o_{n-1} to u_n with lower diagonal $b_n(k)$. Direct substitution gives the exact identity

$$U_n^{[D]} - A_n U_{n-1}^{[D]} + B_n U_{n-2}^{[D]} - F_n U_{n-3}^{[D]} = r_{e,n} + K_n r_{o,n} - G_n r_{o,n-1}. \quad (88.7)$$

The coefficients of K_n are bounded and those of G_n are $O(1/n)$ in the region under consideration. Consecutive carriers have bounded ratios. Therefore the pointwise residual order ε^{D+3} survives this elimination. At row zero K_n reads coordinate zero and G_n is zero, so the same identity covers the boundary.

Choose a fixed smooth function χ , equal to one on $[0, 1/2]$ and zero on $[1, \infty)$, and put

$$\tilde{U}_n^{[D]}(k) = \chi(k/\sqrt{n}) U_n^{[D]}(k).$$

Whenever neighboring-height or lagged cutoff factors at $n, n-1, n-2, n-3$ differ, k is comparable to $\sqrt{n}$ and X to $n^{1/6}$. Every profile multiplied by a bounded carrier ratio is then at most a polynomial in n times $e^{-cn^{1/4}}$. Each cutoff defect is smaller than every fixed inverse power after carrier normalization; cancellation among these defects is unnecessary. The residual support is $k \leq \sqrt{n} + O(1)$.

Write $D_n^{\text{bulk}} = 4^n e^{3an^{1/3}} n^{13/12}$. The X -lattice spacing is ε and the squared envelope sums to $O(\varepsilon^{-1})$. Hence

$$\left\| \tilde{U}_n^{[D]} - A_n \tilde{U}_{n-1}^{[D]} + B_n \tilde{U}_{n-2}^{[D]} - F_n \tilde{U}_{n-3}^{[D]} \right\| = O(D_n^{\text{bulk}} \varepsilon^{D+5/2}). \quad (88.8)$$

On this support S_n^{-1} is uniformly bounded, since its factorial product gives $-\log S_n(k) = O(k^2/n + k/n) = O(1)$. Also $D_n^{\text{bulk}}/\Gamma_n = O(n^{-1/6}) = O(\varepsilon^{1/2})$. Consequently

$$\left\| \Gamma_n^{-1} S_n^{-1} \left[\tilde{U}_n^{[D]} - A_n \tilde{U}_{n-1}^{[D]} + B_n \tilde{U}_{n-2}^{[D]} - F_n \tilde{U}_{n-3}^{[D]} \right] \right\| = O(\varepsilon^{D+3}). \quad (88.9)$$

This is a genuine global norm estimate for every fixed depth, not merely pointwise formal cancellation.

88.4 The leading profile and the all order transfer

The unit leading Airy sample is $q_n(k) = \sqrt{2\varepsilon} \text{Ai}(a + 2X)/\text{Ai}'(a)$, up to normalization $1 + O(\varepsilon^2)$, with $\|\phi_n - q_n\| = O(\varepsilon^2)$ by the companion proof. Equations (85.2) and (88.4) give

$$\left\| \Gamma_n^{-1} S_n^{-1} \tilde{U}_n^{[D]} - \frac{1}{\sqrt{2}K_c} \phi_n \right\| = O(\varepsilon). \quad (88.10)$$

Indeed the finite scalar corrections are $1 + O(\varepsilon)$; higher fixed profiles have relative weighted norm $O(\varepsilon)$; and $S_n^{-1} - 1$ is $O(\varepsilon(1 + X^2))$ in that norm by the bounded logarithm estimate. The cutoff tail is exponentially small. The leading constant in (88.10) is independent of D .

Now let

$$z_n = v_n - \gamma_c \Gamma_n^{-1} S_n^{-1} \tilde{U}_n^{[D]}.$$

Since $\gamma_c = \sqrt{2}K_c a_\infty$, equations (84.11) and (88.10) imply $\|z_n\| = O(n^{-1/3})$. Equation (88.9) supplies the forcing in Lemma 86.1, with $p = (D + 3)/3 > 4/3$. Its endpoint error, multiplied by $\Gamma_n/[4^n e^{3an^{1/3}} n^{3/4}] = \Theta(n^{1/2})$, is $O(n^{-(p-1)}) = O(n^{-D/3})$. Given any desired J , choose $D \geq \max(2, J + 1)$. Combine this with the compatible endpoint expansion of (88.5). This proves Theorem 80.2, with one positive amplitude at every order.

89 Explicit coefficients and the scope of finite certificates

The coefficient ring in Theorem 80.2 follows from a grading, not merely the finite examples below. Let $\omega^3 = 1$, and let $B_\varkappa$ solve $B'' = (2x + \varkappa)B$ with $B(0) = 0, B'(0) = 1$. Uniqueness of this initial-value problem gives

$$B_{\omega\varkappa}(\omega x) = \omega B_\varkappa(x).$$

Both W and $\mathcal{B}$ in (87.4) are invariant under $(t, x, \varkappa) \mapsto (\omega t, \omega x, \omega \varkappa)$. Uniqueness of the formal recursion with its boundary and derivative gauges therefore makes $s(t)$ and the normalized endpoint $F(t, t)/(tA'(0))$ invariant. The carrier equation (88.1), with its zero scalar integration constant, preserves the same invariance. Hence a scalar monomial $\varkappa^r t^j$ has $r + j$ divisible by three. Substituting $\varkappa = 2^{2/3}a$ and $t = 2^{-1/3}\varepsilon$ gives $2^{(2r-j)/3}a^r \varepsilon^j$, with an integer exponent of two. All endpoint coefficients thus belong to $\mathbb{Q}[a]$.

It is convenient to use logarithmic coefficients b_j , defined by the genuine expansion

$$\log c_n = \log(n!) + n \log 4 + 3an^{1/3} + \frac{3}{4} \log n + \log \gamma_c + \sum_{j=1}^J b_j n^{-j/3} + O(n^{-(J+1)/3}). \quad (89.1)$$

The first six are

$$\begin{aligned} b_1 &= \frac{53a^2}{90}, & b_2 &= \frac{271a}{216}, \\ b_3 &= -\frac{41728a^3 - 477495}{1360800}, & b_4 &= -\frac{2053a^2}{9720}, \\ b_5 &= \frac{a(2869124a^3 - 329492565)}{1377810000}, & b_6 &= -\frac{320744a^3 + 10924065}{132269760}. \end{aligned}$$

They are exact rational polynomials in the Airy zero. Exponentiating gives, for example,

$$\begin{aligned} d_1 &= \frac{53a^2}{90}, \\ d_2 &= \frac{a(2809a^3 + 20325)}{16200}, \\ d_3 &= \frac{2084278a^6 + 43365690a^3 + 21487275}{61236000}. \end{aligned}$$

Further multiplicative coefficients through degree six and normalized diagonal-ratio coefficients through degree nine are included in the exact JSON certificates. The ratio begins

$$\frac{c_n}{4nc_{n-1}} = 1 + an^{-2/3} + \frac{3}{4}n^{-1} + \frac{41a^2}{135}n^{-4/3} + \frac{20a}{81}n^{-5/3} + O(n^{-2}).$$

To obtain a ratio remainder of a prescribed order, use the all-order forward expansion deeper than that order before division. Dividing two shallow big- O formulas does not justify extra cancellation of their remainders.

The finite formal certificate verifies both polynomial components of every recurrence coefficient through t^9 , exact boundary and derivative gauges, an independent generic linear-system construction, endpoint integration, and an independent conversion of log corrections into diagonal ratios. It contains both relaxed and compacted data because the same algebraic engine applies to both. The compacted analytic transfer proved above is additional to these finite algebraic identities. Finite tests alone cannot establish asymptotic convergence or amplitude positivity.

The numerical estimate 173.12670485 is inherited from [1]. At $n = 3000$, exact integer dynamic programming followed by high-precision evaluation gives the uncorrected normalization approximately 213.44032023; removing six logarithmic corrections gives approximately 173.1267049959. This slow convergence is consistent with $b_1n^{-1/3}$ but is not a certified enclosure for γ_c . The optional floating-point gauged recurrence independently agrees with the exact-integer diagonal diagnostics to about $4 \cdot 10^{-12}$ relative error at $n = 3000$. None of these decimal comparisons is used in the proof.

[write, 5 October 2026, batch 102] With $a = z$, the coefficients b_1, b_2, b_3 are Part II's logarithmic corrections (16.4) ($477495/1360800 = 393/1120$), and d_1, d_2, d_3 are its relative coefficients c_1, c_2, c_3 . The genuine b_4, b_5, b_6 are new to this report; they are the formal $b_4^{(c)}, b_5^{(c)}, b_6^{(c)}$ of Report 100's (76.3), now proved. The ratio display stops at $n^{-5/3}$; Report 100's formal compacted ratio, printed after (76.3), continues it through n^{-3} and is genuine by Corollary 66.3. The conversions $b \mapsto d$ and $b \mapsto \text{ratio}$ were recomputed symbolically at the write (difference zero). The grading argument is Part II's (Section 25), in this Part's variables. The "exact JSON certificates" are those of Report 100's suite, which Report 101 ships as a byte copy (not shipped again here; Report 100's copies are `data/102-100-relaxed-checks-data-formal_9.json` and `data/102-100-relaxed-checks-data-endpoint_9.json`).

90 Rigorous inverse asymptotics

The distinction between recovering the index of an exact count and inverting an arbitrary threshold matters. Throughout this section $\alpha = 3/4$, and all statements are eventual as their arguments tend to infinity.

[write, 5 October 2026, batch 102] Theorem 90.1 and Corollary 90.2 are Report 100's Theorem 77.1 and Corollary 77.2 at $\alpha = 3/4$, with the same proofs (Report 100 states them for both sequences); they are printed again because Report 101's Newton algorithm below starts from its own (90.1). Corollary 90.3 is the unconditional form of the compacted inverse that Report 100 states only conditionally. All of it is a second route to Part II's inverse section (Section 26) and an instance of the transseries volume's inversion theorems (Part 0).

90.1 A Lambert start that does not need the amplitude value

For large $Y > 0$ define

$$x_0(Y) = \frac{Y}{W_0(4Y/e)}, \quad L_0(Y) = \log(4x_0(Y)), \quad (90.1)$$

$$z(Y) = x_0(Y) - \frac{3ax_0(Y)^{1/3} + (5/4)\log x_0(Y) + \frac{1}{2}\log(2\pi)}{L_0(Y)}, \quad (90.2)$$

where W_0 is the positive real branch of Lambert's W function.

Theorem 90.1 (Eventual exact-value recovery). *One has $z(\log c_n) = n + O(1/\log n)$. Thus nearest-integer rounding recovers n for all sufficiently large exact sequence values c_n .*

Proof. Stirling's formula and (80.4) give $\log c_n = G(n) + O(1)$, where

$$G(x) = x(\log(4x) - 1) + 3ax^{1/3} + \frac{5}{4}\log x + \frac{1}{2}\log(2\pi).$$

Let $G_0(x) = x(\log(4x) - 1)$ and $g = G - G_0$. The Lambert identity gives $G_0(x_0) = Y$ and $G'_0(x_0) = L_0$. With $h = -g(x_0)/L_0 = O(x_0^{1/3}/\log x_0)$, Taylor expansion yields

$$G(x_0 + h) - Y = g'(x_0)h + O(h^2/x_0) = O(x_0^{-1/3}/\log x_0),$$

because $g'(x) = ax^{-2/3} + 5/(4x)$. At $Y = \log c_n$, the mean value theorem and $G'(x) \sim \log x$ imply the assertion; x_0, z, n are asymptotic. The error tends to zero, so eventual rounding is valid. $\square$

The unknown constants in the two-sided bound do not supply an effective cutoff. At $n = 3000$, $z(\log c_n) - n$ is approximately 0.56223, so nearest-integer recovery still fails there. This theorem is not a certified finite-range inversion algorithm.

90.2 Arbitrary thresholds require one exact comparison

The counting sequence is strictly increasing from index one onward. Indeed (80.1) and nonnegativity give, inductively,

$$c_n \geq nc_{n-1} - (n-2)c_{n-2} \geq 2c_{n-1} \quad (n \geq 2),$$

starting with $c_0 = c_1 = 1, c_2 = 3$. Also direct differentiation of (90.2) gives $z'(Y) \sim 1/\log x_0 > 0$.

Corollary 90.2 (Threshold inversion). *For sufficiently large y , let $k(y) = \max\{n : c_n \leq y\}$ and $j_0 = \text{round}(z(\log y))$. Then $j_0 \in \{k(y), k(y) + 1\}$ and*

$$k(y) = \begin{cases} j_0, & c_{j_0} \leq y, \\ j_0 - 1, & c_{j_0} > y. \end{cases}$$

For the least index $j(y)$ with $c_{j(y)} \geq y$, return j_0 if $c_{j_0} \geq y$ and $j_0 + 1$ otherwise. Equality belongs to the first branch in each rule.

Proof. Monotonicity and Theorem 90.1 put $z(\log y)$ between $k(y) + o(1)$ and $k(y) + 1 + o(1)$. Eventual nearest-integer rounding therefore gives only the two stated candidates, and comparison selects the correct one. The least-index rule is analogous; at equality the exact-value theorem applies. $\square$

A threshold arbitrarily close to a sequence value requires an exact or rigorously certified comparison. Unverified floating-point asymptotics do not determine its side of that value.

90.3 Arbitrary order inverse remainders

For fixed $J \geq 0$, put

$$\mathcal{F}_J(x) = \log \Gamma(x + 1) + x \log 4 + 3ax^{1/3} + \frac{3}{4} \log x + \log \gamma_c + \sum_{j=1}^J b_j x^{-j/3}. \quad (90.3)$$

Here $\Gamma(x + 1)$ is the usual gamma function, distinct from the product Γ_n in (84.1). Its derivative satisfies $\mathcal{F}'_J(x) \sim \log x > 0$. Let $x_J(Y)$ be the unique sufficiently large solution of $\mathcal{F}_J(x) = Y$.

Corollary 90.3 (All order compacted inverse). *For every fixed J ,*

$$x_J(\log c_n) = n + O\left(\frac{n^{-(J+1)/3}}{\log n}\right). \quad (90.4)$$

Proof. Equation (89.1) gives $\log c_n = \mathcal{F}_J(n) + O(n^{-(J+1)/3})$. Apply the mean value theorem with $\mathcal{F}'_J(x) \sim \log x$. $\square$

This is an unconditional mathematical consequence of the all-order theorem, with the canonically specified positive constant γ_c . Replacing it by uncertified digits does not make a certified numerical inverse.

90.4 A finite Newton algorithm at each prescribed order

Start at (90.1), put $L = \log(4x_0)$, and iterate

$$x_{k+1} = x_k - \frac{\mathcal{F}_J(x_k) - Y}{\mathcal{F}'_J(x_k)}, \quad K = \lceil \log_2(J + 4) \rceil - 1. \quad (90.5)$$

The derivative is

$$\mathcal{F}'_J(x) = \psi(x + 1) + \log 4 + ax^{-2/3} + \frac{3}{4x} - \sum_{j=1}^J \frac{j}{3} b_j x^{-j/3-1},$$

where ψ is the digamma function. The K th iterate satisfies the same inverse error (90.4). To prove the iteration count, let x_* be the root of $\mathcal{F}_J(x) = Y$. The Airy term gives $x_* - x_0 = O(x_0^{1/3}/L)$. On $x_0 \pm o(x_0)$, $\mathcal{F}'_J \sim L$ and $\mathcal{F}''_J = O(1/x_0)$. Thus exact Newton errors obey

$$|e_{k+1}| \leq C|e_k|^2/(x_0 L), \quad e_k = O\left(\frac{x_0^{1-2^{k+1}/3}}{L^{2^{k+1}-1}}\right).$$

The iterates remain in that neighborhood. Since $2^{K+1} \geq J + 4$, the root error is no larger than $O(n^{-(J+1)/3}/\log n)$, proving the assertion. A single unqualified Newton step does not give arbitrary-order accuracy.

There is a version with finitely many elementary operations in place of the gamma function. Substitute the finite Stirling expression

$$(x + \tfrac{1}{2}) \log x - x + \tfrac{1}{2} \log(2\pi) + \sum_{r=1}^R \frac{B_{2r}}{2r(2r-1)x^{2r-1}}, \quad R = \max\left(0, \left\lceil \frac{J-2}{6} \right\rceil\right),$$

for $\log \Gamma(x+1)$, and differentiate that same expression in every Newton step. Its remainder is $O(x^{-2R-1})$ and $2R+1 \geq (J+1)/3$, so the root changes by no more than the required inverse error. The same K steps suffice. At each fixed order this uses finitely many arithmetic operations, logarithms, real fractional powers, and one principal Lambert- W evaluation, with constants a, γ_c, b_j and Bernoulli numbers. Effective certified evaluation also needs amplitude enclosures and numerical error control, neither supplied by the asymptotic statement alone.

91 Reproducibility and limits of the result

The accompanying source archive contains this article's `.tex` source, a deterministic three-pass build script, explicit exact checks, the formal coefficient certificates and verifier, and the unchanged relaxed companion PDF and source archive. The latter provides the complete proofs behind Lemma 82.1. It is a private mathematical companion prepared for this project, not an externally peer-reviewed publication.

Run `./build.sh` to rebuild this PDF and `./replay.sh` to replay the certificates. The build rejects overfull boxes or unresolved references. The new compacted verifier checks exact one-step elimination identities symbolically, the complete even-time recurrence on exact rational counting arrays, upper-boundary supports, squared gauge ratios, and the canceled small-edge identities. The supplied formal verifier reconstructs the profiles with a triangular solver and compares them to an independent linear-system solution, then checks all local recurrence coefficients through order nine and endpoint coefficient families.

Every acceptance gate raises an explicit exception and remains active under `python -0`. Deliberate negative cases include disabled or reduced coverage, altered boundary or memory coefficients, damaged gauge identities, malformed coefficient data, a boundary-preserving formal perturbation, and corrupted endpoint certificates. The exact replay runs in normal and optimized modes; negative tests must be rejected for the intended reasons. The release manifest records sizes and SHA-256 hashes. The archive is extracted to a fresh directory for a full rebuild and replay before delivery, and all rendered PDF pages are visually inspected.

The distinction between proof and certificate is important. These finite checks validate exact identities, indexing, coefficient computations, and reproducibility. The convergence, uniform

residuals, nonzero amplitude, and all-order error estimates are the analytic arguments in this article and its supplied dependency. No finite matrix sweep or numerical fit proves those statements.

[write, 5 October 2026, batch 102] These paragraphs describe Report 101’s delivered archive (commit 60f54ea06). Its build script, release manifest and manuscript are not shipped; its formal verifier, certificates and negative tests are byte copies of Report 100’s and are shipped once, under Report 100’s prefix. The “private mathematical companion” is Report 100, printed in this Part; Reports 100 and 101 both name their author as “Research prepared for private review”. The README of this report lists the shipped files.

What remains open here. The canonical constant is positive and convergently represented, but its decimal digits and an effective inverse cutoff are not certified. Explicit constants in the spectral, drift, tracking, and endpoint bounds would permit interval tail enclosures. The finite-order expansion does not bound coefficient growth uniformly in the order, establish a Gevrey class, justify optimal truncation, or determine exponentially small Airy sectors. A full transseries requires separate control of those terms and their amplitudes. No such conclusion follows merely from an expansion valid to every fixed algebraic order.

[write, 5 October 2026, batch 102] Re-scoped in Section 105: certified digits and effective cutoffs (Question V.1), coefficient growth and Gevrey bounds (Question V.2), exponentially small sectors (Question V.3).

Report 103: minimal binary automata

Source. Bundle Report 103, archive A331120_Minimal_Binary_Automata_All_Orders_Source.zip, main file `minimal_dfa_amplitude/minimal_dfa_amplitude.tex` (927 lines; 23-page PDF), titled “A positive Airy amplitude for minimal binary automata: Signed memory and all order asymptotics for OEIS A331120”, dated 2 October 2026, author line “Research prepared for Vladimir”. Printed in full except for the restatement of Report 100 in its Section 3, label prefix `air:jp:dfa:`; its Sections 1–13 are Sections 92–104.

Abstract of Report 103.

Let b_n count non-isomorphic minimal deterministic finite automata with n transient states over a fixed binary alphabet, recognizing a finite language. The unique sink gives $n + 1$ total states. Starting from the exact counting triangle of Elvey Price, Fang, and Wallner, we prove

$$b_n = \gamma_D n! 8^n e^{3an^{1/3}} n^{7/8} \left(1 + \sum_{j=1}^m c_j n^{-j/3} + O(n^{-(m+1)/3}) \right)$$

for every fixed $m \geq 0$, with one common constant $0 < \gamma_D < \infty$; a is the largest zero of the Airy function. The signed recurrence has two delayed correction terms. A precise cancellation of its nonsummable scalar drift permits Perron tracking, and a published compacted-tree lower bound rules out a zero amplitude. We then construct profiles to arbitrary depth, prove uniform residual estimates at the exact boundary and all memory lags, and transfer them to the actual counting solution. Four explicit correction coefficients, a convergent amplitude characterization, and arbitrary-order Lambert–Newton inverse estimates are given. The required relaxed-tree spectral theorem is stated completely and imported from the immutable companion proof enclosed with the source package. No convergent infinite asymptotic series, exponentially small sectors, closed-form amplitude, certified amplitude digits, or literature-priority claim is asserted.

92 The result and its published starting point

The sequence is OEIS A331120 [9]:

$$(b_n)_{n \geq 0} = 1, 1, 6, 60, 900, 18480, 487560, 15824880, 612504240, \dots$$

Our indexing follows the counting triangle in Proposition 5 of Elvey Price–Fang–Wallner [2]. The term $n = 0$ is its initial convention; all asymptotic statements concern $n \rightarrow \infty$. Put

$$a = \max\{x \in \mathbb{R} : \text{Ai}(x) = 0\} = -2.338107410459 \dots$$

The numerical value identifies the root only; the constant in every theorem is the exact Airy zero.

Theorem 92.1 (Positive amplitude and all algebraic orders). *There is a constant $\gamma_D \in (0, \infty)$ and a unique compatible sequence of real coefficients $(c_j)_{j \geq 1}$ such that for each fixed integer $m \geq 0$,*

$$b_n = \gamma_D n! 8^n e^{3an^{1/3}} n^{7/8} \left[1 + \sum_{j=1}^m c_j n^{-j/3} + O(n^{-(m+1)/3}) \right]. \quad (92.1)$$

The constant in the error may depend on m . Equivalently, there are compatible logarithmic coefficients d_j such that

$$\log \frac{b_n}{\gamma_D n! 8^n e^{3an^{1/3}} n^{7/8}} = \sum_{j=1}^m d_j n^{-j/3} + O(n^{-(m+1)/3}). \quad (92.2)$$

They are produced by the exact polynomial algorithm of Section 98. The amplitude is characterized by convergent quantities in Section 97.

The published theorem [2, Theorem 1] already establishes the sharp scale

$$b_n = \Theta\left(n! 8^n e^{3an^{1/3}} n^{7/8}\right). \quad (92.3)$$

Thus the factorial, exponential, stretched-exponential, and polynomial factors are prior work. That paper also supplies the exact recurrence and Airy comparison construction. Its lower trial ratio contains the coefficient $29/(12N)$, which reappears in our independently derived formal series. Our contribution here is the proof of convergence to one positive amplitude, the compatible remainder theorem at every algebraic order, and the consequent refined inverses. This is a description relative to the inspected sources, not a worldwide novelty assertion.

The proof does not need (92.3) to establish its positive amplitude. Its published lower input is only

$$2^{n-1} c_n^{\text{tree}} \leq b_n, \quad c_n^{\text{tree}} \geq C n! 4^n e^{3an^{1/3}} n^{3/4} \quad (n \text{ sufficiently large}), \quad (92.4)$$

where c_n^{tree} counts compacted binary trees. The comparison is [2, Equation (1)]; the compacted asymptotic bound is stated on its next page and proved in [1]. No positive-amplitude theorem for compacted trees is required.

The analytic dependency. The finite Jacobi matrices in this problem are exactly those of the companion relaxed-tree report, Report 100. Sections 2–5 of that report (Sections 68–71 here) prove the spectral, localization, gauge, and boundary smoothing estimates used below. We state the complete needed input in Section 94; its proof is available in the enclosed, hash-pinned PDF and source archive. The DFA-specific signed-memory analysis and all-orders transfer are proved here. The L^AT_EX source of the present article compiles on its own without external inputs or bibliography files.

[**write, 5 October 2026, batch 102**] Theorem 92.1 is a second route to Part IV's Theorem 53.1 (a here is Part IV's z , γ_D its γ), by the mechanism of Part IV and of Report 101 in even time: uniform normalization by 2^{-m} (as in (54.1)), domination by the relaxed array, an $O(n^{1/8})$ start, and cancellation of the scalar drift $1/(8n)$. The positivity input is slightly weaker than Part IV's: Part IV uses the automaton Θ theorem (53.2), Report 103 only the structural comparison $b_n \geq 2^{n-1}c_n^{\text{tree}}$ and the compacted lower bound (92.4), from which Elvey Price, Fang and Wallner derive the automaton lower bound. New in Report 103 relative to Part IV: d_4 , c_4 , σ_7 , the ratio expansion (101.3), the explicit inverse term (102.11) and the numerical amplitude estimate near 76.438 (uncertified). Report 103 calls Report 100 a “companion research report prepared for private review”. The “enclosed, hash-pinned PDF and source archive” are byte copies of Report 100's delivered PDF and archive (SHA-256 recorded in Section 104); they are not shipped here, since Report 100 is printed in this Part and its archive survives in commit 60f54ea06.

Meaning of all orders. Equation (92.1) is a Poincaré expansion: every *fixed finite truncation* has its stated remainder. It neither asserts convergence of the infinite series nor permits summing it and subtracting an exact function. Section 102 supplies rigorous inverse expansions in fractional powers and logarithms. Contributions smaller than every algebraic power, sometimes intended by the word transseries, remain outside the proved result.

93 The exact counting array and its startup

Write $b(n, m)$ for the nonnegative integer array from [2, Proposition 5]:

$$\begin{aligned} b(n, m) &= 2b(n, m-1) + (m+1)b(n-1, m) - mb(n-2, m-1), & n \geq m \geq 1, \\ b(n, m) &= 0, & n < m, \\ b(n, 0) &= 1, & n \geq -1. \end{aligned} \quad (93.1)$$

Then $b_n = b(n, n)$. The negative-index seed $b(-1, 0) = 1$ is important: it makes $b(1, 1) = 1$, not 2.

93.1 Uniform normalization and rotation

Define, including the entire $m = 0$ boundary,

$$B(n, m) = 2^{-m}b(n, m).$$

Division of (93.1) gives

$$B(n, m) = B(n, m-1) + (m+1)B(n-1, m) - \frac{m}{2}B(n-2, m-1). \quad (93.2)$$

For admissible integers $0 \leq h \leq N$ with $N-h$ even put

$$e(N, h) = \frac{B((N+h)/2, (N-h)/2)}{((N+h)/2)!},$$

and put $e = 0$ off this domain. Its exact startup is

$$e(0, 0) = 1, \quad e(1, 1) = 1, \quad e(2, 0) = \frac{1}{2}, \quad e(2, 2) = \frac{1}{2}. \quad (93.3)$$

A direct substitution in (93.2) yields, for $N \geq 3$,

$$e(N, h) = \frac{N - h + 2}{N + h} e(N - 1, h - 1) + e(N - 1, h + 1) - \frac{N - h}{(N + h)(N + h - 2)} e(N - 3, h - 1). \quad (93.4)$$

At $h = 0$, both ghost terms vanish exactly and $e(N, 0) = e(N - 1, 1)$. At $h = N$, the memory coefficient is zero and the upper boundary propagates $e(N, N) = 1/N!$. Never evaluate the singular coefficient at $(N, h) = (2, 0)$; use (93.3). The diagonal conversion is

$$b_n = 2^n n! e(2n, 0). \quad (93.5)$$

93.2 Why the normalization is stated explicitly

The nearby transformed arrays on pages 11:6–11:7 of [2] use $b(n, m)/2^{m-1}$ for $m \geq 1$, but leave $m = 0$ unchanged. Their displayed recurrence for that first transformed array is restricted to $m > 1$. If the subsequent rotated recurrence is read uniformly with that piecewise startup, its value at $(3, 1)$ would be

$$1 + \frac{1}{2} - \frac{1}{4} = \frac{5}{4},$$

whereas its defining piecewise normalization gives $\frac{3}{2}$. This discrepancy is visible in the typeset source, not just in text extraction. It does not establish a failure of the published asymptotic theorem. Our construction simply derives every identity from the unambiguous Proposition 5 using uniform division by 2^m . With this convention, $e(3, 1) = 3/4$, and (93.4) agrees with it.

[write, 5 October 2026, batch 102] The values on the uniform side were rechecked at the write: $b(2, 1) = 2^2 - 1 = 3$, so $e(3, 1) = B(2, 1)/2! = \frac{3}{2}/2 = \frac{3}{4}$, while the piecewise normalization $b(2, 1)/2^0/2!$ gives $\frac{3}{2}$. The statement about the printed pages 11:6–11:7 of the AofA 2020 paper, that their recurrence read uniformly gives $\frac{5}{4}$, was not checked: the paper is not in the repository. It stays an unverified remark about a publication, like the remark on Ghosh Dastidar–Wallner in Part 0, and is Question V.4. Part IV uses the same uniform normalization (after (54.1)) and remarks only that a normalization by 2^{m-1} for $m \geq 1$ needs a separate boundary row. Report 103 itself says that the discrepancy is not a claim that the published asymptotic theorem is false.

93.3 A counting comparison that survives the subtraction

Let $R(n, 0) = 1$, $R(n, m) = 0$ for $n < m$, and

$$R(n, m) = R(n, m - 1) + (m + 1)R(n - 1, m). \quad (93.6)$$

This is the relaxed-tree triangle. Nonnegativity of the counting array B , followed by induction in n and m in (93.2), gives

$$0 \leq B(n, m) \leq R(n, m). \quad (93.7)$$

The exceptional lag $B(-1, 0) = 1$ at $(1, 1)$ is nonnegative, so it does not invalidate the induction. This comparison concerns the actual counting solution. We will never infer positivity or contractivity for the signed recurrence on arbitrary initial vectors.

94 The relaxed Jacobi estimates used by the proof

All finite vectors are extended by zero to $\ell^2(\mathbb{N}_0)$. Matrix formulas below are restricted to their actual input and output supports.

[write, 5 October 2026, batch 102] Report 103 continues by restating Report 100's definitions (Section 68): the two-step operator A_n on coordinates $0, \dots, n$ with its exceptional first row, (68.2)–(68.4); the symmetrizer S_n (68.5); the symmetric positive semidefinite $J_n = S_n^{-1} A_n S_n$, whose adjacent coefficient between $k-1$ and k it calls $j_n(k)$ (Report 100's $b_n(k)$, (68.6)); the top eigenvalue λ_n , the positive unit eigenvector ϕ_n , $H_n = J_n/\lambda_n$ and $\Lambda_n = \prod_{j \leq n} \lambda_j$. These are not reprinted. Report 103 writes $R_n^g = S_n^{-1} S_{n-1}$ for Report 100's gauge R_n (68.9) (defined on common support and zero otherwise; the superscript keeps R free for the relaxed triangle (93.6)) and $T_n = H_n R_n^g$, which is Report 100's T_n . Proposition 94.1 is kept, because the rest of Report 103 cites it display by display; its norms are Euclidean ($\|\cdot\|_2$ throughout Report 103).

Proposition 94.1 (Imported relaxed spectral theorem). *For constants independent of sufficiently large n , and for each fixed $p \geq 0$,*

$$\lambda_n = 4 + 4an^{-2/3} + 6n^{-1} + O(n^{-4/3}), \quad (94.1)$$

$$\|(I - \phi_n \phi_n^\top) H_n\|_2 \leq 1 - cn^{-2/3}, \quad (94.2)$$

$$\|(k+1)^p \phi_n\|_2 \leq C_p n^{p/3}, \quad \|\phi_n - \phi_{n-1}\|_2 \leq C/n, \quad (94.3)$$

$$\|Z\phi_n - \phi_n\|_2 \leq Cn^{-1/3}, \quad \phi_n(0) = \sqrt{2}n^{-1/2}[1 + O(n^{-2/3})], \quad (94.4)$$

where $(Zf)(k) = f(k-1)$ and $f(-1) = 0$. The gauge satisfies

$$0 \leq R_n^g \leq I, \quad 1 - R_n^g(k) \leq Ck^2/n^2, \quad (94.5)$$

$$\|(I - R_n^g)\phi_n\|_2 + \|(I - R_n^g)\phi_{n-1}\|_2 \leq Cn^{-4/3}. \quad (94.6)$$

With $\varepsilon = n^{-1/3}$ and $X_k = (k+1/2)\varepsilon$, the normalized samples

$$q_n(k) = \sqrt{2\varepsilon} \frac{\text{Ai}(a + 2X_k)}{\text{Ai}'(a)}$$

after a scalar normalization $1 + O(\varepsilon^2)$ satisfy $\|\phi_n - q_n\|_2 = O(\varepsilon^2)$. The relaxed counting solution $d_n(k) = R(n+k, n-k)/(n+k)!$ obeys

$$v_n^r = \Lambda_n^{-1} S_n^{-1} d_n, \quad \|v_n^r\|_2 \leq 1. \quad (94.7)$$

Finally, for $0 \leq l \leq \lfloor n^{2/3} \rfloor$,

$$\|\delta_0^\top T_n \cdots T_{n-l+1}\|_2 \leq C(l+1)^{-3/4}. \quad (94.8)$$

The proof is in Report 100's Sections 2–5 (Sections 68–71 here); all normalizations here coincide with that proof. In particular, Ai' denotes the derivative with respect to the Airy argument. The half-grid shift corresponds to the Dirichlet point $k = -1/2$, which is forced by the exceptional first row of A_n .

The smoothing bound (94.8) is used later for signed forcing. Once a row norm bound is known, it bounds the row applied to *any signed vector* by Cauchy–Schwarz.

[write, 5 October 2026, batch 102] Report 103 adds an orientation paragraph and a subsection “The smoothing mechanism” that recall Report 100’s proofs: the Dirichlet operator $-\partial_X^2 + 8X$ with eigenvalues $-4a_j$ and unit ground state $\sqrt{2} \text{Ai}(a + 2X)/\text{Ai}'(a)$, normalized by $\int_a^\infty \text{Ai}^2 = \text{Ai}'(a)^2$ ((69.6)); the quasimode, gap, localization and drift (Sections 69–70); and the killed lazy-walk kernel $P = J_\infty/4$ with $P^m(0, 0) \leq C(m+1)^{-3/2}$ and the entrywise domination $0 \leq T_n \cdots T_{n-l+1} \leq (\prod 4/\lambda_j) P^l$ (Section 71). They repeat Report 100 and are not reprinted; the Cauchy–Schwarz remark above is Report 103’s own and is kept. Each line of Proposition 94.1 is a line of Report 100 or of Report 101’s Lemma 82.1 (the shift bound), the sampled profile bound is (69.14), and (94.7) is the contraction stated after (68.12).

95 The exact signed three lag equation

Put

$$u_n(k) = e(2n, 2k) \quad (0 \leq k \leq n), \quad o_n(k) = e(2n-1, 2k+1) \quad (0 \leq k \leq n-1).$$

The even and odd forms of (93.4) are

$$\begin{aligned} u_n(0) &= o_n(0), \\ u_n(k) &= \alpha_n(k) o_n(k-1) + o_n(k) - g_n(k) o_{n-1}(k-1), \\ o_n(k) &= p_n(k) u_{n-1}(k) + u_{n-1}(k+1) - q_n(k) u_{n-2}(k), \end{aligned} \quad k \geq 1, \quad (95.1)$$

where

$$\begin{aligned} \alpha_n(k) &= \frac{n-k+1}{n+k}, & g_n(k) &= \frac{n-k}{2(n+k)(n+k-1)}, \\ p_n(k) &= \frac{n-k}{n+k}, & q_n(k) &= \frac{n-k-1}{2(n+k)(n+k-1)}. \end{aligned}$$

Eliminating odd arrays gives, for $n \geq 4$,

$$u_n = A_n u_{n-1} - \mathcal{B}_n u_{n-2} + \mathcal{F}_n u_{n-3}. \quad (95.2)$$

The two memory matrices are

$$\begin{aligned} (\mathcal{B}_n w)(0) &= \frac{w(0)}{2n}, & (\mathcal{F}_n w)(0) &= 0, \\ (\mathcal{B}_n w)(k) &= P_n(k) w(k-1) + Q_n(k) w(k), & (\mathcal{F}_n w)(k) &= W_n(k) w(k-1), \end{aligned} \quad (95.3)$$

with, for $k \geq 1$,

$$\begin{aligned} P_n(k) &= \frac{(n-k)(2n-2k+1)}{2(n+k)(n+k-1)(n+k-2)}, \\ Q_n(k) &= \frac{2n-2k-1}{2(n+k)(n+k-1)}, \\ W_n(k) &= \frac{(n-k)(n-k-1)}{4(n+k)(n+k-1)(n+k-2)(n+k-3)}. \end{aligned} \quad (95.4)$$

The input support of $\mathcal{B}_n$ is $0, \dots, n-2$; that of $\mathcal{F}_n$ is $0, \dots, n-3$. Outside these supports there is no entry. In particular, a formula that could look negative beyond its support is not a matrix coefficient. Every actual entry in both matrices is nonnegative, but the sign of $\mathcal{B}_n$ in (95.2) is negative.

For completeness, the five elimination identities at $k \geq 1$ are

$$\begin{aligned} L_n(k) &= \alpha_n(k)p_n(k-1), & D_n(k) &= \alpha_n(k) + p_n(k), \\ P_n(k) &= \alpha_n(k)q_n(k-1) + g_n(k)p_{n-1}(k-1), \\ Q_n(k) &= q_n(k) + g_n(k), & W_n(k) &= g_n(k)q_{n-1}(k-1). \end{aligned}$$

At the origin, $p_n(0) = 1$ and $q_n(0) = 1/(2n)$, so

$$u_n(0) = u_{n-1}(0) + u_{n-1}(1) - \frac{1}{2n}u_{n-2}(0). \quad (95.5)$$

These identities prove (95.2) at every row, including the boundary; finite exact checks in the source package supply additional transcription tests.

95.1 Global conjugated memory bounds

Set

$$\tilde{\mathcal{B}}_n = S_n^{-1}\mathcal{B}_nS_{n-2}, \quad \tilde{\mathcal{F}}_n = S_n^{-1}\mathcal{F}_nS_{n-3}.$$

Lemma 95.1. *Uniformly on the full actual supports,*

$$\|\tilde{\mathcal{B}}_n\|_2 \leq C/n, \quad \|\tilde{\mathcal{F}}_n\|_2 \leq C/n^2, \quad \|\tilde{\mathcal{B}}_n\phi_{n-2} - \frac{2}{n}\phi_n\|_2 \leq Cn^{-4/3}. \quad (95.6)$$

Proof. For a diagonal memory entry, the gauge ratio is $R_n^g(k)R_{n-1}^g(k) \leq 1$. A lower-diagonal entry has this two-factor product at $k-1$, divided by $j_n(k)$; for $\tilde{\mathcal{F}}_n$ there is one further factor $R_{n-2}^g(k-1)$. Direct cancellation yields

$$\begin{aligned} \frac{P_n(k)}{j_n(k)} &= \frac{n-k}{n-k+1} \frac{2n-2k+1}{2(n+k-2)\sqrt{(n+k)(n+k-1)}} \leq \frac{C}{n}, \\ \frac{W_n(k)}{j_n(k)} &= \frac{(n-k)(n-k-1)}{n-k+1} \frac{1}{4(n+k-2)(n+k-3)\sqrt{(n+k)(n+k-1)}} \leq \frac{C}{n^2}. \end{aligned}$$

Together with Q_n and the exact row zero, these estimates prove the operator bounds because only finitely many bands occur.

On the actual input support of $\tilde{\mathcal{B}}_n$, write

$$\tilde{\mathcal{B}}_n = \frac{1}{n}(I + Z) - \frac{1}{2n}\Pi_0 + V_n, \quad (95.7)$$

where Π_0 is projection onto coordinate zero. Each band of V_n has row- k entry bounded by $C(k+1)/n^2$. For $k \leq n/2$, expand the displayed rational and square-root coefficients. The gauge error is $O(k^2/n^3)$, which is at most $C(k+1)/n^2$. At zero, the correction $-\Pi_0/(2n)$ gives exactly the required $1/(2n)$. For $k > n/2$, the global $O(1/n)$ bounds already imply the stated estimate, including support-imposed zeros.

The moment estimate gives $\|V_n\phi_{n-2}\|_2 \leq Cn^{-5/3}$. Applying (95.7), the shift estimate, $|\phi_{n-2}(0)| \leq \|Z\phi_{n-2} - \phi_{n-2}\|_2$, and the two-step drift gives the last assertion of (95.6). $\square$

96 Convergence of the leading amplitude

Choose the scalar normalization

$$\begin{aligned} \kappa_n &= \lambda_n - \frac{1}{2n}, & \Gamma_n &= \prod_{j=1}^n \kappa_j, & v_n &= \Gamma_n^{-1} S_n^{-1} u_n, \\ \rho_n &= \lambda_n / \kappa_n, & M_n &= \frac{\tilde{\mathcal{B}}_n}{\kappa_n \kappa_{n-1}}, & N_n &= \frac{\tilde{\mathcal{F}}_n}{\kappa_n \kappa_{n-1} \kappa_{n-2}}. \end{aligned} \quad (96.1)$$

Every κ_n is positive: $\lambda_1 = 3/2$ gives $\kappa_1 = 1$, and the zero-coordinate Rayleigh quotient gives $\lambda_n \geq 1 > 1/(2n)$ for $n \geq 2$. Equations (94.1) and (95.6) imply

$$\kappa_n = 4 + 4an^{-2/3} + \frac{11}{2n} + O(n^{-4/3}), \quad (96.2)$$

$$v_n = \rho_n T_n v_{n-1} - M_n v_{n-2} + N_n v_{n-3}, \quad (96.3)$$

$$\rho_n = 1 + \frac{1}{8n} + O(n^{-5/3}),$$

$$\begin{aligned} \|M_n\|_2 &\leq C/n, & \|N_n\|_2 &\leq C/n^2, \\ \left\| M_n \phi_{n-2} - \frac{1}{8n} \phi_n \right\|_2 &\leq Cn^{-4/3}. \end{aligned} \quad (96.4)$$

The scalar $1/(8n)$ in the last line is exactly the drift to be canceled.

96.1 The initial bound comes from counting

The diagonal gauges and products are positive. Equations (93.7) and (94.7) therefore imply, coordinatewise,

$$0 \leq v_n \leq \frac{\Lambda_n}{\Gamma_n} v_n^r.$$

Moreover,

$$\log \frac{\Lambda_n}{\Gamma_n} = \sum_{j=1}^n -\log \left(1 - \frac{1}{2j\lambda_j} \right) = \frac{1}{8} \log n + O(1).$$

Consequently

$$\|v_n\|_2 \leq Cn^{1/8}. \quad (96.5)$$

This weak polynomial bound is enough to start the argument. It is not obtained by treating the signed evolution as a contraction.

96.2 A damped scalar estimate

We use the following elementary fact. If $x_n \geq 0$ obeys

$$x_n \leq (1 - cn^{-2/3})x_{n-1} + Cn^{-r}$$

for all large n , then $x_n = O(n^{2/3-r})$; exponentially decaying contributions from a fixed initial segment are harmless. Indeed iteration bounds the propagation from j to n by

$$\exp \left(-c \sum_{i=j+1}^n i^{-2/3} \right).$$

For $j \geq n/2$, summing this bound costs $O(n^{2/3})$ and $j^{-r} = O(n^{-r})$. For $j < n/2$, the exponential factor is $O(e^{-c'n^{1/3}})$ and absorbs every fixed polynomial. The same argument applies to a finite sum of polynomial forcings and will be used again for signed vectors after taking norms.

96.3 Perpendicular tracking and scalar cancellation

Write

$$A_n^* = \langle \phi_n, v_n \rangle \geq 0, \quad w_n = v_n - A_n^* \phi_n, \quad \beta_n = \|w_n\|_2.$$

The superscript distinguishes the scalar projection from the matrix A_n . Projection of (96.3) onto the perpendicular complement of ϕ_n , using the gap and absorbing $\rho_n - 1 = O(n^{-1})$ into it, gives

$$\begin{aligned} \beta_n &\leq (1 - cn^{-2/3})\beta_{n-1} + \frac{C}{n}|A_{n-1}^*| \\ &\quad + \frac{C}{n}\|v_{n-2}\|_2 + \frac{C}{n^2}\|v_{n-3}\|_2. \end{aligned} \quad (96.6)$$

The forcing from (96.5) is $O(n^{-7/8})$, so the preceding scalar estimate yields

$$\beta_n = O(n^{-5/24}). \quad (96.7)$$

The crude scalar projection gives $\Delta A_n^* = O(n^{-7/8})$, where $\Delta A_n^* = A_n^* - A_{n-1}^*$.

To obtain a summable increment, use

$$\begin{aligned} \langle \phi_n, R_n^g \phi_{n-1} \rangle &= 1 + O(n^{-4/3}), \\ \|R_n^g \phi_n - \phi_{n-1}\|_2 &\leq C/n. \end{aligned} \quad (96.8)$$

The first equality follows from the gauge estimate and $\langle \phi_n, \phi_{n-1} \rangle - 1 = -\|\phi_n - \phi_{n-1}\|_2^2/2$. Combining (96.8) with (96.4) gives the exact scalar equation

$$\Delta A_n^* = \frac{1}{8n} \Delta A_{n-1}^* + \eta_n, \quad (96.9)$$

where

$$\begin{aligned} |\eta_n| &\leq Cn^{-4/3}(|A_{n-1}^*| + |A_{n-2}^*|) + Cn^{-2}|A_{n-3}^*| \\ &\quad + \frac{C}{n}(\beta_{n-1} + \beta_{n-2}) + \frac{C}{n^2}\beta_{n-3}. \end{aligned} \quad (96.10)$$

All perpendicular contributions may have either sign. In particular the leading positive scalar term $A_{n-1}^*/(8n)$ is canceled by the negative memory projection $-A_{n-2}^*/(8n)$; their difference produces the delayed increment in (96.9).

Substituting (96.5) and (96.7) into (96.10) gives $\eta_n = O(n^{-29/24})$. The crude increment makes the delayed term $O(n^{-15/8})$. Hence $\sum_n |\Delta A_n^*| < \infty$ and $A_n^* \rightarrow A_\infty \geq 0$. This first proves boundedness of v_n . Repeating (96.6) now gives $\beta_n = O(n^{-1/3})$; the crude scalar increment becomes $O(1/n)$, then (96.9) improves it to $O(n^{-4/3})$. Summing its tail proves

$$|A_n^* - A_\infty| + \|v_n - A_\infty \phi_n\|_2 \leq Cn^{-1/3}. \quad (96.11)$$

This order of estimates avoids any circular assumption of boundedness.

96.4 Boundary smoothing and positivity

Set $z_n = v_n - A_\infty \phi_n$. Rewrite the exact equation as

$$z_n = T_n z_{n-1} + h_n,$$

where

$$\begin{aligned} h_n &= (\rho_n - 1)T_n v_{n-1} - M_n v_{n-2} + N_n v_{n-3} \\ &\quad + A_\infty (T_n \phi_{n-1} - \phi_n). \end{aligned}$$

Boundedness of v_n , the memory bounds, and the drift imply $\|h_n\|_2 \leq C/n$. Iterate over $L = \lfloor n^{2/3} \rfloor$ steps and use (94.8):

$$|z_n(0)| \leq C(L+1)^{-3/4} n^{-1/3} + \frac{C}{n} \sum_{l=0}^{L-1} (l+1)^{-3/4} = O(n^{-5/6}).$$

It follows that

$$v_n(0) = \sqrt{2} A_\infty n^{-1/2} + O(n^{-5/6}). \quad (96.12)$$

Summation of $\log \kappa_n$ in (96.2) gives a finite positive K_D with

$$\Gamma_n = K_D 4^n e^{3an^{1/3}} n^{11/8} [1 + O(n^{-1/3})]. \quad (96.13)$$

If $A_\infty = 0$, equations (93.5), (96.12), and (96.13) would give

$$b_n = O\left(n! 8^n e^{3an^{1/3}} n^{13/24}\right).$$

But (92.4) has polynomial power $3/4 = 18/24 > 13/24$, a contradiction. Therefore $A_\infty > 0$ and

$$\gamma_D = \sqrt{2} K_D A_\infty \in (0, \infty). \quad (96.14)$$

Multiplication of (96.12) by $2^n n! \Gamma_n$ proves Theorem 92.1 for $m = 0$.

[write, 5 October 2026, batch 102] The contradiction uses only (92.4). With the compacted amplitude now proved (Report 101 and Part II), the second half of (92.4) is also a consequence of $\gamma_c > 0$; the analytic input is then the published compacted lower bound or, equivalently for this purpose, either compacted amplitude theorem, each of which itself rests on that bound. An intrinsic argument in the manner of Lemma 72.1 is Question V.5.

97 A convergent characterization of the amplitude

The proof provides exact quantities rather than an unspecified subsequential constant. With κ_n defined from finite Jacobi matrices,

$$K_D = \frac{\kappa_1}{4e^{3a}} \prod_{j=2}^{\infty} \frac{\kappa_j}{4 \exp(3a[j^{1/3} - (j-1)^{1/3}]) (j/(j-1))^{11/8}}. \quad (97.1)$$

The logarithm of the j th factor is $O(j^{-4/3})$, so the product converges to a positive finite value. The leading discrepancy is summable because the denominator matches both the $j^{-2/3}$ and j^{-1} terms of $\log \kappa_j$.

For the actual counting solution initialized by Proposition 5,

$$A_\infty = A_3^* + \sum_{n=4}^{\infty} (A_n^* - A_{n-1}^*), \quad A_n^* = \langle \phi_n, \Gamma_n^{-1} S_n^{-1} u_n \rangle. \quad (97.2)$$

This series is absolutely convergent by (96.9)–(96.11). Together (96.14)–(97.2) define γ_D canonically using exact counts and finite-dimensional spectral data.

These formulas establish existence and uniqueness; they are not a closed form. The O -constants in the current proof are not effective numerical enclosures. Truncating these expressions with floating-point spectral computations does not, by itself, certify any digits of γ_D .

98 An exact formal construction at every depth

We now construct an asymptotic solution to *all finite orders*, then prove that it describes the actual counts. Use the variables

$$t = N^{-1/3}, \quad x = (h+1)t, \quad \varkappa = 2^{2/3}a, \quad A(x) = \text{Ai}(a + 2^{1/3}x).$$

Thus $A'' = (2x + \varkappa)A$. Derivatives in this section are with respect to x unless explicitly stated otherwise. Put $t_j = t(1 - jt^3)^{-1/3}$. The two rational coefficients of (93.4) become

$$W(t, x) = \frac{1 - xt^2 + 3t^3}{1 + xt^2 - t^3}, \quad D(t, x) = \frac{t^3(1 - xt^2 + t^3)}{(1 + xt^2 - t^3)(1 + xt^2 - 3t^3)}. \quad (98.1)$$

In particular, the numerator in the second factor is $+t^3$ and the leading scalar is t^3 . Both follow directly by substituting $h = x/t - 1$, $N = t^{-3}$ into the DFA memory coefficient.

Seek compatible formal series

$$s(t) = 2 + \varkappa t^2 + \sum_{j \geq 3} \sigma_j t^j, \quad F(t, x) = \sum_{j \geq 0} t^j f_j(x), \quad f_j = P_j A + Q_j A', \quad (98.2)$$

$$\begin{aligned} P_0 &= 1, \quad Q_0 = 0, \quad P_j, Q_j \in \mathbb{Q}[\varkappa, x], \\ Q_j(0) &= 0, \quad P_j(0) + Q'_j(0) = 0 \quad (j \geq 1). \end{aligned} \quad (98.3)$$

The formal equation is

$$\begin{aligned} s(t)F(t, x) &= W(t, x)F\left(t_1, (x-t)(1-t^3)^{-1/3}\right) \\ &\quad + F\left(t_1, (x+t)(1-t^3)^{-1/3}\right) \\ &\quad - \frac{D(t, x)}{s(t_1)s(t_2)}F\left(t_3, (x-t)(1-3t^3)^{-1/3}\right). \end{aligned} \quad (98.4)$$

If $H_N/H_{N-1} = s(N^{-1/3})$, the denominator in the last line is simply H_{N-1}/H_{N-3} . No recurrence for the actual counting amplitude is being assumed.

Proposition 98.1 (Triangular polynomial algorithm). *Equations (98.2)–(98.4) have a unique solution with the stated gauges at every formal degree. Every coefficient belongs to $\mathbb{Q}[\varkappa, x]$ or $\mathbb{Q}[\varkappa]$ as appropriate. Every profile vanishes exactly at $x = 0$.*

Proof. The degree-two equation is $A'' - (2x + \varkappa)A = 0$. At degree t^k , $k \geq 3$, the new profile $f_{k-2} = PA + QA'$ and scalar σ_k occur as

$$(\partial_x^2 - 2x - \varkappa)(PA + QA') - \sigma_k A + R_0 A + R_1 A' = 0, \quad (98.5)$$

where R_0, R_1 are already determined polynomials. Memory starts at degree three, so the new profile cannot enter it before degree $k + 1$; the new scalar enters its denominator still later. Thus memory does not change the new-unknown operator.

Write $v = 2x + \varkappa$. Direct differentiation gives

$$(\partial_x^2 - v)(PA + QA') = (P'' + 2vQ' + 2Q)A + (2P' + Q'')A'.$$

Eliminating P from (98.5) gives

$$\mathcal{T}Q = 2R_0 - R_1' - 2\sigma_k, \quad \mathcal{T}Q = Q''' - 4(2x + \varkappa)Q' - 4Q. \quad (98.6)$$

On polynomials of degree at most d , this operator is triangular with nonzero diagonal entries $-4(2j + 1)$, $0 \leq j \leq d$. It is therefore invertible over $\mathbb{Q}[\varkappa]$. Define

$$\begin{aligned} \tilde{Q} &= \mathcal{T}^{-1}(2R_0 - R_1'), & \sigma_k &= -2\tilde{Q}(0), \\ Q &= \tilde{Q} + \sigma_k/2, & P(x) &= \int_0^x \frac{-R_1(u) - Q''(u)}{2} du - Q'(0). \end{aligned} \quad (98.7)$$

These equations satisfy both coefficients in (98.5) and both gauges. They uniquely determine the new data. Induction proves the construction at arbitrary depth. Since $A(0) = 0$ and $Q_j(0) = 0$, every $f_j(0) = 0$ exactly. In fact the second gauge also gives $f_j'(0) = 0$ for $j \geq 1$. $\square$

At the first new degree,

$$f_1 = -\frac{2}{3}x^2 A, \quad \sigma_3 = \frac{8}{3} - \frac{1}{4} = \frac{29}{12}. \quad (98.8)$$

The term $-1/4$ is precisely the leading memory term $-D/[s(t_1)s(t_2)]$. The next scalar coefficients, useful for reproduction, are

$$\begin{aligned} \sigma_4 &= \frac{29\varkappa^2}{540}, & \sigma_5 &= -\frac{103\varkappa}{648}, \\ \sigma_6 &= \frac{5341815 - 70264\varkappa^3}{2721600}, & \sigma_7 &= -\frac{767\varkappa^2}{23328}. \end{aligned} \quad (98.9)$$

For example,

$$P_2 = \frac{16\varkappa + 240x^4 + 1641x}{1080}, \quad Q_2 = \frac{x(-4\varkappa + 51x)}{270}.$$

The source package contains all profiles through f_5 and checks the complete two-component formal residual through t^7 . Proposition 98.1, rather than those finite checks, is the proof that the algorithm continues indefinitely.

[**write, 5 October 2026, batch 102**] With $\varkappa = 2^{2/3}a = 2 \cdot 2^{-1/3}a$, the scalars here are twice Part IV's ratio coefficients: $\sigma_m = 2s_m$ in (59.4) for $m = 3, \dots, 6$ (checked at the write; Remark 66.2 proves it for every m). The coefficient σ_7 is new to this report; in Part IV's normalization, where the letter a means $2^{-1/3}$ times the Airy zero, it reads $s_7 = -767a^2/11664$. The formal coefficient D of (98.1) is Part IV's β (59.3).

99 Compatible carriers and genuine residuals

99.1 One carrier normalization at every depth

There is a unique sequence $\ell_j \in \mathbb{Q}[\varkappa]$ satisfying the formal identity

$$\log \frac{s(t)}{2} = \frac{3\varkappa}{2t} \left[1 - (1 - t^3)^{1/3} \right] - \frac{29}{24} \log(1 - t^3) + \sum_{j \geq 1} \ell_j t^j \left[1 - (1 - t^3)^{-j/3} \right]. \quad (99.1)$$

The degree-two and degree-three coefficients agree by (98.8). At degree $j + 3$, the new ℓ_j has nonzero coefficient $-j/3$, proving existence and uniqueness successively.

For fixed $M \geq 2$, take the explicit carrier with multiplicative constant one,

$$H_N^{[M]} = 2^N \exp\left(\frac{3}{2}\varkappa N^{1/3}\right) N^{29/24} \exp\left(\sum_{j=1}^M \ell_j N^{-j/3}\right), \quad (99.2)$$

$$F^{[M]}(t, x) = \sum_{j=0}^M t^j f_j(x).$$

Its exact consecutive ratio agrees with $s(t)$ through degree $M + 3$. If computing ℓ_M requires a profile beyond f_M , Proposition 98.1 supplies it. We do not use independent products of truncated ratios with arbitrary starting values: that would introduce depth-dependent amplitude constants.

In even time set $\varepsilon = n^{-1/3}$ and $X = (k + 1/2)\varepsilon$. Then $t = 2^{-1/3}\varepsilon$ and $x = 2^{2/3}X$. Define

$$U_n^{[M]}(k) = \frac{2^{-29/24}}{\text{Ai}'(a)} H_{2n}^{[M]} F^{[M]}(t, x). \quad (99.3)$$

Here Ai' is the Airy-argument derivative, not A' . The leading expression is

$$\frac{4^n e^{3an^{1/3}} n^{29/24}}{\text{Ai}'(a)} \text{Ai}(a + 2X).$$

At $k = 0$, $\text{Ai}(a + \varepsilon) / \text{Ai}'(a) = \varepsilon + O(\varepsilon^3)$. If

$$B_n^{\text{end}} = 4^n e^{3an^{1/3}} n^{7/8}, \quad D_n^{\text{bulk}} = 4^n e^{3an^{1/3}} n^{29/24}, \quad (99.4)$$

then $U_n^{[M]}(0) = B_n^{\text{end}}[1 + O(\varepsilon)]$, since $29/24 - 1/3 = 7/8$. At every finite order its endpoint Taylor series has compatible coefficients. Indeed a newly retained $t^j f_j(x)$ vanishes at $x = 0$, so its relative endpoint contribution begins no earlier than ε^j ; the new carrier term also begins at that order. The leading coefficient is one at every depth.

99.2 Uniform finite remainders for the one step equation

For every fixed M and C_0 , the exact residual obtained by substituting

$$H_N^{[M]} F^{[M]}(t, (h + 1)t)$$

into (93.4), after division by $H_N^{[M]}$, is bounded on $0 \leq h \leq C_0\sqrt{N}$ by

$$Ct^{M+3}(1+x)^d e^{-cx^{3/2}}. \quad (99.5)$$

Here d, C, c may depend on the fixed depth. This is a finite-remainder estimate, not just a formal coefficient cancellation, as we now verify.

On the region in question, $x \leq Ct^{-1/2}$ and $|xt^2| \leq Ct^{3/2}$. The denominators in (98.1) stay separated from zero. Finite rational expansions thus have remainder bounded by the prescribed power of t times a polynomial in x . The delayed carrier ratios $H_{N-j}^{[M]}/H_N^{[M]}$, $j \leq 3$, are bounded and have ordinary finite-order expansions with bounded remainders: the apparent t^{-1} in their exponential multiplies a difference vanishing at order t^3 . The shifted profile arguments differ from x by $O(t)$ uniformly, since their difference is $\pm t + O(xt^3 + t^4)$.

Every derivative of a fixed profile is a polynomial–Airy pair. On the relevant Taylor intervals it obeys $C(1+x)^d e^{-cx^{3/2}}$; for bounded x this follows from smoothness, and for large x from Airy decay. Products of the finitely many expansions preserve such an envelope. Equation (98.4) cancels all powers through t^{M+2} ; memory begins at order three and creates no further loss. This proves (99.5).

At $h = 0$, both lower evaluations have argument exactly zero, not an approximate zero. Every retained f_j vanishes there before Taylor expansion. Thus this residual estimate includes the actual equation $e(N, 0) = e(N - 1, 1)$, not merely a limiting continuum boundary condition.

99.3 Exact elimination with residuals

Use the same scalar factor $2^{-29/24}/\text{Ai}'(a)$ for the odd approximation formed from $H_{2n-1}^{[M]}F^{[M]}$. Let $r_{\text{ev},n}$ and $r_{\text{od},n}$ be its residuals in (95.1). Let K_n have diagonal one and subdiagonal $\alpha_n(k)$, and let G_n have subdiagonal $g_n(k)$ and row zero zero. Direct elimination, retaining the residuals, gives

$$U_n - A_n U_{n-1} + \mathcal{B}_n U_{n-2} - \mathcal{F}_n U_{n-3} = r_{\text{ev},n} + K_n r_{\text{od},n} - G_n r_{\text{od},n-1}. \quad (99.6)$$

On $k = O(\sqrt{n})$, K_n is bounded and $G_n = O(1/n)$ in its coefficients; the needed adjacent carrier ratios are bounded. Hence (99.5) gives the same ε^{M+3} pointwise envelope after elimination. Row zero is included because K_n retains the origin and G_n vanishes there.

99.4 A cutoff at every memory lag

Choose a fixed smooth χ equal to one on $[0, 1/2]$ and zero on $[1, \infty)$, and put

$$\tilde{U}_n^{[M]}(k) = \chi(k/\sqrt{n})U_n^{[M]}(k). \quad (99.7)$$

Whenever cutoffs at times $n, n-1, n-2, n-3$ or adjacent heights differ, $k \asymp \sqrt{n}$. Then $X \asymp n^{1/6}$, and each fixed profile times a bounded carrier ratio is a polynomial in n times $e^{-cn^{1/4}}$. All cutoff defects are therefore smaller than every prescribed algebraic power. There is no need to cancel different lagged cutoff terms against one another.

The residual is supported at $k \leq \sqrt{n} + O(1)$. Since the X -mesh is ε , the sum of the squared envelope in (99.5) is $O(\varepsilon^{-1})$. The unnormalized residual thus has ℓ^2 norm

$$O(D_n^{\text{bulk}} \varepsilon^{M+5/2}).$$

On its support the exact factorial gauge satisfies

$$-\log S_n(k) = O(k^2/n + k/n) = O(1),$$

so S_n^{-1} is uniformly bounded there. Moreover (96.13) gives

$$D_n^{\text{bulk}}/\Gamma_n = O(n^{-1/6}) = O(\varepsilon^{1/2}).$$

It follows that the genuine normalized defect obeys

$$\|\Gamma_n^{-1} S_n^{-1} [\tilde{U}_n^{[M]} - A_n \tilde{U}_{n-1}^{[M]} + \mathcal{B}_n \tilde{U}_{n-2}^{[M]} - \mathcal{F}_n \tilde{U}_{n-3}^{[M]}]\|_2 = O(n^{-(M+3)/3}). \quad (99.8)$$

In particular, no bound on the inverse gauge has been extended to the distant upper edge where it can be large.

99.5 Matching the same Perron amplitude

Proposition 94.1 and (96.13) imply

$$\left\| \Gamma_n^{-1} S_n^{-1} \tilde{U}_n^{[M]} - \frac{\phi_n}{\sqrt{2}K_D} \right\|_2 = O(\varepsilon). \quad (99.9)$$

To check the constant, the normalized leading expression in (99.3) is $K_D^{-1} \varepsilon^{1/2} \text{Ai}(a + 2X)/\text{Ai}'(a)$, whereas the unit sample has factor $\sqrt{2\varepsilon}$. The finite scalar exponential and higher profiles change this by $O(\varepsilon)$ in the weighted norm. The gauge correction is also $O(\varepsilon)$: on the cutoff support its logarithm is bounded and is at most $C\varepsilon(1 + X^2)$, whose weighted Airy norm is finite. Cutoff tails are exponentially small and the normalized frozen sample error is $O(\varepsilon^2)$. This proves (99.9), with the identical constant at every M .

100 Forced signed memory and the all orders theorem

Lemma 100.1 (Transfer with zero limiting projection). *Suppose finite-support vectors z_n satisfy*

$$z_n = \rho_n T_n z_{n-1} - M_n z_{n-2} + N_n z_{n-3} + f_n,$$

where $\|f_n\|_2 = O(n^{-p})$, $p > 4/3$, and $\|z_n\|_2 = O(n^{-1/3})$. Then

$$\|z_n\|_2 = O(n^{-(p-1)}), \quad z_n(0) = O(n^{-1/2-(p-1)}). \quad (100.1)$$

No sign assumption is imposed on the vectors or forcing.

Proof. Put $\alpha_n = \langle \phi_n, z_n \rangle$ and $\beta_n = \|(I - \phi_n \phi_n^\top) z_n\|_2$. Assume provisionally $\|z_n\|_2 = O(n^{-q})$, starting with $q = 1/3$. The perpendicular gap estimate and the damped comparison give

$$\beta_n = O(n^{-q-1/3} + n^{-p+2/3}). \quad (100.2)$$

The crude scalar equation gives $\Delta \alpha_n = O(n^{-q-1} + n^{-p})$. The refined projection has exactly the same cancellation coefficient as (96.9):

$$\Delta \alpha_n = \frac{1}{8n} \Delta \alpha_{n-1} + \eta_n + \langle \phi_n, f_n \rangle,$$

with

$$|\eta_n| \leq Cn^{-4/3}(|\alpha_{n-1}| + |\alpha_{n-2}|) + Cn^{-2}|\alpha_{n-3}| \\ + \frac{C}{n}(\beta_{n-1} + \beta_{n-2}) + \frac{C}{n^2}\beta_{n-3}.$$

Using (100.2) and the crude increment yields

$$\Delta\alpha_n = O(n^{-q-4/3} + n^{-p}).$$

The initial norm decay implies $\alpha_n \rightarrow 0$. Summing the absolutely convergent tail therefore gives

$$\alpha_n = O(n^{-q-1/3} + n^{-p+1}).$$

Together with (100.2), this replaces q by $\min(q + 1/3, p - 1)$. Finitely many repetitions reach $p - 1$, proving the norm assertion, without a logarithmic loss.

Rewrite the equation as $z_n = T_n z_{n-1} + h_n$. The improved norm estimate and the $O(1/n)$, $O(1/n^2)$ memory bounds give $\|h_n\|_2 = O(n^{-p})$. Over $L = \lfloor n^{2/3} \rfloor$ steps, (94.8) gives

$$|z_n(0)| \leq CL^{-3/4}n^{-(p-1)} + Cn^{-p} \sum_{l=0}^{L-1} (l+1)^{-3/4} \\ = O(n^{-p+1/2}) + O(n^{-p+1/6}).$$

The first is the larger term and equals the endpoint bound in (100.1). □

Completion of Theorem 92.1. Fix $M \geq 2$ and set

$$z_n = v_n - \gamma_D \Gamma_n^{-1} S_n^{-1} \tilde{U}_n^{[M]}.$$

Equations (96.14), (96.11), and (99.9) give the required initial $O(n^{-1/3})$ norm bound. The exact normalized defect (99.8) supplies forcing with $p = (M + 3)/3 > 4/3$. Lemma 100.1, together with $\Gamma_n/B_n^{\text{end}} = \Theta(n^{1/2})$, yields

$$\frac{u_n(0) - \gamma_D \tilde{U}_n^{[M]}(0)}{B_n^{\text{end}}} = O(n^{-M/3}). \quad (100.3)$$

For desired order m , choose $M \geq \max(2, m + 1)$. Expand the compatible endpoint expression of Section 99 through order m ; its omitted terms and (100.3) are $O(n^{-(m+1)/3})$. Since the cutoff equals one at zero, multiplication by $2^n n!$ proves (92.1). Formal logarithm gives (92.2). The amplitude is always the already fixed positive γ_D , not a constant reselected at each depth. Uniqueness of successive asymptotic coefficients follows by subtracting two expansions and taking the first differing order. □

The approximants and transfer errors may be signed. Positivity was needed only for the actual counting solution and for the relaxed comparison propagator; the positive amplitude itself was supplied by the structural compacted-tree lower bound.

101 Explicit coefficients and numerical diagnostics

Formal integration by (99.1) and endpoint evaluation of $F(t, t)$ give the first four logarithmic corrections

$$\begin{aligned} d_1 &= \frac{53a^2}{90}, & d_2 &= \frac{623a}{432}, \\ d_3 &= \frac{4248855 - 166912a^3}{5443200}, & d_4 &= -\frac{5429a^2}{19440}. \end{aligned} \quad (101.1)$$

Exponentiating the finite series yields

$$\begin{aligned} c_1 &= \frac{53a^2}{90}, \\ c_2 &= \frac{a(5618a^3 + 46725)}{32400}, \\ c_3 &= \frac{8337112a^6 + 200508660a^3 + 191198475}{244944000}, \\ c_4 &= \frac{a^2(220933468a^6 + 10228873860a^3 + 53801804475)}{44089920000}. \end{aligned} \quad (101.2)$$

For a direct check of the endpoint conversion, before changing from $t = (2n)^{-1/3}$ to $\varepsilon = n^{-1/3}$ the normalized endpoint profile is

$$\frac{F(t, t)}{2^{1/3} \text{Ai}'(a)t} = 1 + \frac{\varkappa}{6}t^2 + \frac{29}{24}t^3 + \frac{11\varkappa^2}{3240}t^4 + O(t^5).$$

Combining its logarithm with the scalar carrier gives (101.1); the factor $2^{1/3}$ here is the chain-rule derivative of $A(x)$.

Taking sufficiently deep forward expansions before dividing consecutive terms gives the justified ratio expansion

$$\begin{aligned} \frac{b_n}{8nb_{n-1}} &= 1 + an^{-2/3} + \frac{7}{8}n^{-1} + \frac{41a^2}{135}n^{-4/3} + \frac{20a}{81}n^{-5/3} \\ &\quad + \frac{2816a^3 + 108135}{2721600}n^{-2} - \frac{881a^2}{7290}n^{-7/3} + O(n^{-8/3}). \end{aligned} \quad (101.3)$$

Merely shifting a shallow O -remainder does not improve its order. The all-orders theorem allows a deeper truncation, which is why the ratio remainder in (101.3) is legitimate.

[write, 5 October 2026, batch 102] Report 103 swaps the letters of Report 101: here d_j are the *logarithmic* and c_j the *multiplicative* coefficients (Report 101: b_j and d_j ; Part IV: unnamed and $\mathfrak{c}_j$). With $a = z$, d_1, d_2, d_3 are Part IV's (53.4) $(4248855/5443200 = 3497/4480)$ and c_1, c_2, c_3 its $\mathfrak{c}_1, \mathfrak{c}_2, \mathfrak{c}_3$ (62.1). New to this report: $d_4 = -5429a^2/19440$, c_4 , and the ratio (101.3) through $n^{-7/3}$ (Part IV prints no ratio expansion). The conversions $d \mapsto c$ and $d \mapsto$ ratio were recomputed symbolically at the write (difference zero); through $n^{-7/3}$ the ratio involves only a and $d_1, \dots, d_4$, and its $O(n^{-8/3})$ remainder uses Theorem 92.1 at depth $m = 7$.

101.1 What the calculations establish

The exact checker builds Proposition 5 with integer arithmetic, including $b(-1, 0) = 1$. It checks the rotated recurrence and every row of the eliminated memory equation, including the origin and upper supports, using rational arithmetic. Symbolic checks verify the band identities, the completed formal recurrence, the gauges, and endpoint conversions. These are finite corroborations of the algebra, not replacements for the analytic proof above.

The optional numerical producer uses exact integer dynamic programming through $n = 3000$ before evaluating logarithms, factorial logarithms, and Airy constants in floating point. Define

$$G_n = \frac{b_n}{n!8^n e^{3an^{1/3}} n^{7/8}}, \quad G_n^{(4)} = \frac{G_n}{1 + \sum_{j=1}^4 c_j n^{-j/3}}.$$

Representative computed values are

n	G_n	$G_n^{(4)}$
1000	102.077189106	76.4392078929
2000	96.6685122338	76.4384838642
3000	94.0516522622	76.4383234197

These values are compatible with an amplitude near 76.438, but no numerical tail enclosure has been proved. None of these displayed digits is a certified digit of the limit. The theorem and inverse results use the mathematical constant (96.14), never an uncertified decimal substitute.

[**write, 5 October 2026, batch 102**] The value near 76.438 is uncertified evidence for γ_D , which is Part IV's γ ; Part IV prints only uncorrected normalized values, 131.985 at $n = 100$ (in its verification section, just before Section 65), which decrease slowly towards it because $c_1 n^{-1/3}$ is large ($c_1 = 53a^2/90 \approx 3.219$). The intake recomputed $G_{1000} = 102.077189105\dots$ (the table's $\dots 106$ is a last-digit rounding) and $G_{1000}^{(4)} = 76.4392078929$ from exact counts. No enclosure is claimed.

102 Rigorous inverse asymptotics

The forward theorem gives inverse expansions involving Lambert W , logarithms, and fractional powers. We distinguish recovery of the index of an exact sequence value from inversion of an arbitrary threshold. All inverse formulas are for sufficiently large positive inputs.

102.1 An eventual inverse requiring no amplitude value

For $Y > 0$ put

$$X(Y) = \frac{Y}{W_0(8Y/e)}, \quad L(Y) = \log(8X(Y)), \quad (102.1)$$

$$Z(Y) = X(Y) - \frac{3aX(Y)^{1/3} + \frac{11}{8} \log X(Y) + \frac{1}{2} \log(2\pi)}{L(Y)}, \quad (102.2)$$

where W_0 is the principal positive real branch.

Theorem 102.1 (Exact value recovery). *For $Y = \log b_n$,*

$$Z(\log b_n) = n + O(1/\log n). \quad (102.3)$$

Thus nearest-integer rounding recovers n for all sufficiently large exact sequence values, without requiring a numerical value of γ_D .

Proof. Even the two-sided bound (92.3), combined with Stirling's formula, gives

$$\log b_n = G(n) + O(1), \quad G(x) = x(\log(8x) - 1) + 3ax^{1/3} + \frac{11}{8} \log x + \frac{1}{2} \log(2\pi).$$

Put $G_0(x) = x(\log(8x) - 1)$ and $g(x) = G(x) - G_0(x)$. The Lambert identity gives $G_0(X) = Y$ and $G'_0(X) = L$. If $h = -g(X)/L$, then $Z = X + h$ and $h = O(X^{1/3}/\log X)$. Taylor expansion gives

$$G(Z) - Y = g'(X)h + O(h^2/X) = O(X^{-1/3}/\log X).$$

Here $g'(x) = ax^{-2/3} + 11/(8x)$; its further Taylor remainder is smaller than the displayed bound. At $Y = \log b_n$, $X \sim Z \sim n$ and $G'(x) \sim \log x$. The mean value theorem proves (102.3). Since its error tends to zero, eventual nearest-integer recovery follows. $\square$

This theorem has an unspecified eventual cutoff. The proof does not turn an unknown $O(1)$ amplitude term into an effective finite-range rounding certificate.

102.2 Arbitrary thresholds need a comparison

Equation (101.3) gives $b_n/b_{n-1} \sim 8n$, so the sequence is strictly increasing eventually and unbounded. The finitely many earlier values do not affect sufficiently large threshold inverses. Direct differentiation of (102.2) gives $Z'(Y) \sim 1/\log X > 0$.

Corollary 102.2 (One comparison after rounding). *For sufficiently large y , set $j_0 = \text{round}(Z(\log y))$. Then*

$$k(y) = \max\{n : b_n \leq y\} = \begin{cases} j_0, & b_{j_0} \leq y, \\ j_0 - 1, & b_{j_0} > y. \end{cases}$$

The least index $j(y)$ with $b_{j(y)} \geq y$ is

$$j(y) = \begin{cases} j_0, & b_{j_0} \geq y, \\ j_0 + 1, & b_{j_0} < y. \end{cases}$$

The comparison must be exact or rigorously certified; equality belongs to the first branch in each rule.

Proof. If $k = k(y)$, monotonicity puts $Z(\log y)$ between $k + o(1)$ and $k + 1 + o(1)$. For large k , nearest-integer rounding therefore lies in $\{k, k + 1\}$. The comparison selects the correct branch. For the least-index rule the same argument gives $j_0 \in \{j(y) - 1, j(y)\}$; at equality the exact-value theorem applies directly. $\square$

An input arbitrarily close to a sequence value cannot be safely classified by an unverified floating-point asymptotic evaluation.

102.3 An inverse at any prescribed algebraic order

For fixed $J \geq 0$, define the smooth model

$$\mathcal{F}_J(x) = \log \Gamma(x+1) + x \log 8 + 3ax^{1/3} + \frac{7}{8} \log x + \log \gamma_D + \sum_{j=1}^J d_j x^{-j/3}. \quad (102.4)$$

The Γ in this formula is Euler's gamma function, distinct from the product Γ_n in (96.1). For large x , $\mathcal{F}'_J(x) \sim \log x > 0$. Let $x_J(Y)$ be the unique sufficiently large root of $\mathcal{F}_J(x) = Y$.

Theorem 102.3 (All order inverse estimate). *For each fixed $J \geq 0$,*

$$x_J(\log b_n) = n + O\left(\frac{n^{-(J+1)/3}}{\log n}\right). \quad (102.5)$$

Proof. Equation (92.2) says $\log b_n = \mathcal{F}_J(n) + O(n^{-(J+1)/3})$. The derivative is asymptotic to $\log x$, so the mean value theorem gives the claimed inverse error. $\square$

This is an unconditional mathematical theorem with the positive amplitude characterized in Section 97. It is not a certified numerical algorithm if that constant or the function evaluations are replaced by uncontrolled floating-point approximations.

102.4 A finite Newton construction with an explicit iteration count

The implicit root can be replaced, at every prescribed order, by finitely many specified Newton steps. Start from $x_0 = X(Y)$ in (102.1) and iterate

$$x_{k+1} = x_k - \frac{\mathcal{F}_J(x_k) - Y}{\mathcal{F}'_J(x_k)}, \quad K = \lceil \log_2(J+4) \rceil - 1. \quad (102.6)$$

The correct derivative is

$$\mathcal{F}'_J(x) = \psi(x+1) + \log 8 + ax^{-2/3} + \frac{7}{8x} - \sum_{j=1}^J \frac{j}{3} d_j x^{-j/3-1}, \quad (102.7)$$

where ψ is the digamma function. Then

$$x_K(\log b_n) = n + O\left(\frac{n^{-(J+1)/3}}{\log n}\right). \quad (102.8)$$

To prove the finite bound, let x_* be the model root and put $X = x_0$, $L = \log(8X)$. The Airy term gives $x_* - X = O(X^{1/3}/L)$. On a neighborhood $X \pm o(X)$,

$$\mathcal{F}'_J \sim L, \quad \mathcal{F}''_J = O(1/X).$$

Exact Newton errors $e_k = x_k - x_*$ therefore satisfy

$$|e_{k+1}| \leq C|e_k|^2/(XL).$$

For large X , the iterates stay in this neighborhood. Induction yields

$$e_k = O\left(\frac{X^{1-2^{k+1}/3}}{L^{2^{k+1}-1}}\right). \quad (102.9)$$

The first two errors are $O(X^{-1/3}/L^3)$ and $O(X^{-5/3}/L^7)$. Since $2^{K+1} \geq J+4$, the K th error is no greater than the model error in (102.5). This proves (102.8); one Newton step alone is not arbitrarily accurate.

There is also a version using only a finite Stirling polynomial. Replace $\log \Gamma(x+1)$ in (102.4) by

$$(x + \tfrac{1}{2}) \log x - x + \tfrac{1}{2} \log(2\pi) + \sum_{r=1}^R \frac{B_{2r}}{2r(2r-1)x^{2r-1}}, \quad R = \max\left(0, \left\lceil \frac{J-2}{6} \right\rceil\right). \quad (102.10)$$

Its error is $O(x^{-2R-1})$, and $2R+1 \geq (J+1)/3$. Its root therefore differs from x_* by at most the target inverse error. Differentiating this finite expression consistently, the same K Newton steps suffice. For fixed order this uses finitely many arithmetic operations, logarithms, real fractional powers, one Lambert- W evaluation, and the exact constants a, γ_D, d_j and Bernoulli numbers.

102.5 The first explicit fractional power corrections

The first terms can also be written without an implicit root. Set

$$X = \frac{\log b_n}{W_0(8 \log b_n/e)}, \quad L = \log(8X), \quad C_D = \log \gamma_D + \tfrac{1}{2} \log(2\pi).$$

Then

$$\begin{aligned} n = X - \frac{3aX^{1/3} + \frac{11}{8} \log X + C_D}{L} \\ + X^{-1/3} \left[-\frac{d_1}{L} + \frac{3a^2}{L^2} - \frac{9a^2}{2L^3} \right] + O(X^{-2/3}/L). \end{aligned} \quad (102.11)$$

To verify the transcription, write $g_0(X) = 3aX^{1/3} + (11/8) \log X + C_D$ and put $h_0 = -g_0(X)/L$. In the Taylor expansion of $G_0(X+h_0)$, the relevant quadratic term is $h_0^2/(2X)$. The derivative of the Airy term contributes $aX^{-2/3}h_0$, and the first forward correction contributes $d_1X^{-1/3}$. Their coefficient at scale $X^{-1/3}$ is

$$d_1 - \frac{3a^2}{L} + \frac{9a^2}{2L^2}.$$

The second line of (102.11), multiplied by L , cancels it. The remaining residual is $O(X^{-2/3})$, including logarithmic cross terms because $\log X/L$ is bounded. The mean value theorem gives the stated remainder. This establishes the displayed fractional-power and logarithmic structure with a genuine error bound.

[write, 5 October 2026, batch 102] Part IV's explicit inverse (63.2) stops at the remainder $O(X^{-1/3}/\log X)$ (its denominator $1+w$ equals L , since $\log(8X) = 1 + W_0(8Y/e)$ at the anchor). The bracket $X^{-1/3}[-d_1/L + 3a^2/L^2 - 9a^2/(2L^3)]$ of (102.11) is the next term, new to this report. It was rederived at the write from $G_0(X+h) = Y + Lh + h^2/(2X) + O(h^3/X^2)$ with $h = h_0 + h_1$: the terms of $h_0^2/(2X) + aX^{-2/3}h_0 + d_1X^{-1/3}$ at scale $X^{-1/3}$ are $(9a^2/(2L^2) - 3a^2/L + d_1)X^{-1/3}$, and Lh_1 cancels them, as stated.

What is and is not a transseries result. Repeated finite expansion of (102.6) produces compatible algebraic–logarithmic inverse expressions to any prescribed accuracy. This is the rigorous content of the inverse expansion here. It does not establish exponentially small sectors, Stokes constants, Borel summability, or a convergent infinite expansion. Calling the entire forward or inverse object a full transseries would require additional analysis beyond this report.

103 Further questions and boundaries of the proof

[write, 5 October 2026, batch 102] These questions are re-scoped in Section 105, with those of Reports 100 and 101.

Certified numerical amplitudes. The natural next objective is an interval evaluation of the finite Jacobi product and projection, together with explicit bounds for the tails in (97.1) and (97.2). Effective constants for the gap, drift, memory perturbation, and smoothing estimates would turn the observed amplitude near 76.438 into certified digits. They would also yield finite inverse cutoffs.

Uniform depth and coefficient growth. The present argument fixes the truncation depth before taking $n \rightarrow \infty$. Bounds uniform in depth could identify the growth of d_j and an optimal truncation regime. A factorial or Gevrey bound would still need proof; finite lists of coefficients cannot decide it.

Exponentially small Airy modes. The spectral gap suggests a stretched-exponential relaxation mechanism involving other Airy eigenvalues. A genuine beyond-all-orders theory must track excited-mode amplitudes, their coupling under changing matrices and signed memory, and remainders after optimal truncation. An algebraic remainder at every fixed order does not determine those sectors or their amplitudes.

Other alphabet sizes and memory structures. For a larger alphabet, the exact counting recurrence and its combinatorial boundary conditions must first be derived and checked. One may then ask whether a related positive reference operator and a summable scalar correction exist. The coefficient $1/(8n)$ in the present proof is specific to binary minimal automata; simply reusing it would not be justified.

[write, 5 October 2026, batch 102] Largely answered: Part III treats automata over every fixed alphabet of $k \geq 3$ letters, with the exact arity- k recurrence (Part 0, “One family, indexed by the arity”), a positive relaxed reference evolution in a fixed weighted norm, and a delay that is summable for $k \geq 3$, so that no drift cancellation is needed there. As this paragraph cautions, the binary $1/(8n)$ does not carry over. Uniformity in k remains open.

More efficient rigorous inversion. The Newton theorem gives an asymptotic operation count at fixed order, not an optimized finite-input implementation. Combining certified amplitude intervals, rigorous function evaluation, and exact recurrence comparisons could give a practical index-recovery or threshold-search algorithm with a verified stopping test.

Source normalization and independent reproduction. An implementation should test the negative-index seed, $e(2, 0) = 1/2$, and $e(3, 1) = 3/4$ before larger computations. These inexpensive checks prevent a changed startup from silently altering the global amplitude while leaving local Airy coefficients unchanged. The normalization discussion in Section 93 is a local arithmetic reconciliation, not a claim that the original asymptotic theorem is false.

104 Reproducibility and source record

The accompanying package contains the standalone L^AT_EX article, its PDF, a build script, a replay script, explicit exact checker inputs, deliberate-error regression tests, and a separate optional numerical producer. The exact suite is fail-closed: unsuccessful equalities raise explicit exceptions, so checks remain active under optimized Python. Its normal and optimized runs are both tested, and deliberately corrupted cases must fail for the expected reason. Finite checks certify the finite identities that they test, not the analytic limits and remainder estimates.

The imported relaxed proof is included unchanged as a PDF and a source archive. Their SHA-256 digests are:

```
relaxed_tree_amplitude.pdf
4684736e1fd4ccf63962a7c1c21b14
f00e76832ef97d7e808867d97c29d2e3

relaxed_tree_amplitude_sources.zip
72e791418b8b1cf401a2b3b1a0441a24
cb95d2aee1eb2f36fdbcb0f04e05b3f40
```

Each digest is the concatenation of its two displayed lines. The dependency is included for a complete proof chain; it is not re-edited here. In particular, statements about the scope of that earlier report describe its own results at the time of writing. The present positive DFA theorem does not require a subsequent compacted-tree amplitude report.

The primary 2020 DFA paper was checked at pages 11:2, 11:3, 11:6, and 11:7 for the size convention, structural lower comparison, compacted bound, Proposition 5, startup normalization, and trial-ratio coefficient. Its inspected PDF has SHA-256

```
ecdcaa98a839a541f02c9647169371351
60b80156afd9956623448ebff970141
```

The primary-author PDF is linked rather than redistributed. Exact checks and numerical experiments are separate, and no numerical amplitude estimate is used as proof input. No publication, repository modification, or external submission is part of this article.

[**write, 5 October 2026, batch 102**] These paragraphs describe Report 103’s delivered archive (commit 60f54ea06). The two dependency files whose digests are listed above were checked at the intake to be byte copies of Report 100’s PDF and archive; they, the manuscript, its PDF, build script, checksum ledger and manifest are not shipped. The AofA 2020 PDF is not in the repository either, so its digest could not be checked here. Report 103’s author line reads “Research prepared for Vladimir”. The README of this report lists the shipped files.

105 Further questions and research (Part V)

This section was written when Part V was added. It collects every question and every unproved claim of Reports 100, 101 and 103, with its source, its sketch and what is missing, and records which of them other Parts answer. No claim of the three reports was found to be wrong; none is retracted. The questions of the earlier Parts are not repeated; where they coincide, the earlier Part is named.

105.1 Open

- V.1** *Certified digits and effective cutoffs.* (Reports 100, 101, 103, Sections 78, 91, 103; also Parts I, II, IV.) The amplitudes have convergent representations, (73.2)–(73.4), (85.3)–(85.4) and (97.1)–(97.2), whose summands are $O(n^{-4/3})$. Missing: explicit constants for the gap, drift, gauge, memory and smoothing estimates, an interval evaluation of the finite part and a tail enclosure. They would certify digits of $\gamma_r \approx 166.952089$, $\gamma_c \approx 173.126705$ and $\gamma_D \approx 76.438$ (all uncertified) and make the eventual inverse statements effective; at $n = 3000$ the amplitude-free inverse is still off by 0.558 (relaxed) and 0.562 (compacted).
- V.2** *Coefficient growth, Gevrey class, optimal truncation.* (All three reports.) Each expansion is proved at every fixed depth; the envelope arguments are not uniform in the depth. Missing: bounds on the polynomial–Airy recursion uniform in the order. Finite lists of coefficients $(b_1, \dots, b_6, d_1, \dots, d_4, \sigma_3, \dots, \sigma_7)$ cannot decide the growth.
- V.3** *Exponentially small Airy sectors.* (All three reports; Part I’s “Beyond algebraic orders”.) The spectral gap suggests relative corrections of stretched-exponential size $\exp(-cn^{1/3})$, $c > 0$, from the excited Airy modes belonging to $a_2, a_3, \dots$. Missing: evolving excited-mode amplitudes, their coupling under the changing matrices (and, for Reports 101 and 103, under signed memory), and remainders after optimal truncation. No transseries, Stokes constant or Borel summability is claimed.
- V.4** *Report 103’s remark on the printed normalization in the AofA 2020 paper.* Report 103 (Section 93) states that the transformed arrays on pages 11:6–11:7 of [2], normalized by 2^{m-1} for $m \geq 1$ only, give $\frac{5}{4}$ at $(3, 1)$ when their rotated recurrence is read uniformly, against $\frac{3}{2}$ from the piecewise normalization. The uniform-normalization values were rechecked at the write ($e(3, 1) = \frac{3}{4}$; piecewise $\frac{3}{2}$). Missing: a check of the printed pages, which are not in the repository. The remark is about a publication’s display, not about its theorem; Report 103 says so, and neither Report 103 nor Part IV uses that display.
- V.5** *Can Lemma 72.1 replace the published lower bounds in the signed-memory cases?* (New at the write.) Report 100 proves $\gamma_r > 0$ from the exact recurrence alone, by restarting the positive contraction T_n from ϕ_N and comparing $v_N \geq \delta\phi_N$ (Remark 66.1). Reports 101 and 103 and Parts II and IV still take positivity from the compacted or automaton lower bounds of Elvey Price, Fang and Wallner, and Part III from the relaxed lower bound of Ghosh Dastidar and Wallner. The argument does not transfer verbatim: the signed evolutions (84.2) and (96.3) do not preserve the order $v \geq \delta w$, and their tracking estimates start from the counting bound $\|v_n\| \leq Cn^{1/4}$ (resp. $n^{1/8}$), which is proved for the counting solution, not for a restarted one. A possible route: restart in a positive representation instead (Part II’s exact four-state or three-state model, Part IV’s renewal representation, Proposition 54.1), where $X_N \geq \delta Y_N$ propagates; what is missing is a tracking theorem for the restarted positive solution, uniform in the restart time, that shows its scalar projection stays near its initial value. The restart comparison works

wherever the normalized evolution is entrywise nonnegative: bundle Reports 104 and 102 use it for tree-child networks (Parts III and IV of `../a213863-tree-child-networks`). For Part III’s relaxed k -ary evolution, which has nonnegative coefficients, the restart argument may apply directly once its tracking constants are shown to be independent of the restart time.

- V.6** *A positive route to the compacted amplitude.* (Report 100, Section 78, “Compacted trees”; Part II, “Positive effective operators”.) Report 100 suggests a positive multistate encoding of the forbidden local configurations. Such encodings exist (Part II’s four-state, three-state and renewal models) but are not symmetric Jacobi reductions. The amplitude is now proved by signed memory (Report 101, Part II); a proof through a positive encoding, which would also answer V.5 for compacted trees, remains open. Report 100’s caution stands: upper and lower positive comparisons alone can leave different constants.
- V.7** *Efficient rigorous inversion.* (Report 103, Section 103; Parts I, II.) The Newton theorems give an operation count at fixed order, not a finite-input algorithm. Missing: certified amplitude intervals, rigorous function evaluation and a verified stopping test, combined with exact comparisons of neighbouring counts.
- V.8** *Steps given as sketches.* Report 100 sketches the upper min–max bound by sampled Dirichlet trial functions and the lower bound by piecewise-linear interpolation and Rellich compactness (Section 69), and the uniform Taylor envelopes of its residual estimates (Sections 69 and 75); Reports 101 and 103 sketch the same envelopes for the delayed recurrences and import Report 100’s estimates without reproof. The steps are standard and no gap was found at the intake, but they are not written out in full. The same estimates for the one-step matrices are written out in Part I, Sections 2–5, and in Appendix A; a full write-up of the two-step versions, or a derivation of them from the one-step ones, is missing.

105.2 Answered elsewhere in this report

- Report 100’s “Compacted trees” (no compacted amplitude theorem) and its conditional compacted inverse: answered by Report 101 (Theorems 80.1, 80.2, Corollary 90.3) and Part II; its formal compacted expansions are genuine by Corollary 66.3. The positive route stays open (V.6).
- Report 100’s “Higher arity and related recurrences” and Report 103’s “Other alphabet sizes and memory structures”: answered for every fixed $k \geq 3$ by Part III, for relaxed trees, compacted trees and automata; uniformity in k remains open there (Section 37).
- Report 103’s “Source normalization and independent reproduction” is a recommendation, carried out by its own exact checker and at the intake ($b(1, 1) = 1$, $e(2, 0) = \frac{1}{2}$, $e(3, 1) = \frac{3}{4}$ recomputed).
- Report 100’s statement that the inspected repository snapshot `4b874cea0` held no relaxed-amplitude result was true there and is stale now (note in Section 79).

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of Part V, made by the intake after the write (`3e71ff1e6`), reread five items: Report 100’s Lemma 72.1 (restart positivity), the three statements of Section 66.5 with the write’s proofs (Remarks 66.1 and 66.2 and Corollary 66.3), and the conversions between the coefficients printed in Reports 100, 101 and 103 and those of Parts I, II and IV. It found no counterexample and no

mathematical error. Lemma 72.1, both remarks and the conversions are valid. Corollary 66.3 is valid, but its proof left two dependencies unstated: that Report 101's printed $b_1, \dots, b_6$ are the coefficients of its Theorem 80.2 is a finite computation certified by Report 100's exact suite, not a step derived in the text; and $\gamma_c > 0$, needed to take $\log \gamma_c$, comes from the published compacted lower bound (80.4), through Report 101 and Part II. Both are now stated in the proof, in the check's wording, and the opening sentence of Section 66.5, which said that the three proofs use only results printed in this report, is corrected where it stands, with a dated note keeping its first wording. The check's optional precision for Remark 66.2 was adopted: uniqueness of the formal solution is stated within the polynomial–Airy class, outside which a Bi-type solution defeats it, and Part IV takes its uniqueness proof from Part II's paragraph. No other statement or number changed.

The check used its own programs; it did not reuse the write's coefficient script and transcribed every printed value afresh from this report. For Lemma 72.1 it rederived the increment identity (72.3), the barrier for the restarted vector, whose constants depend only on the frozen matrices and not on the restart time N , the summation $|a_\infty(w) - 1| \leq 3CN^{-1/3}$ and the comparison $v_N \geq \delta\phi_N$, and it traced every ingredient (Proposition 69.1, Lemma 68.1 and the positivity of the triangle): none uses $a_\infty > 0$, the bound (67.3) (invoked only in Section 77) or Part I, so the argument is not circular. For Remark 66.1 it checked that Part I derives (F24) before the published lower bound enters, and the normalization $d_{2n,0} = a_n/n!$. For Remark 66.2 it compared the formal equations of the three models symbol by symbol, not only their printed values; it recomputed ℓ_1, ℓ_2, ℓ_3 from the printed $\sigma_2, \dots, \sigma_6$ of each report, which equal the h_1, h_2, h_3 of Parts I, II and IV exactly, and confirmed $\sigma_m = 2s_m$ for $m = 2, \dots, 6$ in all three models and for Report 103's σ_7 against the value $s_7 = -767a^2/11664$ given at the write. For Corollary 66.3 it confirmed that Reports 100 and 101 treat the same triangle with the same normalization, and the logic of the ratio (depth $J = 9$ at n and $n - 1$, with b_7, b_8, b_9 entering only at $n^{-10/3}$). In its own truncated power-series arithmetic, computing the ratio by a route different from the write's (the quotient of the two expansions, not the difference of their logarithms), all 76 conversion identities it tested give difference zero; among them are the relaxed and compacted ratios through n^{-3} , Part I's (F2) and (F3), Report 101's b_j against Report 100's $b_j^{(c)}$ for $j \leq 6$, its d_1, d_2, d_3 , Part II's third logarithmic correction, $\sigma_m = 2s_m$ and $\ell_j = h_j$.

Numerically, it computed both sequences exactly to $n = 3000$; the counts match all 350 terms of the OEIS b-file of A082161 and all 372 of A254789, and the rotated walks of Reports 100 and 101 equal the triangle in exact rationals for $N < 40$. Floating-point runs of the restart (double precision, $n \leq 3000$, restart vectors from ϕ_N at $N = 10, 50, 100, 300, 1000$) keep the restarted projections $a_n(w)$ bounded away from zero, with $1 - a_n(w)$ of the order $N^{-1/3}$ that the lemma predicts, and reproduce the identity $r_n = n!\Lambda_n v_n(0)$ to 2×10^{-12} in the logarithm. The ratio residuals (exact counts minus the printed series through n^{-3}) at $n = 500, 1000, 2000, 3000$ reproduce Report 100's table digit for digit, and their products with $n^{10/3}$ stay bounded (0.126, ..., 0.109 relaxed; 0.233, ..., 0.195 compacted). Fits of the exact counts that take only b_1, b_2, b_3 as input recover the printed b_4 to eight significant digits, b_5 to four or five, and b_6 to three (relaxed) or about two (compacted); the fitted n^{-3} coefficient of the residual is two to four orders of magnitude smaller than an error of 1 % in b_6 would leave. The six-correction quotients at $n = 3000$ and the extrapolated amplitudes 166.95208957422 (relaxed) and 173.1267048501473 (compacted) agree with the printed values; they are uncertified evidence, consistent with the proofs, not enclosures. This was a careful reading with numerical tests, not a formal verification.

Appendix

A Self-contained Jacobi estimates

[write, 2 October 2026] This appendix is source 15's (Part II); source 48 ships the same text with “DFA theorem” in its first sentence. “Section 3” of source 15 is Section 18 here. The appendix serves Parts II and IV, and is a second, self-contained proof of the estimates of Part I, Sections 2–5.

[write, 5 October 2026, batch 102] A third write-up, for the two-step matrices of even time, is Part V's Sections 68–71 (Report 100), which Reports 101 and 103 cite; Part V does not use this appendix.

This appendix proves all shared gauge, gap, profile, projection and spectral product estimates used in the compacted theorem. Its objects are the explicit matrices in Section 18. The relaxed all-orders theorem is not assumed here.

A.1 Exact symmetrization and norm contraction

For the relaxed recurrence define

$$d_{N,j} = \frac{r_{(N+j)/2, (N-j)/2}}{((N+j)/2)!}$$

on the physical support and zero otherwise. Direct substitution gives

$$d_{0,j} = \mathbf{1}_{j=0}, \quad d_{N,-1} = 0, \quad d_{N,j} = \frac{N-j+2}{N+j} d_{N-1,j-1} + d_{N-1,j+1}. \quad (\text{A.1})$$

For $v_N^{(r)}(j) = d_{N,j}/D_N(j)$ the factorial identities

$$\frac{D_N(j)}{D_N(j-1)} = b_{N,j}, \quad \frac{D_{N-1}(j)}{D_N(j)} = Q_N(j)$$

on their required supports prove $v_N^{(r)} = S_N Q_N v_{N-1}^{(r)}$. All edge and diagonal coefficients lie in $[0, 1]$. The finite Jacobi matrix is irreducible, so its largest eigenvalue has a positive unit eigenvector. The matrix exchanges the two parity subspaces, so its spectrum is symmetric about zero and $\|S_N\| = \lambda_N$. Consequently $\|S_N Q_N\| \leq \lambda_N$ and $\|v_N^{(r)}/P_N\| \leq 1$.

A.2 The coarse Airy limit and its gap

Write $\varepsilon = N^{-1/3}$ and $x_j = (j+1)\varepsilon$. For vectors supported on $0, \dots, N+1$, the exact quadratic form is

$$\begin{aligned} \langle v, (2I - S_N)v \rangle &= \sum_{j=0}^N b_{N,j+1} |v_{j+1} - v_j|^2 + |v_0|^2 \\ &\quad + \sum_{j=1}^N (2 - b_{N,j} - b_{N,j+1}) |v_j|^2 + (2 - b_{N,N+1}) |v_{N+1}|^2. \end{aligned} \quad (\text{A.2})$$

The term $|v_0|^2$ is the first Dirichlet-gradient interval, with fictitious value $v_{-1} = 0$. No nonphysical edge coefficient $b_{N,0}$ is needed. For $1 \leq j \leq N+1$,

$$1 - b_{N,j} = \frac{2(j-1)}{(N+j)(1+b_{N,j})} \geq \frac{j-1}{N+j}.$$

Thus for universal $c, C > 0$,

$$\varepsilon^{-2} \langle v, (2I - S_N)v \rangle \geq c \sum_{j=0}^{N+1} x_j |v_j|^2 - C\varepsilon \|v\|^2. \quad (\text{A.3})$$

On every fixed x window the edge weights tend uniformly to one and

$$\varepsilon^{-2}(2 - b_{N,j} - b_{N,j+1}) \longrightarrow 2x_j$$

uniformly there. Interpolate linearly with value $\varepsilon^{-1/2}v_j$ at x_j and zero at the origin; add a zero node after the last finite node and extend by zero. The terminal term of (A.2) controls the final interpolation interval. Bounded scaled energy gives local H^1 bounds. Equation (A.3) gives uniformly small L^2 tails outside large fixed windows. Local compactness followed by this tail estimate gives strong L^2 compactness; the boundary interval enforces zero trace at zero. The limiting lower form is

$$\int_0^\infty (|f'(x)|^2 + 2x|f(x)|^2) dx.$$

Norm convergence follows on a fixed window from the gradient bound and Riemann sums, then globally from (A.3). Sampling smooth, compactly supported functions vanishing at zero supplies the reverse upper form bound. Applying the min-max principle in both directions proves convergence of every fixed low eigenvalue of $\varepsilon^{-2}(2I - S_N)$ to the corresponding eigenvalue of

$$\mathcal{H} = -\frac{d^2}{dx^2} + 2x, \quad f(0) = 0.$$

These eigenvalues are $\mu_k = -2^{2/3}z_k$, where $z_1 = z > z_2 > \dots$ are the negative Airy zeros. Hence

$$\lambda_N = 2 + 2a\varepsilon^2 + o(\varepsilon^2), \quad \lambda_N - \lambda_{2,N} \sim 2^{2/3}(z - z_2)\varepsilon^2, \quad (\text{A.4})$$

where $\lambda_{2,N}$ is the second-largest positive eigenvalue for large N . Between opposite parity subspaces, the singular values of S_N are its positive eigenvalues, possibly with additional zeros. Removing the top singular direction therefore gives norm at most $\lambda_N(1 - cN^{-2/3})$. The negative eigenvalue $-\lambda_N$ is not a second slow direction within occupied parity.

A.3 A finite quantitative quasimode

The Airy equation is $F'' = 2(x+a)F$. Define

$$\begin{aligned} F_2(x) &= \frac{2(-a+2x)}{15}F(x) + \frac{x(2a+x)}{15}F'(x), \\ F_3(x) &= \frac{F(x) - xF'(x)}{3}, \\ f_N(x) &= F(x) + \varepsilon^2 F_2(x) + \varepsilon^3 F_3(x), \\ \hat{\lambda}_N &= 2 + 2a\varepsilon^2 + 3\varepsilon^3 + \frac{13a^2}{15}\varepsilon^4 + \frac{5a}{3}\varepsilon^5. \end{aligned} \quad (\text{A.5})$$

Choose a fixed smooth cutoff χ , equal to one on $[0, 1]$ and zero on $[2, \infty)$, and let $\hat{f}_N$ be the ℓ^2 normalization of $\chi(j/N^{1/2})f_N(x_j)$. Changing or omitting the cutoff in local calculations gives an error smaller than every inverse power of N .

For a direct residual check, the two edge coefficients at a sample point are

$$b_-(x, \varepsilon) = \sqrt{\frac{1 - x\varepsilon^2 + 3\varepsilon^3}{1 + x\varepsilon^2 - \varepsilon^3}}, \quad b_+(x, \varepsilon) = \sqrt{\frac{1 - x\varepsilon^2 + 2\varepsilon^3}{1 + x\varepsilon^2}}.$$

Expand

$$b_-f_N(x - \varepsilon) + b_+f_N(x + \varepsilon) - \hat{\lambda}_N f_N(x)$$

through degree five in ε , reducing derivatives by $F'' = 2(x + a)F$. Every coefficient vanishes. The omitted left sample is zero because $F(0) = F_2(0) = F_3(0) = 0$. On the cutoff support, $x\varepsilon^2 = O(N^{-1/2})$, and Taylor remainders are bounded by $C\varepsilon^6$ times a fixed polynomial times Airy decay. Such envelopes are square-integrable. Cutoff derivatives contribute only $\exp(-cN^{1/4})$ times powers of N . The unnormalized sampled norm is asymptotic to $\varepsilon^{-1/2} \|F\|_{L^2}$, so

$$\|(S_N - \hat{\lambda}_N)\hat{f}_N\| = O(\varepsilon^6). \quad (\text{A.6})$$

The coarse gap identifies the nearby eigenvalue as λ_N ; projecting onto the complementary eigenspaces gives

$$\lambda_N = \hat{\lambda}_N + O(\varepsilon^6), \quad \|\psi_N - \hat{f}_N\| = O(\varepsilon^4). \quad (\text{A.7})$$

The sign follows from the positive leading Airy profile.

The normalized samples have ℓ^2 derivative $O(N^{-1})$ with respect to continuous N , and one-site shift difference $O(N^{-1/3})$. They have bounded polynomial moments on the Airy scale. Together with (A.7), these facts give

$$\|\psi_N - \psi_{N-1}\| = O(N^{-1}), \quad \|(I - Q_N)\psi_N\| = O(N^{-4/3}), \quad \psi_N(0) = \sqrt{2}N^{-1/2}(1 + O(N^{-1/3})). \quad (\text{A.8})$$

For the diagonal estimate use $1 - \sqrt{1-t} \leq t$ on $[0, 1]$, the sampled moments, and (A.7). For the endpoint use

$$I = \int_0^\infty F(x)^2 dx = 2^{-1/3} \text{Ai}'(z)^2, \quad \frac{F'(0)}{\sqrt{I}} = \sqrt{2}. \quad (\text{A.9})$$

The integral identity follows by differentiating $u \text{Ai}(u)^2 - \text{Ai}'(u)^2$ and integrating from z to infinity. The derivative $\text{Ai}'(z)$ is positive for the largest zero. A Riemann-sum estimate with integrable derivative gives relative normalization error $O(\varepsilon)$. The endpoint Taylor error is $O(\varepsilon^2)$, and the normalized quasimode error divided by the leading endpoint is $O(\varepsilon^{5/2})$. This proves the endpoint assertion. The needed Airy decay and polynomially weighted integrability are standard consequences of its decaying ODE solution; see [8].

A.4 Projection and adjoint estimates

Writing $S_N\psi_N = \lambda_N\psi_N$ on its two parity components and taking the corresponding scalar products shows that both components have equal squared norm. Thus g_N, h_N in (18.3) are unit vectors and $S_N h_N = \lambda_N g_N$, $S_N g_N = \lambda_N h_N$. The estimates above imply

$$\|h_N - g_{N-1}\| = O(N^{-1}), \quad \|(I - Q_N)h_N\| + \|(I - Q_N)g_{N-1}\| = O(N^{-4/3}). \quad (\text{A.10})$$

For the second term use the explicit sampled approximation at $N - 1$, rather than the weaker $O(N^{-1})$ profile difference. Its Airy moments with Q_N give the same bound, and its norm error is already $O(N^{-4/3})$.

On the preceding parity,

$$\Pi_N S_N = S_N(I - |h_N\rangle\langle h_N|).$$

The singular gap therefore gives, for every such y ,

$$\|\Pi_N M_N y\| \leq (1 - cN^{-2/3}) \|Q_N y\| \leq (1 - cN^{-2/3}) \|y\|.$$

It is unnecessary to assume $y \perp h_N$. Furthermore

$$\|M_N g_{N-1} - g_N\| = O(N^{-1}), \quad M_N^* g_N = Q_N h_N,$$

so (A.10) also gives the adjoint-ground bound. Finally,

$$t_N = \langle Q_N h_N, g_{N-1} \rangle = 1 + O(N^{-4/3}).$$

Indeed the diagonal defect is $O(N^{-4/3})$, and the unit-vector identity $\langle h, g \rangle = 1 - \|h - g\|^2 / 2$ makes the profile overlap defect $O(N^{-2})$. These prove every projection estimate in Lemma 18.1.

A.5 The positive spectral product constant

From (A.7),

$$\log \lambda_N = \log 2 + aN^{-2/3} + \frac{3}{2}N^{-1} + O(N^{-4/3}).$$

Integral or Euler summation for the first two powers and absolute convergence of the residual series give

$$P_N = \kappa 2^N \exp(3aN^{1/3}) N^{3/2} (1 + O(N^{-1/3})), \quad \kappa > 0.$$

For completeness, κ is the positive limit of the quotient in this formula. Each factor λ_N is positive, and the logarithm of the quotient converges to a finite real number. This completes the proof of Lemma 18.1.

References

- [1] Andrew Elvey Price, Wenjie Fang and Michael Wallner. *Compacted binary trees admit a stretched exponential*. Journal of Combinatorial Theory, Series A **177** (2021), article 105306. arXiv:1908.11181. Source 62: the counting recurrence, the proved Θ estimate, and the conjectural amplitude and all-orders form used in Part I originate in this paper. Source 15: the recurrence (Proposition 2.11), the path interpretation (Definition 2.12), the two-sided bounds (Theorem 1.1) and the fitted amplitude used in Part II are from this paper. Part V (Reports 100, 101, 103; cited by Report 103 as **trees**): published version doi:10.1016/j.jcta.2020.105306, preprint version inspected arXiv:1908.11181v3 (20 August 2020); the recurrences, the Θ theorem, the earlier Airy ansatz and amplitude estimates (Sections 3.3–3.4), and Lemma 5.1 (compacted paths are relaxed paths).
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